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Modeling of Carbon Dioxide Absorption using Aqueous Monoethanolamine, Piperazine and Promoted Potassium Carbonate

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Modeling of Carbon Dioxide Absorption Using Aqueous Monoethanolamine, Piperazine and Promoted Potassium Carbonate

by

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Dedication

Para los que se han ido,

los que siempre han estado,

y los que ahora me acompañan

Abuelitas Otilia e Ines, Abuelitos Máximo y Melitón

Tiitas Inés y Noralba que mi Dios los tenga en su Gloria

Pa, Ma, Andrés y Mamena

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Modeling of Carbon Dioxide Absorption Using Aqueous Monoethanolamine, Piperazine and Promoted Potassium Carbonate

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Rigorous CO_2 absorption models were developed for aqueous 4.5 m K^+/4.5 m PZ, monoethanolamine (7m – 9m), and piperazine (8m) in Aspen Plus® RateSep™. The 4.5 m K^+/4.5 m PZ model uses the Hilliard thermodynamic representation and kinetics based on work by Chen. The MEA (Phoenix) and PZ (5deMayo) models incorporate new data for partial pressure of CO_2 vs. loading and kinetics from wetted wall column data. They use reduced reaction sets based on the more relevant species present at the expected operating loading. Kinetics were regressed to match reported carbon dioxide flux data using a wetted wall column (WWC). Density and viscosity were satisfactorily regressed to match newly obtained experimental data. The activity coefficient of CO_2 was also regressed to include newly obtained CO_2 solvent solubility data.

The models were reconciled and validated using pilot plant data obtained from five campaigns conducted at the Pickle Research Center. Performance was matched within 10% of NTU for most runs. Temperature profiles are adequately represented in all

campaigns. The calculated temperature profiles showed the effect of the *L/G* on the location and magnitude of the temperature bulge. As the *L/G* is increased the temperature bulge moves from near the top of the column towards the bottom and its magnitude decreases. Performance improvement due to intercooling was validated across the campaigns that evaluated this process option.

Absorber intercooling was studied using various solvent rates (L_{min}, 1.1 L_{min} and 1.2 L_{min}). It is most effective at the critical *L/G* where the temperature bulge without intercooling is in the middle of the column. In this case it will allow for higher absorption by reducing the magnitude of the bulge temperature. The volume of packing to get 90% removal with L/L_{min} =1.1 at the critical *L/G* is reduced by 30% for 8m PZ.

For MEA and a solvent flow rate of 1.1 L_{min} packing volume is increased with intercooling at constant *L/G*. This increase is compensated by higher solvent loadings that suggest lower stripping energy requirements.

The critical *L/G* is 4.3 for 8m PZ, 6.9 for 9m MEA and 4.1 for K^+/PZ.

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1

Chapter 1: Introduction

Emissions of greenhouse gases (GHGs) have altered the energy balance of the earth's climate system by increasing atmospheric absorption of the sun's outgoing radiation raising the planet's global average temperature (0.74^{o}C for the 1906-2005 period (Core-Writing-Team, 2007). The increase of various GHGs in the industrial era appears as the dominant factor in climate change (Solomon et al., 2007). Although several of the major GHGs are naturally occurring, their concentration change in the atmosphere during the past 250 years has been strongly linked to human activity. Among the GHGs special attention is given to carbon dioxide (CO_2), nitrous oxide (N_2O), and methane (CH_4) because of their long-term influence on climate due to their chemical stability allowing them to remain in the atmosphere at time scales longer than decades or centuries. In the case of CO_2 atmospheric concentrations have gone from 180 ppm to more than 300 ppm as determined from ice core analysis (Solomon et al., 2007). Its concentration increased around 35% from its pre-industrial value to 379 ppm as of 2005 (Solomon et al., 2007). Most of this increase is related to anthropogenic emissions of which two thirds come from fossil fuel burning. This accounts for approximately 1100 Gt emitted into the atmosphere since the mid 19th century (Sims et al., 2007).

The largest increase in anthropogenic emissions of CO_2 comes from the power generation sector (over 27% by 2004) (Rogner et al., 2007). This sector comprises generation from coal and natural gas fired power plants. Flue gas from these sources contains 13 mol % CO_2 for coal to 3 mol % CO_2 for natural gas. Emissions from this sector are tied to policy and their change will depend on the requirement of nations to obtain energy sources and carriers with long-term security of supply, affordability and minimal environmental impact. Due to their abundance and cost (especially for coal) fossil power plants are expected to maintain a considerable role in the supply of worldwide energy. However, their indiscriminate deployment and continued use is not sustainable. For this reason research has sprung with the intention to reduce their impact on the planet. Among the options available to reduce this sector's emissions is Carbon Capture and Storage (CCS) that consists in removal and later storage of CO_2 from emission points. Its potential has been identified not only for coal, gas, or biomass fired generation or cogeneration but also for major energy intensive industries (i.e. refineries, steel mills), synthetic fuel plants, natural gas fields, and chemical facilities for producing ammonia, hydrogen, cement and coke (Sims et al., 2007). Its future role has been seen as a "transitional technology" (Sims et al., 2007) that will peak as efforts to reduce climate change are increased and will decline as the desired decarbonization of energy sources advances.

Carbon capture technologies have been classified as absorption, adsorption, membrane separation and cryogenic processes (Saxena and Flintoff, 2006). Based on process conditions and current available technology, chemical absorption has been proposed as the most viable to reduce emissions of CO_2 from low CO_2 partial pressure sources (500 mbar). The technology basis for chemical absorption systems has been around for more than 70 years (Bottoms, 1930). It was previously used in the separation

of CO_2 from hydrogen and natural gas, an application less stringent in energy requirements and with a smaller scale than the proposed flue gas removal. The latter requires absorption capabilities at low CO_2 partial pressures, handling oxygen, SO_x, fly ash, soot, NO_x, and high flue gas temperatures (Chapel et al., 1999). The successful implementation of this technology depends on reduction of energy consumption and capital costs. For the former, reduction of steam consumption and absorber pressure drop along with integration of the stripper with the power generation have been identified as promising improvement areas. Capital costs will benefit from cheaper absorber and flue gas cooling vessel materials, absorber size, economies of scale and improved oxidation inhibitors (Chapel et al., 1999). The mentioned process improvements may be accomplished by research and development of new solvents, contactor types, energy optimization, degradation, corrosion, and system simplification (Rochelle et al., 2001). Research presented in this work is intended to contribute in the three initial items focusing in the absorption component of the technology.

A plethora of models have been developed to analyze and design units for carbon capture. Most of the work has been centered on commonly used amines such as MEA, MDEA and K_2CO_3. Models also expand from the use of simple relations and enhancement factors to complex mathematical equations solved simultaneously using process simulators such as HYSYM and Aspen Plus®. This work focuses in developing models using the latest version of Aspen Technology, Inc. Aspen Plus® RateSep™ for the industry standard MEA and for innovative solvents such as concentrated piperazine. The models in this work are also part of an evolutionary approach to modeling absorption for carbon capture. Thermodynamics and kinetics have been better coupled by using Gibb's excess energy, enthalpy of formation, heat capacity and e-NRTL parameters to fit carbon dioxide solubility and heat of absorption. Kinetics were developed regressing data from

concentrated amines and implemented using the activity based power law. Physical properties for these solvents were also regressed to fit data for loaded solutions.

This work presents three solvent models for 4.5m/4.5m K_2CO_3/PZ, MEA and concentrated PZ. The first one follows the methodology by Chen (2007) to obtain activity-based kinetics and uses the same thermodynamic framework by Hilliard (2008). Both the MEA and PZ models presented in this work are an evolutionary step from the work by Chen (2007) and Dugas (2009). These models use newly regressed thermodynamics with recently available high concentration data by Dugas (2009) and Xu (2011). Kinetics are obtained from high concentration WWC data by Dugas (2009).

Similar approaches to this work can be found for MEA by Kvamsdal (2009) using Process Systems Enterprise Ltd. gProms®, Tobiesen (2007) in FORTRAN, Asprion (2006a) built in BASF A.G. proprietary CHEMASIM and Dugas et al. (2008) using Aspen Plus® RateSep™. In the case of PZ this is the first model developed for high concentrations (8m). Comparable models have been developed for low PZ concentration blends (<5m). Cullinane (2005) developed a thermodynamic and kinetic framework for low concentration PZ blends with potassium carbonate in FORTRAN that was later translated into Aspen Plus® by Hilliard (2008) for thermodynamics and Chen (2007) for kinetics. This is the base of the PZ promoted potassium carbonate model presented in this work.

Additionally, this work advances in a better understanding of CO_2 mass transfer with chemical reaction and its adequate modeling. Boundary layer discretization to improve model performance is studied. Diffusion of reactants and products and its influence in CO_2 absorption is also addressed as well as simplification of kinetic expressions to facilitate model development.

Detailed analysis of pilot plant campaigns was part of the scope of this research. It served as a way to validate the developed models as well as assess the provided pilot plant data. Two MEA campaigns were analyzed along with three concentrated PZ campaigns (>5m), which were the first of its kind. Reconciliation of variables allowed closing energy and material balances effectively assessing pilot plant absorption performance and fine-tuning model parameters. Results were used to determine useful process conditions to enhance absorption. Intercooling was evaluated as an important improvement in absorber design. Optimum conditions for intercooling were determined and generalized conclusions on its use are presented.

Pilot plant campaigns provided an additional insight into adequate model development concluding that the use of mixed flow segments better accounts for the imperfect fluid flow dynamics of an absorption unit.

Finally, fine tuned models were used to extrapolate operational conditions for absorption including the use of intercooling to improve performance. Results of this work can provide an approach at estimating requirements for a CO_2 absorption unit for a coal power plant.

2

Chapter 2: Carbon Dioxide Capture Using Absorption/Stripping

Among the technologies proposed for carbon capture, absorption/stripping is currently considered the most viable option. Its final implementation depends on the conditions of the gas to be treated. Pressure, temperature, CO_2 composition and impurities present, determine the selection of solvent, process variables (temperature, pressure, flow rates), construction materials, use of inhibitors and, equipment design and configuration. A system with a high pressure feed can consider a removal solution that uses purely physical solvents, such as UOP's Selexol process. Systems with near atmospheric pressure will require more active solvents that will react with CO_2 to increase their capacity. Amines are usually considered for this application with MEA as the industry standard. This chapter presents fundamental concepts of carbon dioxide using aqueous amines. It describes the analytical models used for its representation and presents the solvents studied in this work.

2.1. Process Description

A number of processes have been conceived for this application (Bottoms, 1930; Reed, 1949; Fujii et al., 1994; Reddy et al., 2006). The basic set up for this process is presented in Figure 2.1. In the case of a coal-fired boiler, particulates, sulfur dioxide and NO_x are removed before reaching the carbon capture unit (Figure 2.1). At this point the

flue gas is at 55°C with approximately 13 mol% CO_2 content, 5 mol% O_2 and 10 ppm of SO_2 (Chapel et al., 1999). The last requirement might be less stringent depending on the tolerance of the solvent to SO_2 and if the process is configured to use reclaiming as a way to reduce SO_2. If the SO_2 content at this stage is above the tolerable limits for the absorption, a SO_2 polisher will be required. A cooling stage is installed before the absorber to reduce its temperature down to 35 – 45° C. This is usually done using a direct contact cooler (DCC).

The flue gas from the DCC enters the bottom of the absorber and contacts the amine solvent that is fed at the top. The CO_2 in the gas reacts with the amine. The lean flue gas leaves the top of the absorber. Due to amine volatility and solvent degradation the lean flue gas goes through a water-wash to reduce emissions of unreacted amine and volatile amine degradation products.

The CO_2 rich solvent goes through a cross heat exchanger where it increases its temperature while cooling the returning lean solvent on its way to the absorber. From there, the rich solvent enters the stripper where it is heated and put in contact with steam. Almost pure CO_2 is produced leaving at the top of the stripper for further compression and later storage or use in enhanced oil recovery (EOR).

2.2. SOLVENT SELECTION

Chen (2011) proposed six key criteria for a good solvent for CO_2 capture: High CO_2 absorption rate, heat of absorption, and cyclic CO_2 capacity, resistance to degradation, and low volatility and corrosiveness.

Monoethanolamine (MEA) has been the industry standard for this application due to its low cost, high heat of absorption, capacity and rates of reaction and low cost per mole of amine. Efforts to reduce solvent costs and/or improve performance have lead to

the study of other options such as potassium carbonate, methyldiethanolamine (MDEA) and piperazine (PZ) among others, as well as blends of amines. This work studied MEA, PZ and PZ promoted potassium carbonate.

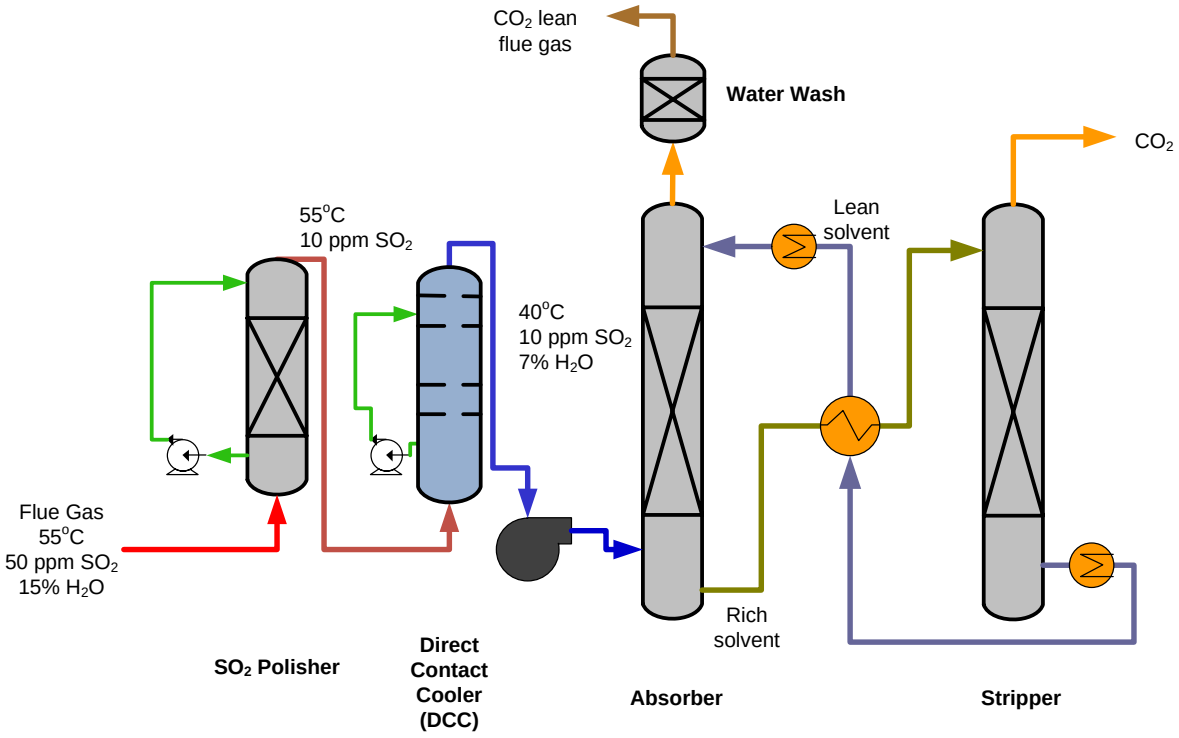


Figure 2.1: Basic configuration of the CO_2 capture system using an amine solvent for a coal-fired boiler with an FGD.

2.2.1. Monoethanolamine

Monoethanolamine, 2-aminoethanol or ethanolamine is a primary amine and primary alcohol (see Figure 2.2). It has a melting point of 10.3°C and a boiling point around 170°C. Aqueous Monoethanolamine (MEA) with a concentration between 15- 30 % has been previously used and is currently considered the state of the art technology for CO_2 absorption/stripping. It is an economic solvent with a high heat of absorption and fast kinetics. It had been proposed for the removal of CO_2 from gas mixtures as early as

1949 (Reed 1949). However, its main application was for natural gas. In the 1970's and 80's Dow Chemical and Union Carbide developed 30 wt.% MEA processes to generate CO_2 for enhanced oil recovery (EOR) (Chapel, Mariz et al. 1999; Butwell and Kubek 1980). In 1989 Dow Chemical sold its process to FLUOR Daniels and it is what is now known as Econamine FG^{SM}. This process has been implemented for removal of CO_2 from natural gas-boiler flue gas for EOR, urea production and the food industry. ABB Lummus Crest/Kerr McGee also have built a CO_2 capture process that uses 20% MEA (Barchas and Davis 1992). Mitsubishi/Kansai Electric have also patented processes that use MEA (Fuji, Hotta et al. 1994; Fujii, Hotta et al. 2001). FLUOR updated its MEA process (Econamine FG Plus^{SM}) and claims to have decreased in 20% the energy requirement of their Econamine FG^{SM} (Reddy, Scherffius et al. 2003; Reddy, Scherffius et al. 2006). Although considered the industry standard, MEA has lower resilience to degradation and lower capacity than some of the newly proposed solvents. Higher concentrations (up to 40 wt.%) have recently been suggested as an improvement to its capacity. On the downside its corrosivity is increased with the additional amine content. (DuPart et al., 1993)

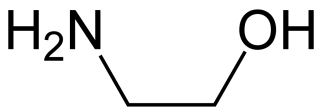


Figure 2.2: Polygonal formula for monoethanolamine (MEA).

2.2.2. Piperazine

Piperazine, hexahydropyrazine or diethylenediamine is a cyclic diamine with a molecular weight of 86.14 (Figure 2.3), a fusion point of 106°C and a boiling point of 146°C. It has been used mainly as a promoter to increase absorption rates and capacity in solvents with recommended application for K_2CO_3 at various concentrations (Cullinane

2005; Chen 2007; Oexmann, Hensel et al. 2008). BASF A.G. has used it as a promoter for their processes for natural gas sweetening blended with MDEA (Appl, Wagner et al. 1982; Asprion and Grossmann 2004). It has also been studied as an accelerator for N,N-diethylethanolamine (presented as a renewable amine because it can be obtained from ethanol) (Vaidya and Kenig 2008). Recently, PZ has been suggested as a solvent at high concentrations (>8m). Dugas (2009) and Freeman (2011) showed that 8 m PZ has a volatility similar to MEA but the advantage of faster kinetics (more than twice) and higher capacity (double) than MEA while being more resistant to oxidative and thermal degradation (degrades at temperatures higher than 150°C). Its process limitation is related to PZ solubility. At room temperature (20°C) 8 m PZ requires a loading of 0.25 mol CO_2/mol equivalent of PZ to stay in solution and will also form solids at high CO_2 loadings. Figure 2.4 shows the solubility envelop for 8m PZ that must be observed when designing and optimizing absorbers to operate with this solvent.

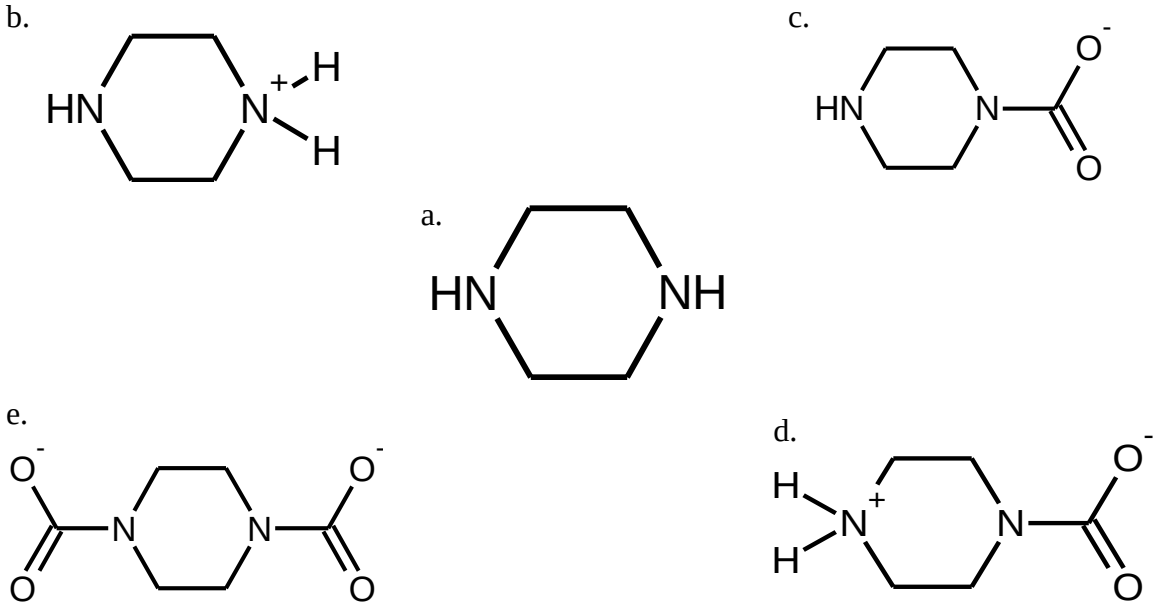


Figure 2.3: Polygonal formula for: a) Piperazine b) Protonated piperazine (PZH^+) c) Piperazine carbamate ($PZCOO^-$) , d) Protonated piperazine carbamate (H^+PZCOO^-), e) Piperazine di-carbamate ($PZ(COO^-)_2$

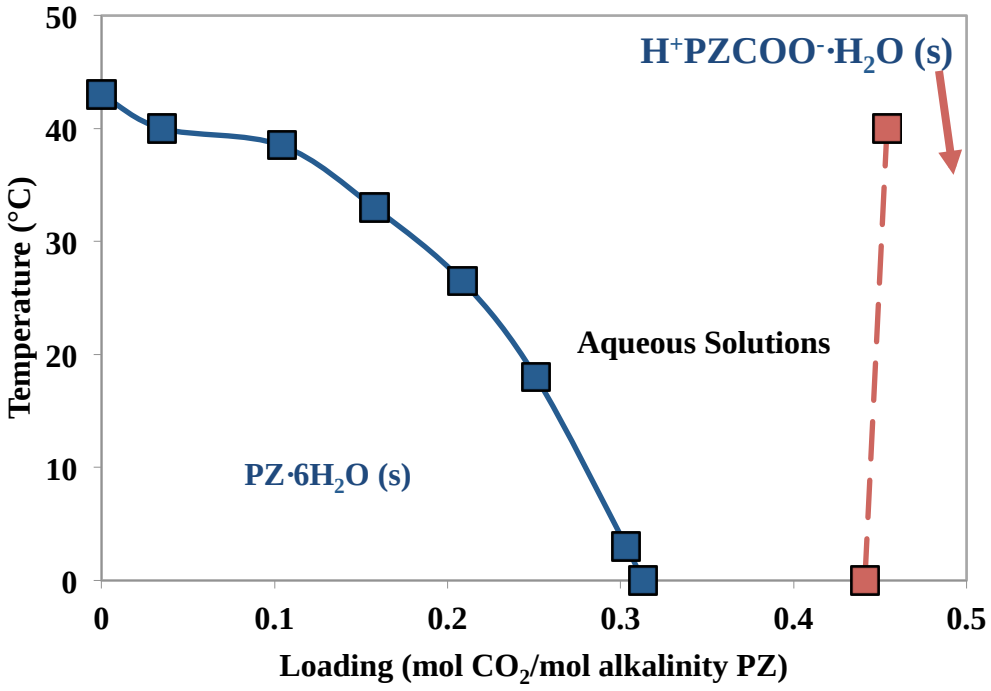


Figure 2.4: Solid Transition Temperature for Aqueous Piperazine (Freeman et al., 2010)

2.2.3. Piperazine promoted potassium carbonate (K^+/PZ)

Potassium carbonate (K^+) has been used as solvent for the removal of carbon dioxide from hydrogen and natural gas (Newman, 1985). It has been preferred due to its low regeneration energy however its low reaction rates make it inadequate for high removal requirement applications.

The K^+/PZ solvent was introduced by Cullinane (Cullinane, 2005) as an alternative to MEA. According to Cullinane, the K^+/PZ solvent has a higher CO_2 capacity than MEA because PZ increases the K^+ absorption capacity. Rates of absorption are increased by the presence of two amine groups in PZ, the high pK_a, and the large quantity of carbonate/bicarbonate that serves as buffer reducing the protonation of the

amine. Cullinane reported a CO_2 absorption rate 1.5 to 3 times faster than that with 30-wt % MEA.

Blends of 5m K^+/ 2.5m PZ and 6.4 m K^+/1.6 m PZ had been suggested (Cullinane, 2005; Chen, 2007) and tested in the pilot plant at The University of Texas at Austin. Results from the pilot plant test showed heat requirements slightly higher than MEA. However, Chen (2007) suggested that the solvent could be tuned to increase performance.

2.3. MASS TRANSFER WITH CHEMICAL REACTION

Carbon dioxide absorption using aqueous amines is a mass transfer process enhanced by chemical reaction. Adequate modeling of this process requires understanding of mass transfer coupled with kinetics. This section is an overview of both phenomena and their implication in carbon dioxide capture

2.3.1. Mass Transfer Theory

In carbon dioxide absorption it is necessary to consider the mass transfer within the bulk liquid and gas phase, into the liquid – gas boundary layer, and from the interface into the adjacent phase. The models that have been developed in this work assume that the bulk liquid is at constant composition (well – mixed) and that diffusion occurs within the boundary layers in both phases.

Models that analyze mass transfer across the boundary layer can be grouped depending on their time dependence into steady or unsteady state. Among the latter are models based on penetration, eddy diffusivity and surface renewal theory (Taylor and Krishna, 1993). The film model is the most well known of the steady state models.

2.3.1.1. Stagnant Film Model

Proposed by Nerst in 1904 (Sherwood et al., 1975), the film model is the simplest and oldest of the available models (Taylor and Krishna, 1993). It is based in the fact that

when fluid flows over a phase boundary its local velocity approaches zero at the interface. Adjacent to the interface, flow occurs in perpendicular direction across a thin film or resistance layer. This normal flux occurs by means of molecular diffusion and is affected by the thickness of the stagnant boundary layer. At the bulk, the level of mixing or turbulence is so high that it is a well-mixed system with uniform temperature and concentration. At the interface, the liquid and the vapor are in equilibrium.

The mass transfer phenomenon can be represented by the following analytical expression:

$$\frac{\partial N_{CO_2}}{\partial x} + \frac{\partial [CO_2]}{\partial t} + R_{CO_2} = 0 \tag{2.1}$$

Where N_{CO2} represents the flux of carbon dioxide and R_{CO2} accounts for the reacting CO_2. The middle term is CO_2 concentration change with time. This term is set to zero since this is a steady state approach. If diffusion is assumed constant in x then:

$$D_{CO_2} \frac{\partial^2 [CO_2]}{\partial x^2} + R_{CO_2} = 0 \tag{2.2}$$

Solving for a system where there is only physical absorption, the flux becomes a function of the concentration gradient and the thickness of the boundary layer:

$$N_{CO_2} = \frac{D_{CO_2}}{\delta} ([CO_2]_i - [CO_2]_b) \tag{2.3}$$

Where δ is the film thickness and the CO_2 concentration sub indexes *i* and *b* refer to the interface and bulk of the solution. According to film theory δ can be calculated using the physical mass transfer coefficient k_l^o:

$$\delta = \frac{D_{CO_2}}{k_l^o} \tag{2.4}$$

A major issue with this model is that it predicts a first-power dependence of the flux on the molecular diffusion coefficient. However, it has been suggested (Sherwood et

al., 1975) that eddy diffusion dominant in the bulk of the fluid disappears as the film layer is approached. Similarly it is assumed that turbulence disappears in the boundary layer such that all mass transfer occurs by molecular diffusion and at the interface $N_A \alpha D^{1.0}$. Thus the dependence of flux on the diffusion coefficient falls between 1 and 0 across the boundary layer.

2.3.1.2. Penetration Model

This model assumes that fluid elements (eddies) arrive at the interface from the bulk fluid. They remain at the interface for a period of time (exposure time, τ) in which mass transfer occurs across the eddy and the interface (Taylor and Krishna, 1993). After this time the element returns to the bulk and is replaced by a fresh one. The Higbie penetration model assumes that all eddies reside at the interface for the same amount of time and the concentration at the interface is maintained constant with the flux changing with time. (Sherwood et al., 1975; Taylor and Krishna, 1993). Starting with equation 2.1 and neglecting any convective flow:

$$N_{CO_2} = 2([CO_2]_i - [CO_2]_e)\sqrt{\frac{D_{CO_2}}{\pi\tau}} \tag{2.5}$$

Where $[CO_2]$ is the concentration of CO_2 and the subscripts *i* and *e* stand for interface and eddy respectively. τ is the exposure time.

The time-average physical mass transfer coefficient (k_l^o) is calculated by the following:

$$k_l^o = 2\sqrt{\frac{D_{CO_2}}{\pi\tau}} \tag{2.6}$$

As equation 2.5 shows, penetration theory predicts a square dependence of the flux on the diffusion coefficient. Similar dependence has been observed by various authors as mentioned in Sherwood et al. (1975) and Danckwerts (1970).

2.3.1.3. Surface Renewal Model

Danckwerts (1951) proposed a modification to the Higbie penetration model that considers various exposure times and degrees of penetration. The fraction of elements with different exposure times is accounted for using a *surface age distribution function, $\psi(t)$.* If the rate of renewal (s) of the area exposed to penetration is constant then:

$$\psi = se^{-s\tau} \tag{2.7}$$

This function represents the probability that any element of an area will have an exposure time (τ) before being replaced by a fresh eddy from the bulk fluid. The following represents the mean steady state flux:

$$N_{CO_2} = ([CO_2]_i - [CO_2]_e)\sqrt{D_{CO_2}s} \tag{2.8}$$

The average physical mass transfer coefficient is given by:

$$k_l^o = \sqrt{D_{CO_2}s} \tag{2.9}$$

The solution of the surface renewal model requires the evaluation of error functions that can make calculations more time consuming.

2.3.1.4. Turbulent Boundary Layer Model

This model accounts for both molecular diffusion that occurs near the interface and eddy diffusion that occurs away from the boundary layer and considers an effect of both at an intermediate region (Sherwood et al., 1975). The mass transfer rate is expressed as:

$$N_{CO_2} = -\left(D_{CO_2} + E_D\right)\frac{d[CO_2]}{dx} \tag{2.10}$$

Where E_D is a function that accounts for the eddy diffusivity. This model assumes that molecular and eddy diffusion take place simultaneously. The usefulness of this

approach is related to the knowledge of E_D. If it can be expressed as a function of x it is possible to integrate this expression and relate the concentration of CO_2 to the position from the interface and the steady state flux. There is no direct approach to obtain E_D however; it can be assessed by its relation to eddy viscosities derived from the velocity profiles. King (1966) relates the eddy diffusivity to a power of the distance (n) and a constant b:

$$E_D = ax^n + b \tag{2.11}$$

The b term allows for more or less eddy diffusion near the interface. Thus, high surface tension may lead to $b \rightarrow 0$. a and n are taken to be independent of the exposure time.

This model uses an age-time distribution to account for the eddy exposure time. Additionally, a is calculated using fluid physical properties and n is proposed to be equal to 4.

2.3.2. Kinetics of Carbon Dioxide Absorption with Aqueous Amines

Equation 2.1 shows a general balance for mass transfer with chemical reaction. The term R_{CO2} accounts for the reactions that occur between the dissolved gas and the amine that enhance absorption. The form of this term changes depending on the type and rate of reaction.

Absorption of carbon dioxide can be irreversible as in the case of sodium hydroxide (NaOH) or reversible as with MEA. The latter coupled with mass transfer is observed for absorption from flue gas.

The reaction of CO_2 with amines is nominally a second order reaction that can be expressed as:

$$CO_2 + zAmH \leftrightarrow P \tag{2.12}$$

Where *AmH* is the reacting amine and *z* is the number of moles reacting with the carbon dioxide (CO_2). The kinetic expression for this reaction is then:

$$R_{CO_2} = k_2[CO_2][AmH]^n - k_r[P] \tag{2.13}$$

Where k_2 is the second order forward rate constant. k_r is the reverse rate constant. $[CO_2]$ and $[AmH]$ are the carbon dioxide and amine concentration respectively. $[P]$ is the product concentration. n is the amine reaction order, which varies depending on the assumed reaction mechanism.

The reaction of CO_2 with primary and secondary amines has been described in literature by two mechanisms. The "zwitterion" mechanism proposes the formation of an intermediate molecule that has a positive and negative charge on each side resulting in a net neutral charge. Initially proposed by Caplow (1968), this mechanism can be represented by the following:

$$CO_2 + AmH \leftrightarrow AmH^+COO^- \tag{2.14}$$

$$AmH^+COO^- + B \leftrightarrow AmCOO^- + BH^+ \tag{2.15}$$

Where equation 2.14 represents the formation of the zwitterion followed by the based (*B*) catalyzed carbamate formation (equation 2.15) (Littel et al., 1992). In the studied systems the base can be represented by another amine molecule, water or a hydroxyl ion (OH^-). The formed carbamate can later undergo hydrolysis to form bicarbonate:

$$AmCOO^- + H_2O \leftrightarrow AmH + HCO_3^- \tag{2.16}$$

A parallel reaction with the hydroxyl ion also contributes to the formation of bicarbonate:

$$CO_2 + OH^- \leftrightarrow HCO_3^- \tag{2.17}$$

Assuming steady state to the zwitterion the CO_2 reaction can be written as:

$$R_{CO_2} = k_f \frac{[CO_2][AmH]}{1+\frac{k_r}{k_B[B]}} \tag{2.18}$$

Where k_f & k_r are the forward and reverse zwitterion rate constants (equation 2.14). k_B is the carbamate formation rate constant using the base (*B*) (equation 2.15). The reaction rate in equation 2.18 is said to be of an order between one and two with respect to amine concentration (Vaidya and Kenig, 2007). If the zwitterion reaction is rate determining and the carbamate formation is almost instantaneous with respect to the reverse zwitterion formation rate ($k_r << k_B[B]$) then the rate expression becomes:

$$R_{CO_2} = k_f[CO_2][AmH] \tag{2.19}$$

When the carbamate formation is rate determining then equation 2.19 becomes:

$$R_{CO_2} = \frac{k_f k_B}{k_r}[B][CO_2][AmH] \tag{2.20}$$

The termolecular mechanism (Crooks and Donnellan, 1989) proposes the formation of a "loosely bound encounter complex" rather than the zwitterion. It can be represented as:

$$CO_2 + AmH \cdots B \leftrightarrow AmCOO^- \cdots BH^+ \tag{2.21}$$

Where the base (*B*) is bound to the amine and then forms the encounter complex. This complex reverts mostly to the reagents with a small fraction reaction with amine or water molecules forming ionic products. This mechanism can be seen as a limiting case

for the zwitterion formation in which ($k_r >> k_B[B]$). If the amine is the dominating base the rate expression can be defined as (da Silva and Svendsen, 2004):

$$R_{CO_2} = k_2[CO_2][AmH]^2 \tag{2.22}$$

2.3.3. Analytical Solution to CO_2 Absorption

As previously stated, the absorption of carbon dioxide using aqueous amines is a case of mass transfer with chemical reaction. In order to find an analytical solution to this problem, Equation 2.1 can be set up for reactants and products. However, its solution is rather complex so assumptions can be made to simplify its analysis. Using film theory two main assumptions can be made: the pseudo first order approximation (PFO) and the interface pseudo first order approximation (IPFO) (Freguia, 2002; Cullinane, 2005; Dugas, 2009). The first order approximation assumes that the concentration of the amine does not change across the boundary layer and can be made constant and equal to the concentration at the bulk of the liquid. The IPFO implies that the absorption reaction rate is fast enough that there is a sub-reaction layer in the boundary layer where most of the reaction occurs and the CO_2 concentration reaches the equilibrium concentration in the bulk. Using equation 2.1 converted to mole fractions:

$$\tilde{\rho}\left(\frac{\partial N_{CO_2}}{\partial z} + \frac{\partial x_{CO_2}}{\partial t}\right) - R_{CO_2} = 0 \tag{2.23}$$

Eliminating the time dependent term (steady state) and defining the rate and flux terms (R_{CO2} & N_{CO2})

$$\tilde{\rho} D_{CO_2} \frac{\partial^2 x_{CO_2}}{\partial z^2} - k_n^x x_{amine}^n \left(x_{CO_2} - x_{CO_2}^*\right) = 0 \tag{2.24}$$

k^x_n is the mole fraction based forward reaction rate constant. The subscript *n* represents the assumed order of the reaction. x_{CO2}^* is the equilibrium concentration.

Using the PFO approximation the amine concentration is made constant and equal to the concentration at the bulk of the liquid. Using this approximation k^{x}_{n} and x_{amine} can be coupled into $k_{n}^{'}$ as follows:

$$\tilde{\rho} D_{CO_2} \frac{\partial^2 x_{CO_2}}{\partial z^2} - k'_n \left(x_{CO_2} - x^*_{CO_2} \right) = 0 \tag{2.25}$$

Rearranging this equation:

$$\frac{\partial^2 x_{CO_2}}{\partial z^2} - \frac{k'_n}{\tilde{\rho} D_{CO_2}} \left(x_{CO_2} - x^*_{CO_2} \right) = 0 \tag{2.26}$$

The diffusion and kinetic terms can be grouped into λ:

$$\lambda^2 = \frac{k'_n}{\tilde{\rho} D_{CO_2}} \tag{2.27}$$

Equation 2.27 can then be expressed as:

$$\frac{\partial^2 x_{CO_2}}{\partial z^2} - \lambda^2 \left(x_{CO_2} - x^*_{CO_2} \right) = 0 \tag{2.28}$$

The solution to Equation 2.28 can be calculated as:

$$x_{CO_2} = C_1 e^{\lambda z} + C_2 e^{-\lambda z} + x^*_{CO_2} \tag{2.29}$$

Considering the conditions at the boundary layer ($x_{CO2} = x^{i}_{CO2}$ at $z = 0$) and that Equation 2.29 is bounded ($C_1 = 0$):

$$x_{CO_2} = \left(x^i_{CO_2} - x^*_{CO_2} \right) e^{-\lambda z} + x^*_{CO_2} \tag{2.30}$$

Equation 2.30 represents the CO_2 profile across the boundary layer. Defining the carbon dioxide flux at the interface ($z = 0$):

$$N_{CO_2} = \tilde{\rho} D_{CO_2} \lambda \left(x^i_{CO_2} - x^*_{CO_2} \right) \tag{2.31}$$

Replacing λ and $k^{'}_{n}$ in equation 2.32:

$$N_{CO_2} = \tilde{\rho} D_{CO_2} \sqrt{\frac{k_n x^n_{amine}}{\tilde{\rho} D_{CO_2}}} \left(x^i_{CO_2} - x^*_{CO_2} \right) \tag{2.32}$$

Equation 2.32 was derived using mole fractions however, the models proposed in this work use activity based kinetics. It is possible to convert the molar kinetic constant k^x_n to an activity based constant k^a_n considering that:

$$R_{CO_2} = k_n^x x_{amine}^n x_{CO_2} = k_n^a a_{amine}^n a_{CO_2} \tag{2.33}$$

$$k_n^a = \frac{k_n^x}{\gamma_{amine}^n \gamma_{CO_2}} \tag{2.34}$$

Here k^x_n refers to the kinetic constant using molar fractions and k^a_n is the activity based kinetic constant.

Assuming that the change in the activity coefficient of amine is negligible across the boundary layer, replacing the kinetic constant and rearranging equation 2.34:

$$N_{CO_2} = \sqrt{\tilde{\rho} D_{CO_2} k_n^a a_{amine}^n \gamma_{CO_2}} \left(x_{CO_{2_i}} - x^*_{CO_2} \right) \tag{2.35}$$

where: N_{CO2} is the flux of CO_2

D_{CO2} is the diffusion coefficient for CO_2

a_{MEA} is the activity of MEA

γ_{CO2} is the activity coefficient of CO_2.

x_{CO2} is the molar fraction of CO_2.

The CO_2 mole fraction driving force can be represented as a pressure driving force using Henry's law:

$$N_{CO_2} = \frac{\sqrt{\tilde{\rho} D_{CO_2} k_n^a a_{amine}^n \gamma_{CO_2}}}{H_{CO_2 - sln}} \left(P_{CO_{2_i}} - P^*_{CO_2} \right) \tag{2.36}$$

where: $H_{CO2\text{-}sln}$ is the Henry's constant for CO_2 in the solution

P_{CO2} is the pressure of CO_2.

The sub index *i* refers to the interface

The asterisk represents the equilibrium pressure at bulk conditions

Equation 2.37 defines a normalized flux, a mass transfer coefficient for the partial pressure across the liquid film (k_g'):

$$k_g' = \frac{N_{CO_2}}{\left(P_{CO_{2_i}} - P^*_{CO_2}\right)} = \frac{\sqrt{\tilde{\rho} D_{CO_2} k_n^a a_{amine}^n \gamma_{CO_2}}}{H_{CO_2-sln}} \tag{2.37}$$

This mass transfer coefficient is obtained through wetted wall column experiments and can be used to obtain kinetic values for amine systems, as will be discussed later in this work.

3

Chapter 3: Absorber Model Development

This work will focus on the absorber section of the carbon capture system. The main component of this system is the CO_2 absorber. Here the flue gas comes into contact with amine solution that absorbs carbon dioxide. This application is different from basic liquid-gas absorption in that the mass transfer is enhanced by chemical reaction requiring additional considerations for model development. This chapter describes the modeling approach used in this work. It includes a description of the thermodynamic framework and kinetics developed as well as the fit for physical properties. The resulting models are used to assess the effects of process variables (i.e. temperature, loading,) on k_g'.

3.1. Absorption Modeling Fundamentals

Absorption modeling has been usually based on the equilibrium stage models that assume that, at a stage or packing segment, the outbound gas and liquid streams are in thermodynamic equilibrium (Kenig and Seferlis, 2009). However, chemical absorption requires consideration of the reaction kinetics of the system. Various approaches can be taken to solve this problem varying on their level of complexity. Figure 3.1 shows the range of complexity available for reactive separations (Noeres et al., 2003). Mass transfer and hydrodynamics can also be approached with various levels of rigor. The literature search shows that the level of complexity used in these models is also tied to

time of development as more computing capacity has become available in recent years Table 3.1 shows reviewed rate based models. Additional models are found in Appendix A.

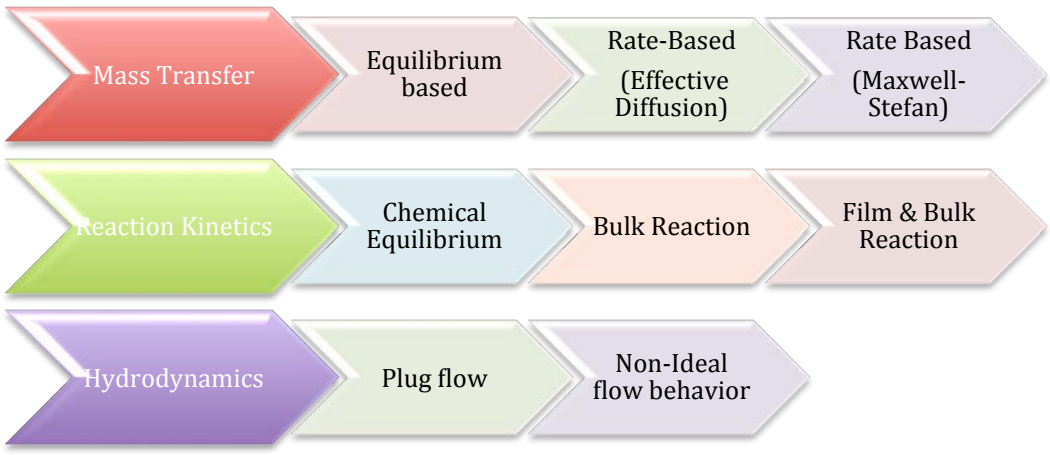


Figure 3.1: Chemical Absorption Modeling Approaches. Complexity increases from left to right (Modified from Noeres et al. 2003)

The more complex and rigorous treatment of absorption processes entails the use of Maxwell-Stefan diffusivities, additional considerations at liquid-vapor interface and process hydrodynamics that include correlations for hold-up, pressure drop and interfacial area. This is the "rate-based" approach that aims to describe actual rates for mass and heat transfer and chemical reactions (Kenig et al., 2003).

Table 3.1: Reviewed Rate-Based Absorber Models

Reference	Conditions	Kinetics	VLE	Framework	Description
Kucka et al.(2003)	2.1–4.4 m MEA 2.5%-15.1% CO_2	Kucka et al. (2002, 2003)	Austgen et al. (1989)	Aspen Custom Modeler	Validation against pilot plant data. Includes analysis of film discretization
Asprion (2006a)	Study of optimum film discretization in reactive systems. This reference is cited when describing the MEA model in BASF proprietary CHEMASIM framework.				
Tobiesen et al.(2007)	7 m MEA	Equilibrium & kinetics matching the Castor report B.	Model described by Hoff et al. (2004) using own data, Jou et al. (1995) and Ma'mun et al.(2005)	FORTRAN 90	Model uses penetration theory. A comparison with pilot plant data is presented.
Dugas et al. (2008)	7 m MEA 11-13 % CO_2	Versteeg et al. (1996) with activity coefficient corrections	Freguia (2002)	Aspen Plus® RateSep™	Model developed to study the results of the CASTOR pilot plant campaign #2
Kvamsdal et al. (2009)	8 m MEA	Freguia and Rochelle (2003) with adjusted parameters to match pilot plant data in Kvamsdal et al. (2008)	Equilibrium speciation model by Hoff et al. (2004)	gPROMS	Dynamic model to study start up and load reduction.
PROMAX	Commercial package Developed by Bryan Research and Engineering. It the replacement for TSWEET and PROSIM. It has a limited number of amines available: MEA, DEA, DGA, MDEA, DIPA, TEA, and AMP as well as its blends. (from http://www.bre.com)				
PROTREAT™	Commercial package developed by Optimized Gas Treating, Inc. Its creators present it as a rate-based model however it uses enhancement factors to account for reaction kinetics in the liquid film. It can model the following amines: MEA, DEA, DGA, DIPA, MDEA, and Piperazine. (http://www.ogtrt.com). It has two thermodynamic packages: Kent-Eisenberg or Lee Mather Luo et al. (2008)				
Plaza et al. (2010)	9 m MEA 12 mol% CO_2	Fitted from raw data from Aboudheir (2002)	Hilliard (2008)	Aspen Plus® RateSep™	Model accuracy was assessed comparing with pilot plant data. Lean loading optimization for a novel stripper configuration

Reaction rates in the film and their effect on enhancing mass transfer need to be considered to properly model the absorption process. When the rates are very slow it is possible to simplify the problem by using a single film segment or eliminating the film all together. However, in many cases it is necessary to differentiate the reaction regime present in the film and its enhancement on physical mass transfer. The Hatta number (Ha) represents a ratio of the maximum possible reaction and mass transfer rates (Vaidya and Kenig, 2007). It serves as an indicator of the rate of reaction. A low Ha number represents slow kinetics. If Ha>3, the diffusion of unreacted CO_2 is negligible and the flux of CO_2 is controlled by reaction kinetics near the gas liquid interface. The Hatta number is defined:

$$Ha = \frac{\sqrt{D_{CO_2} k_f [AM]}}{k^o_{CO_2}} \tag{3.1}$$

Where D_{CO2} is the binary diffusion of CO_2 in the solvent

k_f is the forward rate constant for the controlling absorption reaction

[AM] represents the amine concentration

k^o_{CO2} is the physical mass transfer coefficient

This approximation is commonly used in the models that employ enhancement factors to account for reactions at the boundary layer. However, a more rigorous approach using the film model consists in discretizing the boundary layer to account for nonlinear concentration profiles. This discretization will be affected by the reaction rates. As mentioned earlier very slow rates (small Ha) will require one film segment for its representation. In the other extreme, instantaneous reactions will not be considerably affected by the number of film segments because enhancement becomes a function of reactant and product diffusivities, and the concentrations of CO_2 at the interface and

amine at the bulk liquid (Asprion, 2006a). Efficient film discretization will assure a proper representation of the phenomena in the boundary layer without sacrificing computing time or accuracy. Discretization schemes have been suggested by various authors and will also be explored in this work. (Kucka, Muller, Kenig, et al., 2003; Asprion, 2006a, 2006b; Chen, 2007; Tobiesen et al., 2007)

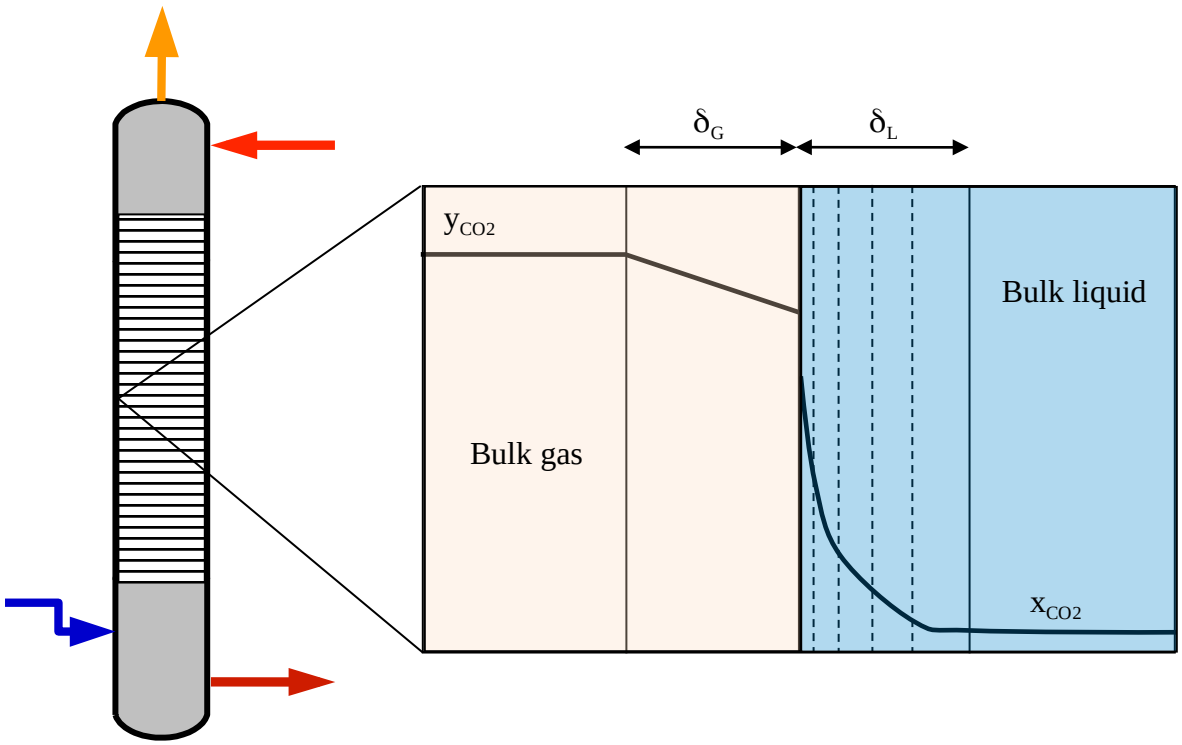


Figure 3.2: Absorber discretization for rigorous modeling. Packing is divided into segments and within each segment the liquid film (δ_L) is discretized to account for the rapid changes across the boundary layer. Here a representation of the expected CO_2 concentration profiles is also included. Since no reaction occurs in the gas film, the change in concentration from the bulk to the interface will only depend on diffusion through the film (δ_G). The concentrations at the gas-liquid boundary are in equilibrium and from there reaction kinetics enhance mass transfer. The concentration at the bulk will be determined by the rate of reaction

Discretization is also used for the height of the column. If trays are used, the number of segments is the same as the number of trays but for random and structured packing this is a parameter that can be optimized. In the majority of cases segments are equally divided adding enough segments to assure proper process representation. This work will evaluate adequate absorber packing discretization. The main goal is to reduce computing time and convergence issues while maintaining accuracy. Figure 3.2 shows the absorber model discretization.

The models in this work were developed in Aspen Plus® RateSep™. This is an extension to Aspen Plus® RadFrac™ with rate-based capabilities. It allows for the modeling of mass transfer and electrolytic, equilibrium, and kinetic reactions. It has built-in in mass and heat transfer and column hydraulics correlations for a number of trays and packing types. RateSep™ allows the user to carry out liquid and gas film discretization to better represent the absorption phenomenon.

Model development in Aspen Plus® RateSep™ was conducted implementing a thermodynamic framework, followed by regression of key physical properties. Kinetics in the developed models are activity based and use the power law to represent the behavior of the reaction constant as follows:

$$k = k_o \left(\frac{T}{T_o}\right)^n e^{\frac{-E}{R}\left(\frac{1}{T}-\frac{1}{T_o}\right)} \tag{3.2}$$

Where: k_o is the pre-exponential factor (independent of temperature),

n is the temperature exponent,

E is the activation energy; R is the gas constant,

T_o is the reference temperature (298.15K)

The regressed thermodynamic and physical properties framework was used to develop contactor models (wetted wall column) to extract the necessary parameters for

the amine kinetics (Equation 3.2). This was done using raw data from the contactor experiments as will be discussed later in this chapter.

Aspen Plus® RateSep™ has four available flow models to determine bulk properties to calculate mass and energy fluxes and reactions kinetics (Figure 3.3). In the mixed flow model the bulk properties for each phase are assumed to be the same as the outlet conditions for that phase leaving that stage. This model is recommended for tray contactors. The VPlug flow model uses the outlet conditions for the liquid and the average between inlet and outlet conditions for the vapor except for the pressure that uses the outlet value. The VPlug – Pavg flow model uses all the outlet conditions for the liquid except for the pressure where the average of inlet and outlet is used. The gas uses all the average conditions. The fourth model is the Counter-current model. It uses average conditions for gas and liquid phases. It is recommended for structured packing. (*Aspen Plus® Manual*). Models in this work were initially developed using the counter-current option. However, reconciliation of pilot plant data showed the need to account for flow non-idealities such as back mixing. As it will be shown in later chapters the countercurrent model is less adequate to account for this phenomenon in pilot plant conditions.

3.2. ABSORPTION WITH PIPERAZINE PROMOTED POTASSIUM CARBONATE (K^+/PZ)

Cullinane (2005) measured the thermodynamics and kinetics of potassium carbonate, piperazine, and carbon dioxide using a wetted wall column and developed a rigorous thermodynamic model in FORTRAN using the electrolyte non-random two-liquid (e-NRTL) theory. The model predicted vapor-liquid equilibrium (VLE) and speciation for the H_2O-K_2CO_3-PZ-CO_2 system. The equilibrium constants and interaction parameters were regressed using experimental data and input into the FORTRAN model.

Additionally, a rigorous kinetic model was developed that determined the rate constants and diffusion coefficients based on experimental data.

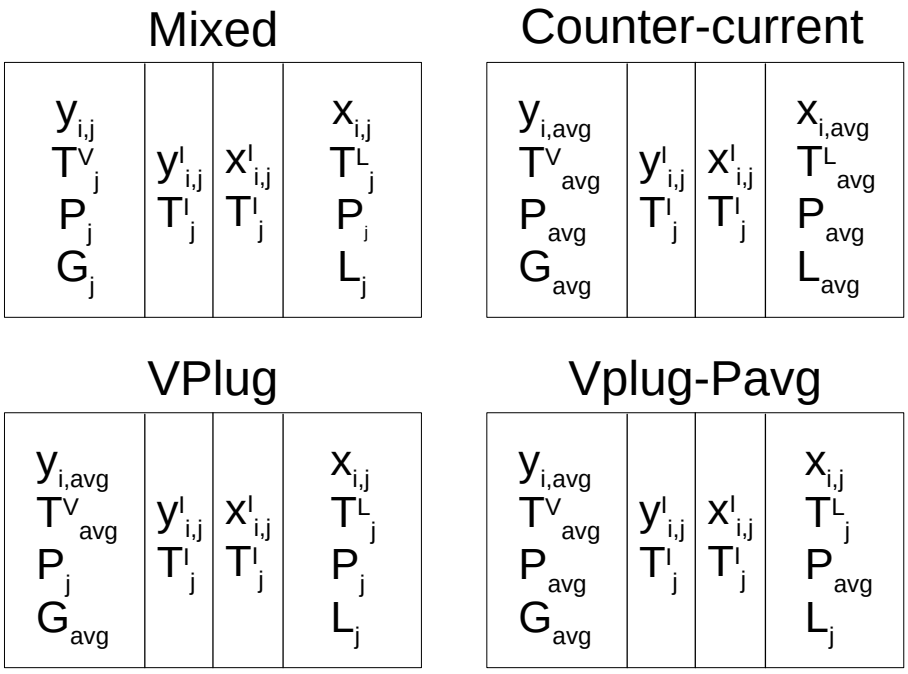


Figure 3.3: Flow model options available in Aspen Plus® RateSep™

Cullinane conducted experiments with 0.45–3.6 m PZ and 0–3.1 m potassium carbonate at 25–110 °C. The absorption rate of CO_2 was determined using the eddy diffusivity model developed by Bishnoi and Rochelle (2002) and rate constants were regressed from the experimental data using the model. The reaction of CO_2 with piperazine was modeled with a termolecular, base-catalyzed mechanism.

The following amine reactions were used in the Cullinane model for the formation of piperazine carbamate ($PZCOO^-$)

$$PZ + CO_2 + \begin{cases} OH^- \overset{k_{PZ-OH^-}}{\longleftrightarrow} PZCOO^- + H_2O & (3.3) \\ H_2O \overset{k_{PZ-H_2O}}{\longleftrightarrow} PZCOO^- + H_3O & (3.4) \\ PZ \overset{k_{PZ-PZ}}{\longleftrightarrow} PZCOO^- + PZH^+ & (3.5) \\ CO_3^{-2} \overset{k_{PZ-CO_3^{-2}}}{\longleftrightarrow} PZCOO^- + HCO_3^- & (3.6) \\ PZCOO^- \overset{k_{PZ-PZCOO^-}}{\longleftrightarrow} H^+PZCOO^- + PZCOO^- & (3.7) \end{cases}$$

The second set of reactions included in the model represent the formation of piperazine dicarbamate ($PZ(COO^-)_2$)

$$PZCOO^- + CO_2 + \begin{cases} H_2O \overset{k_{PZCOO^- - H_2O}}{\longleftrightarrow} PZ(COO^-)_2 + H_3O & (3.8) \\ PZ \overset{k_{PZCOO^- - PZ}}{\longleftrightarrow} PZ(COO^-)_2 + PZH^+ & (3.9) \\ CO_3^{-2} \overset{k_{PZCOO^- - CO_3^{-2}}}{\longleftrightarrow} PZ(COO^-)_2 + HCO_3^- & (3.10) \\ PZCOO^- \overset{k_{PZCOO^- - PZCOO^-}}{\longleftrightarrow} H^+PZCOO^- + PZ(COO^-)_2 & (3.11) \end{cases}$$

All of the buffering reactions were considered to be in equilibrium and reversible rate expressions for CO_2 with PZ and $PZCOO^-$ were developed. The catalysis of the formation of bicarbonate ion by hydroxide, piperazine, and piperazine carbamate was also included in the Cullinane model. The reactions to form bicarbonate ion were included to properly model equilibrium in the boundary layer and do not affect the CO_2 absorption rate. The three reversible reactions are:

$$CO_2 + OH^- \overset{k_{OH^-}}{\longleftrightarrow} HCO_3^- \quad (3.12)$$

$$PZ + CO_2 + H_2O \overset{k_{PZ}}{\longleftrightarrow} PZH^+ + HCO_3^- \quad (3.13)$$

$$PZCOO^- + CO_2 + H_2O \overset{k_{PZCOO^-}}{\longleftrightarrow} H^+PZCOO^- + HCO_3^- \qquad (3.14)$$

Hilliard (2005) developed a VLE model in Aspen Plus® using the thermodynamic data by Cullinane and the Data Regression System® (DRS) to simultaneously regress the interaction parameters and equilibrium constants to be used in the built-in electrolyte-NRTL model.

Later, Chen (2007) developed a rate-based model using Aspen Plus® RateSep™. It incorporates the Hilliard VLE model to predict equilibrium and speciation, and the rate constants developed by Cullinane to predict kinetics. The model calculates heat and mass transfer using Onda (1968) and the Chilton-Colburn Analogy and physical properties using correlations specified by the user within the Aspen Plus® framework.

Initial absorber modeling by Chen predicted an unexpected temperature profile indicating that the heat of absorption for CO_2 was not being correctly predicted by Aspen Plus®. This was in part due to the fact that the simultaneous regression of the interactions parameters by Hilliard did not incorporate heat capacity data for the K_2CO_3-PZ-CO_2-H_2O system. Therefore, the temperature dependence of the regressed binary interaction and enthalpy parameters may not have been adequately captured.

The heat of absorption calculated by Aspen Plus® is derived from an enthalpy balance using the heats of formation, heat capacities, and heats of vaporization of the various species. The heats of formation of the piperazine species (PZH^+, $PZCOO^-$, $PZ(COO^-)_2$, and H^+PZCOO^-) were adjusted to provide the same heat of CO_2 absorption as that predicted from equilibrium constants used in the chemistry model. The heats of formation (liquid) at 298.15K were calculated using the parameters from the equilibrium constants and the Van't Hoff equation. The equilibrium equations and results obtained for the four piperazine species can be found in Chen (2007).

Since the Hilliard VLE model does not contain heat capacity parameters for the PZ species (PZH^+, $PZCOO^-$, $PZ(COO^-)_2$, H^+PZCOO^-) regressed entropy reference values for the four PZ species were used to calculate heat capacities. Multi-parameter heat capacity correlations were developed using the equilibrium constants. Aspen Plus® does not account for the existence of net-neutrally charged zwitterions which were included in the Hillard K^+/PZ VLE model. The H^+PZCOO^- ion was given a net charge of 0 and was thus treated as a molecule. This created a number of issues such as the skewed predictions of the heats of absorption. Therefore, Chen set the charge for the H^+PZCOO^- ion to 0.0001.

The equilibrium constants were activity-based while the rate constants developed by Cullinane used concentration-based units. Chen implemented activity-based kinetics within the model, taking advantage of the fact that the new version of RateSep™ has the capability to handle activities, in terms of mole gamma. This was done using the following equation:

$$k_n^a = \frac{k_n^c [PZ][CO_2][b]}{a_{PZ} a_{CO_2} a_b \tilde{\rho}} \tag{3.15}$$

Where: k^a_n is the activity-based rate constant,

k^c_n is the concentration-based rate constant,

[b] is the concentration of species *b* in units of mol/L,

a_{PZ},a_{CO2} and a_b are the mole activities of PZ, CO_2, and the base (*b*)

ρ is the total molar density in moles per liter of solvent and will be specific for a particular solvent composition and loading. Therefore, a representative total molar concentration was selected and assumed to be constant across the column.

Kinetics developed by Cullinane contain a correction for ionic strength. However, in Aspen Plus®, this correction cannot be directly implemented. Therefore, a representative ionic strength at 50°C and 0.5 loading (mol CO_2/total alkalinity) was selected over the temperature and loading range of this study (the calculated ionic strength varied less than 5%)

Chen (2007) reported results for the forward and reverse activity-based rate parameters for piperazine, piperazine carbamate, and bicarbonate reactions as input into Aspen Plus® RateSep™ for systems with 5 m K^+/2.5 m PZ, 6.4 m K^+/1.6 m PZ. The power law constants for the 4.5 m K^+/4.5 m PZ, were calculated using the methodology used by Chen and are presented in Tables 3.2 through 3.4.

Table 3.2: Activity-Based Rate Parameters for formation of PZCOO- in 4.5m K+/4.5m PZ.

b*	**Forward**			**Reverse**		
	***k* x 10^{10}**	***E*** **(kJ/kmol)**	***n***	***k***	***E*** **(kJ/kmol)**	***n***
OH^-	28.11	-67,847	34.75	2.12 x 10^{-2}	246,966	-38.02
H_2O	0.0127	-42,414	23.48	2.93 x 10^{12}	160,611	-26.81
PZ	3.25	-155,841	53.66	6.12 x 10^2	325,276	-66.41
CO_3^{-2}	17.64	-105,880	53.25	3.42 x 10^3	200,502	-32.53
$PZCOO^-$	10.40	-51,821	28.72	6.12 x 10^2	325,276	-66.41

*****b corresponds to PZ + CO2+b↔$PZCOO^-$ + bH^+

Table 3.3: Activity-Based Rate Parameters formation of $PZ(COO^-)_2$ in 4.5m K^+/4.5m PZ.

*b**	Forward			Reverse		
	***k* x 10^{12}**	***E* (kJ/kmol)**	***n***	***k***	***E* (kJ/kmol)**	***n***
H_2O	0.0039	61,606	-1.46	2.22 x 10^{13}	78,135	-1.46
PZ	2.02	-51,821	28.72	9.45 x 10^3	242,800	-41.06
CO_3^{-2}	5.39	-1,860	28.31	2.59 x 10^4	118,027	-7.18
$PZCOO^-$	0.0065	52,199	3.78	2.07 x 10^4	116,084	3.78

*****b corresponds to PZ + CO2+b↔$PZCOO^-$ + bH^+

Table 3.4: Activity-Based Rate Parameters for the formation of HCO_3^- in 4.5m K^+/4.5m PZ.

*a**	*b**	Forward			Reverse		
		***k* x 10^{12}**	***E* (kJ/kmol)**	***n***	***k***	***E* (kJ/kmol)**	***n***
--	**OH^-**	5.28 x 10^{-6}	54,758	5.24	2.18 x 10^{-3}	66,014	19.54
H_2O	**PZ**	2.27 x 10^{-8}	-30,856	24.15	2.35	146,702	-8.85
H_2O	**$PZCOO^-$**	1.04 x 10^{-8}	73,163	-0.79	7.39	19,986	35.99

*a, b correspond to a+CO_2+b ↔HCO_3^- + bH^+ . (OH^- does not form bH^+)

3.3. ABSORPTION WITH AQUEOUS MONOETHANOLAMINE (MEA) - PHOENIX

A new model has been developed for aqueous MEA. "Phoenix" was developed entirely in this work. Thermodynamics were regressed using various literature sources and kinetics were obtained from wetted wall column (WWC) data by Dugas (2009).

3.3.1. Density and viscosity in the Phoenix model

3.3.1.1. Density calculation in the Phoenix model

Density in the Phoenix model uses a FORTRAN subroutine that calculates the molar volume of the solution using the total molar fraction of MEA, CO_2 and water:

$$V_{sln} = x^T_{MEA} V_{MEA} + x^T_{H_2O} V_{H_2O} + x^T_{CO_2} V_{CO_2} + x^T_{MEA} x^T_{H_2O} V^* \quad (3.16)$$

Where: V is the molar volume in mL/mol

x^T_i is the total mol fraction of i

V^* is the volume associated with the interaction between water and MEA. It is equal to -1.8218 mL/mol.

The molar volume of CO_2 (V_{CO2}) is a dissolved volume equal to 0.04747 mL/mol

V_{MEA} is the MEA molar volume calculated as follows:

$$V_{MEA} = \frac{MW_{MEA}}{aT^2 + bT + c} \quad (3.17)$$

MW_{MEA} is the molecular weight of MEA (61.09 g/mol)

a = -5.35162 x 10^{-7}

b = -4.51417x10^{-4}

c = 1.19451

Figures 3.4 – 3.6 compare model results for 4m, 7m, and 10m MEA against experimental data by Amundsen et al. (2009). Results show an adequate fit at the compared loadings and amine concentrations, however the fit is less accurate with increased MEA concentration.

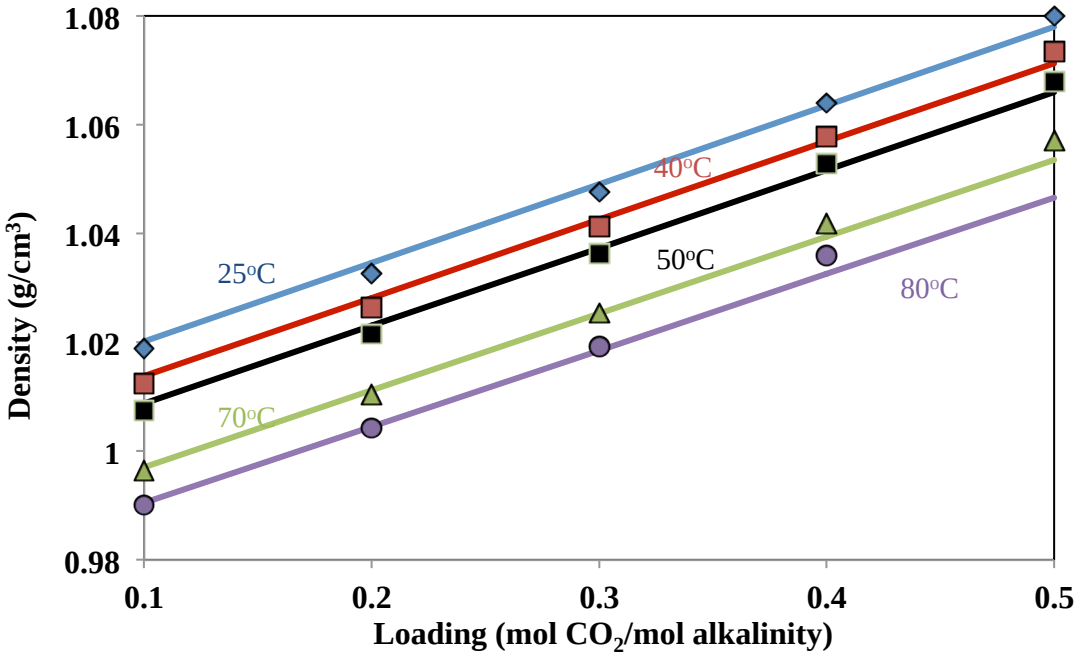


Figure 3.4: Density fit for the Phoenix model for 4m MEA. Lines generated in Aspen Plus®. Points by Amundsen et al. (2009).

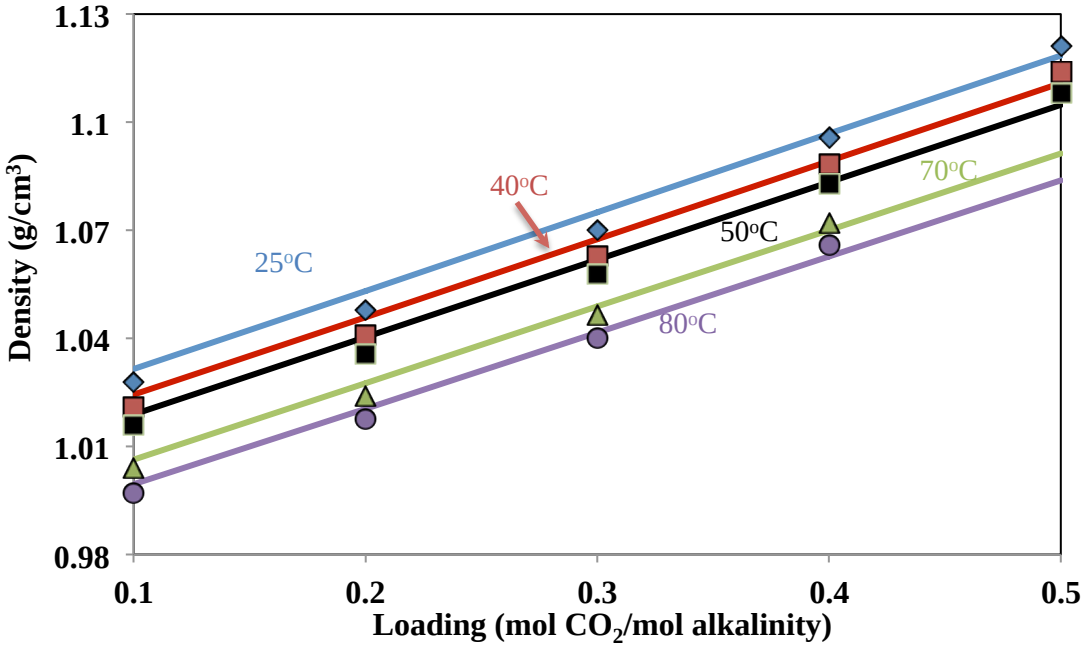


Figure 3.5: Density fit for the Phoenix model for 7m MEA. Lines generated in Aspen Plus®. Points by Amundsen et al. (2009).

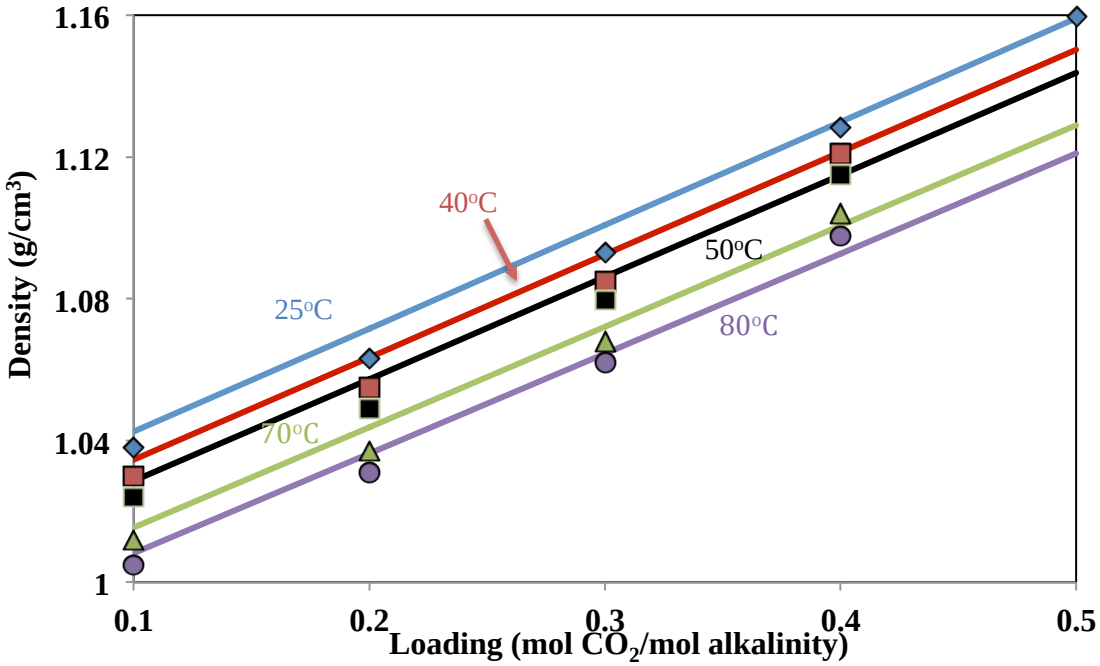


Figure 3.6: Density fit for the Phoenix model for 10m MEA. Lines generated in Aspen Plus®. Points by Amundsen et al. (2009).

3.3.1.2. Viscosity calculation in the Phoenix model

The viscosity of the solution was also implemented in the model using a FORTRAN subroutine that calculates the viscosity using the following equation from Weiland et al. (1998):

$$\mu_{sln} = \mu_{H_2O} exp\left(\frac{(a\bar{x}_{MEA} + b)(\alpha(c\bar{x}_{MEA} + dT + e) + 1)\bar{x}_{MEA}}{T^2}\right) \qquad (3.18)$$

Where: μ is the viscosity of the solution (sln) and water (H_2O) in mPa-s

$\bar{x}_{MEA}$ is the mass percent of MEA in the solution

T is the temperature in Kelvin

α is the CO_2 loading (mol CO_2/ mol MEA)

a = 21.186

b = 2373

c = 0.01015

d = 0.0093

e = -2.2589

Figures 3.7 to 3.9 show the resulting fit comparing to data by Amundsen et al. (2009). As with density the fit is adequate but more accurate at lower amine concentrations.

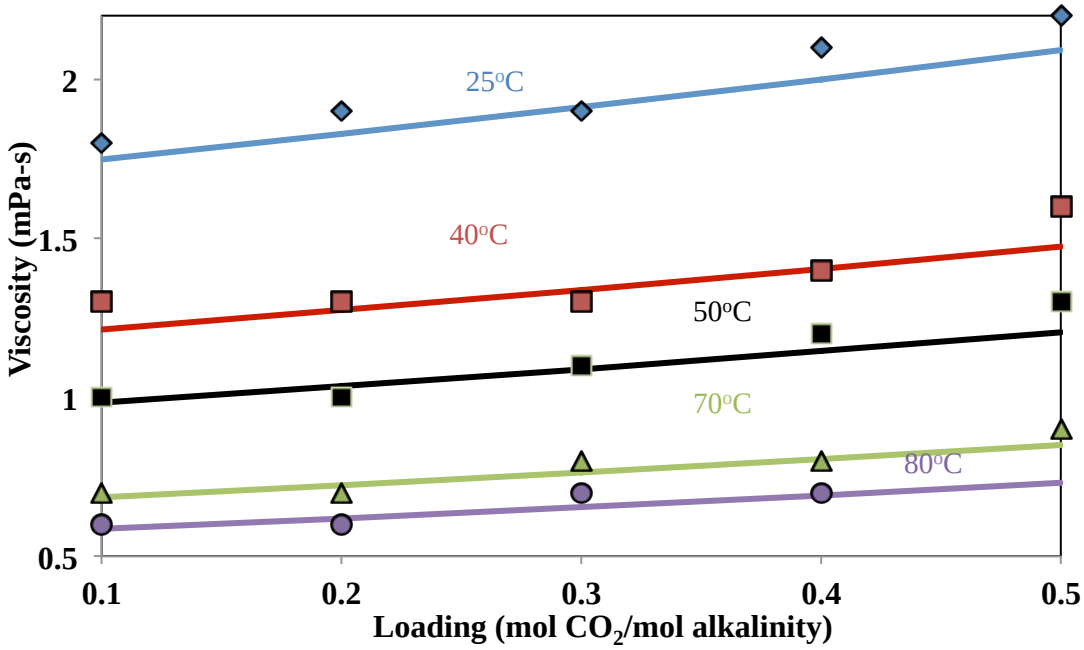


Figure 3.7: Viscosity fit for the Phoenix model for 4m MEA. Lines generated in Aspen Plus®. Points by Amundsen et al. (2009).

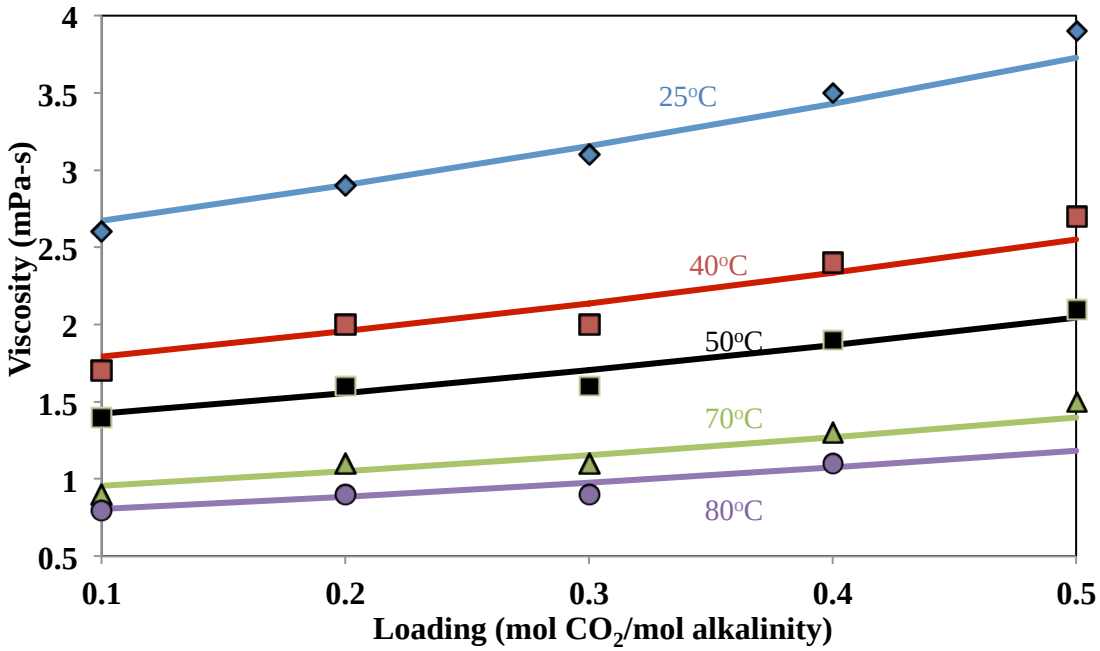


Figure 3.8: Viscosity fit for the Phoenix model for 7m MEA. Lines generated in Aspen Plus®. Points by Amundsen et al. (2009).

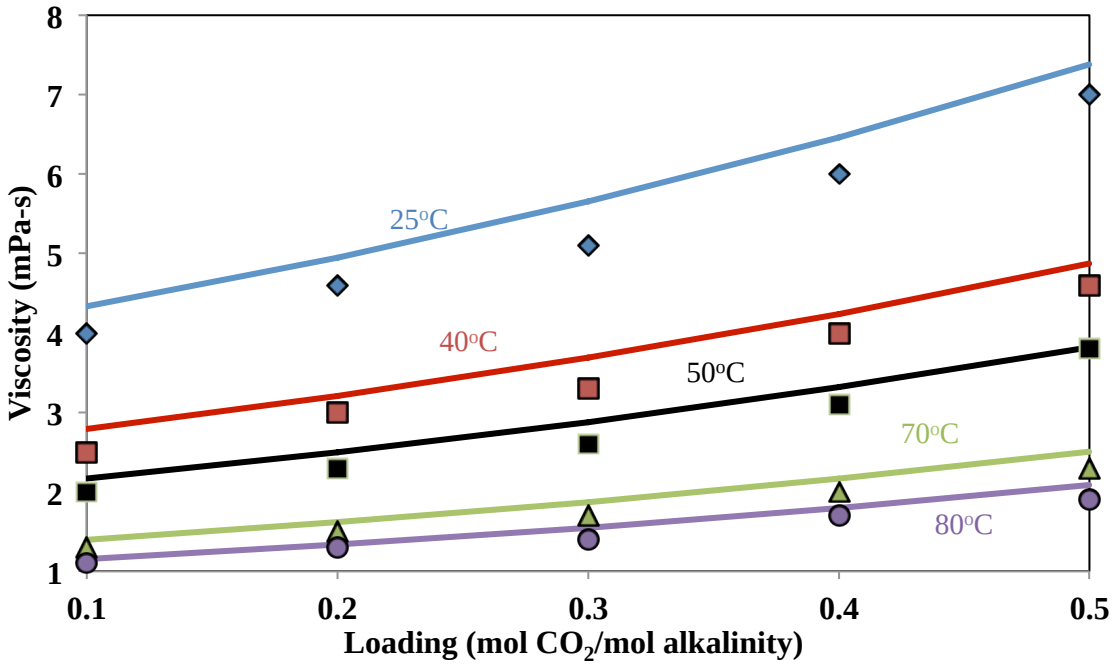


Figure 3.9: Viscosity fit for the Phoenix model for 10m MEA. Lines generated in Aspen Plus®. Points by Amundsen et al. (2009).

3.3.2. Thermodynamics

3.3.2.1. Shortcomings of the Hilliard Framework

Hilliard (2008) used the electrolyte nonrandom two-liquid (e-NRTL) activity coefficient model in Aspen Plus® to develop a rigorous and consistent thermodynamic representation of mixtures of MEA – H2O – CO2. This model differs from previous work in that it represents additional data on amine vapor pressure, enthalpies of absorption, heat capacity, and NMR speciation. It uses Gibbs free energy and enthalpy values within Aspen Plus® to maintain thermodynamic consistency. Details of this model and the fitted data can be found in Hilliard (2008).

However, in the Hilliard framework the MEA activity coefficient behavior was inadequately represented. MEA activity coefficients calculated by Dugas (2009) are lower than those generated by the Aspen Plus® model, have a lower dependence on loading, and have an inverse dependence on temperature as shown on Figure 3.10.

Since the reaction rates in the Aspen Plus® kinetics are defined using the activities it is expected that the behavior of the activity coefficient of MEA will have an effect on the rate constant. This issue could only be solved by re-regression of the MEA thermodynamic framework.

3.3.2.2. Phoenix Thermodynamic Framework

In addition to correcting the activity coefficients of MEA Phoenix includes additional VLE and heat capacity data by Xu (2011) and Nyugen (Rochelle et al., 2011) respectively. The Phoenix thermodynamic framework was regressed using VLE data by Goldman (1959), Lee (1976), Jou (1995), Ma'mun (2005), Hilliard (2008), Dugas (2009) and Xu (2011); and heat capacity and loaded amine volatility by Hilliard (2008). Data by

Bates and Pinching (1951) and Hamborg and Versteeg (2009) for the pKa of the amine protonation was also fitted.

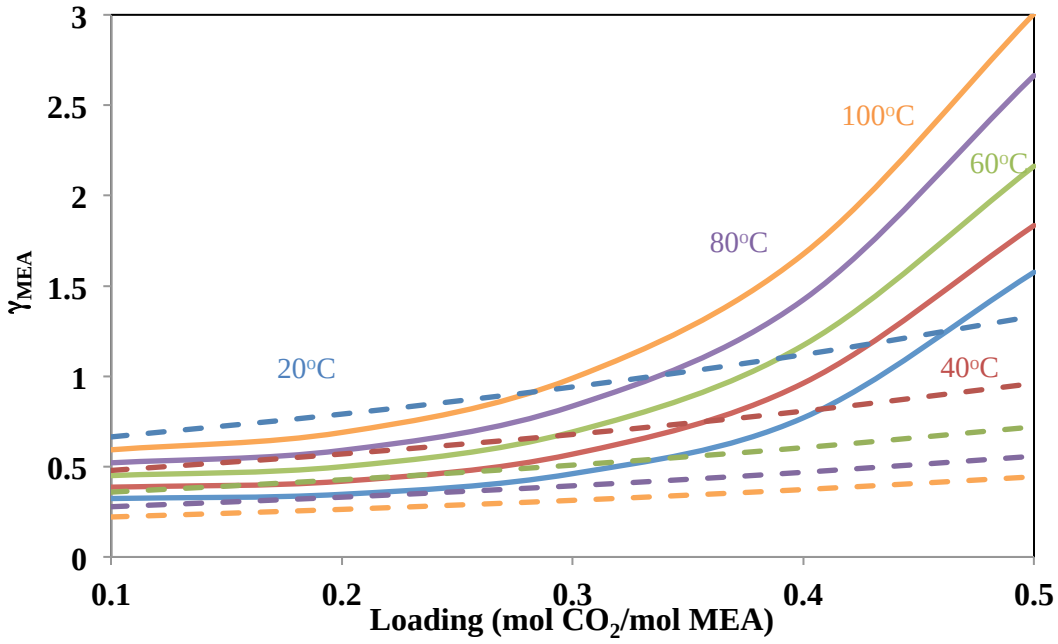


Figure 3.10: MEA activity coefficient for 7 m MEA. Dashed lines are generated using correlations found in Dugas (2009). Continuous lines are generated from the Aspen Plus® model by Hilliard (2008). Asymmetric reference state.

MEA was modeled as a Henry's component. This allowed the use of additional parameters such as the NRTL MEA/H_2O, and the Henry's constant parameters to fit the data and correct the behavior of the activity coefficient of MEA. The resulting values do not match the values calculated from Dugas (2009) but the temperature behavior has been corrected (Figure 3.18). The activity of CO_2 was also fit using correlations found in Dugas (2009). Figures 3.11 to 3.19 show the resulting fit. Fit parameters can be found in Table 3.5. Some of the parameters used were not eliminated even though they present a

considerable standard deviation due to the fact that removing them affected negatively the data fit.

Figure 3.11 shows an adequate fit for the solubility of CO_2 in MEA over the regressed temperature range (40–160 ^oC).

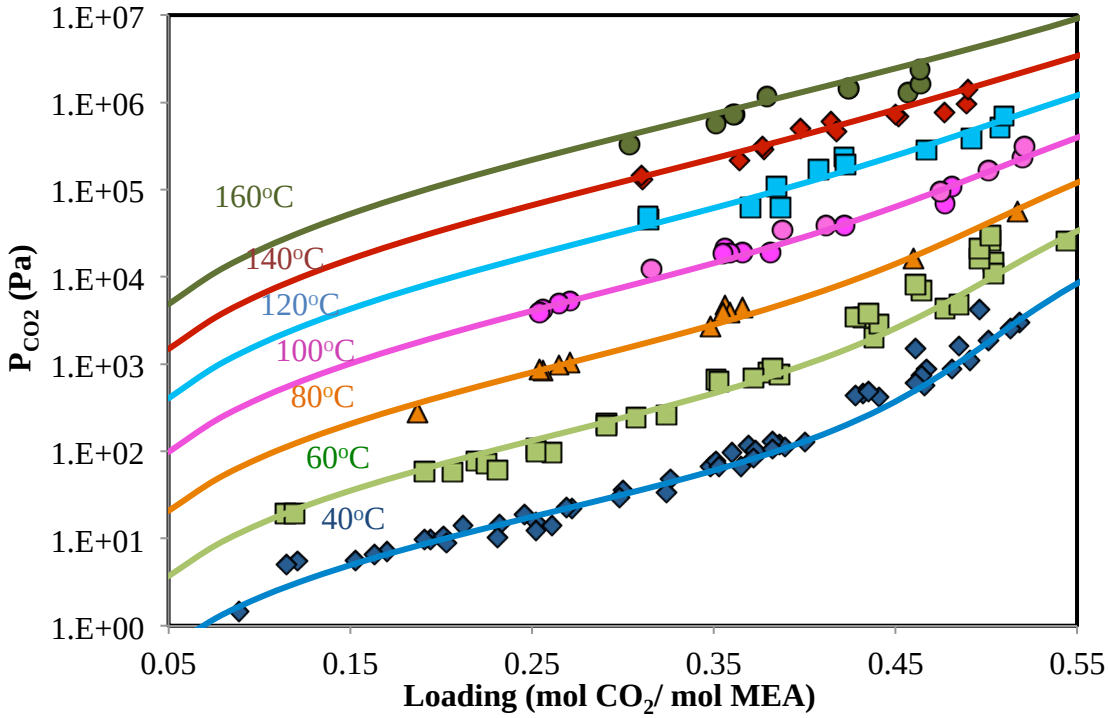


Figure 3.11: CO_2 Solubility in 3.5-12 m MEA. Lines are Phoenix fit with 7m MEA. Points by Jou et al. (1995), Hilliard (2008), Dugas (2009), Xu (2011)

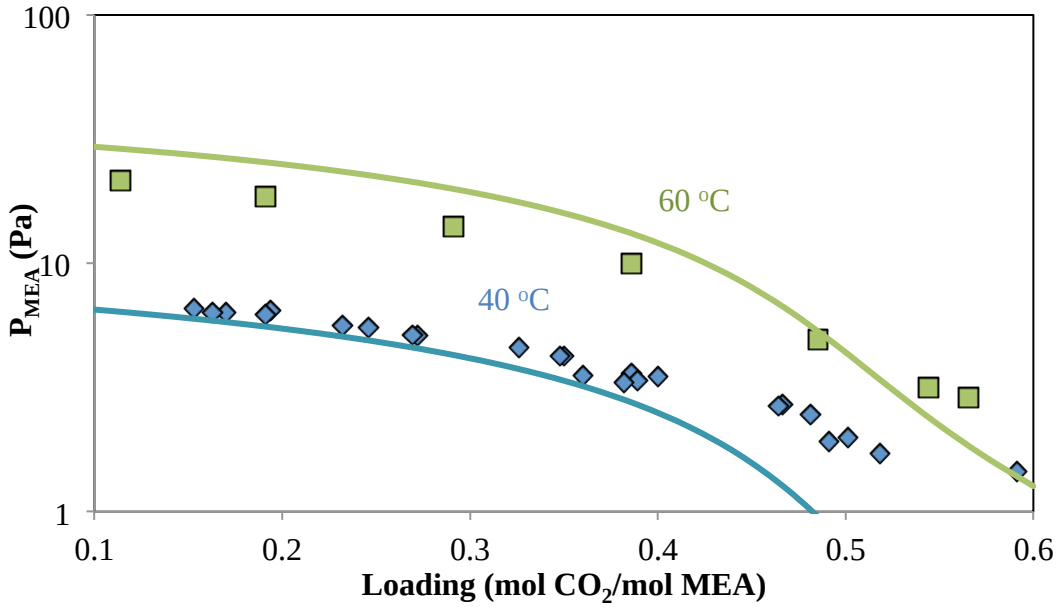


Figure 3.12: 7 m MEA volatility. Lines are the Phoenix fit. Points by Hilliard (2008)

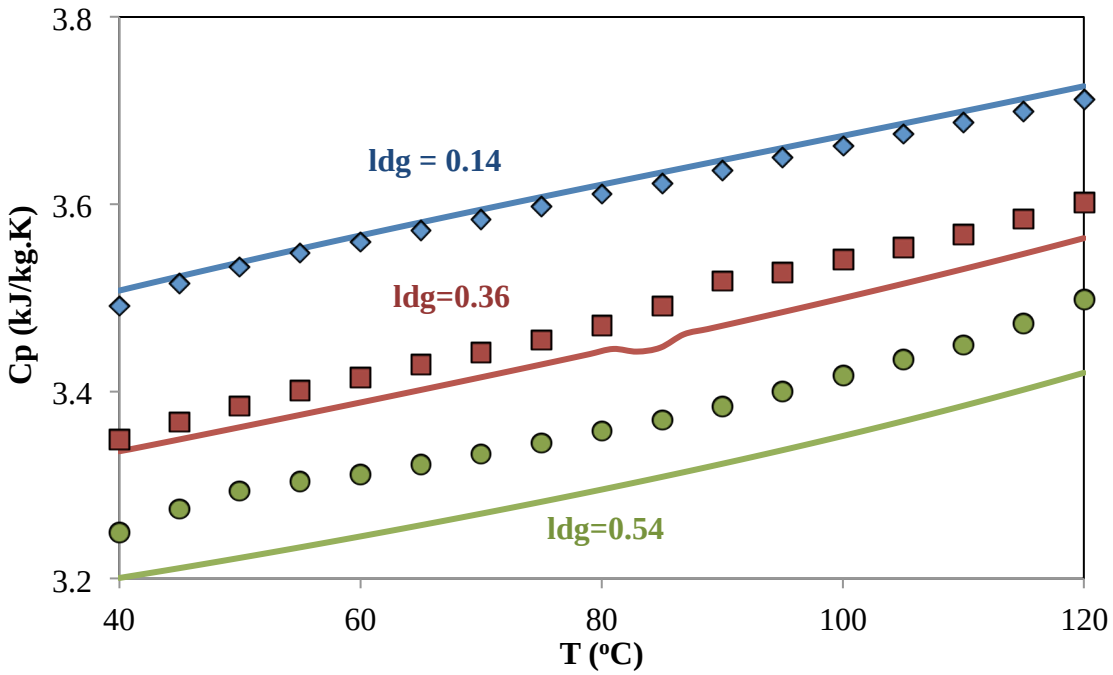


Figure 3.13: Specific heat capacity for 7 m MEA. Lines are the Phoenix fit. Points by Hilliard (2008)

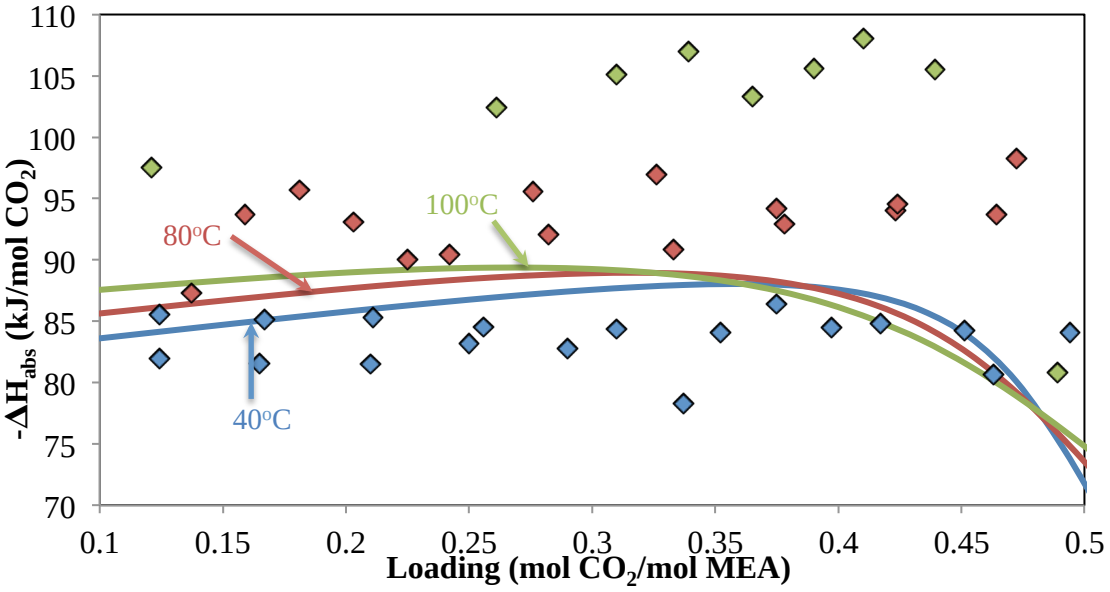


Figure 3.14: CO_2 Heat of absorption for 7 m MEA predicted by Phoenix. Points by Kim and Svendsen (2007)

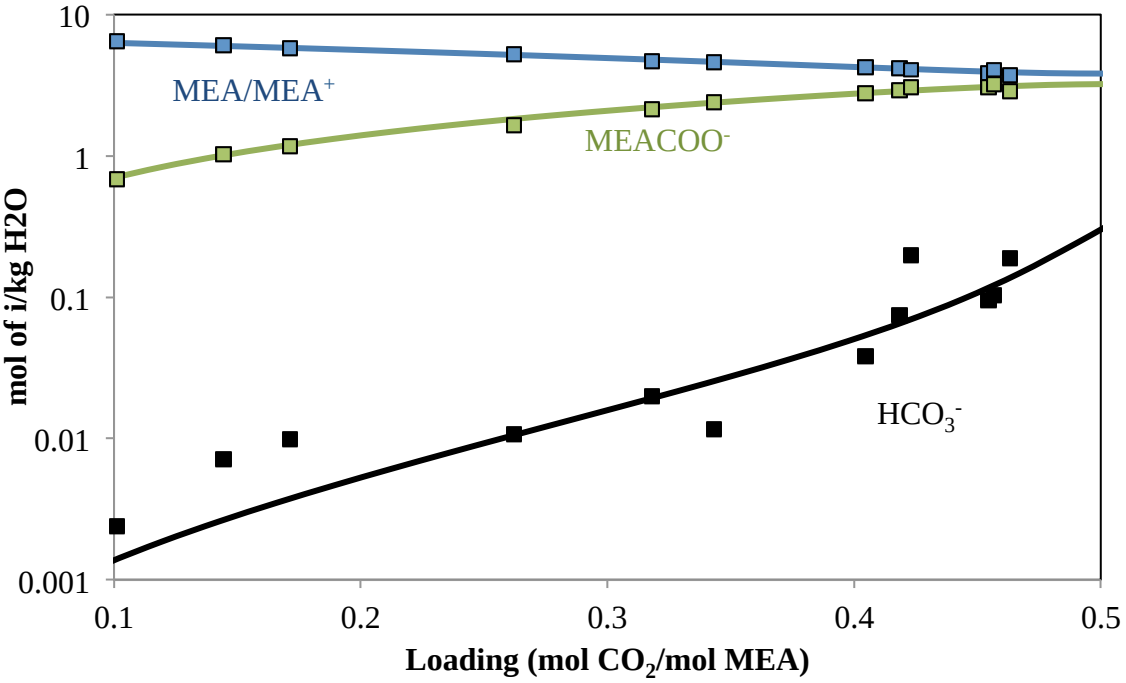


Figure 3.15: NMR speciation for 7 m MEA T=27°C. Lines are from Phoenix. Points from Hilliard (2008)

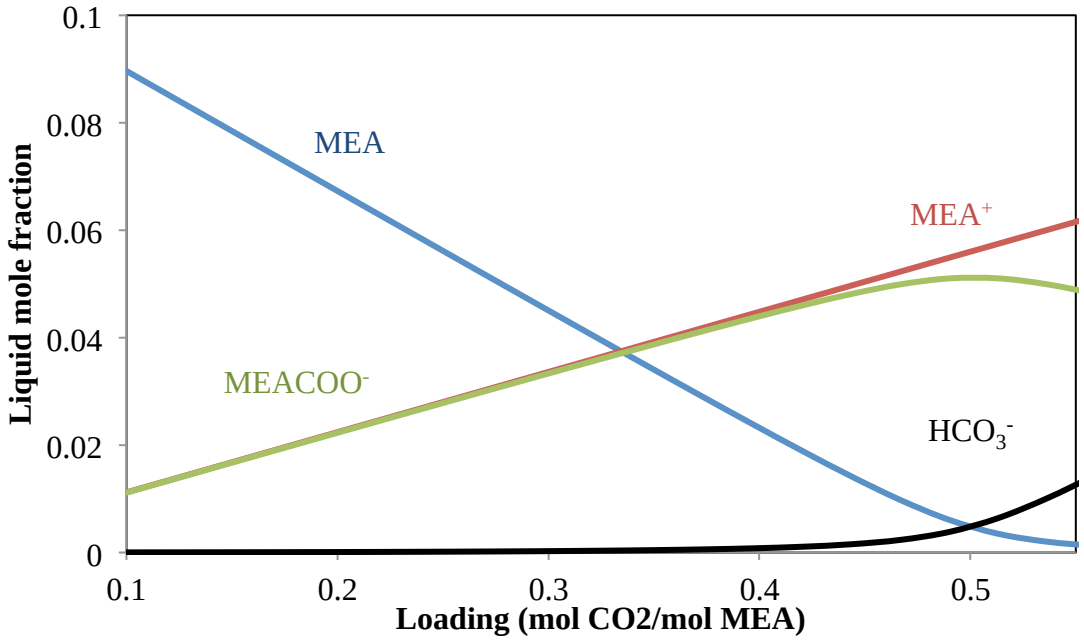


Figure 3.16: Phoenix predicted speciation for 7 m MEA T=27°C

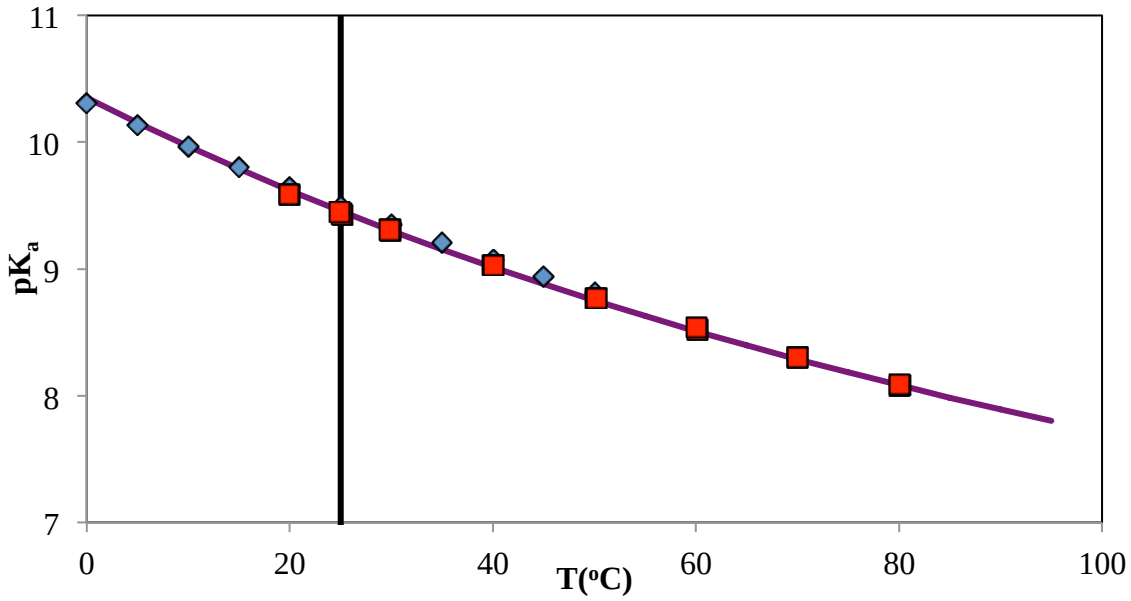


Figure 3.17: Asymmetric, molality based pK_a for MEA. Line is Phoenix calculated. (♦) Bates and Pinching (1951); (■) Hamborg and Versteeg (2009)

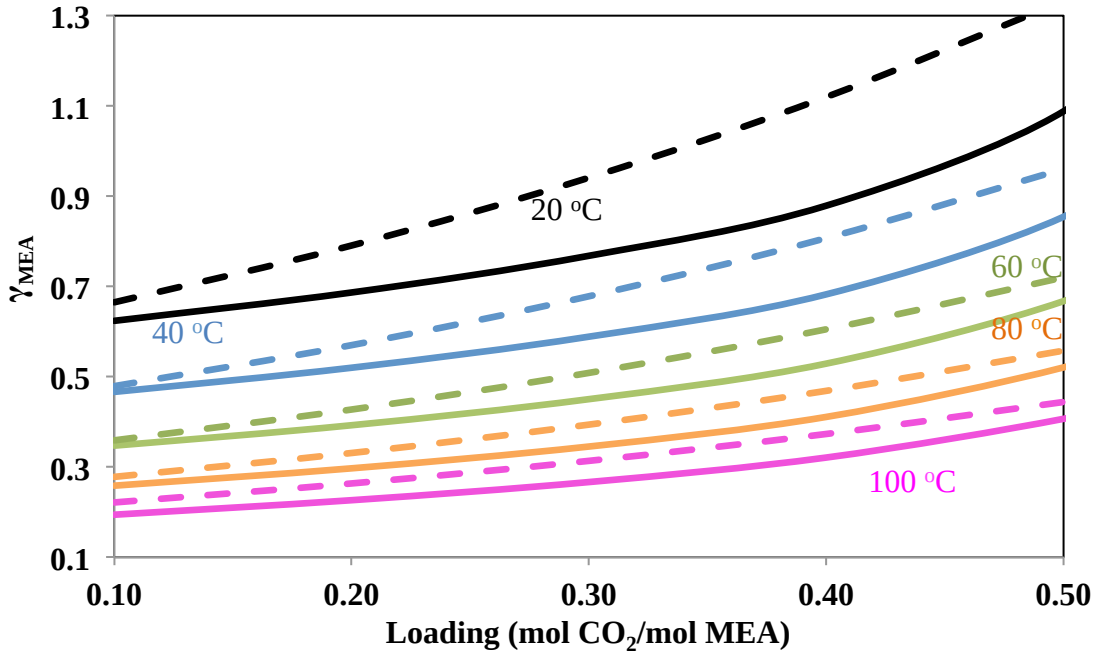


Figure 3.18: **Activity coefficient for 7 m MEA. Solid lines are from the Phoenix model. Dashed lines calculated using correlations in Dugas (2009). Asymmetric reference state**

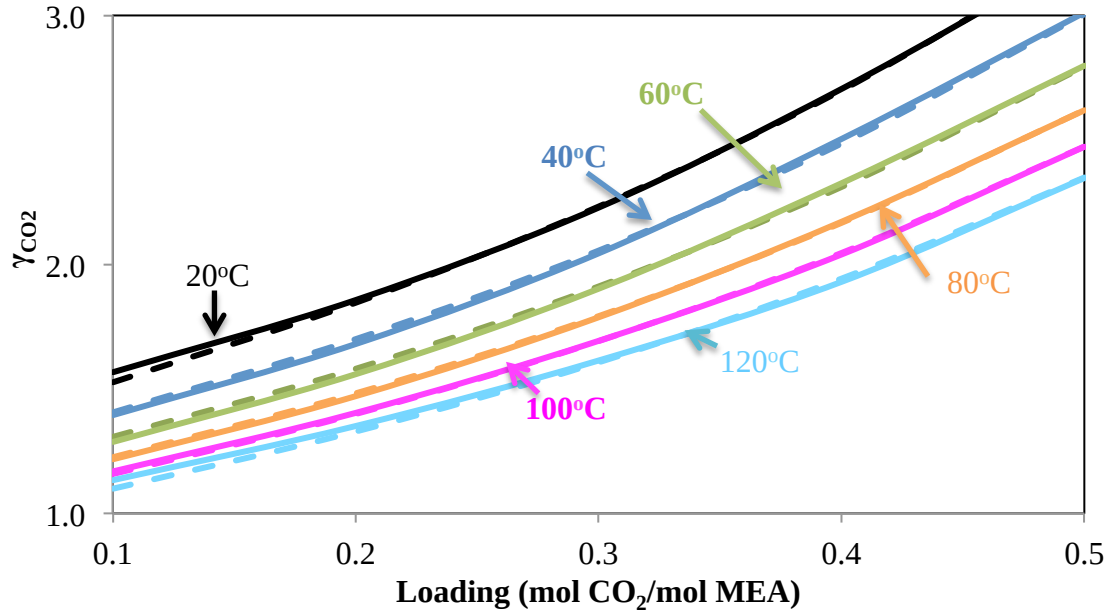


Figure 3.19: **Activity coefficient fit for CO_2. Solid lines Phoenix. Dashed lines using correlations in Dugas (2009)**

Amine volatility (Figure 3.12) is under predicted especially for 40 ^{o}C at loadings below 0.4. This may be attributed to the detection limits of the analysis. Data at 60 ^{o}C is over predicted up to loading of 0.4.

Heat capacities (Figure 3.13) are within the reported values with decreasing accuracy with increasing loading. However, considering the experimental error and the range of the values, the obtained fit is deemed adequate.

The heat of absorption (Figure 3.14) is within the range of the data by Kim (2007) although lower as temperature is increased. However, the model is still adequate since the heat of absorption generated by the model is within the experimental error.

Phoenix adequately matches the NMR data by Hilliard (2008) (Figure 3.15) and the resulting speciation is consistent with that presented in Hilliard (Figure 3.16).

The pK_a for MEA protonation was matched initially at 25 ^{o}C by changing the delta aqueous Gibbs energy of formation for the protonated amine (MEA^{+}) – DGAQFM. The delta aqueous enthalpy of formation – DHAGFM – was determined by fitting the data ±20 ^{o}C from 25 ^{o}C. The initial values for the heat capacity parameters for the protonated amine were determined by the fit of the pKa and were then bounded in the regression.

Following the fit of the VLE and all the other parameters mentioned above the activity of CO_2 was fit (Figure 3.19) using values obtained from correlations presented in Dugas (2009) fitting data by Browning (1994) and Hartono (2009) for the activity of CO_2 in MEA solutions using the N_2O analogy. The CO_2 related eNRTL parameters were modified to fit the activity coefficients of CO_2 using a FORTRAN subroutine provided by Aspentech. Modification these parameters did not affect the VLE fit since the concentration of CO_2 is low throughout the loading range compared to other species present in the solvent.

Table 3.5: Phoenix Regressed Parameters

Component (s)	Parameter	Value	Deviation
$MEACOO^-$	ΔG^o_f	-492827713	-230952.9%
	ΔH^o_f	-700000000	-162600.0%
	$Cp^{\infty,aq}$ /1	-162334.1	-70%
	$Cp^{\infty,aq}$ /2	-18.00	-274%
	$Cp^{\infty,aq}$ /3	1.8637	58%
MEA^+	ΔG^o_f	-187099000	a
	ΔH^o_f	-338703500	a
	$Cp^{\infty,aq}$ /1	26187.5	32%
	$Cp^{\infty,aq}$ /2	8.46	323%
	$Cp^{\infty,aq}$ /3	-0.5625	-152%
MEA	Cp^o /1	-132569.3	-29%
	Cp^o /2	687.01	26%
	Cp^o /3	0.0939	245%
	Cp^o /4	-0.0002814	-208%
H_2O/ MEA^+,$MEACOO^-$	$\tau_{m/ca}$/1	8.7394	0.55%
MEA/ MEA^+,$MEACOO^-$	$\tau_{m/ca}$/1	5.8612	3.94%
H_2O/ MEA^+,$HCO3^-$	$\tau_{m/ca}$/1	9.2484	1.05%
MEA/ MEA^+,$HCO3^-$	$\tau_{m/ca}$/1	16.2	30.82%
MEA/ H_2O	HENRY/1	8	19.94%
	HENRY/2	-8650	-2.33%
	HENRY/3	4.6967	5.35%
	HENRY/4	-0.007396	-40.81%
H_2O/MEA	τ_{mm} /1	-155.1749	-10.83%
	τ_{mm} /2	2350	0
	τ_{mm} /5	28.3986	11.45%
	τ_{mm} /6	-0.03887	-19.07%
MEA/H_2O	τ_{mm} /1	-0.6530	-41.50%
	τ_{mm} /2	-184.5171	-45.26%
	τ_{mm} /5	0.03	38.41%
MEA^+,$MEACOO^-$/CO_2	$\tau_{ca/m}$ /1	13.89	22.5%
	$\tau_{ca/m}$ /2	-5629	-21.7%
CO_2/ MEA^+,$MEACOO^-$	$\tau_{m/ca}$/1	38.07	16.4%
	$\tau_{m/ca}$/2	-8292	16.2%
MEA^+,$HCO3^-$/ CO_2	$\tau_{ca/m}$ /1	9.00	48.3%
	$\tau_{ca/m}$ /2	-1000	0.0%
CO_2/ MEA^+,$HCO3^-$	$\tau_{m/ca}$/1	12.95	12.8%
	$\tau_{m/ca}$/2	-592.8	-190.0%
CO_2/MEA	τ_{mm} /1	17.18	18.8%
	τ_{mm} /2	-4977	-15.7%
MEA/CO_2	τ_{mm} /1	4.18	50.5%
	τ_{mm} /2	-1467.63	-72.9%

[a] Parameter fitted independently using the pK_a of the protonation of MEA.

3.3.3. Phoenix Kinetics

Kinetics for CO_2 absorption using MEA have been widely studied using various contactors such as the stirred cell reactor (Kucka et al., 2003), the laminar jet (Astarita, 1961; Aboudheir, 2002) and the wetted wall column (Dugas, 2009). The latter two are suitable to emulate process conditions due to their exposure times: the wetted wall column for packed columns and the laminar jet for bubble plates (Danckwerts, 1970).

3.3.3.1. The Wetted Wall Column (WWC) model

The kinetics for this work were obtained by Dugas (2009) using a wetted wall column (WWC). In this contactor the liquid flows in a film down a vertical tube or rod. The gas enters an outer cylinder surrounding the vertical tube. In this apparatus it is possible to assess solvent absorption rates as well as CO_2 equilibrium partial pressure at various loadings.

Dugas (2009) generated rate data for CO_2 absorption in MEA at various concentrations (7 and 9 molal), CO_2 loading, and temperature. This data was used to evaluate the forward rate constants for the formation of carbamate. A model of the wetted wall column (WWC) used by Dugas (2009) was developed in Aspen Plus® RateSep™. It uses the Phoenix thermodynamic framework and FORTRAN subroutines for liquid and gas mass transfer coefficients, and interfacial area. These subroutines include correlations found in Cullinane (2005). For the gas film transfer coefficient (k_g):

$$k_g = \frac{1.075}{RT} D_{CO_2}^{0.15} d^{0.7} \left(\frac{v}{h}\right)^{0.85} \tag{3.19}$$

where: R is the gas constant

D_{CO2} is the diffusion coefficient of CO_2

d is the hydraulic diameter of the WWC (0.44 cm)

v is the linear velocity of the gas

h is the height of the contact cylinder in the WWC (0.091 cm)

The liquid film transfer coefficient is defined taking into account that the WWC experiments are conducted at conditions for which the dimensionless penetration distance (η) is less than 0.01. The penetration distance is defined as:

$$\eta = D_{CO_2} h \left(\frac{2}{3^{4/3}}\right)\left(\frac{W}{Q}\right)^{4/3}\left(\frac{\rho g}{\mu}\right)^{1/3} \tag{3.20}$$

where: W is the circumference of the contact cylinder (0.040 cm)

Q is the volumetric flow rate of the liquid

ρ is the liquid density

g is the gravity constant

μ is the liquid viscosity

For these conditions the liquid mass transfer coefficient (k_L^0) is

$$k_L^0 = \frac{Q}{A}\sqrt[3]{\frac{\eta}{\pi}} \tag{3.21}$$

Here A is the gas-liquid contact area. It is equal to 38.52 cm^2 and it is included, scaled up, as a constant in the interfacial area subroutine.

Figure 3.20 shows the experimental set up used by Dugas. Figure 3.21 shows the PFD of the corresponding Aspen Plus® simulation. Detailed gas and liquid specifications were taken from the experimental data in Dugas (2009). Individual CO_2, water, and PZ streams are fed to a flash tank (saturator) that saturates the inlet gas stream with water at the experimental pressure and temperature. On the solvent side a mixer (MX1) is used to

generate a single stream matching experimental conditions. Solvent temperature is specified using a heat exchanger before the WWC. Two additional flash tanks (EQ1, EQ2) are used to calculate equilibrium constants and the CO_2 partial pressure in equilibrium with the solvent.

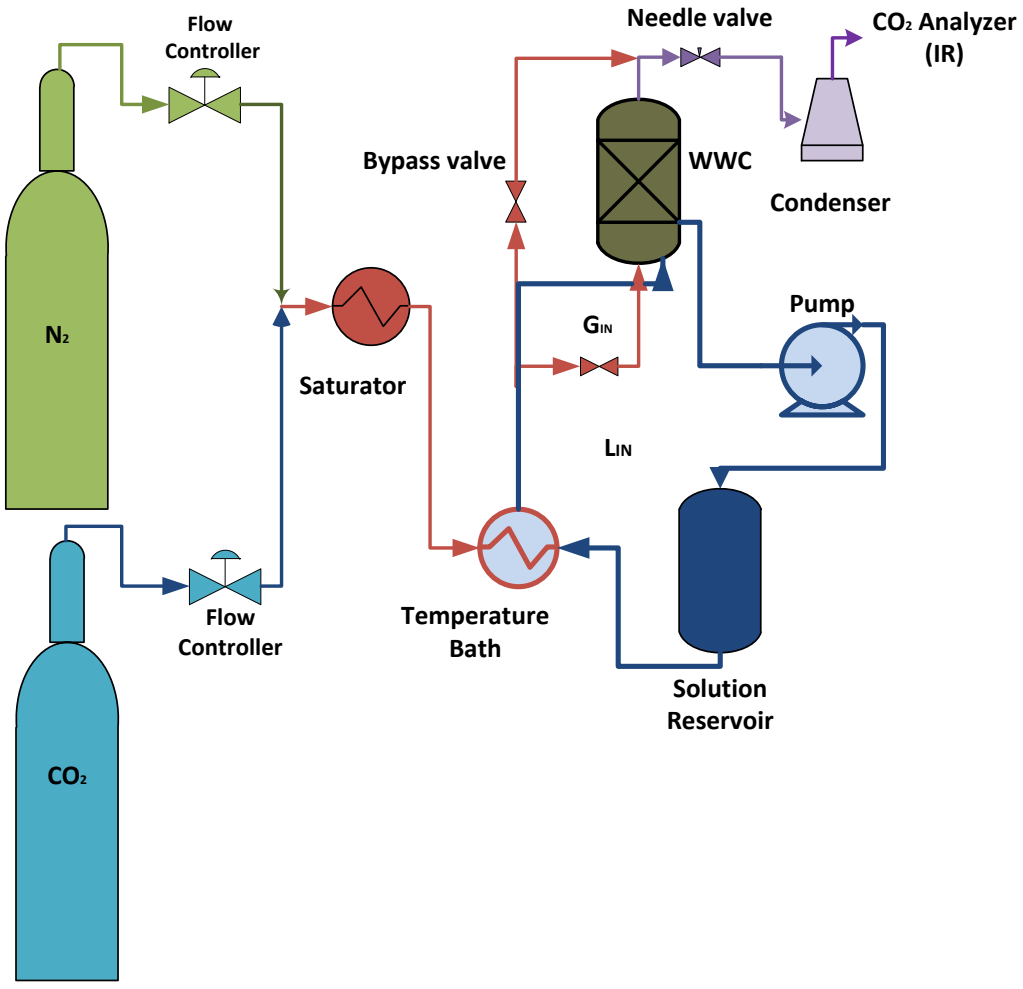


Figure 3.20: Experimental WWC set up. Dugas (2009)

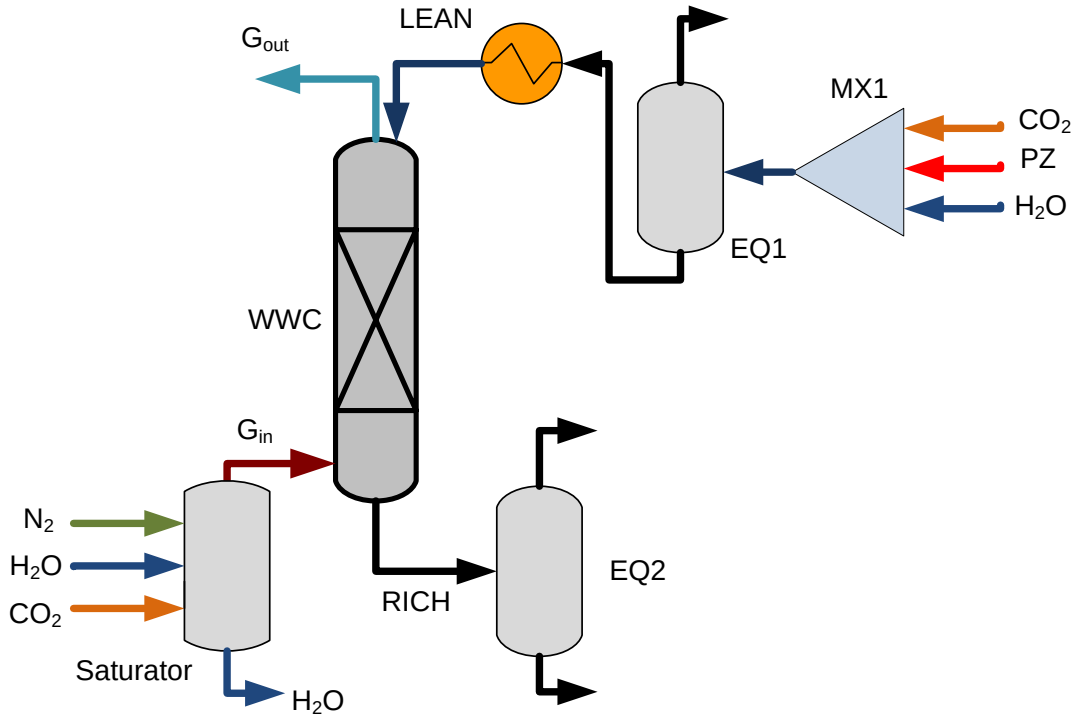


Figure 3.21: PFD for the WWC Aspen Plus® simulation.

3.3.3.2. Phoenix reactions

Kinetics were represented using the following set of reversible reactions:

$$2\,MEA + CO_2 \leftrightarrow MEA^+ + MEACOO^- \tag{3.22}$$

$$MEA + CO_2 + H_2O \leftrightarrow MEA^+ + HCO_3^- \tag{3.23}$$

This is a reduction of the commonly used reaction set based on the work by Hilliard (2008) that showed that the concentration levels of H^+, OH^- and CO_3^{-2} at the process conditions are negligible. The reactions involving these components were eliminated or combined to obtain the proposed reaction set, simplifying the model. The inclusion of equations 3.22 and 3.23 in Aspen Plus® RateSep™ requires the

representation of the forward and reverse rates separately. Rates are included in the model using activities (*a*) as follows:

$$r^{f}_{MEACOO^-} = k^{f}_{MEACOO^-} a_{MEA} a_{CO_2} \tag{3.24}$$

$$r^{r}_{MEACOO^-} = k^{r}_{MEACOO^-} \frac{a_{MEACOO^-} a_{MEA^+}}{a_{MEA}} \tag{3.25}$$

$$r^{f}_{HCO_3^-} = k^{f}_{HCO_3^-} a_{MEA} a_{CO_2} \tag{3.26}$$

$$r^{r}_{HCO_3^-} = k^{r}_{HCO_3^-} \frac{a_{HCO_3^-} a_{MEA^+}}{a_{H_2O}} \tag{3.27}$$

In these expressions the superscripts *f* and *r* represent forward and reverse rate respectively. The subscript *MEACOO*$^-$ refers to the reaction for the formation of carbamate (Equation 3.22). The subscript HCO_3^- is for the equilibrium for the formation of bicarbonate (Equation 3.23).

The forward bicarbonate reaction constant (Equation 3.26) was evaluated using data for the apparent second order constants at 25°C for reaction of amines and CO_2. This data was presented against the base dissociation constant (pK_b) in Rochelle et al. (2001). The values of the reaction constant for tertiary amines were fit as a function of pK_b (Figure 3.22) obtaining the following expression:

$$k^{f}_{HCO_3^-} = 3.278\, e^{-1.11\, pK_b} \tag{3.28}$$

The forward rate constant for MEA (pK_b=4.45) was extracted from this fit. This result was converted to activity basis using values from Aspen generated from Phoenix for the activity coefficients for CO_2 and MEA as follows:

$$k^{a}_{f} = \frac{k^{c}_{f} \rho_s^2}{\gamma_{MEA} \gamma_{CO_2}} \tag{3.29}$$

Where: k_f^a is the forward reaction rate in activity basis, kmol/m^3-s

k_2^c is the forward reaction rate concentration based, m^3/gmol-s

ρ_s is the molar density of the MEA solvent in kmol/m^3 at 25°C

γ_{MEA} , γ_{CO2} are the activity coefficients of MEA and CO_2 at 25°C and a loading of 0.30.

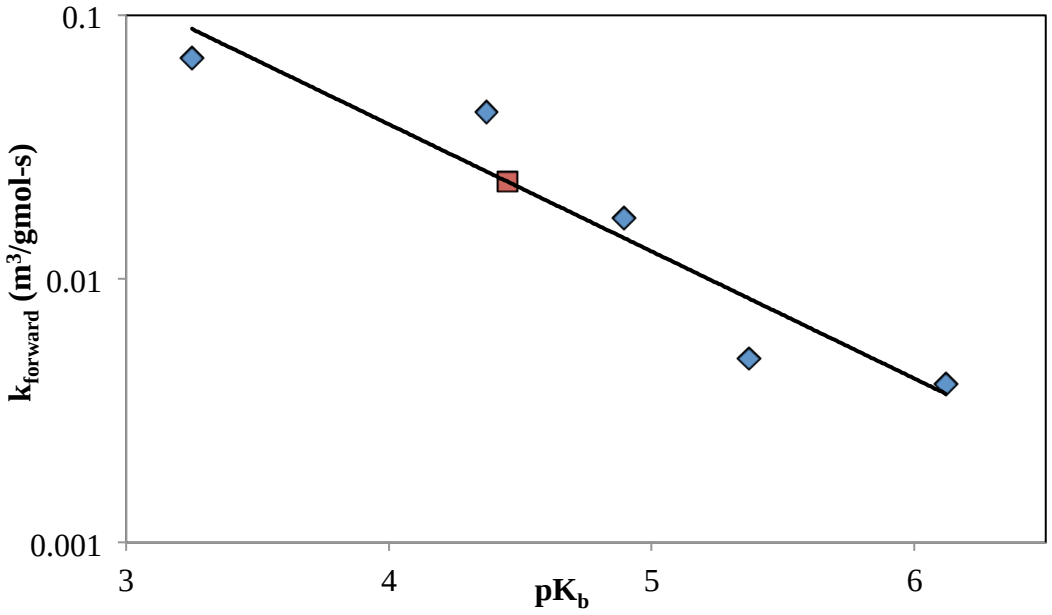


Figure 3.22: Second order rate constants for the reaction of tertiary amines & CO_2 at 25 ^{o}C . Points from Rochelle et al. 2001. (♦) Tertiary amines (■). Line fitted Equation 3.23 (R^2 = 0.8994)

This result was used as the reference value for the power law equation (Equation 3.2). The energy of activation (*E*) was approximated using the data for MDEA (49 kJ/gmol) reported by Pacheco et al. (1998) and presented in Rochelle et al. (2001). Thus, the form of the forward reaction rate constant for the bicarbonate reaction is:

$$k_{HCO_3^-}^f = 22991.13\, e^{-\frac{49000}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} \qquad (3.30)$$

Where: k^{f}_{HCO3-} is the forward reaction rate, kmol/m^3-s

22991.13 kmol/m^3-s is the reference state rate constant from equation 3.28

T is the temperature of the point.

298.15 is the reference temperature. It corresponds to the temperature for the reference state constant

The reference state constant was used along with the equilibrium constants to determine the reverse reference constant for the bicarbonate reaction. Equilibrium was previously evaluated using a flash calculation block in Aspen Plus® to extract the activity coefficients and liquid equilibrium concentrations. The reverse reference rate constant was computed as follows:

$$k^{r}_{HCO_3^-}\Big|_{298.15} = \frac{22991.13}{K_{HCO_3^-}} \tag{3.31}$$

Where: $k^{r}_{HCO3}|_{298.15}$ is the reverse bicarbonate formation rate constant at 298.15 K (kmol/m^3-s).

K_{HCO3-} is the equilibrium constant for the bicarbonate reaction evaluated at 298.15 K

$$K_{HCO_3^-} = \frac{a_{MEA^+} a_{HCO_3^-}}{a_{MEA} a_{CO_2} a_{H_2O}} \tag{3.32}$$

Where a is the activity for each specie. $a_i = \gamma_i x_i$ at equilibrium.

The energy of activation for the reverse rate was obtained using the forward rate at a loading of 0.30 and 60°C and the reference state constant from Equation 3.31.

$$k^{r}_{HCO_3^-} = 18.35\, e^{-\frac{96230}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} \tag{3.33}$$

The calculated bicarbonate reaction rate constants were entered into the WWC model in the following form:

$$r^{f}_{HCO_3^-} = 22991.13\, e^{-\frac{49000}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} a_{MEA} a_{CO_2} \tag{3.34}$$

$$r^{r}_{HCO_3^-} = 18.35\, e^{-\frac{96230}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} \frac{a_{HCO_3^-} a_{MEA^+}}{a_{H_2O}} \tag{3.35}$$

Where: r_{HCO3-} is the carbamate *f* forward or *r* reverse rate in kmol/s

a_i is the activity of species *i*

3.3.3.3. Diffusion Coefficients in the Phoenix model

Dugas (2009) showed that at higher loadings and temperatures carbon dioxide absorption is limited by the diffusion of reactants and products from and to the interface. Hence, the data at lower loadings and temperatures is more sensitive to kinetic values than the higher loading and temperature data and less affected by the diffusion coefficients. The MEA model includes FORTRAN subroutines to modify the diffusion coefficients for CO_2, and amine and amine products. The diffusion matrix is modified to account for the differences in diffusivity of CO_2 and other dissolved gases and amine and amine products.

For MEA the diffusion coefficients were calculated based on correlations presented by Versteeg et al.(1988) for CO_2 in water:

$$D_{CO_2-water} = 2.35 * 10^{-6} e^{\frac{-2119}{T}} \tag{3.36}$$

Where: $D_{CO2\text{-}water}$ is the diffusivity of CO_2 in water in m^2/s

T is the temperature in Kelvin.

The diffusivity coefficient in MEA is then calculated using a modified Stokes-Einstein relation:

$$D_{CO_2-sln} = D_{CO_2-water}\left(\frac{\mu_{water}}{\mu_{sln}}\right)^{0.8} \tag{3.37}$$

where: $D_{CO2\text{-}sln}$ is the diffusivity of CO_2 in the amine solution in m^2/s;

μ is the viscosity in Pa-s. The subscript *sln* represents solvent properties.

Equations 3.36 and 3.37 were also used for N_2. The diffusion coefficients of MEA, MEA^+, $MEACOO^-$, and HCO_3 were calculated using the equation for MEA in water proposed by Hayduk and Laudie (1974):

$$D_{MEA-water} = \frac{13.26 * 10^{-5}}{\mu_{water}^{1.4} V_{b-MEA}^{0.589}} \tag{3.38}$$

where: $D_{MEA\text{-}water}$ is the diffusivity of MEA in water in cm^2/s;

μ_{water} is the viscosity of water in cP;

$V_{b\text{-}MEA}$ is the molar volume of MEA at its normal boiling point in $cm^3/gmol$.

The diffusivity of the amine species in the solvent is calculated by the following:

$$D_{MEA-sln} = D_{MEA-water}\left(\frac{\mu_{water}}{\mu_{sln}}\right)^{0.8} \tag{3.39}$$

Where: $D_{MEA\text{-}sln}$ is the diffusivity of MEA and MEA species in the solution;

μ_{sln} is the viscosity of the solution.

According to the analysis of mass transfer with fast chemical reaction (See section 2.2.3) the flux of CO_2 and k_g' depend on the square root of the diffusion of CO_2. However, a sensitivity analysis of k_g' with respect to diffusion using the WWC model in

Aspen Plus® (Figure 3.23) showed a combined effect of the diffusion of reactants and products such that the product of both gives a square root dependence on $k_g^{'}$. This is due to the way that the Maxwell- Stefan diffusivities are implemented in Aspen Plus® RateSep™. Details of the solution of the Maxwell-Stefan equations can be found in in Krishna and Standart (1976). In summary, the approach uses the mass transfer coefficient for the WWC as defined in Equations 3.19 and 3.21 in place of the diffusion coefficient when solving the Maxwell-Stefan matrices. This solution method proves to be adequate when there is no reaction in the film. However, for the absorption of CO_2 in amines it confounds the effect of the CO_2 diffusion in the enhancement of mass transfer due to chemical reaction leading to a fictitious effect of products at lean conditions and low temperatures.

A workaround to this issue has been proposed for the Phoenix model. It consists in defining the diffusion coefficient of CO_2 as a function of the diffusion of MEA and MEA products. Additionally, all diffusion coefficients have been centered at 40°C and a loading of 0.30. This allowed determination of the kinetic constants using the data from the WWC at 40°C and 60°C and adjustment of the diffusion coefficients using the data at 100°C. The diffusion coefficient of CO_2 was squared and divided by the diffusion coefficient of the amine to force the square dependence of the diffusion coefficient of CO_2 in the amine solution. These considerations lead to the following expressions for the diffusion of CO_2 in the amine solvent:

$$D_{CO_2-sln} = D^{o}_{CO_2} \left(\frac{\mu_{water}}{\mu_{sln}}\right)^{-1.2} \left(\frac{T}{313.15}\right)^{2-n} \tag{3.40}$$

Where: $D_{CO2\text{-}sln}$ is the diffusion of CO_2 and other gases in the solvent,

n is a regressed exponent calculated by fitting WWC data at 100°C

D^o_{CO2} is the reference diffusion constant calculated as:

$$D^o_{CO_2} = \left.\frac{D^2_{CO_2}}{D_{MEA}}\right|_{40℃} \tag{3.41}$$

where the diffusion of CO_2 and MEA is calculated at 40°C and a loading of 0.30 using equations 3.39 and 3.41 respectively. It is equal to 1.05×10^{-9} for 7m MEA and 8.29×10^{-10} for 9m MEA.

For MEA the diffusion coefficient is calculated using the following:

$$D_{MEA-sln} = D^o_{MEA}\left(\frac{\mu_{water}}{\mu_{sln}}\right)^{-0.8}\left(\frac{T}{313.15}\right)^n \tag{3.42}$$

where: D^o_{MEA} is the diffusion coefficient of MEA in solution calculated at 40°C and a loading of 0.30. 9.85×10^{-10} for 7m MEA and 8.27×10^{-10} for 9m MEA

n is a temperature correcting factor fitted using the WWC data at 100°C.

3.3.3.4. Carbamate formation in the Phoenix model

Work by Plaza et al. (2010) showed that an additional MEA activity in the carbamate kinetic expression could be useful to account for the ionic strength variation with loading. This approach was considered in the development of the Phoenix kinetics. As a result Equations 3.24 and 3.25 were modified:

$$r^f_{MEACOO^-} = k^f_{MEACOO^-} a^2_{MEA} a_{CO_2} \tag{3.43}$$

$$r^r_{MEACOO^-} = k^r_{MEACOO^-} a_{MEACOO^-} a_{MEA^+} \tag{3.44}$$

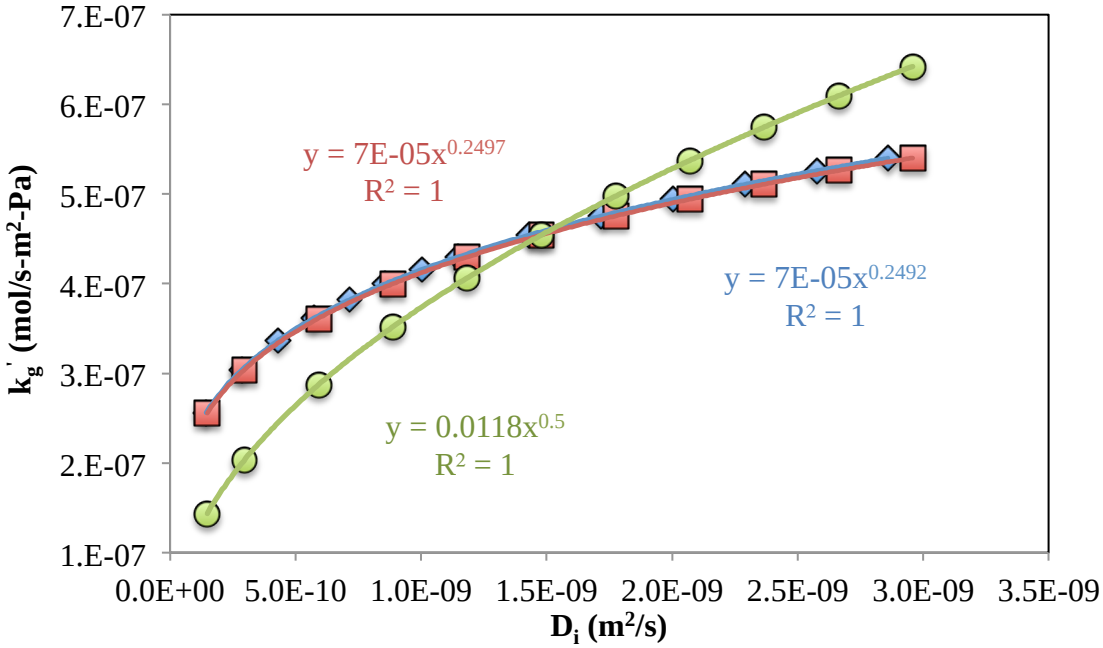


Figure 3.23: Diffusion sensitivity analysis for the WWC. T = 60 °C, ldg = 0.25. Effect of: (♦) D_{CO2} , (■) D_{MEA}, (●) Simultaneous change in D_{CO2} & D_{MEA}. Solid lines are trendlines that correspond to the power law fit equations of the same color.

Determination of k_o and E for the kinetic constants in Equations 3.43 and 3.44 was done using the measured values of CO_2 flux (N_{CO2}) with desorption and absorption data from Dugas (2009). The energy of activation was set to zero for the forward and reverse carbamate formation. The pre-exponential factor for the forward rate was adjusted to match the reported flux. The reverse k_0 was then obtained using the equilibrium constant.

The data in Dugas (2009) required two loading adjustments. The first one is related to the VLE fit. Dugas (2009) reported CO_2 flux as a function of a mean logarithmic average of the CO_2 partial pressure in the WWC defined as:

$$LMPD_{CO_2} = \frac{\left(P_{CO_2|in} - P^*_{CO_2}\right) - \left(P_{CO_2|out} - P^*_{CO_2}\right)}{\ln\left(\frac{P_{CO_2|in} - P^*_{CO_2}}{P_{CO_2|out} - P^*_{CO_2}}\right)} \tag{3.45}$$

Where: $P_{CO2|in}$, $P_{CO2|out}$ are the inlet and outlet CO_2 partial pressures,

P^*_{CO2} is the equilibrium pressure for CO_2 at the experiment conditions.

The equilibrium pressure in Dugas is determined by fitting the resulting data to the flux data such that at a flux of zero the driving force is equal to zero (see Figure 3.24). Since the equilibrium data fit in the model is not exact, it is necessary to adjust the reported loading to the model by changing the loading.

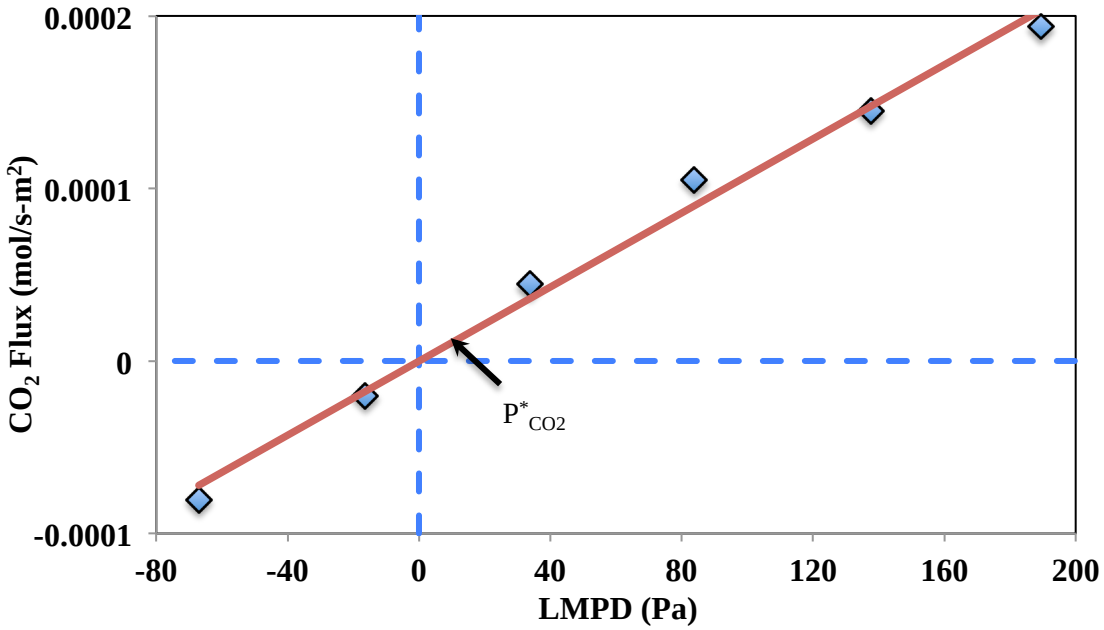


Figure 3.24: WWC raw data curve fit for 7m MEA, loading = 0.351, T= 40°C. (Dugas, 2009)

The second loading adjustment is necessary to match absorption and desorption points. The largest flux absorption and desorption points reported by Dugas (2009) were selected to extract the carbamate kinetic constants since points closer to the equilibrium have smaller fluxes with higher statistical error. However, points at the extreme may not lie directly over the fitted line in Figure 3.24 generating the need for a second loading correction.

The pre-exponential factors for data at 40°C and 60°C were then re-regressed to obtain the energy of activation. These values were used to match the data at 100°C and to find the exponential factor (*n*) in equations 3.40 and 3.42. The final carbamate rate expressions are:

$$r^{f}_{MEACOO^-} = 8.5616\text{x}10^{10} e^{\frac{-3963.9}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} a^{2}_{MEA} a_{CO_2} \quad (3.46)$$

$$r^{r}_{MEACOO^-} = 24800 e^{\frac{-59600}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} a_{MEACOO^-} a_{MEA^+} \quad (3.47)$$

Where: $r_{MEACOO-}$ is the carbamate ^{f} forward or ^{r} reverse rate in kmol/s

a_i is the activity of species *i*

T is the temperature in Kelvin

The fitted exponential factor (*n*) for diffusivity is equal to 22.6 for 7m MEA and 17.55 for 9m MEA.

. Figure 3.25 shows the resulting fit for the WWC CO_2 flux. The fit is adequate up to a loading of 0.50. Past that value the model is not capable of matching the flux data with the regressed kinetics. This issue does not affect process development since operating conditions are never over this value, as it will be shown in later in this document.

3.3.4. $k_g^{'}$ in the Phoenix model

The WWC Phoenix model can be used to asses the behavior of $k_g^{'}$ to process variables. Figures 3.26 and 3.27 shows model predictions for $k_g^{'}$ as a function of loading at various temperatures and compares it to data by Dugas (2009). Results show that the behavior of $k_g^{'}$ is adequately represented by the model with better agreement with experimental data for absorber operational temperatures (40°C – 80°C) and for 9m MEA which are the conditions that will studied later in this work.

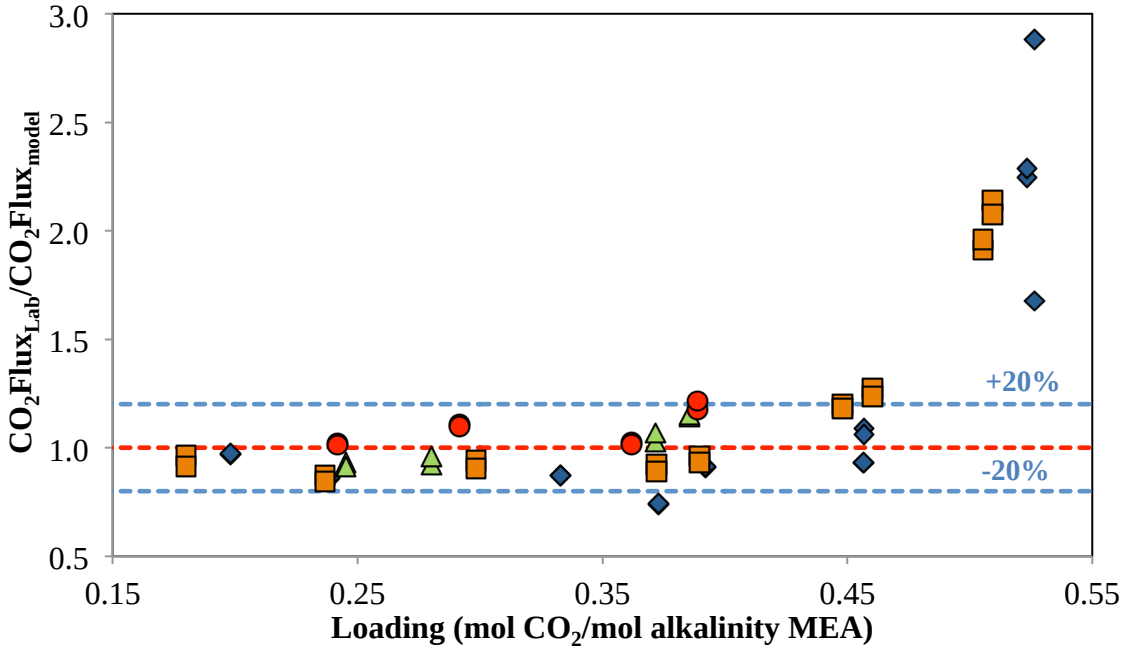


Figure 3.25: WWC Carbon dioxide flux comparison for 7m and 9m MEA. (♦) 40°C, (■) 60°C, (▲) 80°C (●) 100°C.

Figures 3.26 and 3.27 also show a drop in k_g' with loading. Dugas (2009) observed this and defined k_g' as a series combination of reaction and diffusion resistances represented as:

$$\frac{1}{k_g'} = \frac{1}{k_g''} + \frac{1}{k_L^o}\left(\frac{\Delta P_{CO_2}^*}{\Delta[CO_2]_T}\right) \tag{3.48}$$

Here k_g'' represents the mass transfer coefficient accounting for the enhancement thanks to the reaction in the liquid film, up to now designated as k_g'. The second term is related to the physical mass transfer for reactants and products. As k_g'' becomes smaller k_l^o gains more significance. Figure 3.28 shows the relative change of k_g' versus the relative change in the physical mass transfer coefficient (k_L^o). The significance of the physical mass transfer for reactants and products changes with the temperature and loading. At low temperature (40°C) it accounts for less than 20% of the resistance to

mass transfer. However, at higher temperatures and loadings it becomes significant to the point of controlling the rate of mass transfer. This is tied to the increase in reaction kinetics depleting the reactants at the interface so the reaction rate becomes limited by the capability for reactants and products to diffuse in and out of the boundary layer.

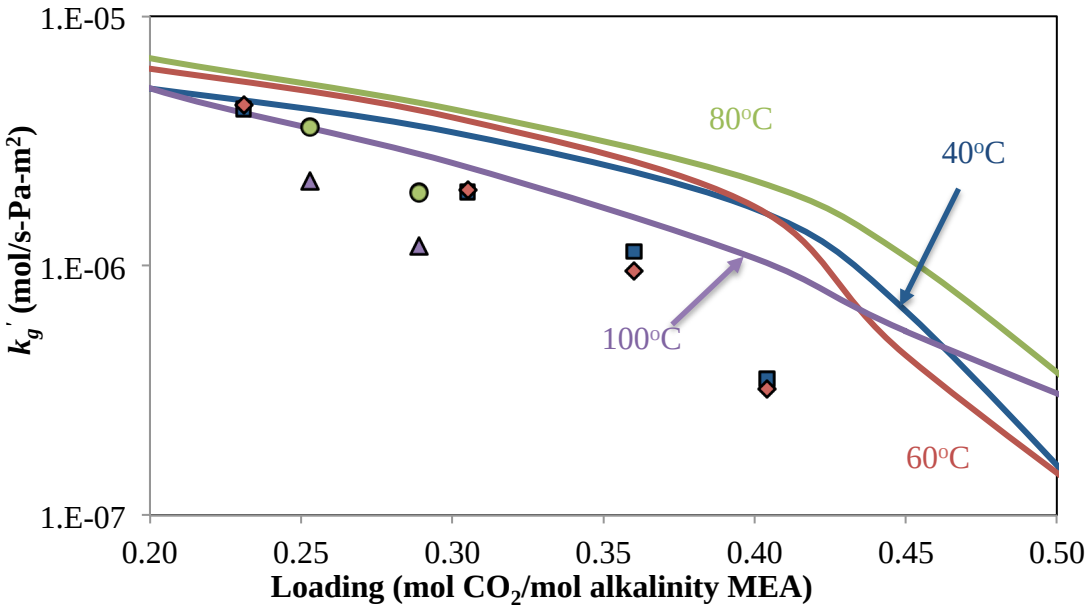


Figure 3.26: k_g' data for 7m MEA. Points are from Dugas (2009). Lines are Phoenix predictions.

3.3.5. Bicarbonate Formation and Phoenix

The Phoenix WWC was also used to look at the importance of the bicarbonate reaction in the absorption of CO_2. For this purpose activities were calculated by Phoenix and used along with the reaction rate expressions for 9m MEA for the carbamate (Equation 3.46) and the bicarbonate (Equation 3.34) formation. Looking at both forward rates at the interface and at the bulk it is possible to observe that the bicarbonate formation has very little influence in the CO_2 absorption at low loading and temperature (Figure 3.29). At high temperature and loading bicarbonate formation gains some

significance especially at the interface for the higher loading. However, at its highest value it is less than 0.3% of the carbamate formation rate.

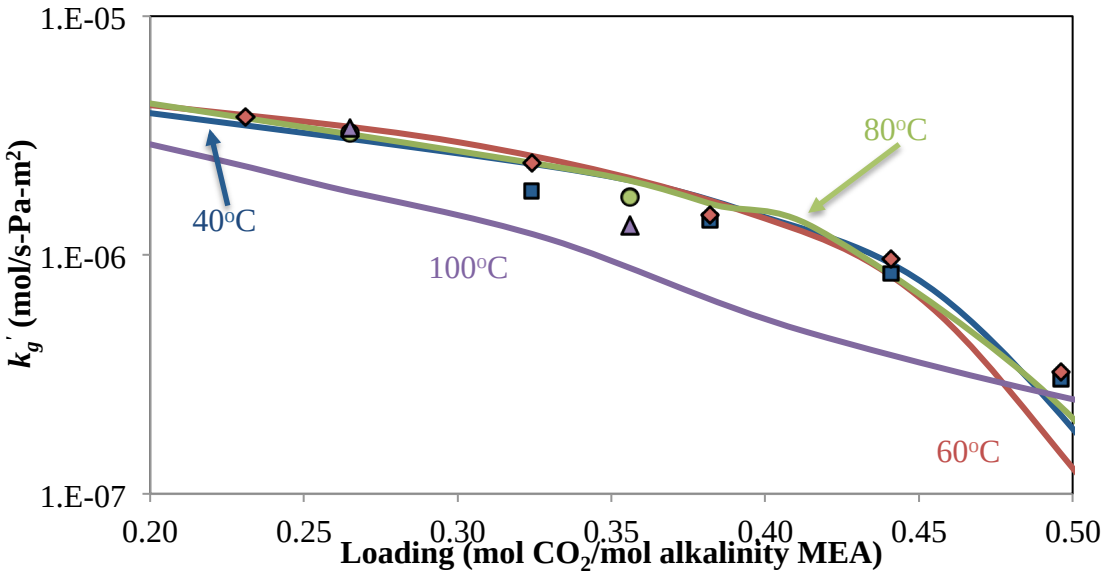


Figure 3.27: k_g' data for 9m MEA. Points are from Dugas (2009). Lines are Phoenix WWC predictions.

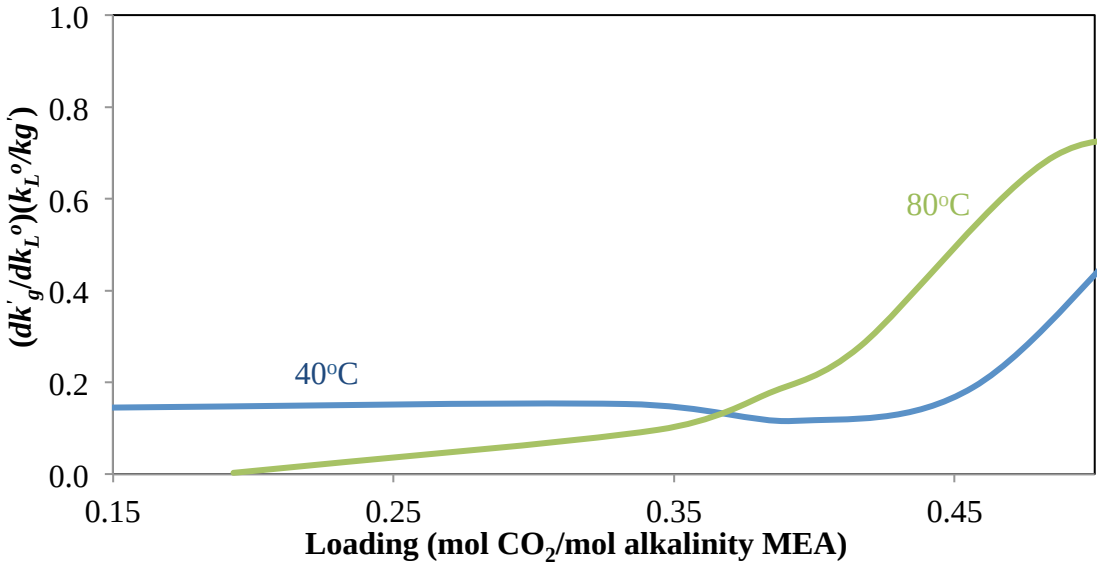


Figure 3.28: Relative change in k_g' with respect to the relative change in k_L^o at various loadings at 9m MEA using Phoenix WWC.

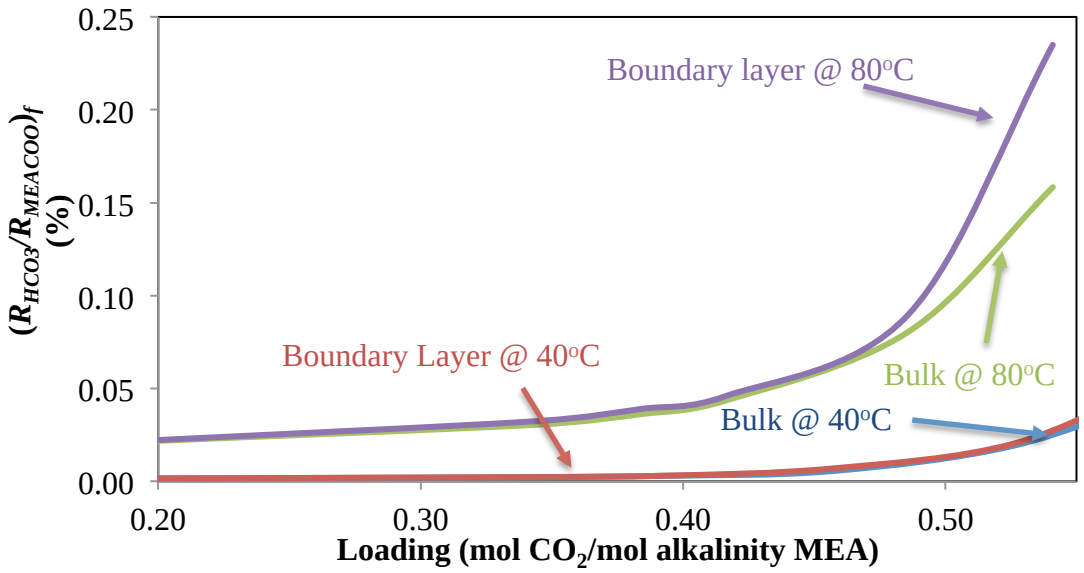


Figure 3.29: Ratio of Forward Carbamate and Bicarbonate formation for 9m MEA using Phoenix WWC.

3.4. ABSORPTION WITH CONCENTRATED PIPERAZINE (PZ) – 5DEMAYO

A new model for absorption of carbon dioxide using concentrated PZ has been developed. Named after the day of its finalization "5deMayo" includes a thermodynamic framework based on work by Hilliard (2008) and added high concentration (above 5m PZ) data by Freeman (2010), Xu (2011) and Dugas (2009). It represents PZ and the protonated PZ carbamate (H^+PZCOO^-) as Henry's components. This approach aided in fitting the PZ volatility and resolved issues previously observed with treatment of the zwitterion (Chen, 2007). Figures showing the adequate fit of the model will be presented in this section along with the resulting parameters.

3.4.1. Density and viscosity in the 5deMayo model

Density and viscosity were fit to data generated by Freeman (2011) for PZ from 5 m to 9 m PZ.

3.4.1.1. Density calculation in 5deMayo

Density data for 5 m, 7 m, 8 m, and 9 m PZ and from 0.20 to 0.46 mol CO_2/mol equivalent PZ was regressed in Aspen Plus® using the Clarke model for liquid molar volume for electrolyte solutions. Because this model uses detailed speciation, care should be used when using it with future thermodynamic models that may speciate the PZ solution differently. The liquid molar volume for the solution (V_{sln}^{l}) is calculated using the following:

$$V_{sln}^{l} = V_{solv} + \sum x_{CA} V_{CA} \tag{3.48}$$

where:

V_{solv} is the liquid molar volume for the solvent;

V_{ca} is the electrolyte effective partial molar volume;

x_{ca} is the apparent electrolyte mole fraction.

The liquid molar volume for the solvent is calculated using the following expression:

$$V_{solv} = x_w V_w^* + x_{nws} V_{nws}^{l} \tag{3.49}$$

where:

V_w^* is the water volume from the steam tables;

x_{nws} is the sum of the mole fractions of all non-water solvents;

V_{nws}^{l} is the liquid molar volume for the mixture of all non-water solvents calculated using the Rackett equation.

The Rackett expression was used for PZ and CO_2 and H^+PZCOO^-

$$V_{nws}^{l} = \frac{RT_c (Z_m^{RA})^{\left[1+(1-T_r)^{2/7}\right]}}{P_c} \tag{3.50}$$

Where:

$$T_c = \frac{\sum_i \sum_j x_i x_j V_{ci} V_{cj} \left(T_{ci} T_{cj}\right)^{0.5} \left(1 - k_{ij}\right)}{V_{cm}^2} \tag{3.51}$$

$$\frac{T_c}{P_c} = \sum_i x_i \frac{T_{ci}}{P_{ci}} \tag{3.52}$$

$$Z_m^{RA} = \sum_i x_i Z_i^{RA} \tag{3.53}$$

$$V_{cm} = \sum_i x_i V_{ci} \tag{3.54}$$

$$T_r = \frac{T}{T_c} \tag{3.55}$$

The electrolyte contribution in the molar volume is determined using the apparent mole fraction (x_{ca}) calculated from the true ionic composition based on the proposed system chemistry.

$$V_{ca} = V_{ca}^{\infty} + A_{ca} \left(\frac{\sqrt{x_{ca}}}{1 + \sqrt{x_{ca}}} \right) \tag{3.56}$$

The parameters V_{ca}^{∞}, and A_{ca} for Equation 3.56 were regressed for the cation PZH^+ and the other ionic species along with the values for Z_i^{RA} in Equation 3.53 for CO_2, PZ and the zwitterion using Aspen Plus® Data Regression System (DRS). (Table 3.6)

The regressed values adequately represent the change in density with loading at the selected amine concentrations at operating condition (loading < 0.45) (Figures 3.30 – 3.33).

Table 3.6: 5deMayo Regressed density parameters.

Parameter	Component(s)	Value
Z_i^{RA}	PZ	0.2646
	CO_2	0.1679
	H^+PZCOO^-	0.1944
V_{ci}	PZ	0.31
	H^+PZCOO^-	0.5595
V_{ca}^{∞}	$PZH^+ - PZCOO^-$	0.1342
	$PZH^+ - PZ(COO^-)_2$	0.4700
	$PZH^+ - HCO_3^-$	0.4275
A_{ca}	$PZH^+ - PZCOO^-$	0.1530
	$PZH^+ - PZ(COO^-)_2$	-0.9665
	$PZH^+ - HCO_3^-$	-2.3356

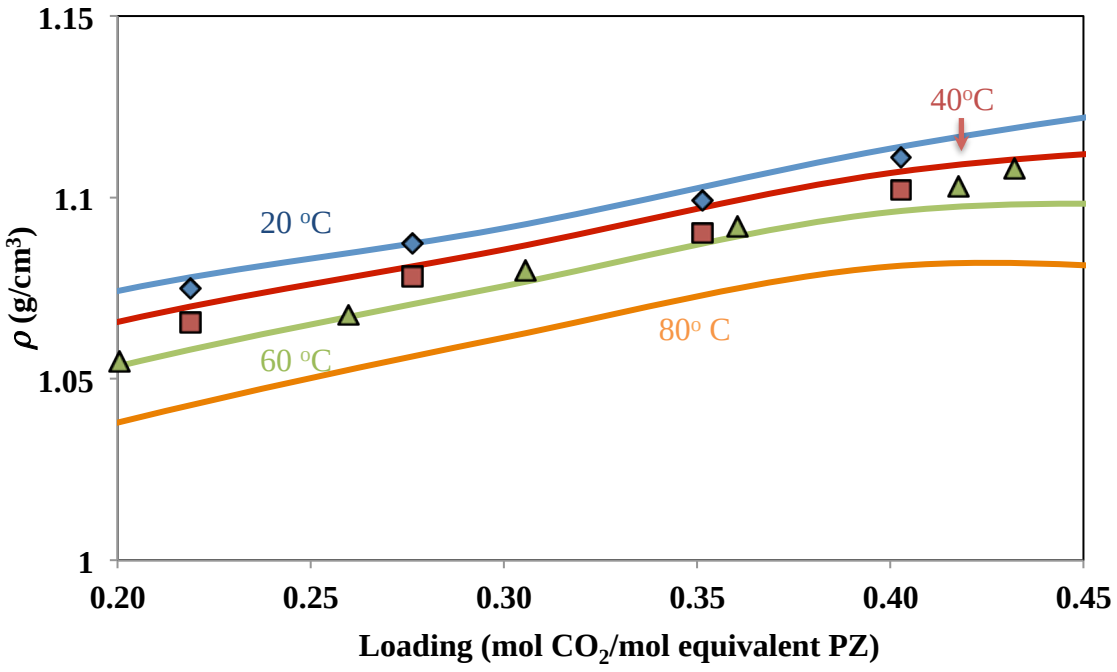


Figure 3.30: Density fit for 5m PZ. Lines 5deMayo model. Points Freeman (2011)

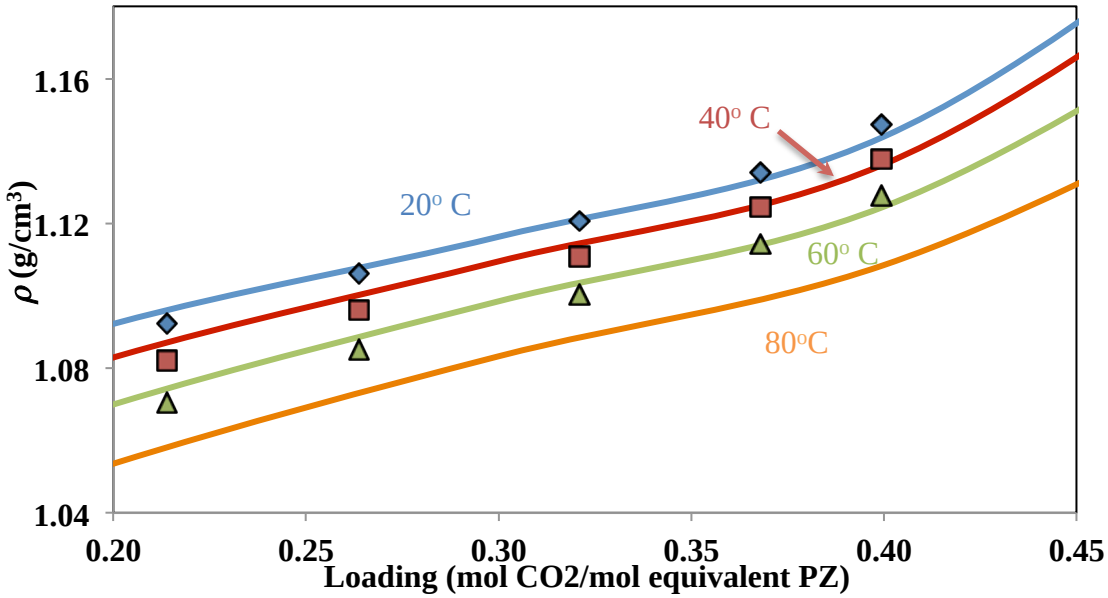


Figure 3.31: Density fit for 7m PZ. Lines 5deMayo model. Points Freeman (2011).

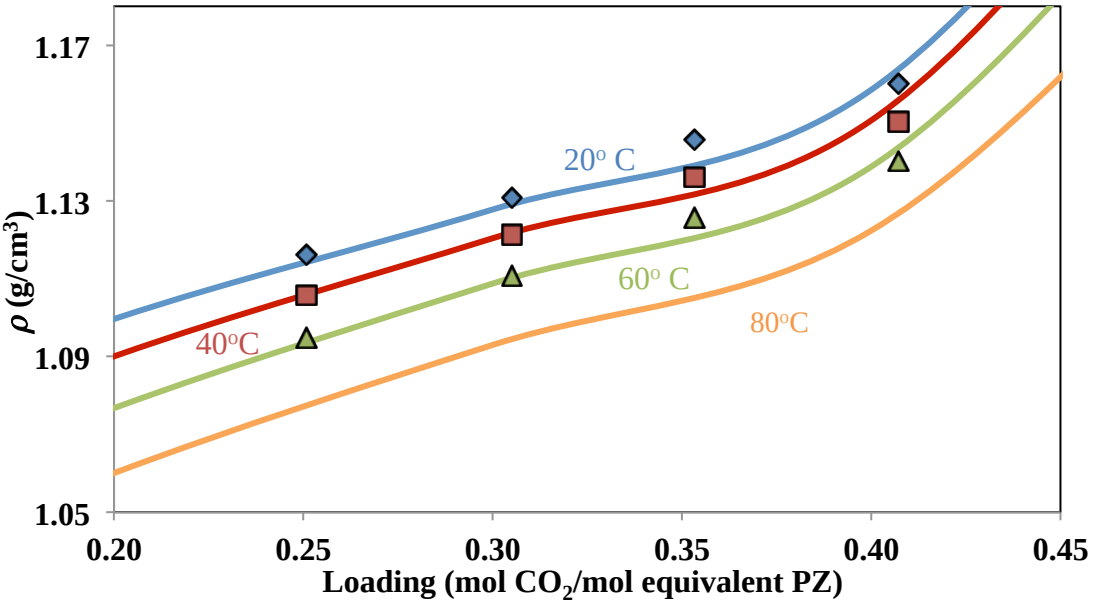


Figure 3.32: Density fit for 8m PZ. Lines 5deMayo model. Points Freeman (2011).

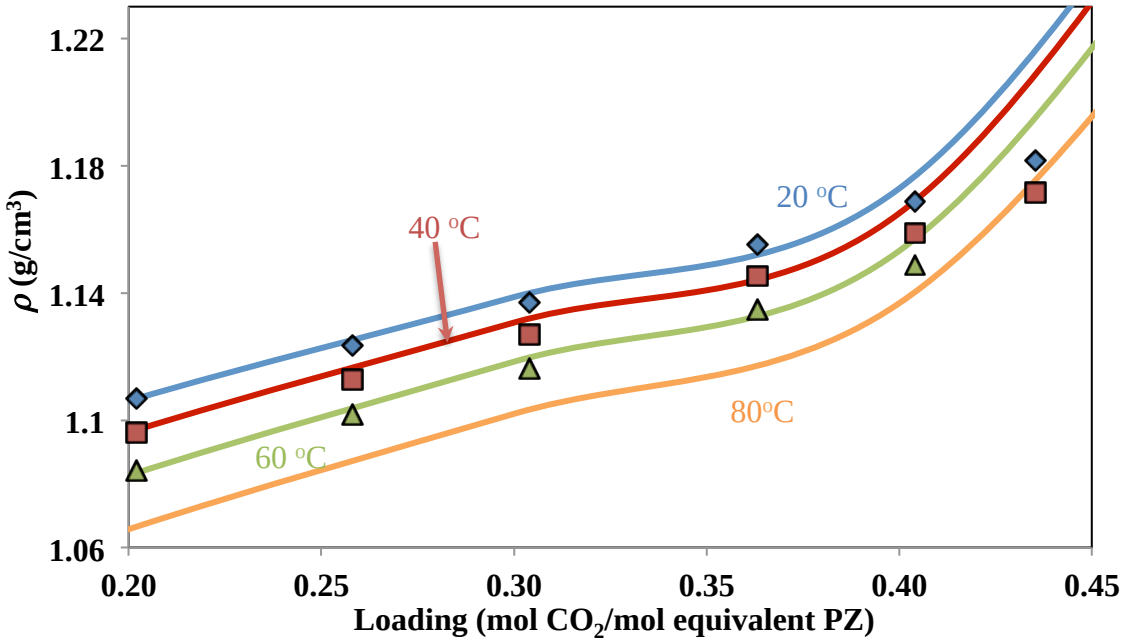


Figure 3.33: Density fit for 9m PZ. Lines 5deMayo model. Points Freeman (2011).

3.4.1.2. Viscosity calculation in 5deMayo

Data available for 5 m, 7 m, and 9 m PZ was fit using the expression of viscosity as a function of loading, amine concentration, and temperature proposed by Weiland et al. (1998)

$$\mu_{sln} = \mu_w exp\left(\frac{[(a\Omega + b)T + (c\Omega + d)][\alpha(e\Omega + fT + g) + 1]\Omega}{T^2}\right) \qquad (3.57)$$

where:

μ_{sln} ,μ_w are the viscosity of the solution and water, respectively (cP);

Ω is the weight fraction of PZ;

α is the loading of the solution (mol CO_2/mol alkalinity);

T is the temperature (K).

Table 3.7 shows the regressed parameters for this correlation. This expression was selected because it was not possible to adequately fit the viscosity data with the

available correlations in Aspen Plus®. It was implemented in Aspen Plus® using a FORTRAN subroutine. The obtained results are presented in Figures 3.34 – 3.37.

Table 3.7: Viscosity parameters used in Equation 3.57

Parameter	a	b	c	d	e	f	g
Value	487.52	1389.31	1.58	4.50	8.73	-0.0038	-0.30

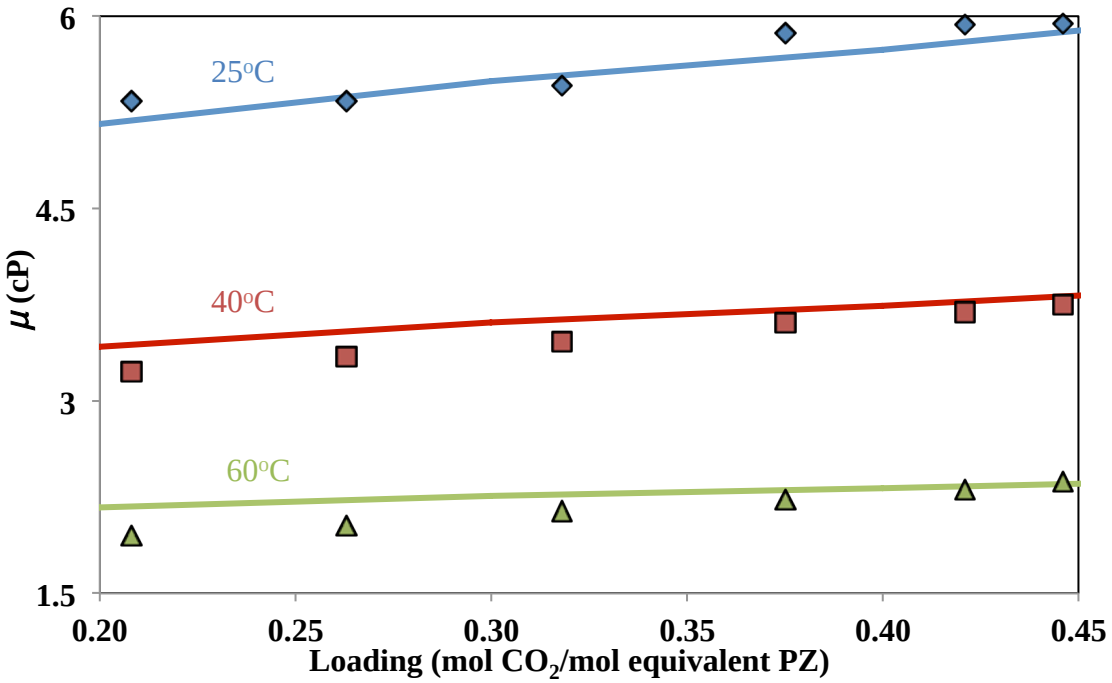


Figure 3.34: Viscosity fit for 5m PZ. Lines 5deMayo model. Points Freeman (2011).

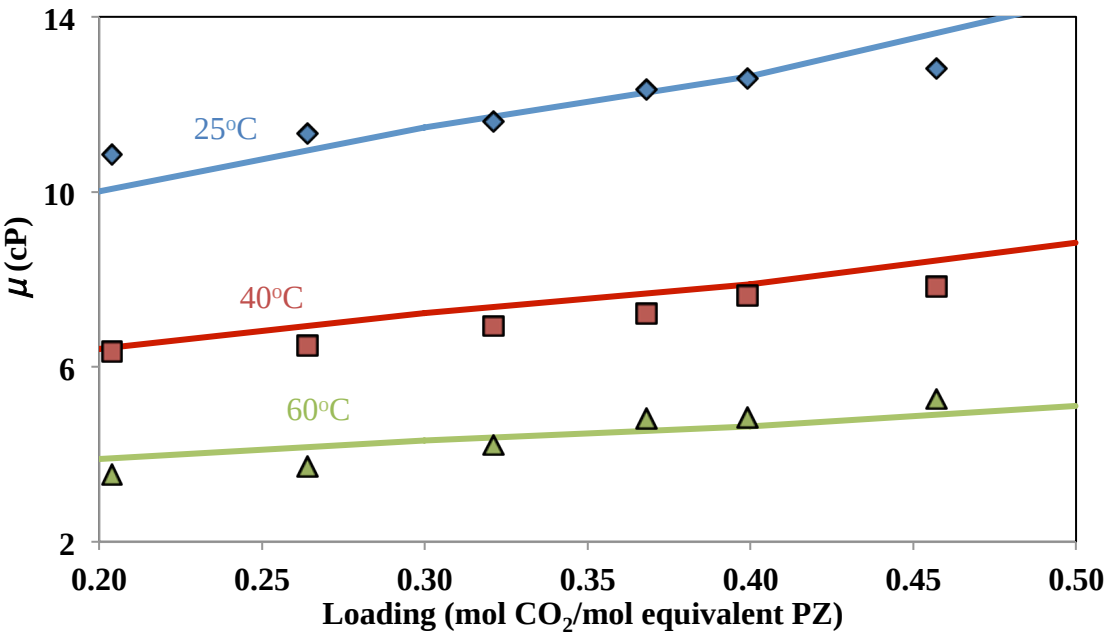


Figure 3.35: Viscosity fit for 7m PZ. Lines 5deMayo model. Points Freeman (2011).

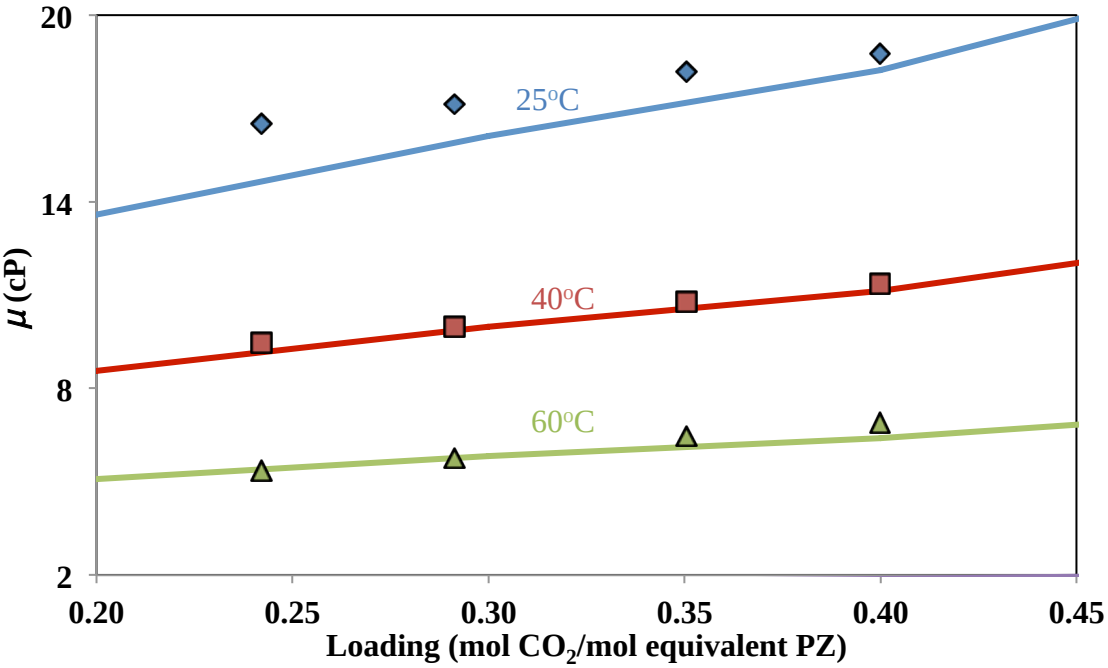


Figure 3.36: Viscosity fit for 8m PZ. Lines 5deMayo model. Points Freeman (2011).

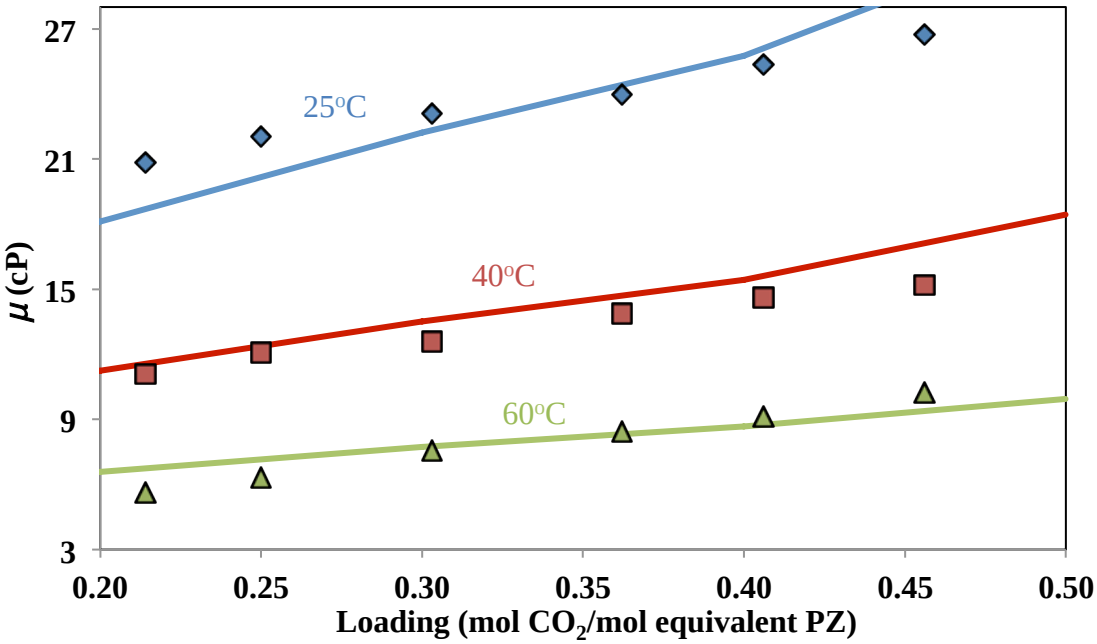


Figure 3.37: Viscosity fit for 9m PZ. Lines 5deMayo model. Points Freeman (2011).

3.4.2. 5deMayo Thermodynamics

Work presented in this section was conducted with collaboration of Peter Fraile and led to a publication (Frailie et al., 2011) describing the framework development and is part of the model license 5841ROC Copyright© The University of Texas at Austin, 2011. This section demonstrates the adequate fit of the model. Figure 3.38 shows the VLE fit for CO_2 loaded PZ for 0.9m – 12m PZ. Loading is defined as moles of CO_2 per mol equivalent of PZ. Mol equivalents of PZ are defined based on the number of amine groups in the molecule (2 for PZ):

$$Loading = \frac{moles\ CO_2}{2\ moles\ PZ} \tag{3.58}$$

Table 3.8 shows the regressed parameters along with their standard deviations. Parameter values are part of the model license 5841ROC Copyright © The University of Texas at Austin, 2011.

Figure 3.38 shows that the model adequately represents the solubility of CO_2 in PZ for the operating loading range (0.2-0.45) and from 40 to 160°C. It is an improvement in the predictions by the Hilliard (2008) model.

Amine volatility was slightly over predicted across the analyzed loading range and temperature. (Figure 3.39)

Heat capacities were not matched (Figure 3.40). However, model values are within the range of the experimental data and within the expected experimental error. They are deemed adequate for process modeling purposes considering process temperatures.

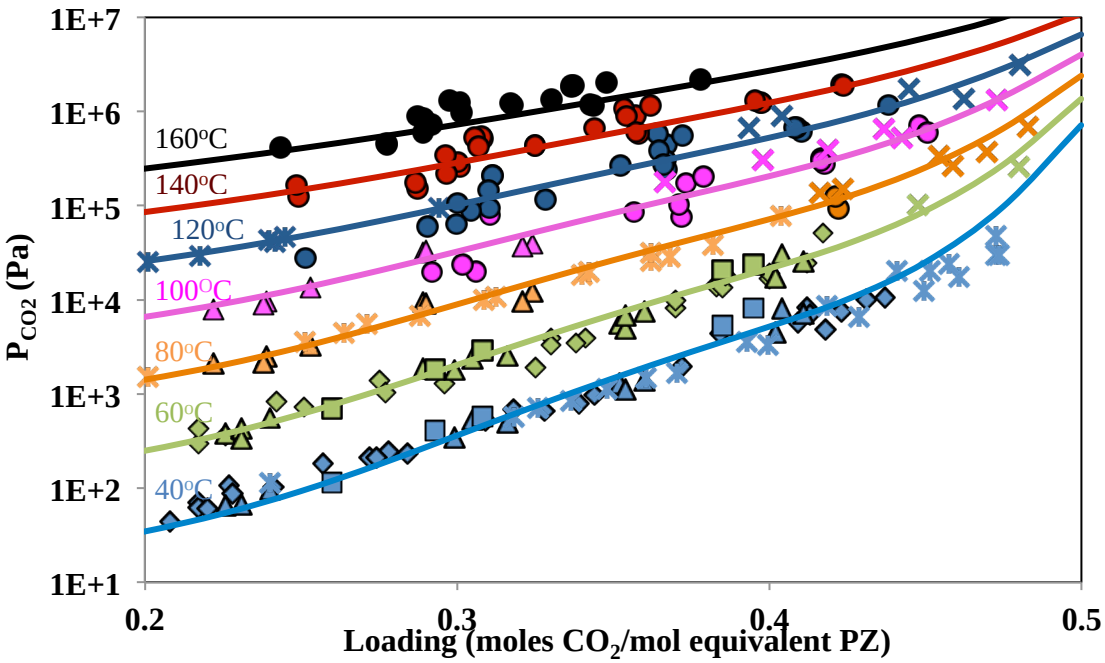


Figure 3.38: CO_2 Solubility in 0.9m -12 m PZ. Lines are 5deMayo fit for 8m PZ. Points are experimental data by (♦) Hilliard (2008), (■) Nguyen (Rochelle et al., 2011), (▲) Dugas (2009) (●) Xu (2011), (∗) Ermatchkov (2006), (×) Kemps (2003)

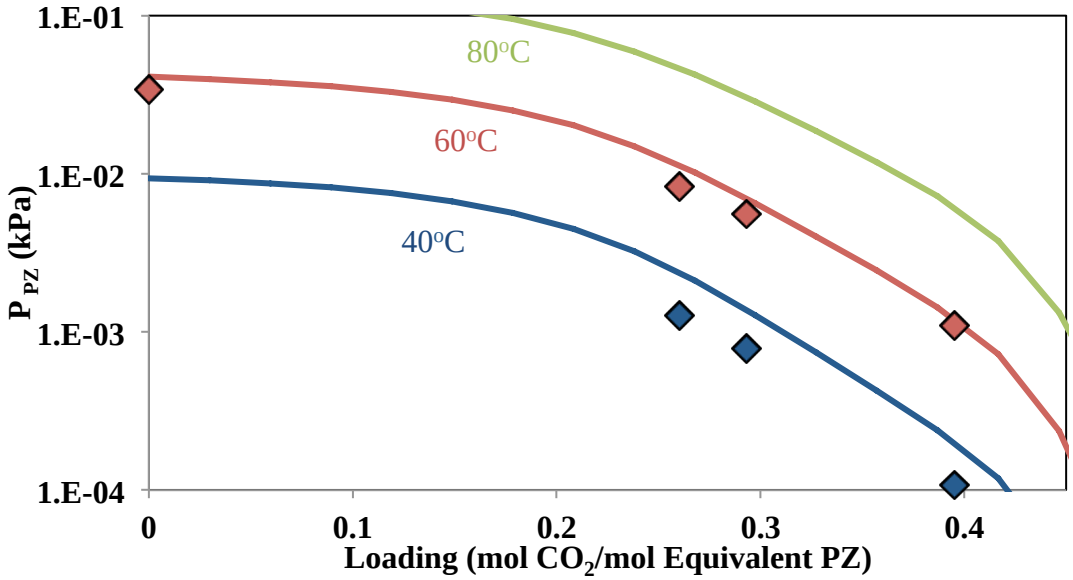


Figure 3.39: 8 m PZ volatility. Lines are the 5deMayo model. Points by Nguyen (Rochelle et al., 2011)

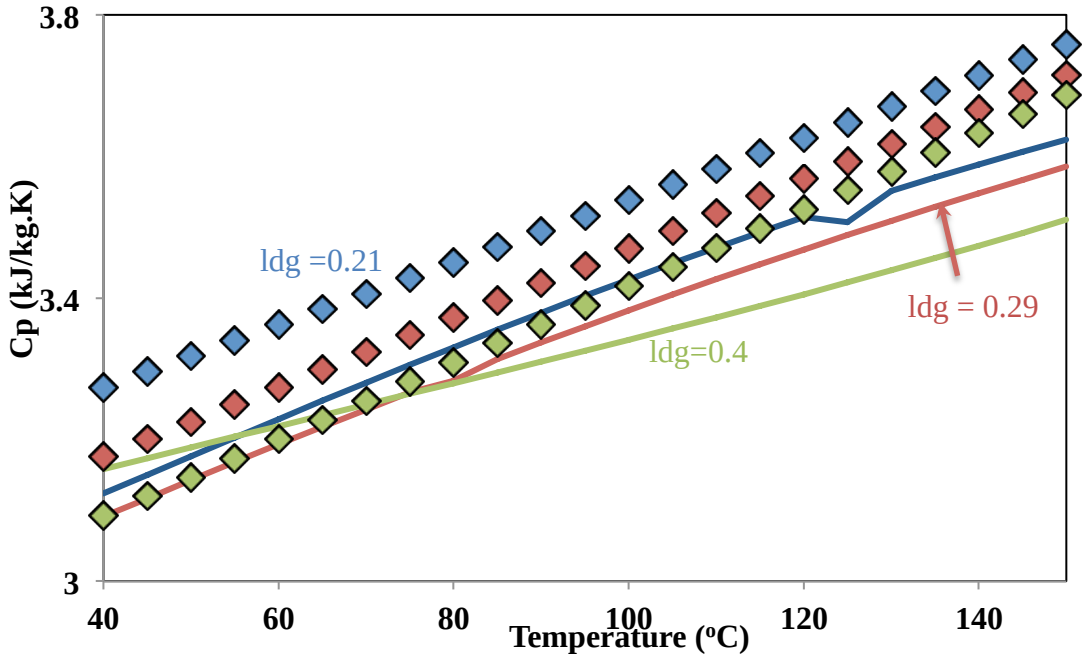


Figure 3.40: Specific heat capacity for 8 m PZ. Lines are the 5deMayo fit. Points by Nguyen (Rochelle et al., 2011)

Carbon dioxide heat of absorption for 8m PZ is approximately 65 kJ/mol CO_2 calculated using Aspen Plus® and the Gibbs-Helmholtz relation:

$$\frac{dP_{CO_2}}{d\left(\frac{1}{T}\right)} = \frac{-\Delta H^{lv}}{R} \tag{3.59}$$

where: P_{CO2} is the partial pressure of CO_2

ΔH^{lv} is the heat of absorption for CO_2 in the solvent

R is the gas constant

T is the temperature

This value is lower compared to the reported data by Freeman (2011). The later were obtained following the method by Kim (2007) thus it presents similar uncertainty so the model results are considered adequate.

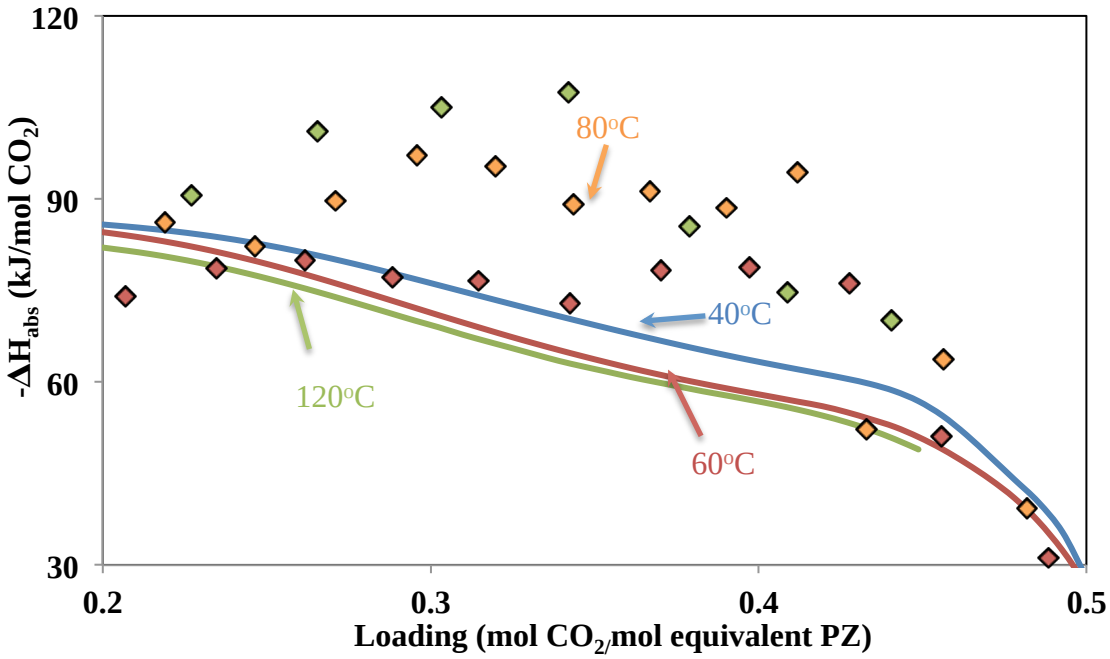


Figure 3.41: CO Heat of absorption for 8m PZ predicted by 5deMayo using the Gibbs-Helmholtz relation. Points by Freeman (2011)

Speciation results (Figure 3.42) are slightly lower than the data by Nguyen (Rochelle et al., 2011) but within the experimental error. However, results might have been affected by the fit for pK_a, which is considerably lower (Figure 3.44). The lower model pK_a values are due to a misinterpretation of the reference state of the literature data. Hetzer (1968) reported asymmetric molality based equilibrium constant values for the protonation of PZ. In order to fit model values to literature data the following conversion was used:

$$\ln K^*_{PZH^+} = \ln K_{PZH^+} + \ln\left(\frac{1000}{MW}\right) + \ln \gamma^{\infty}_{PZ} \tag{3.60}$$

where: MW is the molecular weight of water

K_{PZH+} is obtained from Aspen Plus®

Equation 3.60 implied a correction to molality based and to the asymmetric reference state. However by defining PZ as Henry's component the Aspen Plus® extracted equilibrium constant has the same reference state as the literature data and no additional correction is needed.

This error will need to be solved but 5deMayo is deemed acceptable for modeling since speciation is adequate. Nevertheless, it will affect any pH predictions. Kinetics will not be affected since the model uses a reaction set with no hydrogen ions explicitly present as will be seen in the next section.

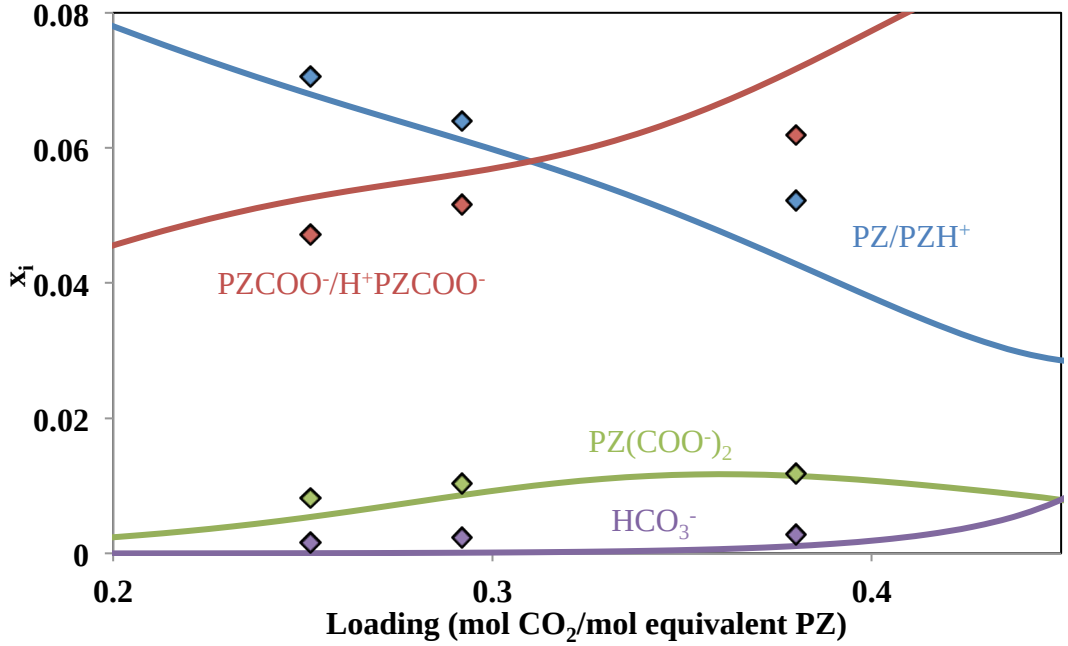


Figure 3.42: NMR speciation for 8 m PZ T=40°C. Lines are from 5deMayo. Points from Nguyen (Rochelle et al., 2011)

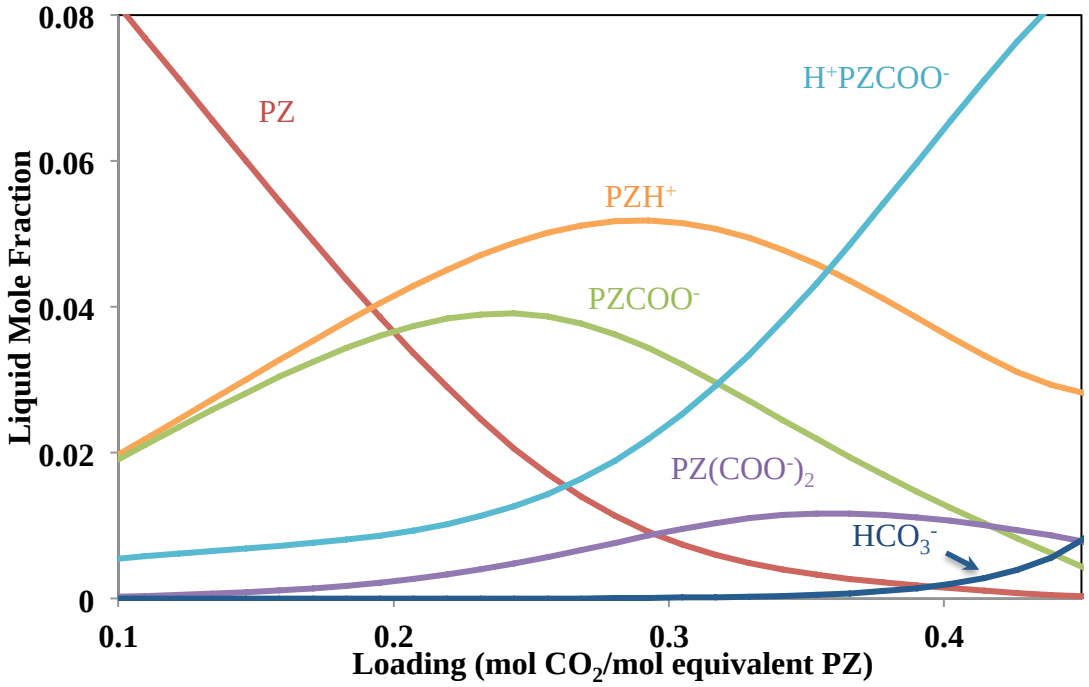


Figure 3.43: 5deMayo predicted speciation for 8 m PZ at 40°C.

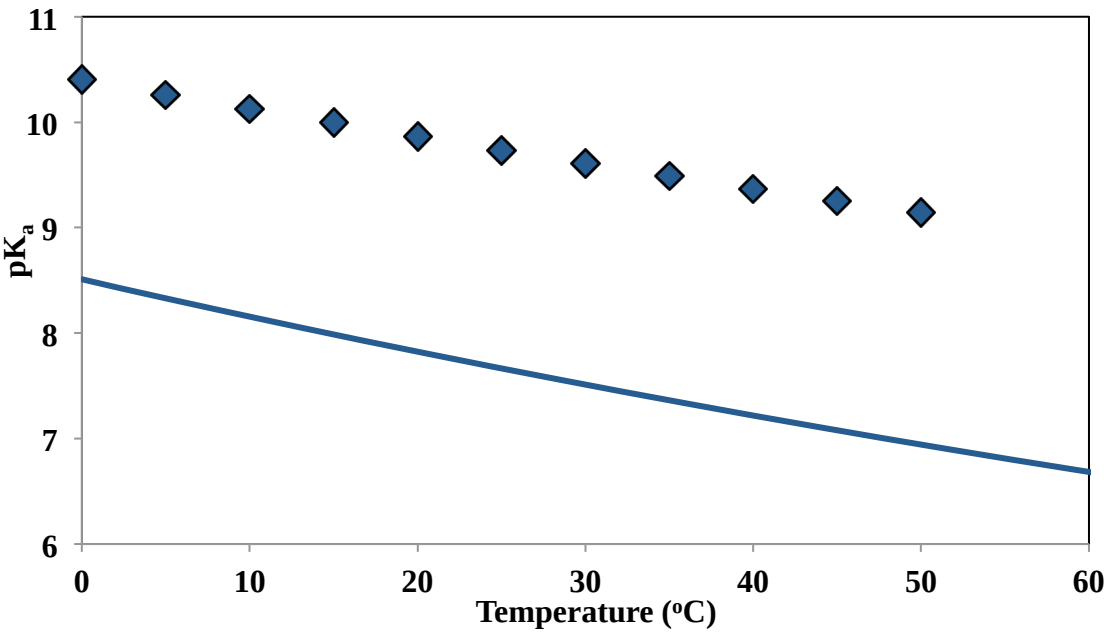


Figure 3.44: Asymmetric, molality based pK_a for PZ. Line is 5deMayo calculated. Points by Hetzer (1968)

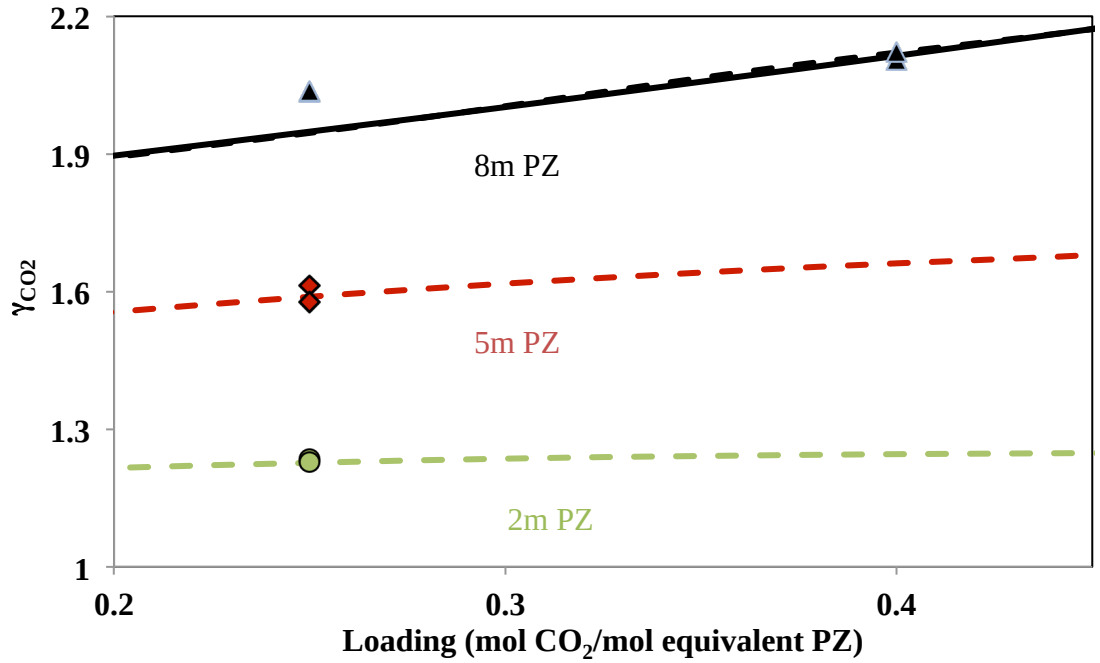


Figure 3.45: Activity coefficient fit for CO_2 at 25°C. Solid lines using Equation 3.40. Dashed lines using 5deMayo. Points from NTNU.

The activity of CO_2 was fit to data provided by Svendsen's research group at The Norwegian University of Science and Technology (NTNU) for the Henry's constant of N_2O in 2m , 5m and 8m PZ. Using the N_2O analogy to CO_2 and the Henry's constant for both gases in water the following equation was regressed:

$$\ln \gamma_{CO_2} = a + b[PZ] + cX^T_{CO_2} + d\left(\frac{1}{T} - \frac{1}{298.15}\right) \quad (3.59)$$

Where: [PZ] is the concentration of PZ in molality

X^T_{CO2} is the total concentration of CO_2. Calculated as the CO_2 present in the solvent as CO_2, $PZCOO^-$, $PZ(COO^-)_2$, HPZCOO and HCO^-_3

T is the temperature in K

a = 0.04477

b = 0.06081

c = 0.03393

d = 364.19

Figures 3.45 and 3.46 show the fit to the limited available data. This data was extrapolated for loading from 0.1 to 0.5 and used to fit the 5deMayo model in Aspen Plus®

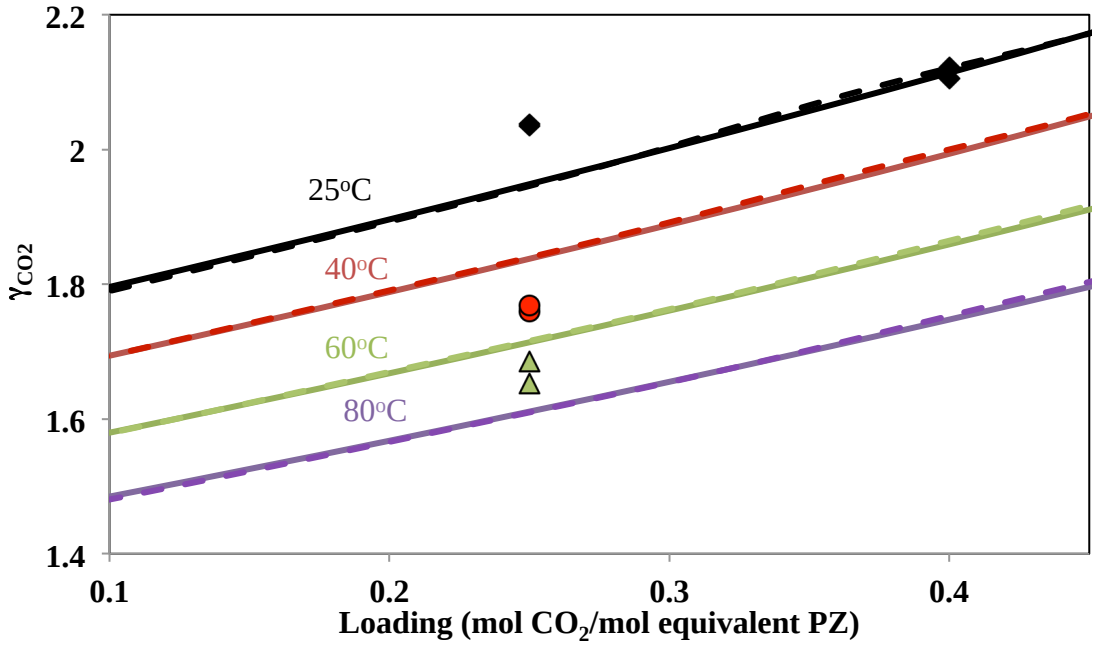


Figure 3.46: Activity coefficient fit for CO_2 for 8m PZ. Solid lines using Equation 3.40. Dashed lines using 5deMayo. Points from NTNU.

3.4.3. 5deMayo Kinetics

The "5deMayo"' thermodynamic framework was used to set up a wetted wall column (WWC) model to obtain kinetic constants for 8 m PZ. The WWC was implemented as described in section 3.3.3.1. Table 3.9 shows the kinetic expressions included in the 8 m PZ model. The values for the bicarbonate formation reaction were determined using a method similar to the one used for MEA. Equation 3.25 was used to obtain the reference rate constant (k_o) using the pK_b for $PZCOO^-$ (4.49). A similar conversion for the constant is required:

$$k_f^a = \frac{k_f^c \rho_s^2}{\gamma_{PZCOO^-}\gamma_{CO_2}} \tag{3.60}$$

Where: k_f^a is the forward reaction rate in activity basis, kmol/m^3-s

k_2^c is the forward reaction rate concentration based, m^3/gmol-s

ρ_s is the molar density of the PZ solvent in kmol/m^3 at 25°C

γ_{PZCOO^-} , γ_{CO2} are the activity coefficients of piperazine carbamate and CO_2 at 25°C and a loading of 0.30.

Equation 3.60 is defined in terms of the carbamate activity coefficient due to the proposed bicarbonate formation reaction.

$$k^f_{HCO_3^-} = 21951.76\, e^{-\frac{49000}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} \tag{3.61}$$

Where: $k^f_{HCO3^-}$ is the forward reaction rate, kmol/m^3-s

21951.76 kmol/m^3-s is the reference state rate constant from equations 3.25 and 3.41

T is the temperature of the point.

298.15 is the reference temperature. It corresponds to the temperature for the reference state constant

As in the Phoenix model, the reference state constant was used along with the equilibrium constants to determine the reverse reference constant for the bicarbonate reaction.

$$k^r_{HCO_3^-}\Big|_{298.15} = \frac{21951.76}{K_{HCO_3^-}} \tag{3.62}$$

Where: $k^r_{HCO3}|_{298.15}$ is the reverse bicarbonate formation rate constant at 298.15 K (kmol/m^3-s).

K_{HCO3^-} is the equilibrium constant for the bicarbonate reaction evaluated at 298.15 K

$$K_{HCO_3^-} = \frac{a_{H^+PZCOO^-}a_{HCO_3^-}}{a_{PZCOO^-}a_{CO_2}a_{H_2O}} \tag{3.63}$$

Table 3.8: 5deMayo Regressed Parameters and standard deviations Fraile (2011)

Parameter	Species	Std. Dev.	Units
$Cp^o/1$	PZ	1.20E+05	J/kmol-K
$Cp^o/2$		350	J/kmol-K
$\tau_{mm}/1$	H_2O/PZ	1.80	N/A
$\tau_{mm}/1$	PZ/H_2O	0.10	N/A
$\tau_{mm}/3$	H_2O/PZ	0.018	N/A
Henry/1	PZ/H_2O	0.96	N/A
Henry/2		310	K
ΔG^o_f	H^+PZCOO^-	1.20E+05	J/kmol
ΔH^o_f		3.70E+10	J/kmol
$Cp^o/1$	H^+PZCOO^-	8,610	J/kmol-K
$Cp^o/2$	H^+PZCOO^-	0.0028	J/kmol-K
ΔG^o_f	$PZCOO^-$	2.60E+05	J/kmol
ΔH^o_f	$PZCOO^-$	3.70E+10	J/kmol
$Cp^{\infty,aq}/1$	$PZCOO^-$	8,278	J/kmol-K
$Cp^{\infty,aq}/2$	$PZCOO^-$	0.0052	J/kmol-K
ΔG^o_f	$PZ(COO^-)_2$	3.70E+10	J/kmol
ΔH^o_f	$PZ(COO^-)_2$	3.73E+10	J/kmol
$Cp^{\infty,aq}/1$	$PZ(COO^-)_2$	5,002	J/kmol-K
$Cp^{\infty,aq}/2$	$PZ(COO^-)_2$	0.0054	J/kmol-K
$Cp^{\infty,aq}/1$	PZH^+	11,700	J/kmol-K
$Cp^{\infty,aq}/2$	PZH^+	37.1	J/kmol-K
$\tau_{m/ca}/1$	$H^+PZCOO^-/(PZH^+,PZCOO^-)$	0.46	N/A
$\tau_{ca/m}/1$	$(PZH^+,PZCOO^-)/ H^+PZCOO^-)$	0.04	N/A
$\tau_{m/ca}/1$	$H^+PZCOO^-/(PZH^+,PZ(COO^-)_2)$	0.57	N/A
$\tau_{ca/m}/1$	$(PZH^+,PZ(COO^-)_2)/ H^+PZCOO^-)$	0.2	N/A
$\tau_{m/ca}/1$	$H^+PZCOO^-/(PZH^+,HCO_3^-)$	0.44	N/A
$\tau_{ca/m}/1$	$(PZH^+,HCO_3^-)/ H^+PZCOO^-$	0.29	N/A
$\tau_{mm}/1$	H_2O/ H^+PZCOO	0.018	N/A
$\tau_{mm}/1$	H^+PZCOO^-/H_2O	0.0029	N/A
$\tau_{mm}/2$	H_2O/ H^+PZCOO	5.9	K
$\tau_{mm}/2$	H^+PZCOO^-/H_2O	4.0	K
$\tau_{mm}/3$	H_2O/ H^+PZCOO	0.0083	N/A
$\tau_{mm}/5$	H_2O/ H^+PZCOO	0.021	N/A
$\tau_{mm}/5$	H^+PZCOO^-/H_2O	0.015	N/A

Where *a* is the activity for each specie. $a_i = \gamma_i x_i$ at equilibrium.

Following the same methodology in the Phoenix model the reverse rate energy of activation was defined. The final bicarbonate expressions are as follows:

$$r^f_{HCO_3^-} = 21951.76\, e^{-\frac{49000}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} a_{PZCOO^-} a_{CO_2} \tag{3.64}$$

$$r^r_{HCO_3^-} = 947.54\, e^{-\frac{84471}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} \frac{a_{H^+PZCOO^-} a_{HCO_3^-}}{a_{H_2O}} \tag{3.65}$$

Table 3.9: Kinetic expressions used in the 5deMayo model.

Reaction	Rate Expression
$2\,PZ + CO_2 \rightarrow PZH^+ + PZCOO^-$	r= $k_1 a_{PZ} a_{CO_2} a_{PZCOO^-}$
$PZH^+ + PZCOO^- \rightarrow 2\,PZ + CO_2$	r= $\frac{k_1}{K_{PZCOO^-}} \frac{a_{PZH^+} a^2_{PZCOO^-}}{a_{PZ}}$
$2\,PZCOO^- + CO_2 \rightarrow PZ(COO^-)_2 + H^+PZCOO^-$	r= $k_2 a^2_{PZCOO^-} a_{CO_2}$
$PZ(COO^-)_2 + H^+PZCOO^- \rightarrow 2\,PZCOO^- + CO_2$	r= $\frac{k_2}{K_{PZ(COO^-)_2}} a_{H^+PZCOO^-} a_{PZ(COO^-)_2}$
$PZCOO^- + CO_2 + H_2O \rightarrow HCO_3^- + H^+PZCOO^-$	r= $k_3 a_{PZCOO^-} a_{CO_2}$
$HCO_3^- + H^+PZCOO^- \rightarrow PZCOO^- + CO_2 + H_2O$	r= $\frac{k_3}{K_{HCO_3^-}} \frac{a_{H^+PZCOO^-} a_{HCO_3^-}}{a_{H_2O}}$
$PZ + H^+PZCOO^- \leftrightarrow PZH^+ + PZCOO^-$	Equilibrium*

* **Reaction 4 is included as an equilibrium reaction.**

3.4.3.1. Diffusion Coefficients in the 5deMayo model

The diffusion of ion species in Aspen Plus® is calculated using the Nernst-Hartley model.

$$D_i = \left(\frac{RT}{z_i F^2}\right)(l_1 + l_2 T) \sum x_k \tag{3.66}$$

Where: D_i is the diffusion coefficient of ion *i*

R is the gas constant

F is Faraday's number the (9.65×10^7 C/kmol)

z_i is the charge number of specie i

x_k is the mole fraction for any molecular specie present.

T is the temperature (K)

l_1 and l_2 are the regressed parameters. (IONMOB/1 and IONMOB/2 in Aspen Plus®). These parameters are also used in the Jones-Dole electrolyte correction for viscosity. However, since viscosity in the model is calculated using a FORTRAN subroutine, these parameters are not used and are set at the default value (l_1=5 , l_2=0). In Equation 3.47 the charge (z_i) is in the denominator so a small charge number such as the one used for H^+PZCOO^- in Chen (2007) generated a large D_i for this component in the diffusion matrix generating inaccuracies in the final model. This issue was solved in the 5deMayo model by defining the zwitterion as a Henry's component with charge zero. Aspen Plus® defines molecule diffusion coefficients using the Wilke-Chang model with Vignes correction for molecules (Poling et al., 2001):

$$D_i^o = 1.17282 \times 10^{-16} \frac{(\phi M)^{0.5} T}{\mu_{sln} V_i^{0.6}} \tag{3.67}$$

Where: D^o_i is the diffusion of specie i in the liquid

μ_{sln} is the viscosity of the solution (Pa-s)

V_i is the molar volume of i (m^3/mol)

T is temperature (K)

ϕM is calculated using the following equation:

$$\phi M = \frac{\sum_{j \neq i} x_j \phi_j M_j}{\sum_{j=i} x_j} \tag{3.68}$$

x_j is the mole fraction of j (solvent non diffusing component)

ϕ_j is the association factor of the solvent. 2.26 for water. 1 for all other solvents.

M_j molecular weight of j

Dugas (2009) studied diffusion of loaded 8 m PZ using a diaphragm diffusion cell. He reported an integral diffusion coefficient value for loadings between 0.25-0.29 ($1.2x10^{-10}$ m^2/s) at 30°C. This value is an effective diffusion of all species in solution. Evaluation of the values predicted by the model using the described correlations determined them adequate considering the available data thus no further modifications were made.

3.4.3.2. Carbamate and di-carbamate formation in the 5deMayo model

Table 3.9 shows the reactions for the formation of carbamate and di-carbamate included in the 5deMayo model. The rate constants for these reactions were determined using the data by Dugas (2009) using a method similar to the described for the carbamate formation in the Phoenix MEA model (Section 3.3.3.4). However, to reduce the number of unknowns (2 per forward and reverse reaction for a total of 8) a ratio between carbamate and the dicarbamate kinetics was used based on regression work by Cullinane (2005). The di-carbamate formation constant was made equal to 95% of the carbamate formation value. This was entered into the WWC model in Aspen Plus® and the flux was matched following the procedure outlined in section 3.3.3.4 including the loading corrections. The data was regressed as a single data set that includes high loading and high temperature points. No additional diffusion correction was applied. The resulting kinetic expressions are as follows:

$$r^f_{PZCOO^-} = 3\times10^9\, e^{-\frac{32042}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} a_{PZ} a_{PZCOO^-} a_{CO_2} \qquad (3.69)$$

$$r^{r}_{PZCOO^-} = 1843\, e^{-\frac{102455}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} \frac{a_{PZH^+} a^2_{PZCOO^-}}{a_{PZ}} \quad (3.70)$$

$$r^{f}_{PZ(COO^-)_2} = 2.85\times10^9\, e^{-\frac{32042}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} a^2_{PZCOO^-} a_{CO_2} \quad (3.71)$$

$$r^{r}_{PZ(COO^-)_2} = 1854\, e^{-\frac{940594}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} a_{H^+PZCOO^-} a_{PZ(COO^-)_2} \quad (3.72)$$

where: r_{PZCOO-}, $r_{PZ(COO-)2}$ are the carbamate and di-carbamate rates respectively in kmol/s

Sub indexes *f* and *r* stand for the direction of the reaction (forward or reverse)

a_i is the activity of species *i*

T is the temperature in Kelvin

Figure 3.47 shows the final fit of the 5deMayo model against the WWC data by Dugas (2009). The model matches the WWC data within 20% of the reported values. As with the Phoenix model the fit is affected at high loadings (ldg >0.40) evidenced by the discrepancies between the fit in absorption and desorption points.

3.4.4. k'_g and 5deMayo

As with the Phoenix model, the WWC with 5deMayo was used to evaluate the behavior of kg' for 8m PZ. Figure 3.48 show kg' as a function of loading at various temperatures compared to data by Dugas (2009). Experimental data was well matched and the k_g' behavior is as expected: reduced with the increase in loading.

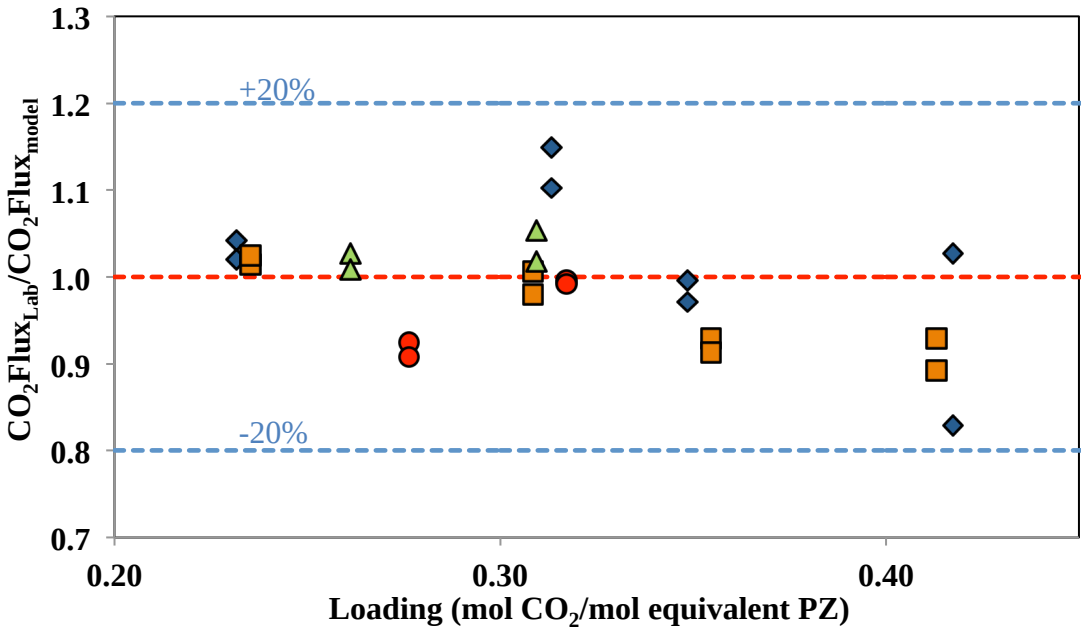


Figure 3.47: WWC Carbon dioxide flux comparison for 8 PZ. (♦) 40ºC, (■) 60ºC, (▲) 80ºC (●) 100ºC.

This phenomenon also results in an increase in the importance of the physical mass transfer (k^o_L) in the CO_2 absorption mechanism as loading and temperature are increased. Figure 3.49 shows the relative change in k_g' with respect to the change in k^o_L. PZ rates make diffusion of reactants and products important for this solvent for broader loading and temperature conditions than for MEA.

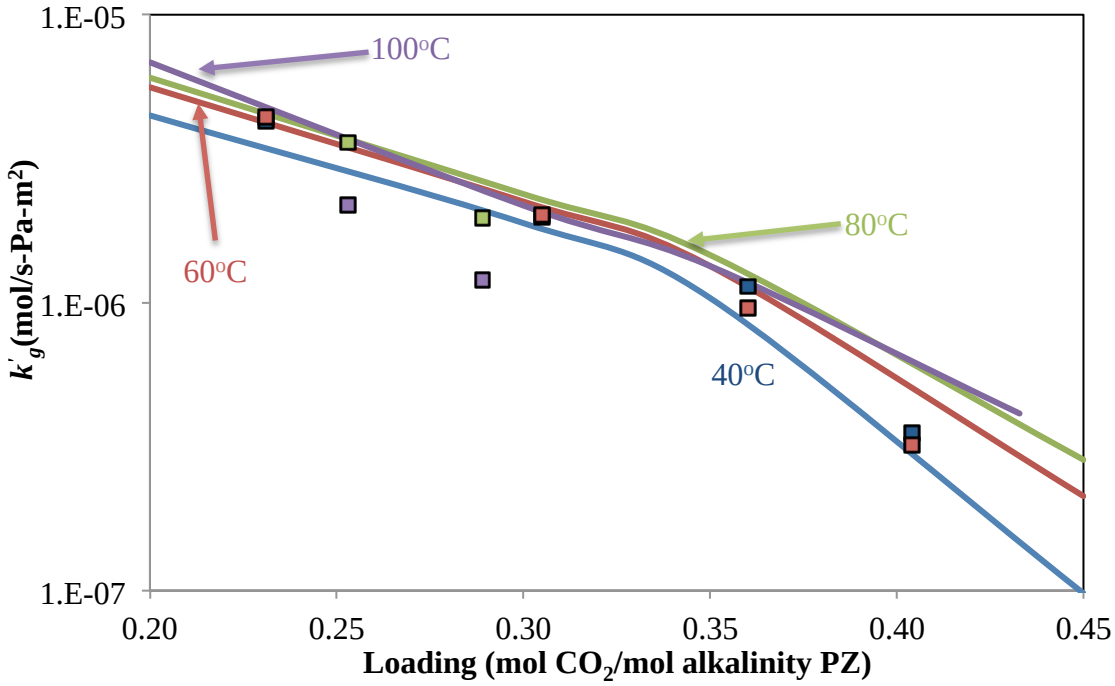


Figure 3.48: $k_g^{'}$ **data for 8m PZ. Points are from Dugas (2009). Lines are 5deMayo WWC predictions .**

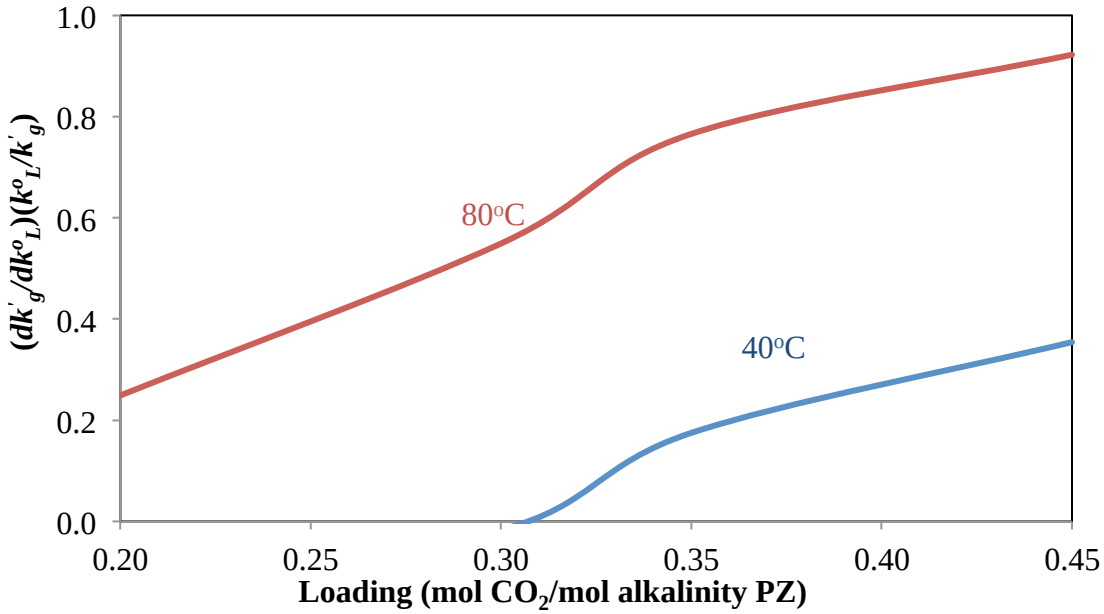


Figure 3.49: Relative change in $k_g^{'}$ with respect to the relative change in k_L^{o} at various loadings for 8m PZ using 5deMayo WWC.

4

Chapter 4: Pilot Plant Studies

A number of pilot plant studies have been used to validate model performance for CO_2 absorption systems (Wilson et al., 2004; Idem et al., 2006; Chen, 2007; Thiele et al., 2007; Luo et al., 2008; Plaza et al., 2008). The models developed in this work will be tested and validated using pilot plant data generated at the J.J. Pickle Research Center of The University of Texas at Austin.

This section presents the results of 5 pilot plant campaigns for concentrated MEA and PZ. These campaigns served to demonstrate solvent performance (in the case of PZ) and to study absorption performance using intercooling.

4.1. Pilot Plant Description

Figures 4.1 – 4.3 show the basic PFD of the J.J Pickle Pilot Plant. The system consists of two twin columns with a diameter of 42.7 cm and a packing height of 6.10 m divided into two 3.05m sections. The system allows for testing of various packing, flow, temperature and loading conditions Two major modifications were included at the plant during this work. An intercooling stage was installed in the absorber between the two packing sections and a two stage flash system replaced the stripper. Figure 4.4 shows the intercooling loop. Illustrations 4.1 and 4.2 show images of the pilot plant and the installed intercooling stage.

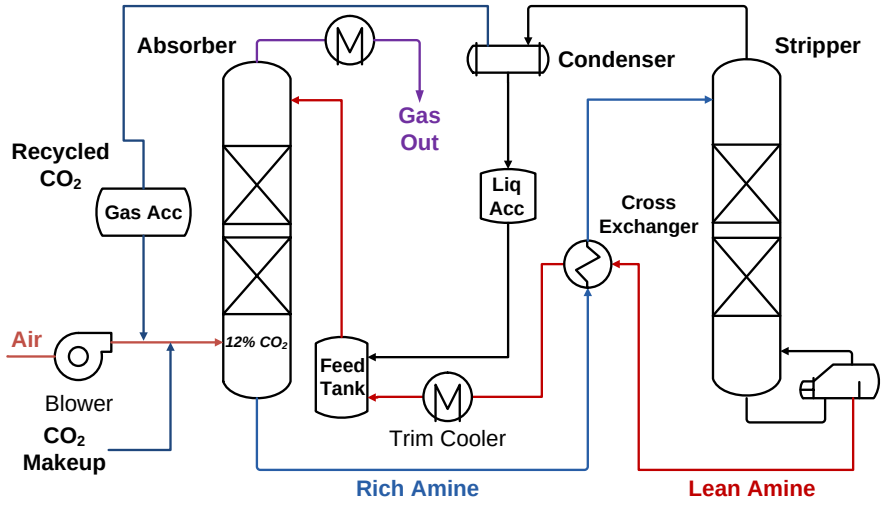


Figure 4.1: PFD for the J.J Pickle Research Center Pilot Plant for the October'07 and November'08 Campaigns. ***(Courtesy of Dr. Eric Chen)***

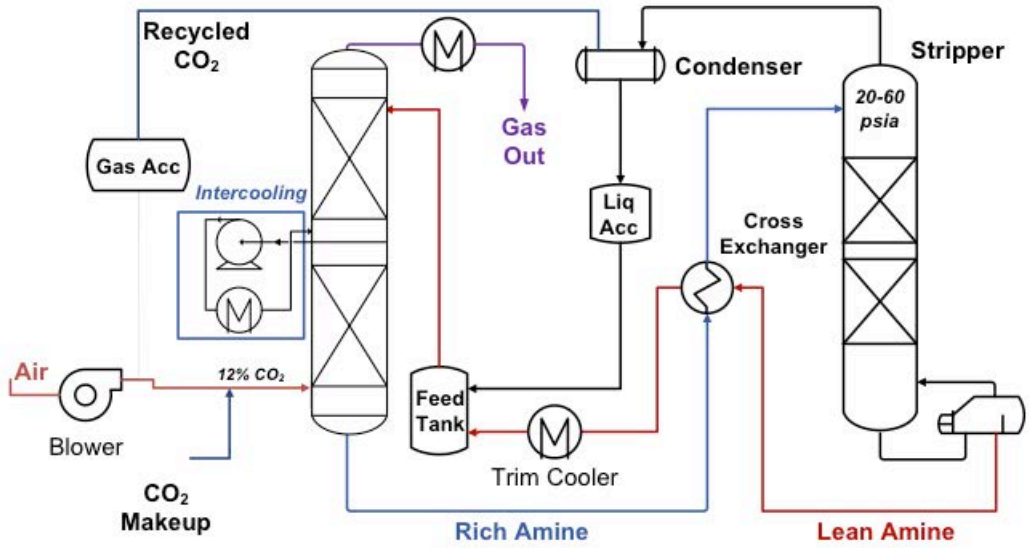


Figure 4.2: PFD for the J.J Pickle Research Center Pilot Plant for the September'10 Campaigns. ***(Courtesy of Dr. Eric Chen)***

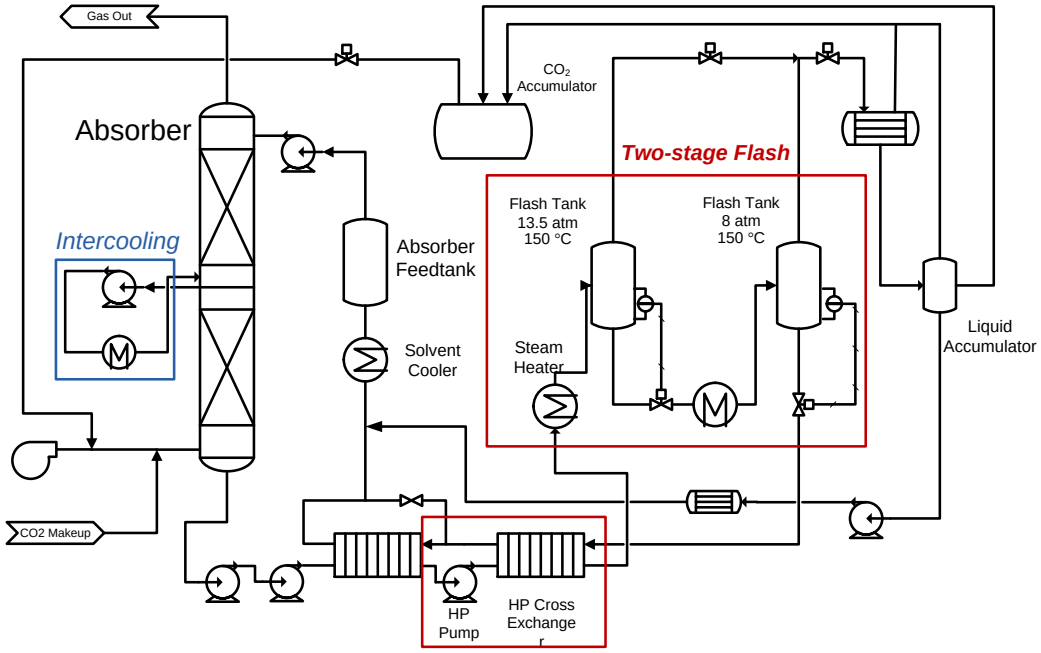


Figure 4.3: PFD for the J.J Pickle Research Center Pilot Plant for the December'10 Campaign. ***(Courtesy of Dr. Eric Chen)***

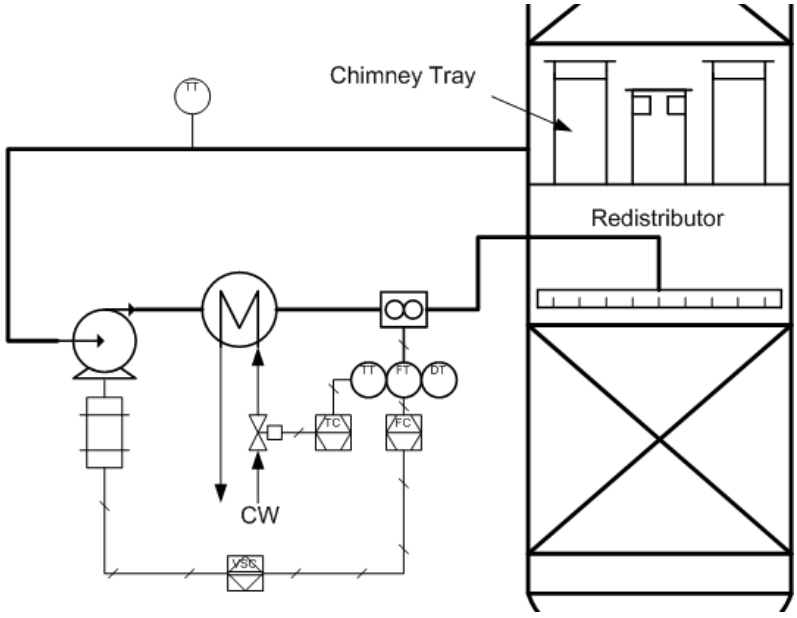


Figure 4.4: PFD for the absorber intercooling loop for the September'10 and December'10 Campaigns. ***(Courtesy of Dr. Eric Chen)***



Illustration 4.1: J.J Pickle Research Center Pilot Plant. ***(Courtesy of Dr. Eric Chen)***



Illustration 4.2: Absorber Intercooling loop ***(Courtesy of Dr. Eric Chen)***

In the pilot plant air is mixed with pure CO_2 coming from a storage tank (make up) and from the stripper to create a synthetic flue gas with 12% CO_2. The gas enters the bottom of the column and contacts the solvent coming down from the top. The exiting lean gas is cooled to remove water and any volatile amine. The rich solvent leaves the bottom of the absorber and is pumped through a cross heat exchanger (plate and frame) to be preheated before entering the stripper (Figure 4.1 - 4.2). The December'10 campaign regenerates the solvent using a two-stage flash (Figure 4.3) so the solvent pressure is increased using two pumps and goes through a low and high-pressure cross heat exchanger. The pressurized solvent enters a steam heater on its way to the flash tank train.

The returning stripped solvent passes through the cross heat exchanger (two of them in the December campaign) enters a solvent trim cooler and is collected in a tank before returning to the absorber.

The September'10 and December'10 campaigns also included an intercooling loop in the absorber (Figures 4.2 to 4.4). Solvent was collected in a modified chimney tray and pumped into a plate and frame heat exchanger to reduce its temperature to 40°C (Illustration 4.2). During these two campaigns solvent flowed through the intercooling loop at all times. Opening the cooling water valve to the heat exchanger controlled intercooling.

4.2. PILOT PLANT MEASUREMENTS

Temperature, pressure, flow rates and CO_2 and amine concentration were measured and reported by the pilot plant for gas and liquid streams. The first four are connected to the Delta V control system at the pilot plant (density for CO_2 concentration) and are collected over a period of time at steady state.

Temperature is measured using Platinum Resistance Temperature Detectors (RTDs) located through out the column. Their measuring capabilities are limited by how much they can penetrate the column without perforating the packing. This places them closer to the absorber wall. The first campaign had 6 temperature measurement points located as shown in Figure 4.5(a) (Dugas, 2006). For the rest of campaigns additional RTD's were installed in the column. (Figures 4.5(b) and Figure 4.6)

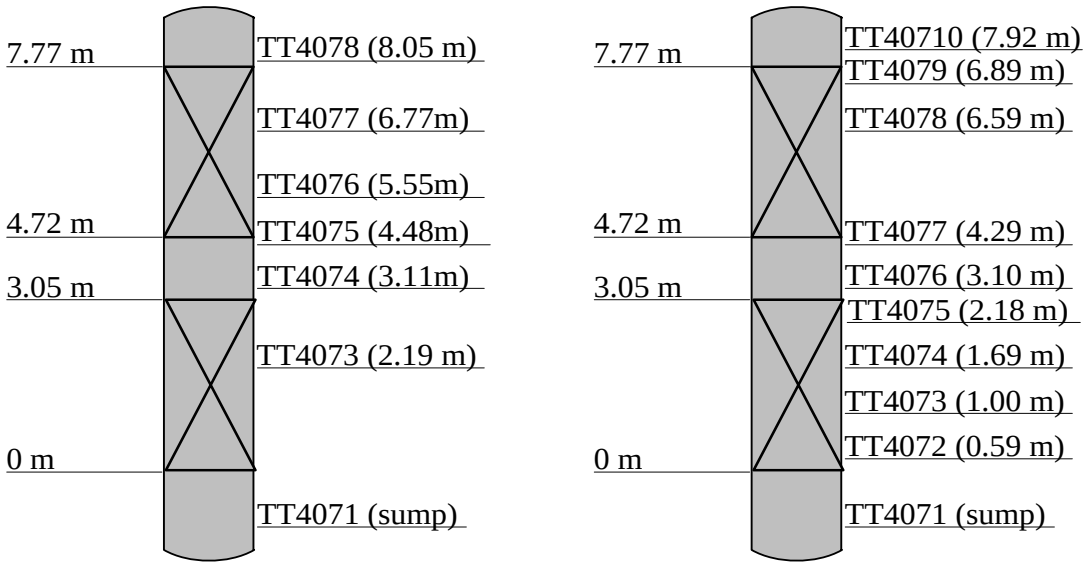


Figure 4.5: Absorber temperature measurement locations: a) October'07 campaign (Dugas, 2006). b) November'08 campaign.

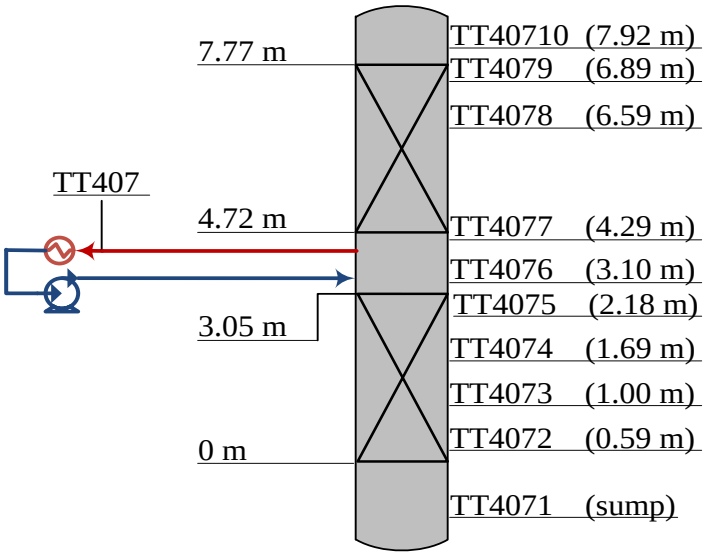


Figure 4.6: Absorber temperature measurement locations for the September'10 and December'10 campaigns.

Liquid flow was measured using Micro Motion® coriolis flow meters upstream the absorber and downstream the stripper for the lean solvent flow and downstream of the absorber for the rich solvent. Density and temperature were also taken at these points and used along with correlations as a function of temperature and specific gravity to estimate loading. These correlations were determined through outside experiments and pilot plant operation and are as follows:

For 35wt % MEA:

$$x_{CO_2} = \big((a\,SG - b)c + dT - e\big)f - g \tag{4.1}$$

For 42wt% PZ:

$$x_{CO_2} = \left(-a + bSG - (cSG^2) - dT + e\frac{T}{SG}\right)f \tag{4.2}$$

where: x_{CO2} is the weight fraction of CO_2 in the solvent

SG is the specific gravity of the solvent measured by the flow meter at the location of analysis

T is in degree Fahrenheit

The constants for equations 4.1 and 4.2 can be found in Table 4.1. There is a slight adjustment depending on the location of the sample.

Loading was also determined by titration using methanol and potassium hydroxide. Samples were taken from the absorber lean feed and rich outlet. In the September'10 PZ campaign liquid samples were also taken from the middle of the column.

Gas CO_2 content was determined using two online Vasala analyzers. The inlet measurement is located downstream of the air blower and the outlet downstream of the cooler and water knockout drum.

Table 4.1: Constants in Eq. 4.1 and 4.2 to measure CO_2 content in solvent.

Amine	Location	*a*	*b*	*c*	*d*	*e*	*f*	*g*
MEA (35 wt.%)	Lean	1028.5	1031.7	0.001	0.00036	0.0374	0.9	0.001
	Rich							0.0015
	Stripper							0.001
PZ (42 wt.%)	Lean	-4.32304	6.536	-2.329	-0.00173	0.00229	0.915	----
	Rich						0.94	
	Stripper						0.91	

Pressure drop across each packing section was measured using Rosemount pressure transmitters. There are three transmitters in the absorber. Two measure the

lower packing pressure drop with different measurement range so one reading is selected based on its value.

4.2.1. Loading Measurement

As mentioned in the previous section, loading is obtained at various locations in the pilot plant using the readings from the flow meter and through sampling followed by titration. Of these methods the latter is considered the most accurate. The online measurement is used more as an operation guide. Figures 4.7 to 4.10 show the parity plots for each of the campaigns. There are 4 lean loading determinations: two calculated from titration measurements, one from the absorber side and one downstream from the stripper; and two calculated using Equation 4.1 and data from density meters upstream of the absorber and downstream of the stripper (Tables 4.2, 4.4, 4.6 and 4.8). Rich loading is calculated from a sample taken downstream of the absorber and calculated from data from a density meter downstream of the absorber (Tables 4.3, 4.5, 4.7 and 4.9). All of these were taken into consideration to specify the variations allowed to loading during the data reconciliation.

In the November'08 campaign the last four runs had concentrations lower than 8m PZ (approximately 5 m). Equation 4.2 was developed for 8 m PZ so a higher deviation from titration data was observed.

Figure 4.7 and Tables 4.2 and 4.3 show that for the PZ November'08 campaign most measurements are within ± 10% of the absorber titration results. Stripper lean titration shows the most deviation from the absorber lean data.

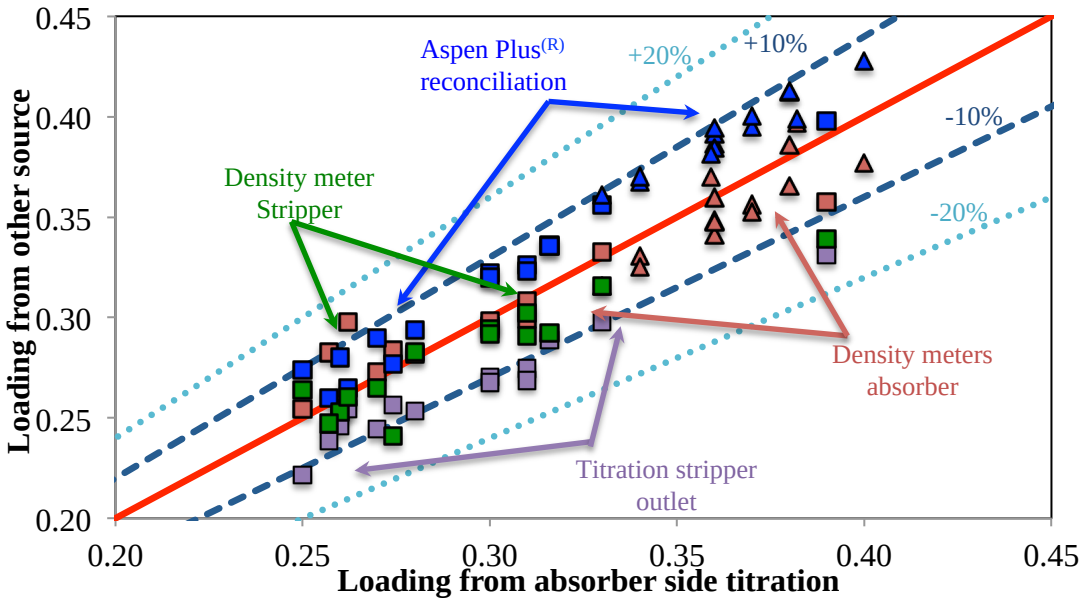


Figure 4.7: Parity Plot loading for the PZ November'08 Campaign. (■) Lean loading (▲) Rich loading

Table 4.2: Lean loading deviations from absorber titration for PZ Nov'08

Run	Titration Absorber Loading	Deviation (measured – titration)			
		Titration Stripper	Density meter		Aspen Reconciliation
			Absorber	Stripper	
1	0.26	-0.01	-0.01	-0.01	0.02
2	0.31	-0.04	-0.002	-0.01	0.02
3	0.25	-0.03	0.005	0.01	0.02
4	0.39	-0.06	-0.03	-0.05	0.01
5	0.28	-0.03	0.002	0.003	0.01
6	0.30	-0.03	-0.004	-0.01	0.02
7	0.31	-0.04	-0.01	-0.02	0.01
8	0.30	-0.03	-0.002	-0.01	0.02
9	0.27	-0.03	0.003	-0.01	0.02
10	0.33	-0.03	0.003	-0.01	0.03
11	0.32	-0.03	0.02	-0.02	0.02
12	0.27	-0.02	0.01	-0.03	0.003
13	0.26	-0.02	0.03	-0.01	0.003
14	0.26	-0.01	0.04	-0.002	0.003
Average		-0.03	0.003	-0.01	0.02
Std. Deviation		0.01	0.02	0.02	0.01

In the MEA September'10 campaign the density meter loading measurements show the largest deviation when compared to the titration results (Figure 4.8 and Tables 4.4 and 4.5. This might be indicative of a calibration issue with the correlation being used for determining loading for MEA (Equation 4.1). Since the MEA inventory had also been heavily used in other campaigns, the density offset may be indicative of solvent degradation.

Table 4.3: Rich loading deviations from absorber titration PZ Nov'08

Run	Titration Absorber Loading	Deviation (measured – titration)	
		Density Absorber	Aspen Reconciliation
1	0.34	-0.01	0.03
2	0.37	-0.01	0.02
3	0.33	-0.01	0.03
4	0.4	-0.02	0.03
5	0.36	-0.01	0.03
6	0.36	-0.02	0.03
7	0.36	-0.01	0.03
8	0.37	-0.02	0.03
9	0.34	-0.02	0.03
10	0.38	-0.01	0.03
11	0.38	0.01	0.03
12	0.36	-0.0003	0.02
13	0.359	0.01	0.02
14	0.382	0.01	0.02
Average		-0.01	0.03
Std. Deviation		0.01	0.005

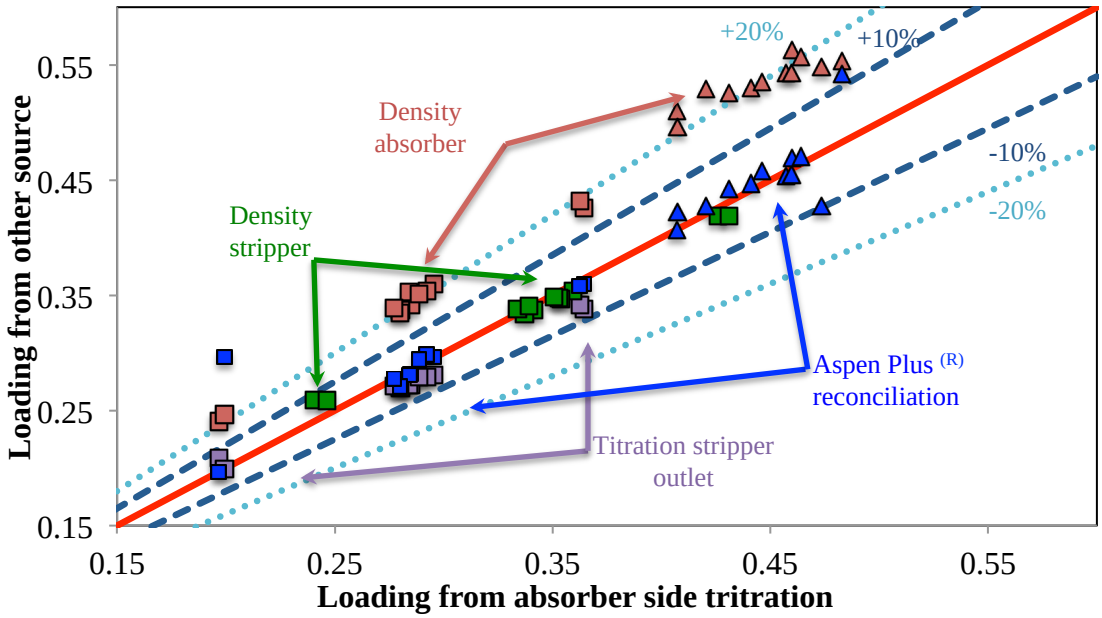


Figure 4.8 Parity Plot loading for the MEA September'10 Campaign. (■) Lean loading (▲) Rich loading

Table 4.4: Lean loading deviations from absorber titration MEA Sept'10

Run	Titration Absorber Loading	Deviation (measured – titration)			
		Titration Stripper	Density		Aspen Reconciliation
			Absorber	Stripper	
1	0.28	-0.01	0.06	0.06	0.001
2	0.28	-0.01	0.06	0.05	0.01
3	0.27	-0.01	0.06	0.05	-0.003
4	0.27	-0.01	0.06	0.05	-0.01
5	0.27	-0.01	0.07	0.06	-0.004
6	0.28	-0.01	0.06	0.06	0.01
7	0.27	-0.01	0.05	0.06	-0.01
8	0.27	-0.01	0.06	0.06	0.0004
9	0.34	-0.03	0.06	0.05	-0.005
10	0.34	-0.02	0.07	0.06	-0.004
11	0.20	-0.0003	0.05	0.06	0.10
Average		-0.01	0.06	0.06	0.01
Std. Deviation		0.01	0.01	0.004	0.03

Table 4.5: Rich loading deviations from absorber titration MEA Sept'10

Run	Titration Absorber Loading	Deviation (measured – titration)	
		Density meter Absorber	Aspen Reconciliation
1	0.42	0.11	0.01
2	0.43	0.09	0.01
3	0.46	0.09	-0.004
4	0.46	0.08	-0.01
5	0.41	0.10	-0.001
6	0.41	0.09	0.01
7	0.44	0.09	0.01
8	0.45	0.09	0.01
9	0.46	0.10	0.01
10	0.46	0.09	0.01
11	0.47	0.07	-0.05
Average		0.09	0.001
Std. Deviation		0.01	0.02

Comparison of the PZ September10 and December'10 campaigns (Figures 4.9 - 4.10 and, Tables 4.6 -4.9) show an increase of the deviation between titration results and flow meter readings with the later campaign. Deviations for September were still within ±10% while the December campaigned showed deviations in the rich side within ±20 and more data dispersion. The latter campaign had an additional density meter that provided data to calculate the loading of the solvent reaching the stripping system for regeneration (Figure 4.10 and Table 4.9)

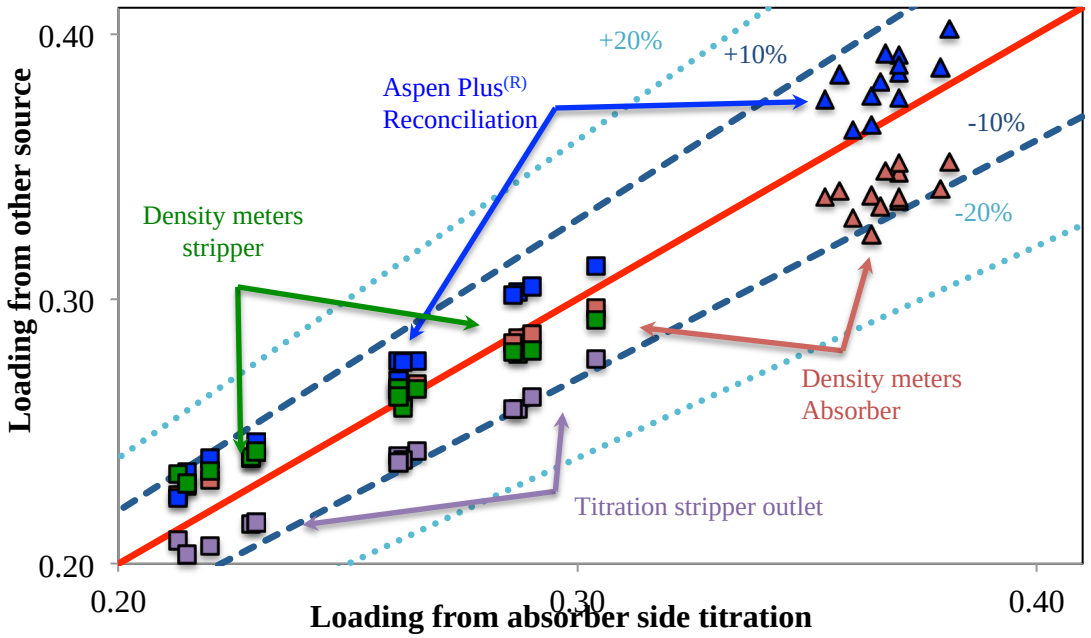


Figure 4.9 Parity Plot loading - PZ September'10 Campaign (■) Lean loading (▲) Rich loading

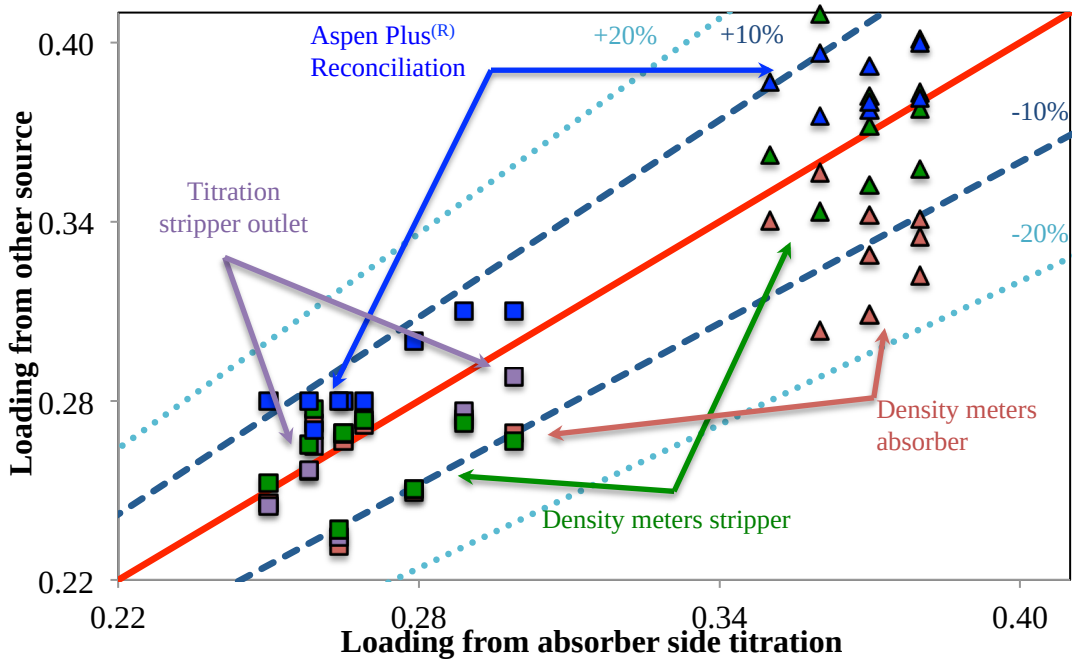


Figure 4.10 Parity Plot loading for the PZ December'10 Campaign (■) Lean loading (▲) Rich loading

Table 4.6: Lean loading deviations from absorber titration PZ Sept'10

Run	Titration Absorber Loading	Deviation (measured – titration)			
		Titration Stripper	Density meter		Aspen Reconciliation
			Absorber	Stripper	
1	0.29	-0.03	-0.002	-0.01	0.02
2	0.29	-0.03	-0.003	-0.01	0.02
3	0.29	-0.03	-0.003	-0.01	0.01
4	0.30	-0.03	-0.01	-0.01	0.01
5	0.27	-0.02	0.003	0.001	0.01
6	0.26	-0.02	0.01	0.01	0.02
7	0.26	-0.02	0.001	-0.003	0.01
8	0.26	-0.02	0.005	0.002	0.01
9	0.21	0.00	0.01	0.02	0.01
10	0.22	-0.01	0.01	0.02	0.02
11	0.22	-0.01	0.01	0.02	0.02
12	0.23	-0.01	0.01	0.01	0.01
13	0.23	-0.01	0.01	0.01	0.02
Average		-0.02	0.005	0.003	0.01
Std. Deviation		0.01	0.01	0.01	0.004

Table 4.7: Rich loading deviations from absorber titration PZ Sept'10

Run	Titration Absorber Loading	Deviation (measured – titration)	
		Density meter Absorber	Aspen Reconciliation
1	0.37	-0.03	0.01
2	0.37	-0.02	0.02
3	0.36	-0.02	0.01
4	0.37	-0.02	0.02
5	0.36	-0.03	0.004
6	0.37	-0.02	0.03
7	0.36	-0.02	0.03
8	0.35	-0.02	0.02
9	0.37	-0.03	0.02
10	0.38	-0.03	0.02
11	0.37	-0.03	0.02
12	0.36	-0.04	0.00
13	0.38	-0.04	0.01
Average		-0.03	0.01
Std. Deviation		0.01	0.01

Table 4.8: Lean loading deviations from absorber titration PZ Dec'10

Run	Titration Absorber Loading	Deviation (measured – titration)			
		Titration Stripper	Density meter		Aspen Reconciliation
			Absorber	Stripper	
1	0.27	0.004	0.002	0.004	0.02
2	0.26	0.01	0.01	0.02	0.01
3	0.25	-0.01	-0.004	0.002	0.03
4	0.26	-0.001	-0.002	0.01	0.02
5	0.26	-0.03	-0.03	-0.03	0.02
6	0.28	-0.03	-0.03	-0.03	0.02
7	0.30	-0.01	-0.03	-0.03	0.01
8	0.29	-0.01	-0.02	-0.02	0.02
9	0.27	0.005	0.003	0.004	0.01
Average		-0.01	-0.01	-0.01	0.02
Std. Deviation		0.01	0.02	0.02	0.01

Table 4.9: Rich loading deviations from absorber titration PZ Dec'10

Run	Titration Absorber Loading	Deviation (measured – titration)		
		Density meter		Aspen Reconciliation
		Absorber	Stripper	
1	0.35	-0.01	0.01	0.04
2	0.36	-0.004	0.05	0.04
3	0.37	-0.04	0.002	0.02
4	0.38	-0.04	0.003	0.02
5	0.36	-0.06	-0.02	0.02
6	0.37	-0.06	-0.02	0.01
7	0.38	-0.06	-0.02	0.001
8	0.38	-0.04	0.00	0.02
9	0.37	-0.03	0.01	0.01
Average		-0.04	0.002	0.02
Std. Deviation		0.02	0.02	0.01

Comparison of lean loadings obtained through titration with the values obtained using density measurements and Equations 4.1 and 4.2 show the need to improve the correlations to determine loading. Discrepancies between the two methods are as high as 0.06 in some of the runs. On the rich side for MEA, discrepancy between titration and

density loading determination is much higher with an average deviation of 0.09 ± 0.01. For PZ all campaigns showed a negative deviation from titration that increased with time reaching -0.04 ± 0.01 for the PZ December campaign.

Lean loading titration of the stripper side samples show a negative deviation from the absorber inlet sample across all campaigns and solvents. This might be related to the difficulties in handling hot solvent samples without losing CO_2 content.

The lean loadings obtained through titration are fed into the model for data reconciliation in this study. Based on the loading analysis this work uses a multiplying factor to account for an initial deviation across all runs in each campaign as will be described in later sections. Summary of the loading deviations for the pilot campaigns studied are included in Tables 4.10 and 4.11

Table 4.10: Summary of lean loading deviations from pilot plant campaigns

Campaign	Titration	Density meter		Aspen
	Stripper	Absorber	Stripper	Reconciliation
PZ Nov'09	-0.03	0.003	-0.01	0.02
PZ Sept'10	-0.02	0.005	0.003	0.01
PZ Dec'10	-0.01	-0.01	-0.01	0.02
MEA Sept'10	-0.02	0.005	0.003	0.01

Table 4.11: Summary of rich loading deviations from pilot plant campaigns

Campaign	Density meter		Aspen
	Absorber	Stripper	Reconciliation
PZ Nov'09	-0.01	NA	0.03
PZ Sept'10	-0.03	NA	0.01
PZ Dec'10	-0.04	0.002	0.02
MEA Sept'10	0.09	NA	0.001

4.2.2. CO_2 Material Balance

Mass balance is important when evaluating the quality of the pilot plant data. All though effort is taken to close the material and energy balance of each run, losses are left unaccounted for and can be observed when comparing the CO_2 removal using gas side measurements to the removal calculated using loading and liquid flow rate. Losses can be attributed to water evaporation, equipment leaks and solvent volatility but are generally thought to be insignificant. Figures 4.11 to 4.14 show the parity plots obtained for CO_2 removal for the studied pilot plant campaigns. Values are compared against gas side removal calculated from gas composition and gas flow measurement considered the more trusted method. The absorbed CO_2 is calculated using the loadings from titrations of lean and rich solvent samples taken from the absorber side along with the flow rates and amine concentration Absorbed CO_2 is also determined using the loadings calculated using density measurements and Equations 4.1 and 4.2. The Aspen Plus® reconciled values assure mass and energy balance closure as it will be later explained. The energy balance will be studied in later sections using the model temperature results.

In the PZ November'08 campaign liquid side removal matches the gas side results within ±15% (Figure 4.11 and Table 4.12) except for runs 4 and 10. The deviation for run 4 may indicate issues with the loading calculation using titration for that run. The liquid side results are within ±15% (Figure 4.12 and Table 4.13) of the liquid side values for the MEA September'10 campaign. However, in the case of the other two PZ campaigns, the deviation from the gas side calculated removal is higher (within ±20%). Flow meter values present larger discrepancies from the titration data, mostly lower than the latter. There are higher discrepancies between gas side and the liquid titration removal and the flow meter values. Even higher were the discrepancies between new

data available from two density meters on the flash tank side. Values are all above 20% of the gas side removal.

CO_2 pilot plant data mass balance closure is on average overestimated 9± 4% for PZ using absorber side titration data against gas side calculated values. For MEA closure is underestimated 5±4% for the MEA September'10 campaign.

In this work the gas side removal was used as the input parameter for the data reconciliation as it will be explained in later sections.

Table 4.12: CO_2 removal deviations from gas side calculated value PZ Nov'08

Run	CO_2 removed Gas side (mol/hr)	Liquid Side Deviation (liquid – gas) -%		
		Liquid flowrate and loadings from:		Aspen Reconciled
		Titration	Eq. 4.1	
1	2629	8.7	16.1	0.2
2	2120	6.0	0.5	1.4
3	2901	13.3	0.6	0.7
4	964	-59.8	6.1	0.5
5	2225	7.3	4.1	1.8
6	2605	5.8	-6.2	0.4
7	2233	-14.9	-1.4	0.1
8	1956	7.4	-1.4	-0.02
9	2796	11.1	-3.9	0.6
10	1525	20.9	-1.7	5.6
11	1953	2.4	-9.1	-0.2
12	2865	-4.3	-5.7	-1.5
13	2658	0.7	-5.8	-1.5
14	2184	11.4	-0.3	-1.1
Average		1.1	-0.6	0.5
Std. Deviation		19.5	6.3	1.8

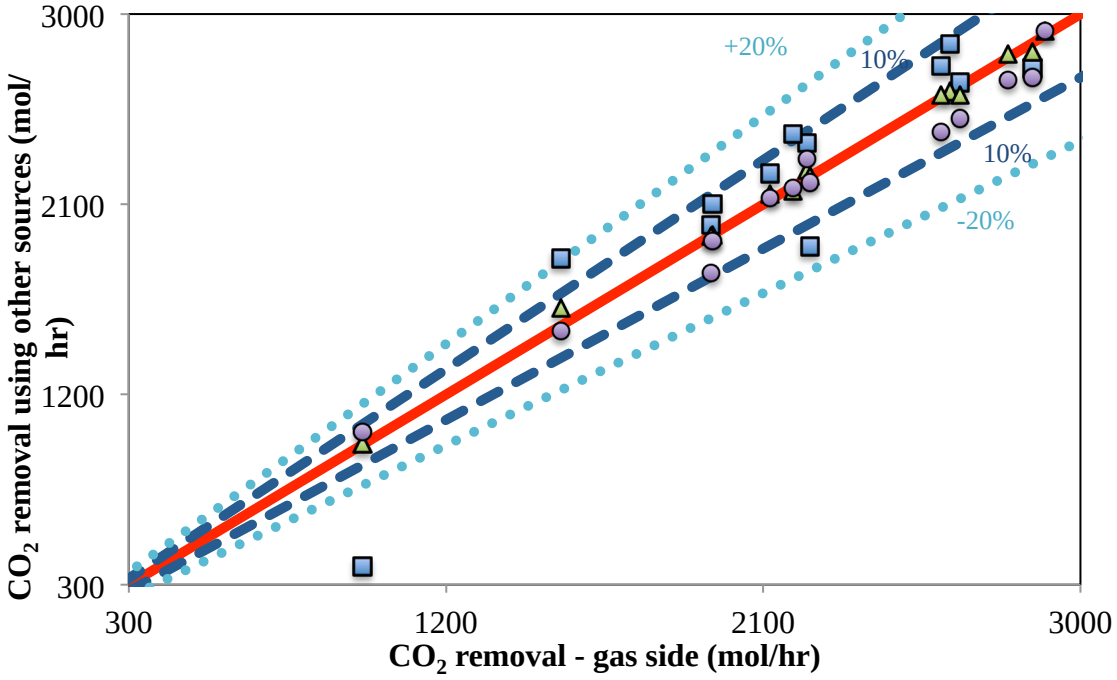


Figure 4.11 Parity plot removal for the PZ November'08. Compared to calculated value from gas side pilot plant data. (■) Liquid side titration, (●) Liquid Density meter values (▲) Aspen Plus® reconciliation

Table 4.13: CO_2 removal deviations from gas side calculated value MEA Sept'10

Run	CO_2 removed Gas side (mol/hr)	Liquid Side Deviation (liquid – gas) -%		
		Liquid flowrate and loadings from:		Aspen Reconciled
		Titration	Density meter	
1	2893	-4.1	9.5	0.2
2	3110	-2.3	1.9	0.2
3	2494	-0.7	-3.7	0.2
4	2645	-0.9	-4.4	0.3
5	2868	-0.3	10.7	0.6
6	2874	-5.4	0.2	0.1
7	3453	-9.0	-7.2	0.1
8	3528	-6.9	-8.0	0.1
9	2329	-10.9	5.1	2.2
10	2349	-5.9	-4.7	4.4
11	3474	-7.7	-14.5	12.5
Average		-4.9	-1.4	1.9
Std. Deviation		3.4	7.4	3.6

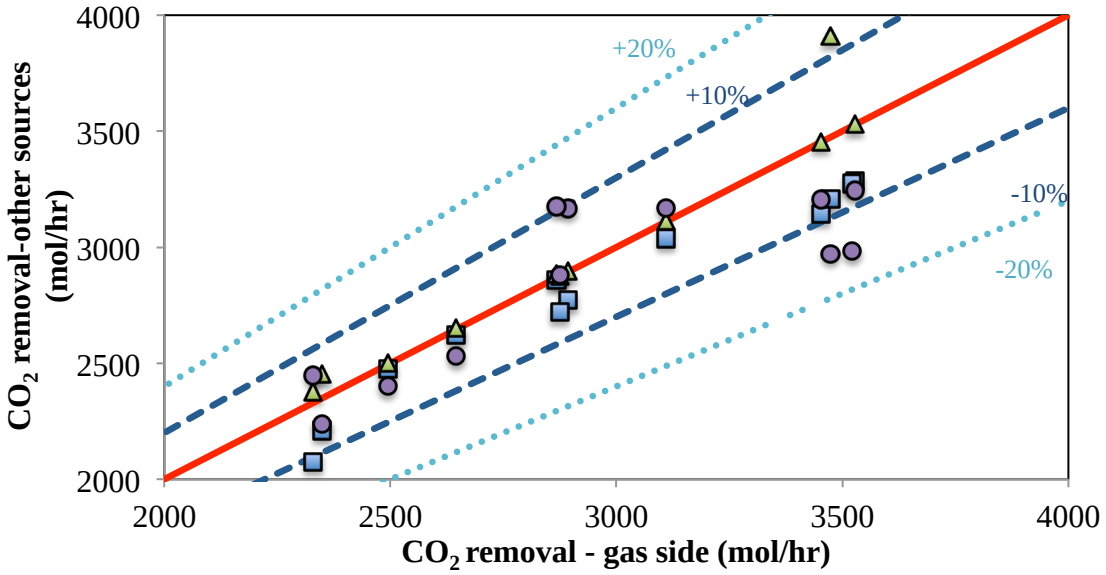


Figure 4.12 Parity plot removal for the MEA September'10. Compared to calculated value from gas side pilot plant data. (■) Liquid side titration (●) Liquid Density meter values (▲) Aspen Plus® reconciliation

Table 4.14: CO_2 removal deviations from gas side calculated value PZ Sept'10

Run	CO_2 removed Gas side (mol/hr)	Liquid Side Deviation (liquid – gas) -%		
		Liquid flowrate and loadings from:		Aspen Reconciled
		Titration	Density meter	
1	2731	27.3	-16.1	-0.5
2	2660	4.4	-15.5	-0.8
3	3816	15.3	-13.8	-0.8
4	4187	0.0	-13.9	-1.4
5	2660	19.4	-23.0	-0.1
6	2652	0.1	-27.0	0.3
7	3677	-3.7	-21.2	-0.8
8	5351	-4.2	-22.8	-1.0
9	2786	9.8	-15.8	-0.4
10	2754	17.8	-2.9	0.6
11	3747	26.7	8.9	0.5
12	5821	23.8	-11.3	-0.1
13	5045	23.8	-5.4	-0.1
Average		12.3	-13.8	-0.4
Std. Deviation		11.4	9.3	0.6

Table 4.15: CO_2 removal deviations from gas side calculated value PZ Dec'10

Run	CO_2 removed Gas side (mol/hr)	Liquid Side Deviation (liquid – gas) -%			
		Liquid flowrate and loadings from:			Aspen Reconciled
		Titration	Density meter Absorber	Density meter Stripper	
1	2645	-5.6	-11.4	21.4	0.4
2	2248	-7.0	-21.7	41.5	1.2
3	3814	21.1	-12.9	38.8	1.3
4	3636	12.1	-22.2	27.1	0.6
5	2918	9.0	-16.4	37.0	-0.7
6	2771	29.4	-12.3	60.0	-1.3
7	2680	37.3	-9.8	72.8	0.7
8	2328	12.8	-25.2	49.1	0.8
9	3999	17.9	-13.3	46.1	-0.6
Average		14.1	-16.1	43.8	0.3
Std. Deviation		13.7	5.2	14.9	0.9

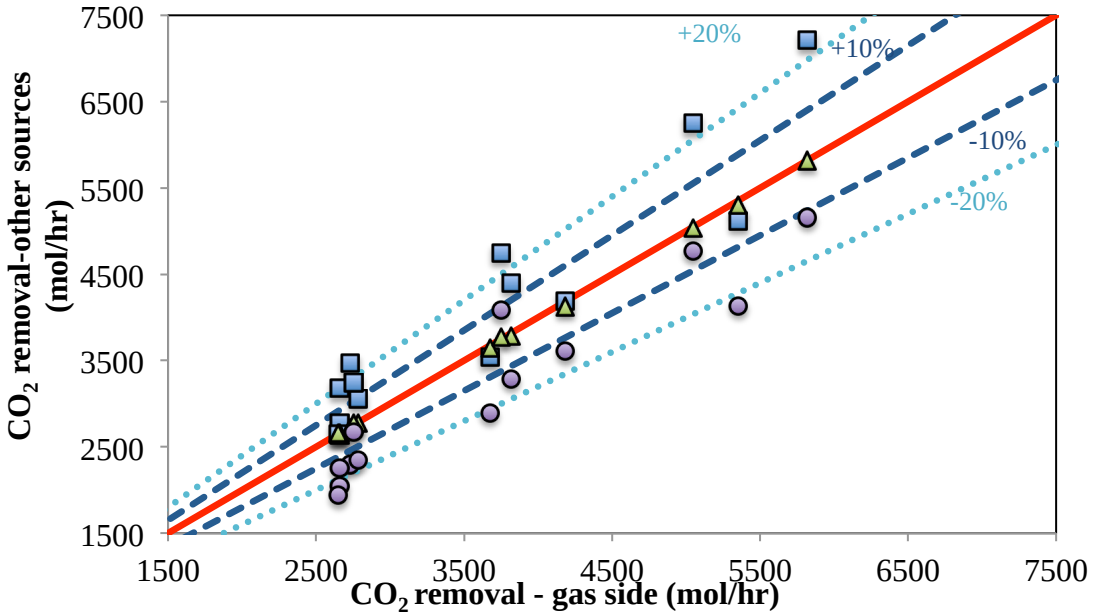


Figure 4.13 Parity plot removal for the PZ September'10 Campaign. Compared to calculated value from gas side pilot plant data. (■) Liquid side titration (●) Liquid Density meter values (▲) Aspen Plus® reconciliation

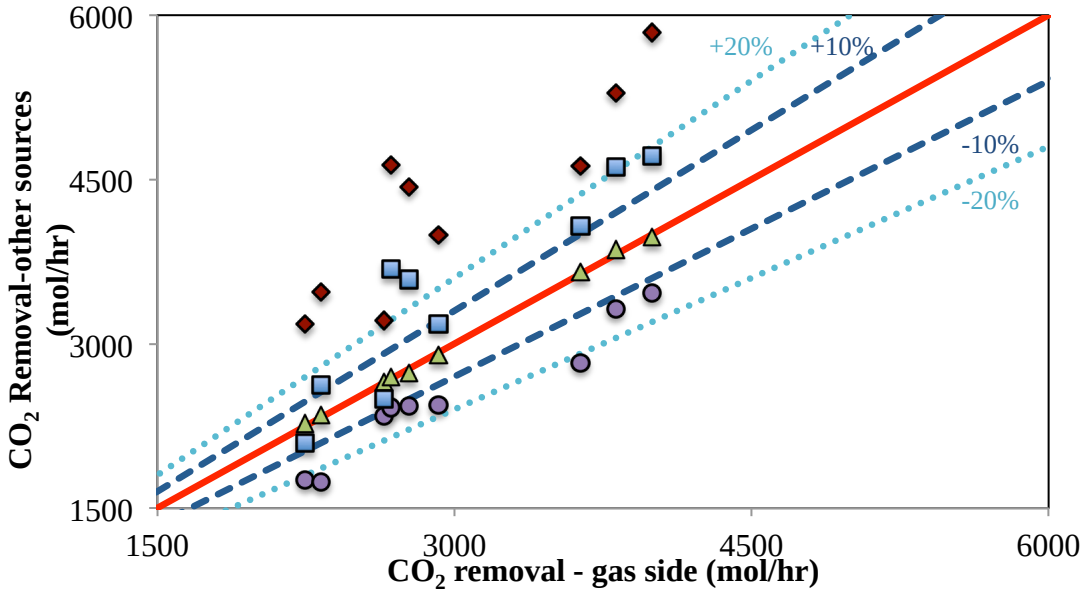


Figure 4.14 Parity plot removal for the PZ December'10 Campaign. Compared to calculated value from gas side pilot plant data. (■) Liquid side titration (●) Absorber density meters (◆) Stripper density meters (▲) Aspen Plus® reconciliation

Table 4.16: Summary of CO_2 removal deviations from gas side calculated value.

Campaign	Liquid Side Deviation (liquid – gas) -%			
	Liquid flowrate and loadings from:			Aspen Reconciled
	Titration	Density meter		
		Absorber	Stripper	
PZ Nov'09	1.1	-0.6	NA	0.5
PZ Sept'10	12.3	-13.8	NA	-0.4
PZ Dec'10	14.1	-16.1	43.8	0.3
MEA Sept'10	1.1	-0.6	NA	0.5

4.3. PILOT PLANT ABSORBER MODEL

4.3.1. Model Description

The PFD of the pilot plant model is the same for all runs (Figure 4.15) but the specifications are changed to account for the introduction of the intercooling loop.

The gas and liquid inlet streams are calculated using individual component streams that go into a mixer. The solvent loading reported by the pilot plant is used to calculate the mass streams that enter the mixer (LMIX). The stream of CO_2 passes by a multiplier before reaching the mixer. This multiplier is used to adjust loading across all runs as it will be explained later in this chapter. On the gas side, the CO_2 flow is specified using reported pilot plant compositions. The water content is based on the air humidity reported around the time of the run. These values are obtained from the website: *http://www.wunderground.com*. This site contains historic climate records for the US. The air composition is defined as nitrogen and oxygen (79%, 21% respectively). For the liquid side, the individual gas streams are fed to a mixer (GMIX). There is a heat exchanger block after each of the mixers to control inlet pressure and temperature conditions based on pilot plant reported data.

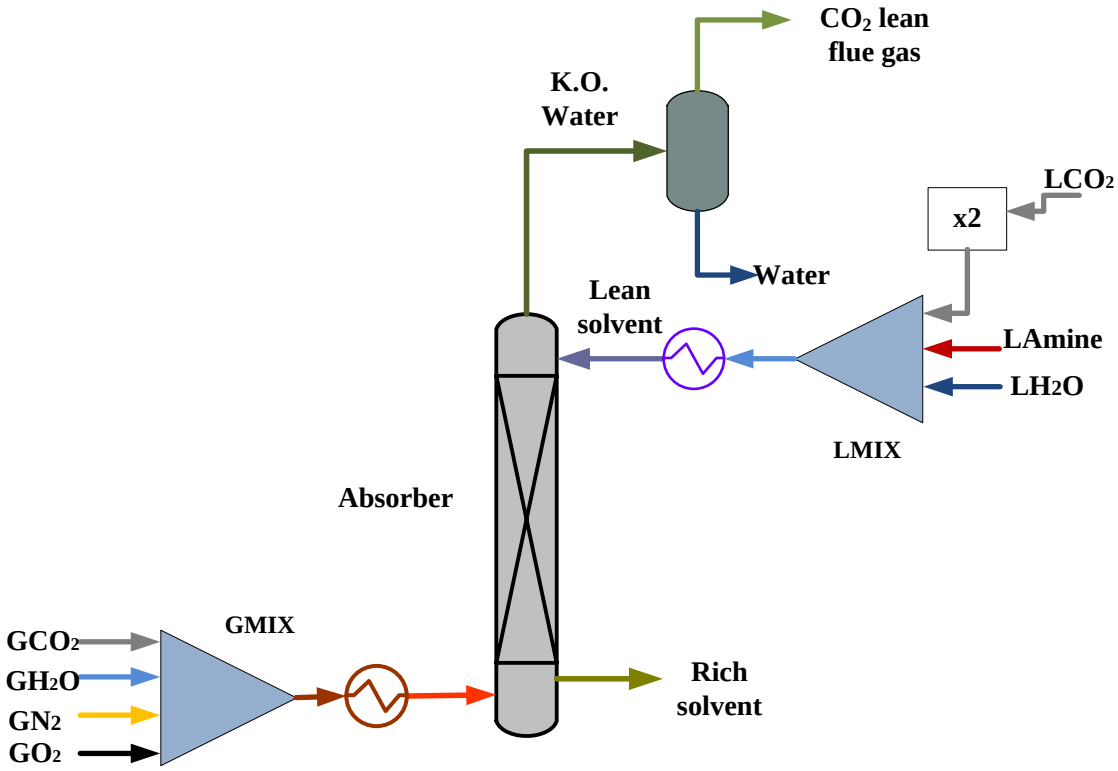


Figure 4.15: Aspen Plus® Pilot Plant PFD

4.3.2. Model Discretization

In Chapter 3 kinetics were obtained from WWC data for MEA and PZ. As mentioned previously the models used for this task required adequate discretization of the boundary layer to properly represent the enhancement effect of the carbamate reaction in the absorption of CO_2. Figure 4.16 shows the CO_2 concentration profiles obtained for different film segmentations. Use of only a few segments may not properly represent the concentration profile across the liquid film. The limitation on the number of segments is related to an increase in computation time. It may also be convenient to have smaller size segments where noticeable concentration change occurs and larger segments where concentration gradients are less dramatic. Convergence may also be affected by

inadequate film segmentation. Figure 4.17 shows a plot of two segmentation schemes that use the same number of segments, but the less optimized scheme generates calculation instabilities that lead to convergence issues and resulting errors.

Final liquid film discretization was determined by evaluating the resulting CO_2 mol fraction profiles using 40 segments and establishing key locations for segment placement. More segments were placed towards the interface and at the inflexion point before the profile tends to flatten out to the bulk concentration. Final segmentation for MEA and PZ is shown in Figure 4.16. It was reduced to 20 segments and as the figure shows it matches closely the segmentation scheme composed of double the segments. Table 4.17 includes the final segmentation used in the models in this work.

Table 4.17: Segmentation used in the models in this work.

Segment	δ/δ_{Total}	Segment	δ/δ_{Total}
Interface	0	11	0.035
1	0.001	12	0.04
2	0.0025	13	0.05
3	0.005	14	0.06
4	0.0075	15	0.08
5	0.01	16	0.1
6	0.0125	17	0.2
7	0.015	18	0.4
8	0.02	19	0.6
9	0.025	20	0.8
10	0.03	**Bulk**	1

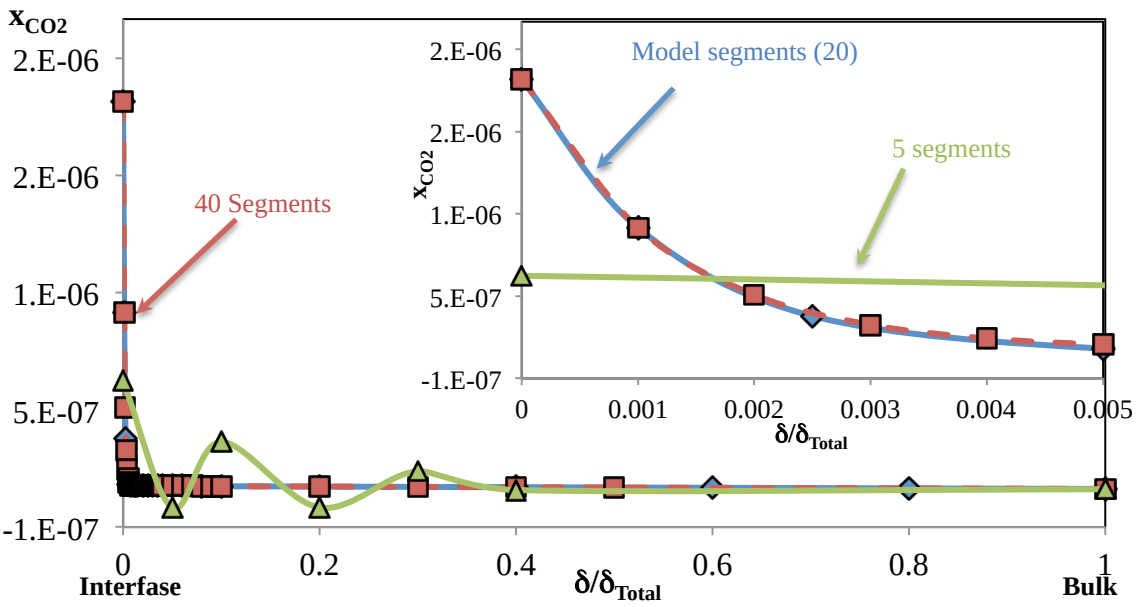


Figure 4.16: CO_2 concentration profiles for various segmentation schemes in the liquid film for 9 m MEA, α = 0.29 and T = 45°C (Points represent segment location) 5 seg. (–), 20 seg. (–), 40 seg. (– –).

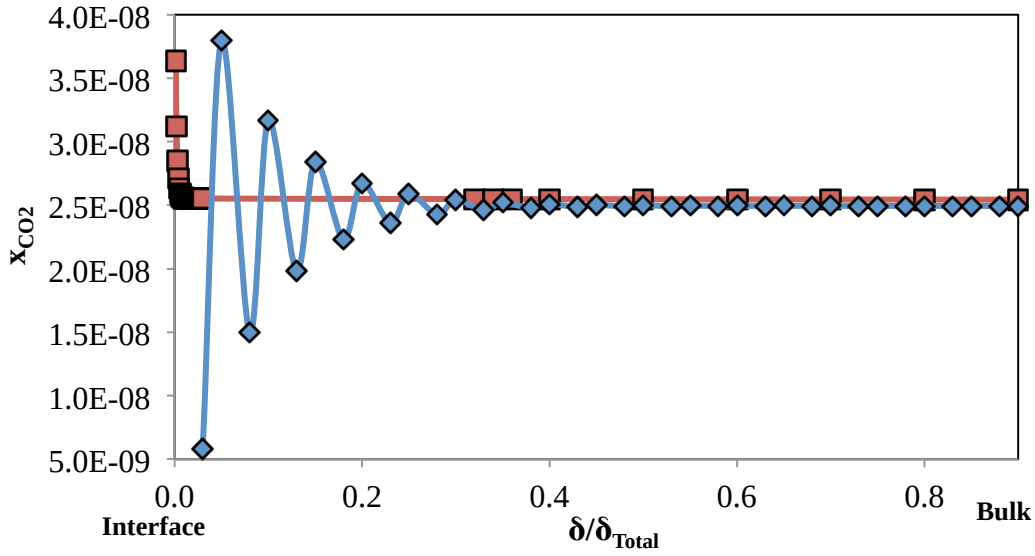


Figure 4.17: CO_2 concentration profiles for 39 segments. Points represent segment location. Blue curve represents initial segmentation. Red line is the final corrected segmentation.

Packing height discretization in absorber modeling is related to the selected flow model (Figure 3.3). In this work consists in using the mixed flow model and the packing element height to determine the segment size. Chen (2007) used the counter-current flow model to represent the pilot plant absorber. He demonstrated how the counter-current model with small segment size approaches a plug flow model. However, this can be seen as the ideal case in that there is no consideration of back mixing throughout the packing height. Figure 4.19 shows the temperature profiles obtained using the four available flow models. As the figure shows, mixed flow with fewer segments gives overall lower temperatures. The larger segments imply more averaged conditions across the column height. Fewer segments will also generate a lower removal performance prediction for the absorber using mixed flow (Figure 4.18). Some flow models are affected by the segment size generating convergence issues and profile inconsistencies as seen for the counter-current model with 9 segments. The V-Plug and Vplug-Pavg model behave similarly converging to a slightly higher performance and lower column temperatures with fewer segments.

There is no clear way to determine the extent of back mixing in the pilot plant. The approximation used in this work consists in using the mixed flow model and the packing element height to determine the segment size. This allows accounting for this phenomenon across a reasonable length. Thus, the number of segments for a column is determined by:

$$N_m = \frac{h_s}{h_p} \tag{4.3}$$

Where: N_m is the number of segments to represent a packing section.

h_s is the height of packing for a section

h_p is the characteristic element height of the installed packing.

The pilot plant campaigns studied in this work used three types of packing for the absorber. Table 4.18 shows the characteristics of each type of packing and the resulting required mixed segments. The intercooling loop was modeled as an additional segment as it will be described in the following section.

4.3.3. Pilot Plant Absorber Intercooling

The absorber is modeled with two specifications based on the presence of the intercooling loop. For the campaigns with intercooling four configurations were tested to represent this process improvement. In this work intercooling was implemented using the *pumparound* option with a single packing section and a middle stage with no reaction.

The simplest option was using a heat stream to set the temperature of the intercooled stage to 40°C. This option converges easier but is less rigorous since it does not consider the flow of solvent through the intercooling loop and adds additional mass transfer at the intercooling stage that is not available at the pilot plant. It also lowers the temperature of the column more than expected since the cooling is done on the stage creating additional heat transfer lowering the gas temperature and thus the whole column.

Another option evaluated to represent intercooling is shown in Figure 4.20. It consists in setting up two columns with the packing height of a section connected by a liquid stream that goes through a heat exchanger, representing the intercooling loop, and a gas stream connected to the bottom of the upper column. This option showed considerable convergence time and difficulty compared to the other alternatives and presented a similar mass transfer representation as the other simpler options.

Table 4.18: Characteristics of the Packing Used in the Pilot Plant Campaigns

Campaign	October'07	November'09	Sept'10/Dec'10
Packing Type	Flexipac AQ Style 20/ 1Y[a]	Mellapak 2X	RSP-250
Packing Element Height (m)	0.267[b]	0.223	0.232
Surface Area (m^2/m^3)	213[c]	205	250
Corrugation Angle	50°[c]	60°	45°
Channel Side (m)	0.009	0.02	0.014
Base (m)	0.0127	---	---
Void Fraction	0.91	0.92	0.99
Area factor (C_A)[d]	N/A	0.658	0.724
h_{lp} average[d]	N/A	5%	8%
Modeling segments/section	11	14	13

[a] Flexipac AQ Style 20 was modeled as Flexipac 1Y except for the surface area.
[b] Petre et al. (2003)
[c] Chen (2007)
[d] Area Factor is used in the interfacial area correlation by Wang (2011)

Intercooling was also represented by dividing the column into three sections. The first and third one represent the pilot plant packing sections and the second section accounts for the intercooling stage. Since there is minimum mass transfer in the intercooling loop, the section has a very small height (h<0.001). This stage uses the pump around option in RateSep™ specifying the cooled solvent temperature and flow rate and setting reactions rates to zero.

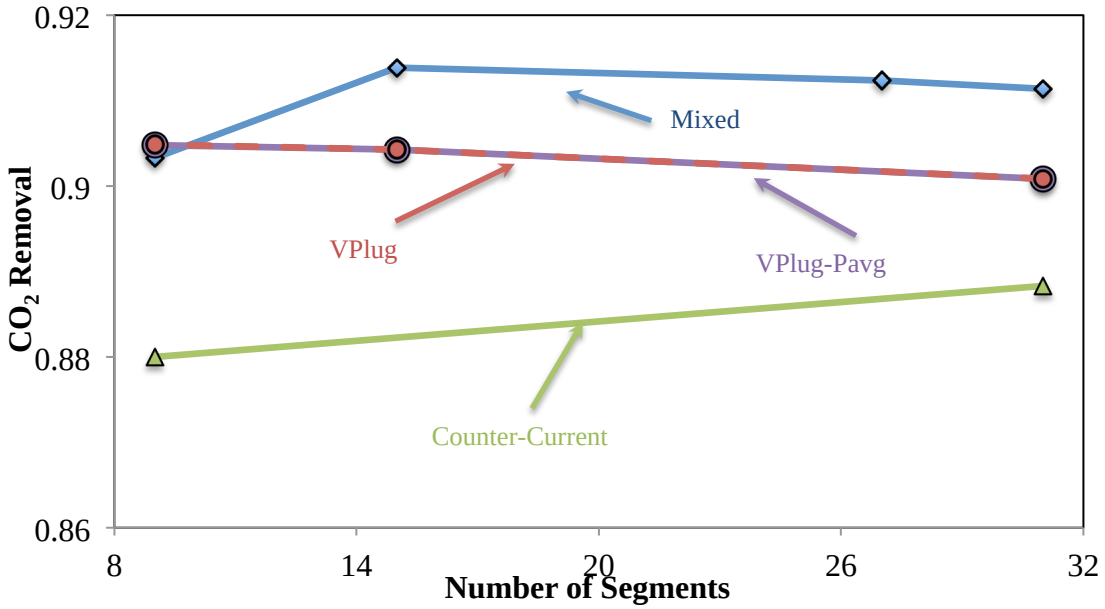


Figure 4.18: Predicted CO_2 removal as a function of number of segments and flow model. Run 1. Campaign September'10 MEA: 9.2 m MEA, ldg =0.30 L/G = 5.85 Removal = 0.865

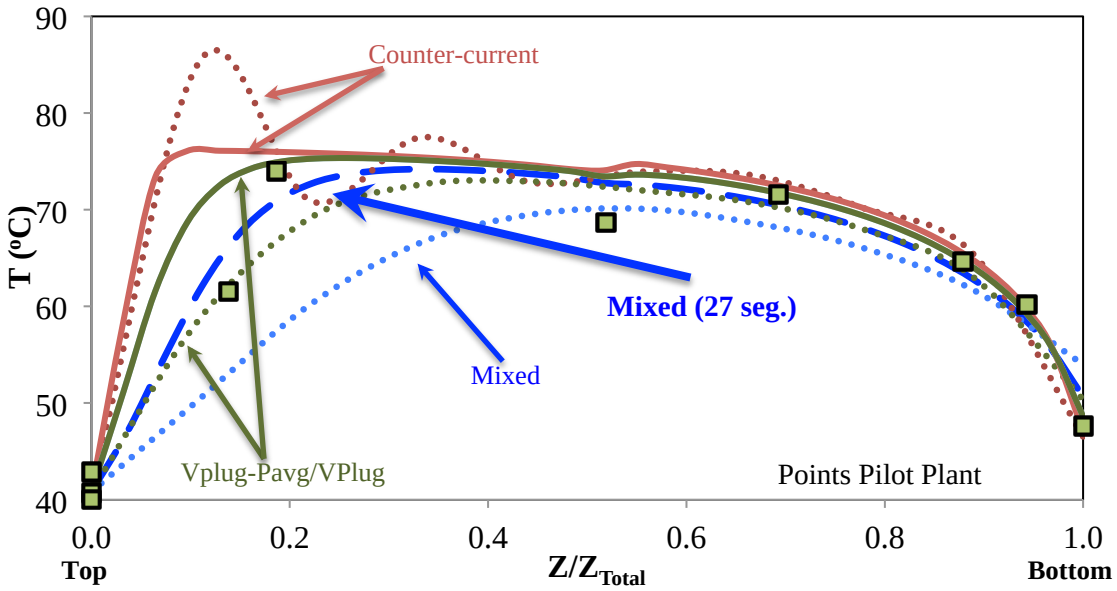


Figure 4.19: Liquid T profiles per number of segments and flow model. Run 1. Campaign September'10 MEA: 9.2 m MEA, ldg =0.30 L/G = 5.85 mol/mol. Removal = 0.865. Pointed line: 9 segments, Solid line: 31 Segments

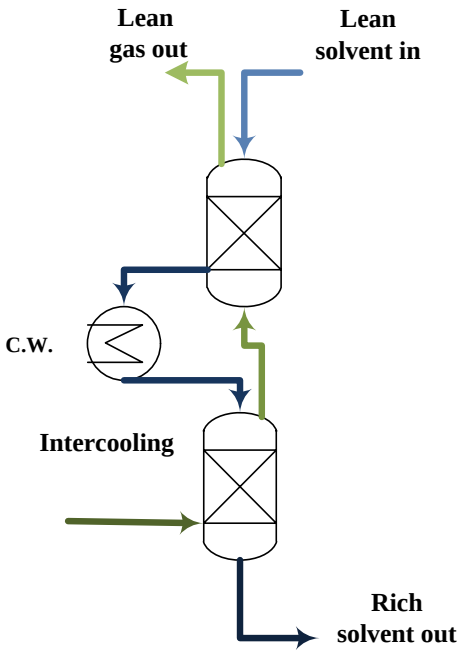


Figure 4.20: Aspen Plus® PFD for an intercooled absorber using two columns for packing beds.

The *pumparound* option was implemented using a single packing section with a middle stage with no reaction. This minimized the CO_2 mass transfer but not the water and heat transfer in this section. Furthermore, an additional stage height is added to the total packing height to account for the lost packing height.

The two latter options showed very similar mass transfer rate behavior (Figure 4.21) and adequate temperature representation (higher for the 3 sections option). They both show a dip in the mass transfer rates around the intercooling but in reality performance is similar to the 2 column option since in the single section pump around an additional segment is added and in the 3 section option the second section is not accounted for in the total packing height. This differs from the heat stream option where the additional mass transfer observed in Figure 4.21 is not deducted anywhere resulting in

performance overestimation of the column and the intercooling loop. Nevertheless, Figure 4.21 shows that the mass transfer performance is fairly similar for all possible representations.

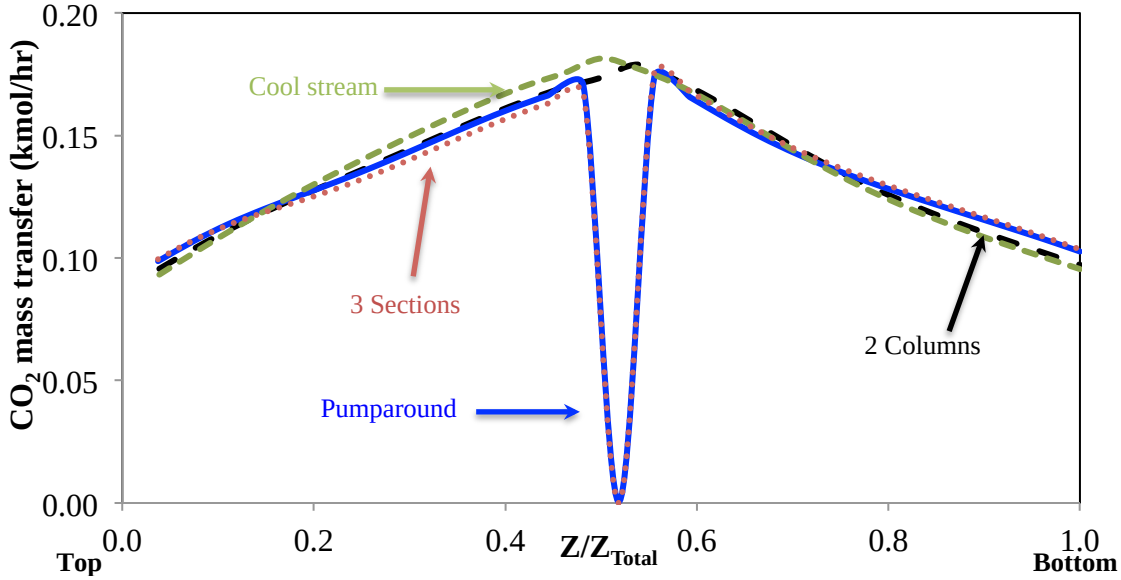


Figure 4.21: CO_2 intercooling absorber configurations mass transfer profiles. Run 8. Campaign Sept'10 MEA: 8.7 m MEA, ldg =0.28 L/G = 4.31 mol/mol.

4.3.4. Mass transfer coefficients and interfacial area

CO_2 absorption using aqueous amines is a process of mass transfer with fast chemical reaction. It has a strong dependence on the interfacial area and, as the solvent loading increases, liquid mass transfer gains more importance. Satisfactory representation of the pilot plant operation requires adequate liquid and mass transfer coefficients as well as a correct representation of the effective interfacial area.

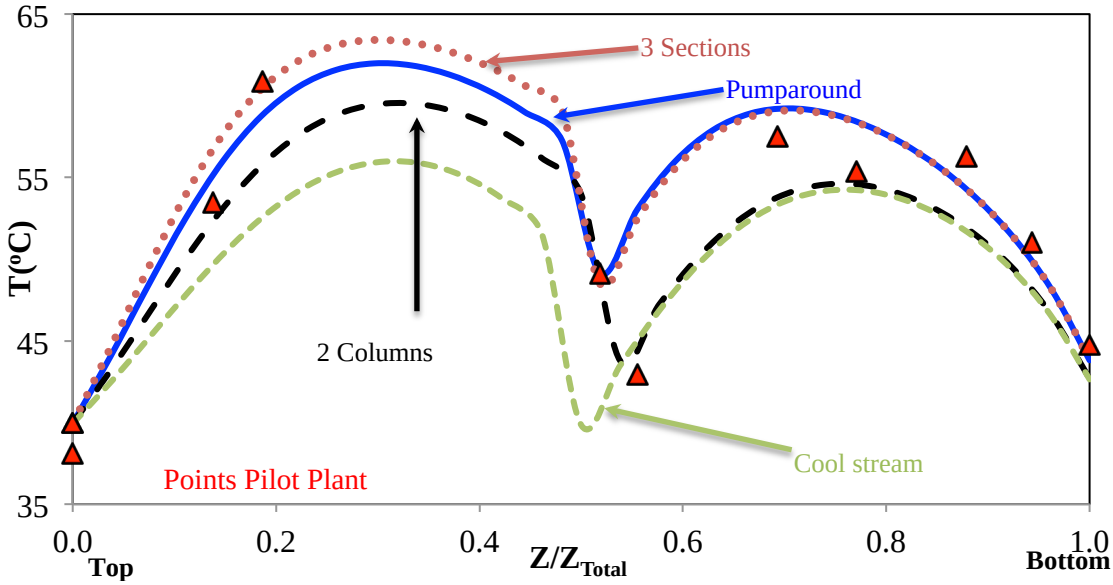


Figure 4.22: CO_2 intercooling absorber configurations liquid T profiles. Run 8. Campaign September'10 MEA: 8.7 m MEA, ldg =0.28 L/G = 4.31 mol/mol.

The pilot plant campaigns analyzed in this work used three types of packing in the absorber (Table 4.18). Different correlations for mass transfer coefficients and effective area were selected based on the packing type.

The absorber for the first MEA campaign (October'07) was packed with Flexipac AQ Style 20, new Koch-Glitsch Inc. high capacity packing with a specific area of 213 m^2/m^3 (Chen, 2007). This packing has less surface area, and lower liquid holdup, but its higher capacity allows operation at high gas rates with reduced pressure drop (Chen, 2007).

Aspen Plus® RateSep™ contains a number of correlations for mass transfer and interfacial area for various types of packing and manufacturers. However, Flexipac AQ Style 20 is not available in the program database. Chen (2007) had used Flexipac 2Y in modeling this packing. This work uses Flexipac 1Y with the specific area of the AQ Style

20 type. A new correlation included in Aspen Plus® v.7.3 was selected to represent the mass transfer coefficient and interfacial area. Hanley and Chen (2011) developed correlations for sheet metal structured packing of the Mellapak/Flexipac type for liquid and gas mass transfer and interfacial area. For Flexipac 1Y the equations are as follows:

$$k_L^o = 0.33 Re_L Sc_L^{1/3} \left(\frac{\tilde{\rho}_L D_L}{d_e}\right) \tag{4.4}$$

where: k_L^o is the liquid side physical mass transfer coefficient.

D_L is the liquid phase binary diffusivity (m^2/s)

ρ_L is the liquid molar density (mol/m^3)

d_e is the equivalent diameter (m)

$$k_G = 0.0084 Re_G Sc_G^{1/3} \left(\frac{\tilde{\rho}_G D_G}{d_e}\right) \left(\frac{\cos\theta}{\cos\left(\frac{\pi}{4}\right)}\right)^{-7.15} \tag{4.5}$$

where: k_G is the gas side mass transfer coefficient

D_G is the gas phase binary diffusivity (m^2/s)

ρ_G is the gas molar density (mol/m^3)

θ is the corrugation inclination angle in radians measured from the vertical

$$\frac{a_e}{a_p} = 0.539 Re_G^{0.145} Re_L^{-0.153} We_L^{0.2} Fr_L^{-0.2} \left(\frac{\tilde{\rho}_G}{\tilde{\rho}_L}\right)^{-0.033} \left(\frac{\mu_G}{\mu_L}\right)^{0.090} \left(\frac{\cos\theta}{\cos\left(\frac{\pi}{4}\right)}\right)^{4.078} \tag{4.6}$$

where: a_e is the effective mass transfer area (m^2/m^3)

a_p is the specific area of packing (m^2/m^3)

The dimensionless numbers are defined as follows:

$$Re_f = \frac{d_e v_f \rho_f}{\mu_f} \tag{4.7}$$

where: Re_f is the Reynolds number of the phase *f*.

The Schmidt number of the phase (Sc_f) is defined as:

$$Sc_f = \frac{\mu_f}{\rho_f D_f} \tag{4.8}$$

where for these equations v_f is the superficial velocity (m/s)

μ_f is the viscosity (kg/m-s)

ρ_f is the density of the phase (kg/m^3)

The sub index *f* represents the phase (liquid or gas)

$$We_L = \frac{d_e v_L^2 \rho_L}{\sigma} \tag{4.9}$$

where: We_L is the liquid Weber number

σ is the surface tension (N/m)

$$Fr_L = \frac{v_L^2}{g d_e} \tag{4.10}$$

where: Fr_L is the liquid Froude number

g is the acceleration of gravity (m/s^2)

The equivalent diameter (d_e) is calculated as:

$$d_e = \frac{4\varepsilon}{a_p} \tag{4.11}$$

where: ε is the void fraction of packing

a_p is the specific area of packing (m^2/m^3)

For the two other packing types used in the pilot plant campaigns (Table 4.18) correlations by Wang (2011) for interfacial area and mass transfer coefficients were implemented in FORTRAN and included in the model as user subroutines (Appendix C) These were developed based on measurements in the air/water column available at the J.J Pickle Research Center of The University of Texas at Austin. For interfacial area:

$$\frac{a_e}{a_p} = C_A \left(\frac{\rho_L}{\sigma}\right)^{0.116} v_L^{0.155} \tag{4.12}$$

where: C_A is the area constant for the packing found in Table 4.18

For the gas side mass transfer coefficient Wang (2011) fitted a form of the modified Rocha-Bravo-Fair equation (1996):

$$\frac{k_G}{a_p D_G} = 0.0317 \left(\frac{v_{GE}\rho_G}{a_p \mu_G}\right)^{0.88} \left(\frac{\mu_G}{\rho_G D_G}\right)^{0.33} \tag{4.13}$$

where: v_{GE} is calculated using the following:

$$v_{GE} = \frac{v_G}{\varepsilon\left(1 - h_{lp}\right)\sin\theta} \tag{4.14}$$

where: h_{lp} is the packing liquid hold-up. In the model implementation the average liquid hold up for each packing type was used (Wang, 2011)

The liquid side physical mass transfer coefficient ($k_L{}^o$) was calculated using a modified version (Wang, 2011) of the equation by Brunazzi (2011):

$$Sh_L = 829.85 \frac{Gz^{0.5}}{Ka^{0.09}} \tag{4.15}$$

where the dimensionless numbers are defined by the following set of equations. For the Sherwood number (Sh_L):

$$Sh_L = \frac{k_L^o S}{D_L} \tag{4.16}$$

where: S is the packing channel side (see Table 8)

For the Graetz number (Gz):

$$Gz = Re_L Sc_L \left(\frac{\delta \sin\theta}{h_p}\right) \tag{4.17}$$

where: h_p is the characteristic element height of the installed packing

Sc is the Schmidt number defines as Equation 4.7

δ is the liquid film thickness defined as:

$$\delta = \left(\frac{3\mu_L}{\rho_L g \sin\theta} \frac{v_{LE}}{h_{lp} \sin\theta} \right)^{0.5} \tag{4.18}$$

where v_{LE} is defined as:

$$v_{LE} = \frac{v_L}{\varepsilon\left(1 - h_{lp}\right) \sin\theta} \tag{4.19}$$

The Kapitsa number (*Ka*) is defined:

$$Ka = \frac{\sigma^3 \rho_L}{\mu_L^4 g} \tag{4.20}$$

4.3.5. Reconciliation and Validation

Analysis of pilot plant data and model validation has used Aspen Plus® and its parameter estimation and reconciliation tool. This tool uses pilot plant measurements along with model parameters and inputs, and an assigned standard deviation based on equipment accuracy, data uncertainty and/or engineering judgment and minimizes an objective function that includes the sum of the squared measurement errors:

$$f = \sum_{i=1}^{n} \frac{(PP_i - m_i)^2}{\sigma_i} \tag{4.21}$$

where: PP_i represents the experimental data,

m_i is the model prediction of the experimental data,

σ_i is the assigned standard deviation.

The standard deviation serves as boundary for the values that the model prediction can use. Reliable measurements are given small standard deviations; unreliable measurements are given large standard deviations. This allows for small offsets between the plant and model variables for the reliable measurements and large offsets for the unreliable measurements. (*Aspen Plus Help file*). Thiele, Faber et al. (2007), present a

similar approach to data reconciliation but use a different form of the objective function due to large deviations in some of their data sets.

The reconciliation of the pilot plant campaigns in this work was done using 6 manipulated variables. These are fit as a common parameter in all of the runs of a campaign. The manipulated variables are the interfacial area factor, the CO_2 multiplier and four parameters to account for heat loss in the liquid and in the gas side.

The interfacial area factor multiplies the effective mass transfer area (a_e). This helps tune the mass transfer models and kinetics to the pilot plant results. The CO_2 multiplier is used to give loading a uniform initial variation to account for differences in the model VLE and the observed results. The four remaining manipulating variables are related to heat-loss in the absorber column. The absorber column is not insulated so it is necessary to account for this variable to close the energy balance. Chen (2007) estimated that the heat-losses for the pilot plant absorber could be of the order of 51 kW. Aspen Plus® RateSep™ allows the user to enter values for heat-loss for the liquid and vapor side. This heat loss is distributed evenly throughout the column packing. For the reconciliation the reported ambient temperature for each run was considered to calculate the heat loss:

$$Q_f = A_f + B_f T_{amb} \tag{4.22}$$

where: Q_f is the heat-loss assigned to a phase (liquid of gas)

T_{amb} is the run average ambient temperature reported by *http://www.wunderground.com* in Kelvin

A_f, B_f are the manipulated variables for a phase (liquid or gas)

The adequate fit of each run was evaluated using the Number of Transfer Units (*NTU*) calculated as:

$$NTU = -\ln\left(1 - R_{CO_2}\right) \tag{4.23}$$

where: R_{CO2} is the removal fraction of carbon dioxide.

NTU standard deviation was set to 1% to force the reconciliation to give priority to matching the measured CO_2 removal.

Pressure drop was assumed linear and taken from the pilot plant measurements. Since it is not part of the scope of this work to evaluate pressure drop models, pressure drop was made constant by setting its possible deviation to zero. Pressure drop measurements were subtracted from the measured inlet gas pressure to determine top of column pressure and solvent inlet pressure.

Reported column temperatures were specified as interface temperature in the model when it was not possible to determine if the reported data corresponded to the liquid or vapor phase. This allowed the model to attempt to match the reported value using both liquid and gas conditions.

4.4. MEA CAMPAIGNS

Aqueous monoethanolamine (MEA) has been the industry standard for CO_2 removal from flue gas. Its base application is using 30 wt.% (7m) however, Reddy et al. (Reddy et al., 2003) have suggested up to 40 wt.%. In this work the results of two campaigns are presented: a single run campaign testing a high capacity low surface area packing and a second one month campaign to test intercooling as a process optimization alternative. Table 4.19 summarizes the conditions for both campaigns.

Table 4.19: MEA Campaigns Conditions

Campaign	October '07	September '10
MEA (m)	8.8	8.4 – 9.3
Gas rate (ACFM)	500	350 -500
Liquid Rate (gpm)	17.5	8 – 26
Absorber Packing	Flexipac AQ Style 20	RSP-250
Intercooling	No	Yes/No
Runs	1	11

4.4.1. October 2007

This campaign consists of a single point measurement that was taken at the end of a larger private company campaign at the Pickle Research Center. The absorber was packed with Flexipac AQ Style 20 a high capacity, low hold-up structured packing. It served to test the performance of the greater MEA (9 m). The pilot plant configuration can be seen in Figure 4.1 and Figure 4.15 shows the PFD used to model this campaign. Table 4.20 presents the reconciliation results for this case.

The reconciled temperature profile for this campaign is presented in Figure 4.23 along with the predicted CO_2 mass transfer rate profile. Model liquid and gas temperature profiles are adequately matched although slightly lower than the reported pilot plant data. Area was adjusted 12%. Lean and rich loading were reduced less than 7%. NTU was matched with less than 1% deviation. The adjusted model set the heat-loss parameters to zero. It is possible that this single point does not provide enough information to assess heat-loss.

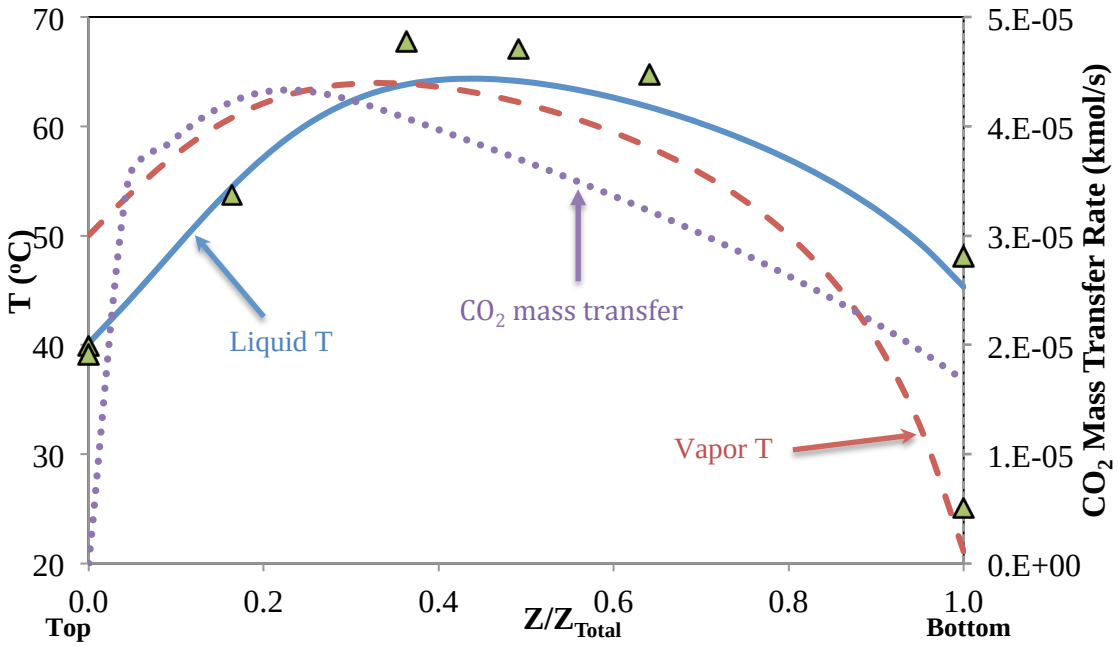


Figure 4.23: Temperature profiles. MEA October'07 Campaign (▲) measured, (—) reconciled, (···) Model predicted CO_2 mass transfer

4.4.2. September 2010

The goal of this campaign was to test intercooling as a process improvement to increase absorber performance. Intercooling was used on/off expecting an increase of NTU with the use of intercooling.

Table 4.21 shows the results for the manipulated variables and Table 4.22 the resulting deviations for this campaign. The interfacial area was reduced 27%. The CO_2 multiplying factor is close to one. This implies that there was no need for a across the board shift in loading to match pilot plant conditions. The obtained loading deviations (Table 4.22) corroborate this as they are below the allowed ±4%. Figure 4.8 show the close fit between loading determined using titration and the reconciled values.

Table 4.20: Pilot Plant Reconciliation Results, MEA October'07 Campaign

Variable	Pilot Plant Value	Specified Standard Deviation	Reconciled Value	Actual Deviation
Area factor	1.0	---	0.8792	---
L/G (mol/mol)	4.86	[a]	4.86	---
Lean ldg	0.36	10%	0.34	-6.8%
Rich ldg	0.48	10%	0.45	-6.1%
y_{CO2} - in	0.1192	[a]	0.119	-0.05%
y_{CO2} - out	0.047	3%	0.046	-1.1%
T_G – in (^{o}C)	25.11	4	21.11	4.0
T_G – out (^{o}C)	42.0	8	50	-7.9
T_L – in (^{o}C)	39.9	4	40.0	0.1
T_L – out (^{o}C)	48.11	8	45.23	-0.9
***Tamb* (^{o}C)**	20.3	0	---	---
Column T (^{o}C)				
TT4077	53.7	2	55.9	-2.1
TT4076	67.8	2	63.8	4
TT4075	67.1	2	63.9	3.1
TT4073	64.7	2	61.7	3.0
NTU	1.035	1.0%	1.034	-0.1%
Q_{Loss} (kW)	---	---	0.0	---

[a] Deviation specified for individual streams. Variable is calculated after the reconciliation.

Table 4.21: Pilot Plant Reconciliation Results, MEA September'10 Campaign.

Variable	Values
Area factor	0.73±0.02
CO_2 Multiplier	0.99±0.01
A_L (W)	0.0035±0.003
B_L (W/K)	7.30±0.05 x10^{-9}
A_G (W)	0.0044±0.002
B_G (W/K)	1.86±2 x10^{-8}

Figures 4.24 – 4.27 show the temperature profiles for the campaign runs. Profiles are overall well fit although better fit for the non-intercooled cases. Intercooling profiles seem to be better matched for lower L/G cases (< 5 mol/mol). It is also possible to

observe, even for intercooled runs, how the temperature bulge shifts from the top of the column towards the bottom of the column with higher L/G. Temperature profiles for the cases without intercooling display a higher temperature bulge than the intercooled cases ($T > 65^oC$)

Figure 4.12 shows the resulting model CO_2 loading compared to the reported value from the pilot plant. Adjusted loadings allowed closure of the CO_2 mass balance. For large L/G runs (>7.0) closure was achieved with higher deviation from the reported outlet gas CO_2 mol fraction especially for runs 5 and 10 (deviation = -20%). However, this value is still considered adequate since it accounts for a change in y_{CO2} of less than 0.005.

Runs 1 and 3 have similar NTU (2.0) even though run three has lower loading and L/G. This is reflected in the CO_2 mass transfer rate profiles (Figure 4.28) that show both with a similar approach to a pinch at the lean end while Run 1 has a larger CO_2 mass transfer rate at the bottom of the column than Run 3. However, Run 1 has a larger gas flow (415 ACFM vs. 350 ACFM).

Run 5 has the best performance of the runs without intercooling (NTU = 3.8) matching the best intercooled run. However, requires a high L/G (7.1) and a fairly lean loading (0.28). The mass transfer profile shows a high mass transfer rate at the bottom of the column and an approach to a lean end pinch. The temperature increases but does not reach higher values ($T<65^oC$) thanks to the L/G.

Table 4.22: Pilot Plant Reconciliation Result Deviations, MEA September'10 Campaign

		Deviation[a]										
	Specified	Actual (%) – except for T										
Run		1	2	3	4	5	6	7	8	9	10	11
L/G (mol/mol)	b	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.2	0.0	0.0	0.1
Lean ldg	4.0%	0.4	2.3	-1.1	-2.1	-1.2	2.0	-3.2	0.1	-1.3	-1.2	-4.6
Rich ldg	4.0%	1.6	2.6	-0.8	-1.1	-0.2	3.5	1.3	2.6	2.0	1.3	0.5
Δ ldg	N.A.	4.4	3.1	-0.2	0.3	2.2	7.1	9.0	6.6	14.3	10.3	4.2
MEA *(m)*	b	0.0	-0.7	0.0	0.2	-0.1	-0.5	0.2	0.9	-0.1	-0.1	0.8
y_{CO2} - in	3.0%[c]	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
y_{CO2} - out	3.0%	-1.0	-2.1	-1.2	-3.3	-20.5	-3.2	0.3	-0.3	-9.5	-20.4	1.6
T_G – in (°C)	2.0	-2.0	-2.0	-2.0	-2.0	0.0	-1.9	-2.0	-2.0	-0.0	0.0	-0.9
T_G – out (°C)	5.0	9.0	-1.5	8.9	3.9	0.6	3.1	6.4	7.5	2.0	1.0	4.4
T_L – in (°C)	3.0	0.1	-0.7	-0.4	-0.7	0.0	-0.2	-0.5	0.0	0.0	0.0	0.3
T_L – out (°C)	4.0	2.4	4.1	4.1	4.4	7.1	5.4	-0.3	-1.1	3.1	4.4	0.3
NTU	1.0%	2.0	2.7	2.0	2.7	3.8	3.6	1.7	1.9	1.8	2.0	1.7
(Dev)[d]		(0.5)	(0.8)	(0.7)	(1.3)	(6.5)	(1.0)	(0.2)	(0.2)	(6.0)	(13.4)	(-0.9)
Q_{Loss} (W)[e]	---	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0	8.0

[a] Positive value indicates that reconciled value is greater than measured .
[b] Specified using individual component streams.
[c] Specified using the lean CO_2 stream and the CO_2 multiplier (see Figure 4.15)
[d] This row contains the resulting NTU and the deviation in parenthesis.
[e] Calculated heat loss from reconciliation

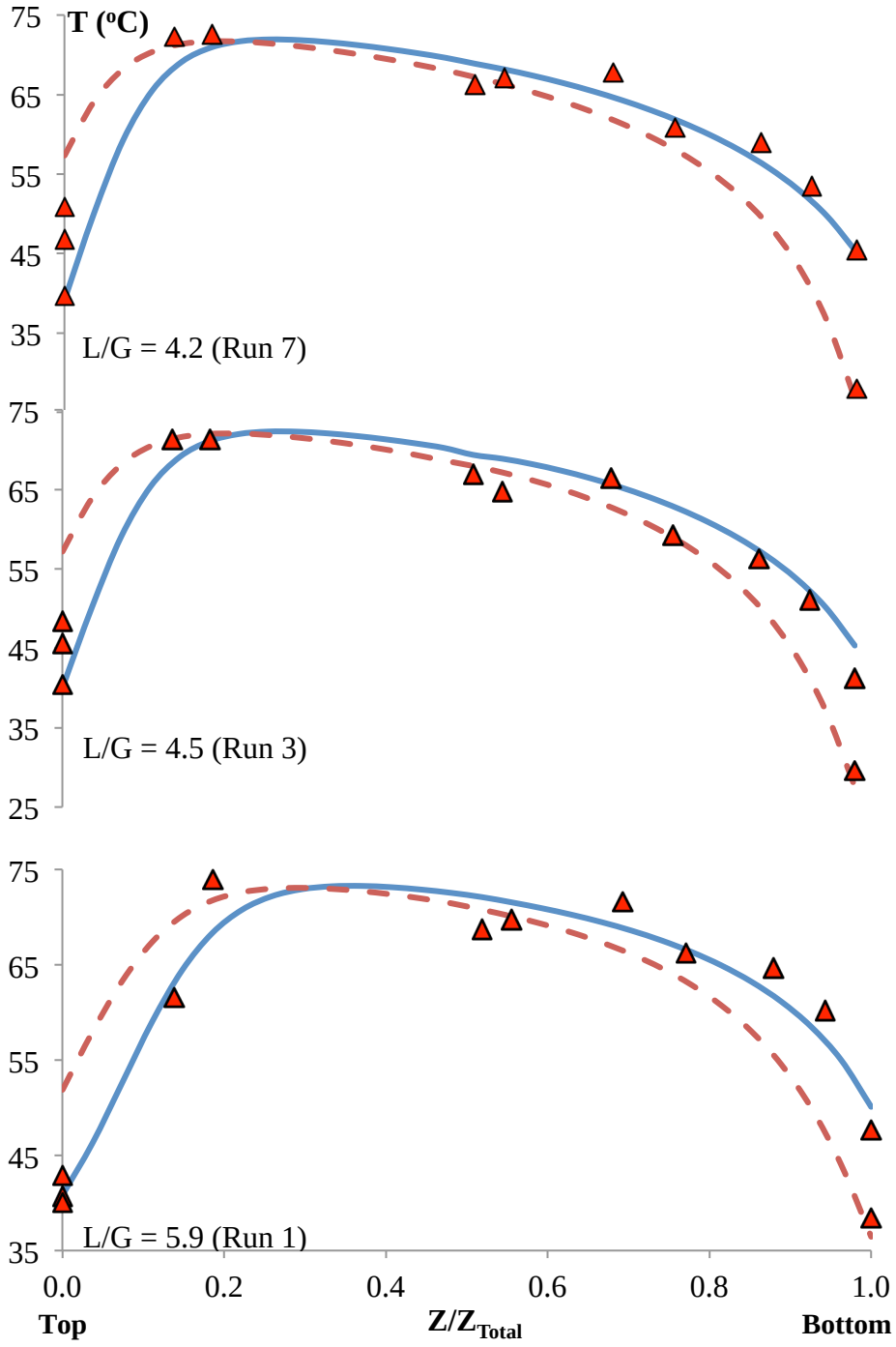


Figure 4.24: T profiles. MEA Sept'10 Campaign – No intercooling. (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G in mol/mol)

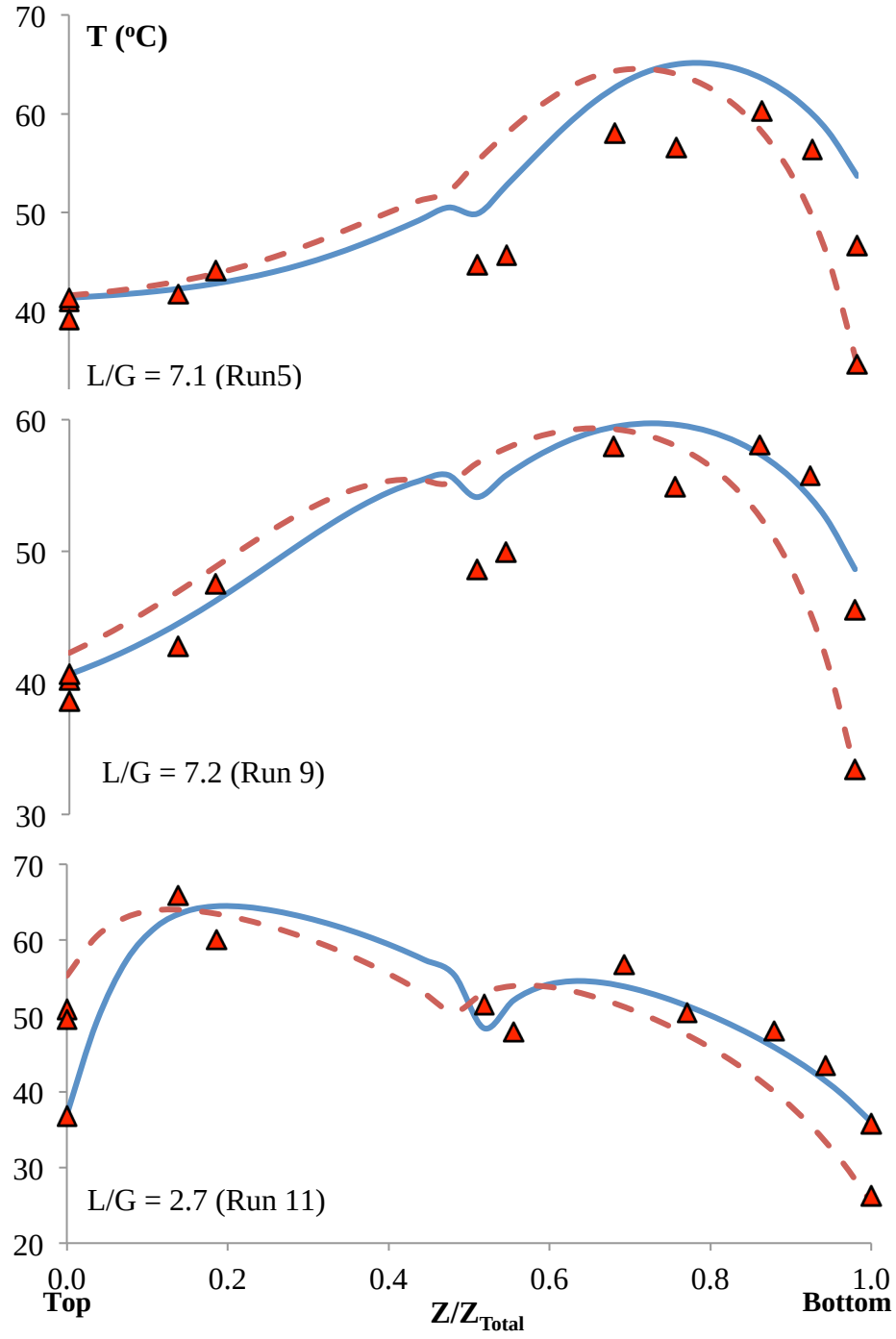


Figure 4.25: T profiles. MEA Sept'10 Campaign – Top two no intercooling. Run 11 intercooled (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G in mol/mol)

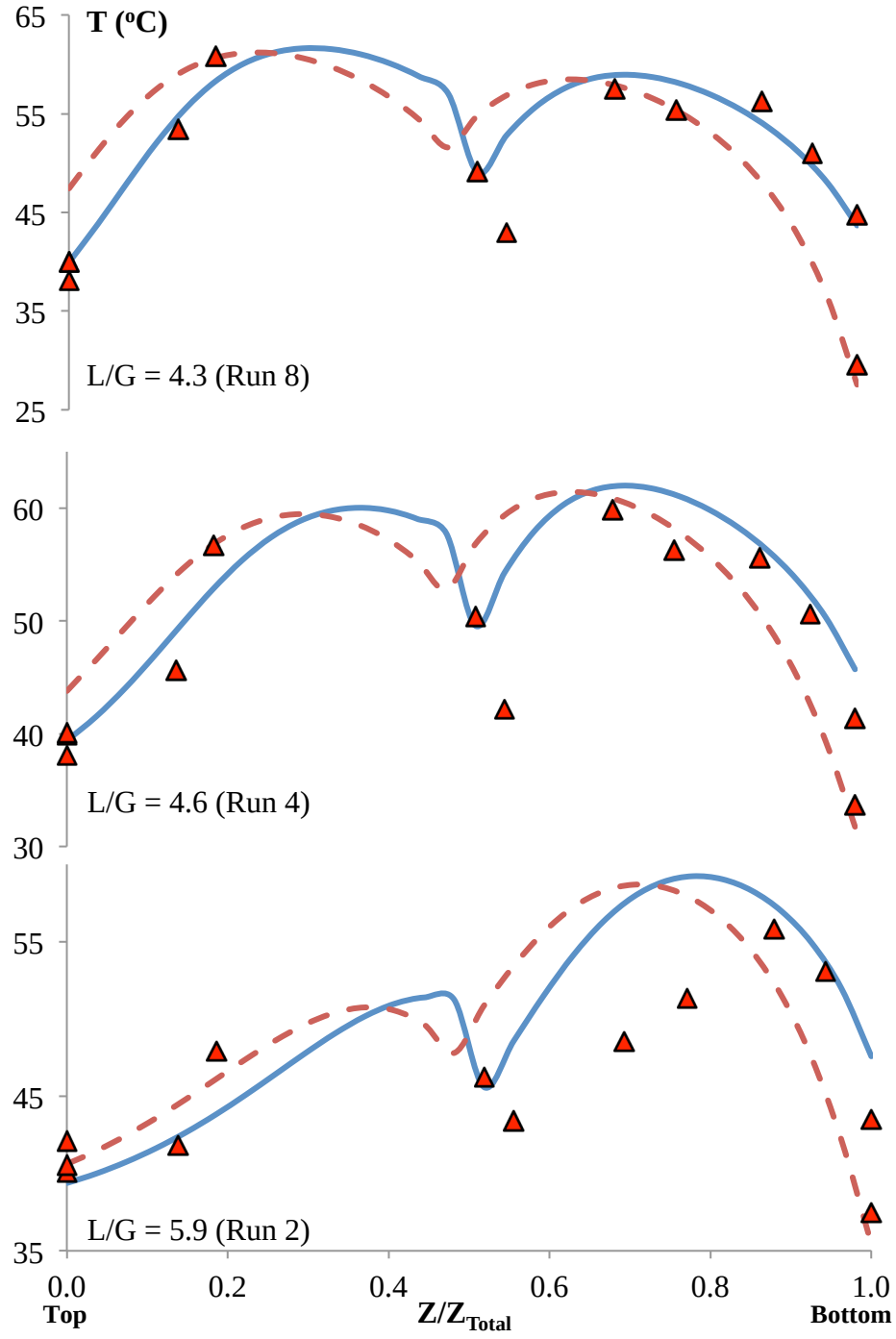


Figure 4.26: T profiles. MEA Sept'10 Campaign – Intercooled: (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G in mol/mol)

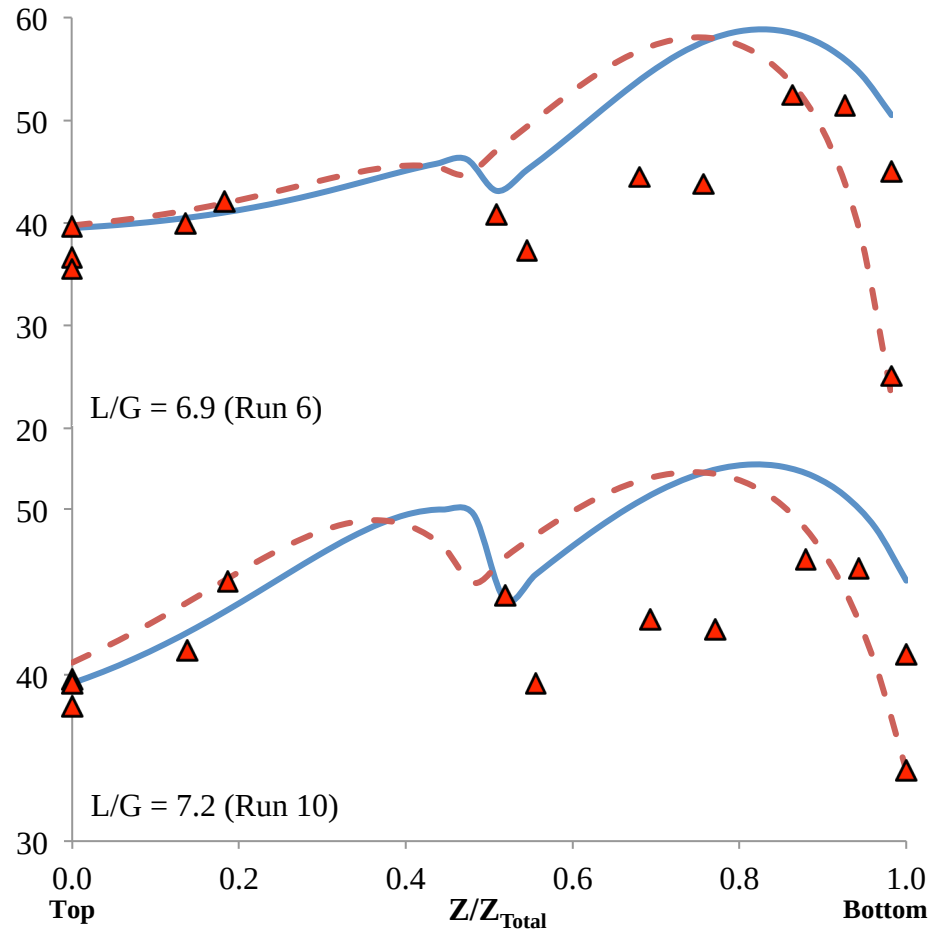


Figure 4.27: T profiles. MEA Sept'10 Campaign – Intercooled: (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G in mol/mol)

The largest L/G was tested in Run 9 but the high loading of the feed solvent (ldg=0.36) limits the column performance as reflected on the mass transfer rates in the column (Figure 4.28)

The highest temperature bulge is displayed in Run 7 (75°C) that has the lowest L/G (4.2 mol/mol) for the runs without intercooling and one of the lowest performance values for this campaign (NTU = 1.7) yet the rates in the CO_2 rate profile are not as low as expected since it has a higher gas flow rate (500 ACFM).

Run 8 shows a well-distributed CO_2 mass transfer rate (Figure 4.29) across the column with higher rates around the intercooling section and a lower temperature bulge that can also be attributed to the intercooling.

Run 4 has a NTU of 2.7 with an approach to a lean side pinch (Figure 4.29). Temperatures are even through the column but removal is slightly higher in the lower section of the column and with the highest values around the intercooling section.

The CO_2 mass transfer profile for run 2 show a lean end pinch with the highest removal around the intercooling section and towards the bottom of the column (Figure 4.30). The temperature bulge in this section is reduced due to the high L/G (5.9 mol/mol) and intercooling (Temperature at the intercooling section around 45°C).

Run 10 with a high L/G (7.2 mol/mol) has low CO_2 mass transfer rates through out the column with the highest values around the intercooling section. However, even with a rich lean solvent (0.36) the NTU is of 2.0 (86% removal). This is mainly due to the high L/G and the limited temperature increase thanks to the L/G and intercooling.

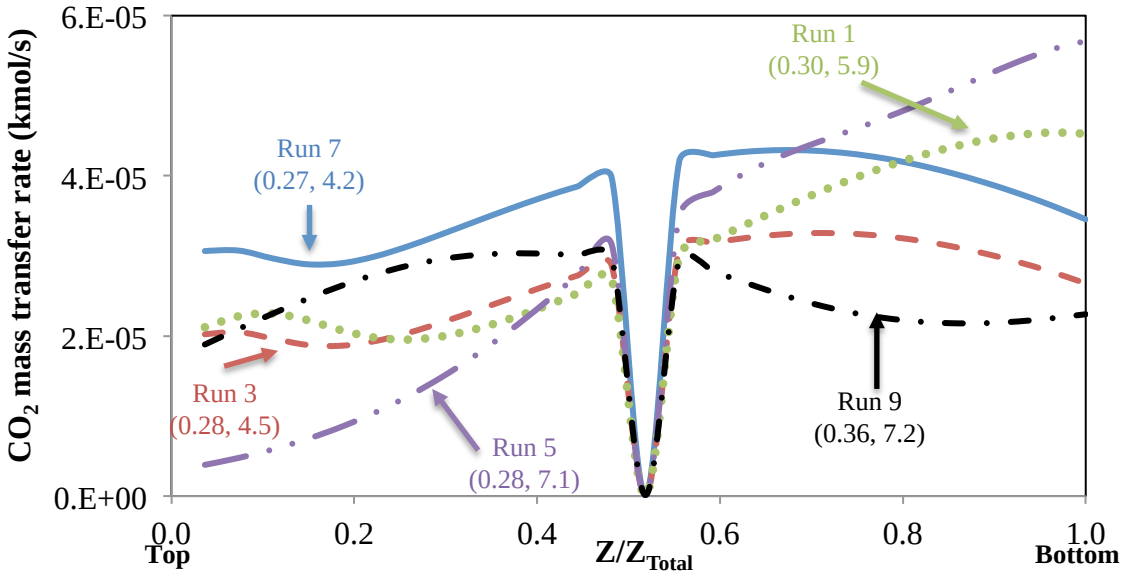


Figure 4.28: Predicted CO_2 mass transfer rate profiles. MEA Sept'10 Campaign. No Intercooling (lean loading, L/G).

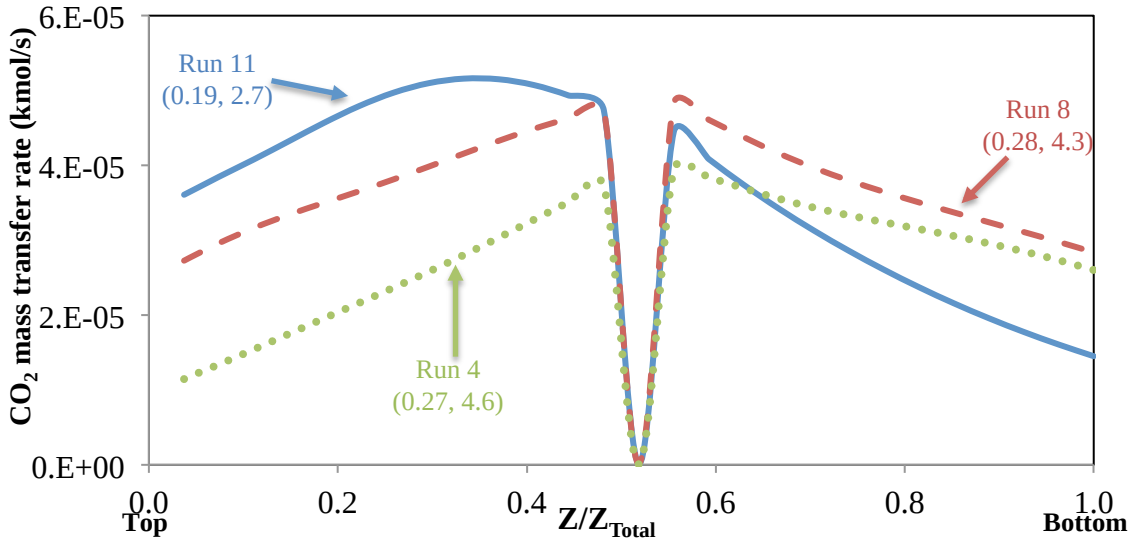


Figure 4.29: Predicted CO_2 mass transfer rate profiles. MEA Sept'10 Campaign. Intercooled (lean loading, L/G).

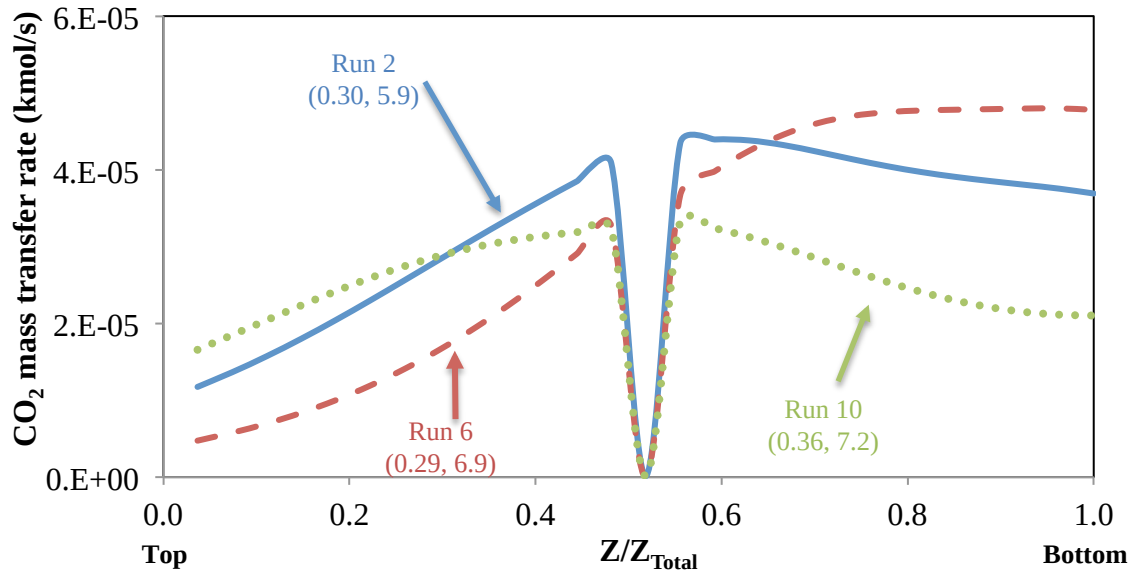


Figure 4.30: Predicted CO_2 mass transfer rate profiles. MEA Sept'10 Campaign. Intercooled (lean loading, L/G).

4.5. PZ CAMPAIGNS

Concentrated piperazine (PZ) has been proposed as an evolutionary step to MEA in CO_2 capture systems. Its volatility is similar to MEA but it provides greater CO_2 absorption rate, capacity and resistance to oxidative and thermal degradation than MEA. In order to demonstrate these benefits a one-month pilot plant campaign was conducted in November 2009 at The University of Texas at Austin with 5 m, 8 m and 9 m PZ.

A second pilot plant campaign in Summer 2010 evaluated the performance of 8 m PZ with an intercooled absorber and a third campaign coupled the absorber to a high-temperature two-stage flash. Table 4.23 summarizes the conditions for the PZ campaigns and Table 4.24 includes the resulting values for the manipulated variables.

Table 4.23: PZ Campaigns conditions

Campaign	November '09	September '10	December'10
PZ (m)	5, 8, 9	7.8 – 8.4	7.3 – 8.4
Gas rate (ACFM)	350	350 - 750	350 -650
Liquid Rate (gpm)	12, 15, 18	8-26	8 – 26
Absorber Packing	Mellapak 2X	RSP -250	RSP-250
Intercooling	No	Yes/No	Yes
Runs	14	12	9

Table 4.24: Resulting Manipulated Variables for PZ Pilot Plant Campaigns

Variable	PZ Campaign		
	Nov'09	Sept'10	Dec'10
Area factor	1.17±0.15	1.02±0.16	0.72±0.13
CO_2 Multiplier	1.05±0.03	1.05±0.03	1.06±0.04
A_L (W)	0	50.7±8460	2.41±239
B_L (W/K)	0	0.0	0.00
A_G (W)	1647±0.20	95.11±73.5	1983±551
B_G (W/K)	29.2±4.6	25.9±0.16	61.4±0.29

4.5.1. November 2008

This was the first campaign with concentrated PZ. Its goal was to verify the benefits of the solvent and to gain knowledge on the operational aspects of the solvent. Special attention was required for the preparation of the solvent. PZ is solid at normal conditions and must be loaded with CO_2 to achieve greater concentration. The pilot plant configuration is given in Figure 4.1. The absorber was packed with Sulzer Chemtech Mellapak 2X. Table 4.24 presents the resulting manipulated variables and Table 4.25 shows the resulting deviations for each run. Figures 4.31 – 4.36 show the resulting absorber temperature profiles.

The interfacial area was adjusted by 16.5% to match pilot plant results. CO_2 composition required a final positive loading adjustment of 5%. This adjustment seems to concur with the CO_2 mass balance offset with respect to the gas side using the titration results (Figure 4.11). This shift can also be observed in the loading parity plot (Figure 4.7). Piperazine concentration was reduced for all the runs, with the highest adjustments for the nominal 5m PZ campaigns (Runs 11 -14). This probably added to the positive shift in all campaign loadings.

The previously mentioned adjustments result in adequate CO_2 mass balance closure as seen in Figure 4.11.

Temperature profiles were adequately matched for all runs except Run 13 where fit might have been more difficult due to the low inlet gas temperature (-5.2°C).

Heat loss is very similar across the campaign with the highest value around 10 kW. Ambient temperature effects are noticeable (average ambient temperature 12.5°C ± 7°). It is worth mentioning that the heat loss is attributed to the gas side in reconciliation results (Table 4.24)

Table 4.25: Pilot Plant Reconciliation Result Deviations PZ November '08 Campaign

		Deviation[a]													
	Specified	Actual (%) – except for T													
Run		**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**
L/G (mol/mol)	**b**	-0.3	0.3	-0.3	0.0	-0.1	0.1	0.1	0.2	-0.2	-0.1	0.1	0.3	0.3	0.1
Lean ldg	15%	7.8	5.1	9.5	1.9	4.9	7.4	4.2	6.6	7.4	7.9	6.1	1.1	1.1	1.2
Rich ldg	15%	8.1	6.7	9.4	6.9	7.4	8.7	9.6	8.3	8.9	8.5	8.6	6.7	6.2	4.5
Δ ldg	N/A	9.0	14.5	9.1	202	16.0	15.2	43.3	15.3	14.7	12.9	20.7	24.7	19.2	11.5
PZ *(m)*	**b**	-0.9	-2.2	-4.6	-4.2	-5.9	-8.8	-9.1	-10.4	-13.5	-13.6	-14.5	-15.8	-17.1	-18.3
y_{CO2} - in	3.0%[c]	1.0	1.5	2.6	-0.3	2.1	-0.4	-0.3	0.6	1.4	2.6	0.2	-2.6	-2.6	-0.9
y_{CO2} - out	NA	2.2	-1.4	-2.5	0.2	-2.0	-1.8	-0.5	0.7	-6.3	-3.1	0.7	9.8	2.8	1.3
T_G – in (°C)	2.0	0.5	0.4	-0.2	0.1	0.2	0.0	0.0	0.4	0.4	0.2	0.3	0.1	0.1	0.5
T_G – out (°C)	2.0	4.1	3.2	7.0	0.0	5.2	3.3	2.0	3.5	1.6	0.8	2.8	3.1	0.4	2.4
T_L – in (°C)	2.0	0.9	-0.1	0.9	-0.5	0.7	-1.5	-1.4	-1.5	-1.0	-1.2	-0.7	-1.4	-0.1	0.3
T_L – out (°C)	4.0	-2.0	-3.6	-4.1	-4.7	-4.6	-4.3	-3.0	-2.5	-3.2	-2.8	-2.5	-3.3	-2.4	0.0
NTU	1.0%	2.1	1.2	2.3	0.4	1.3	1.6	1.4	1.0	2.8	0.8	1.0	2.1	1.6	1.2
(Dev)[d]		(0.7)	(2.7)	(2.5)	(0.6)	(3.6)	(0.9)	(0.2)	(0.0)	(3.1)	(8.6)	(-0.4)	(-5.5)	(-3.6)	(-2.0)
Q_{Loss} (kW)[e]	---	10.0	9.1	9.8	9.1	9.9	6.7	10.2	8.7	10.2	8.0	9.9	7.7	9.7	9.7

[a] Positive value indicates that reconciled value is greater than measured .
[b] Specified using individual component streams.
[c] Specified using the lean CO_2 stream and the CO_2 multiplier (see Figure 4.15)
[d] This row contains the resulting NTU and the deviation in parenthesis.
[e] Calculated heat loss from reconciliation

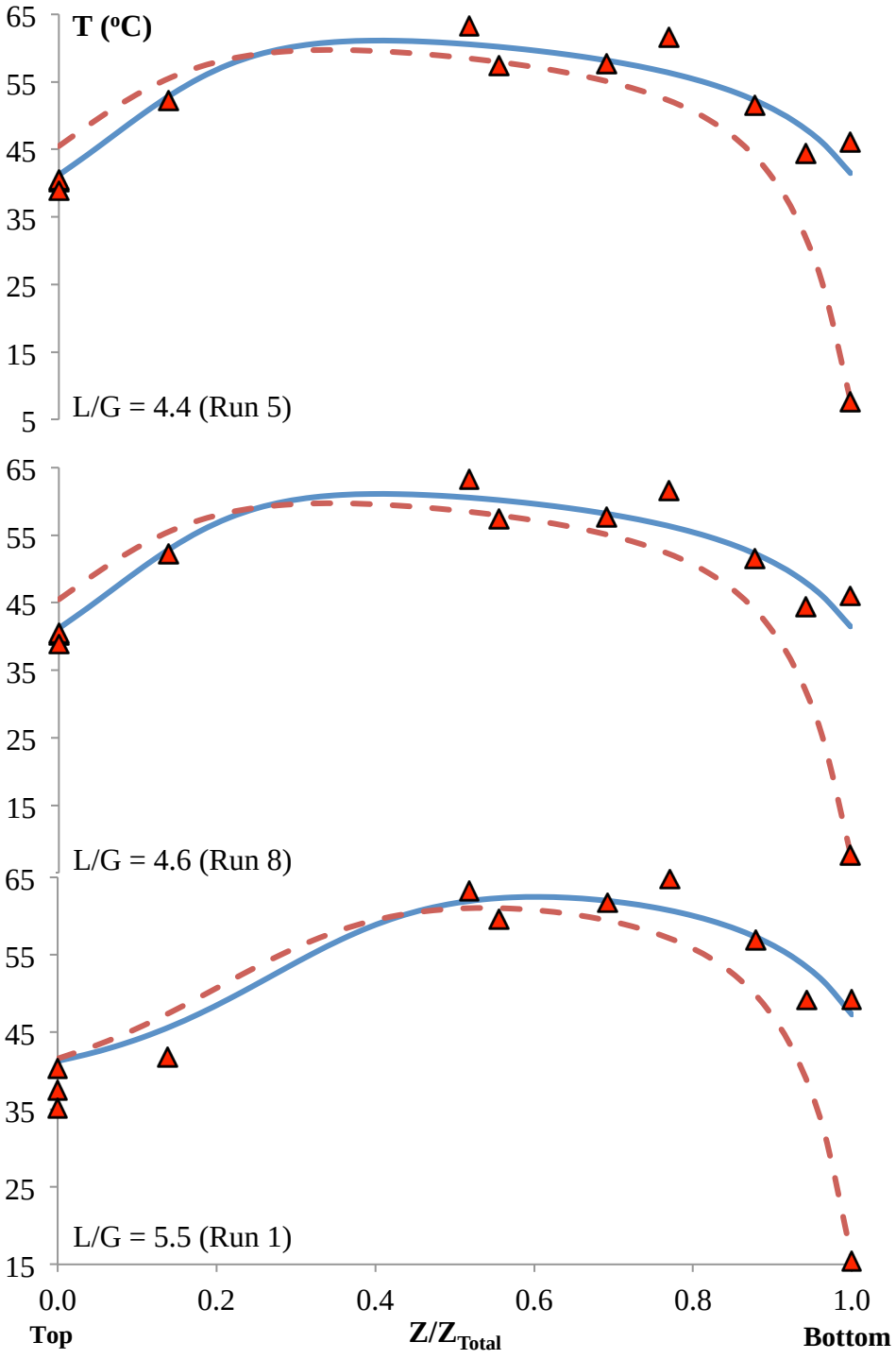


Figure 4.31: T profiles. PZ Nov. '08 Campaign. 7m PZ (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G in mol/mol)

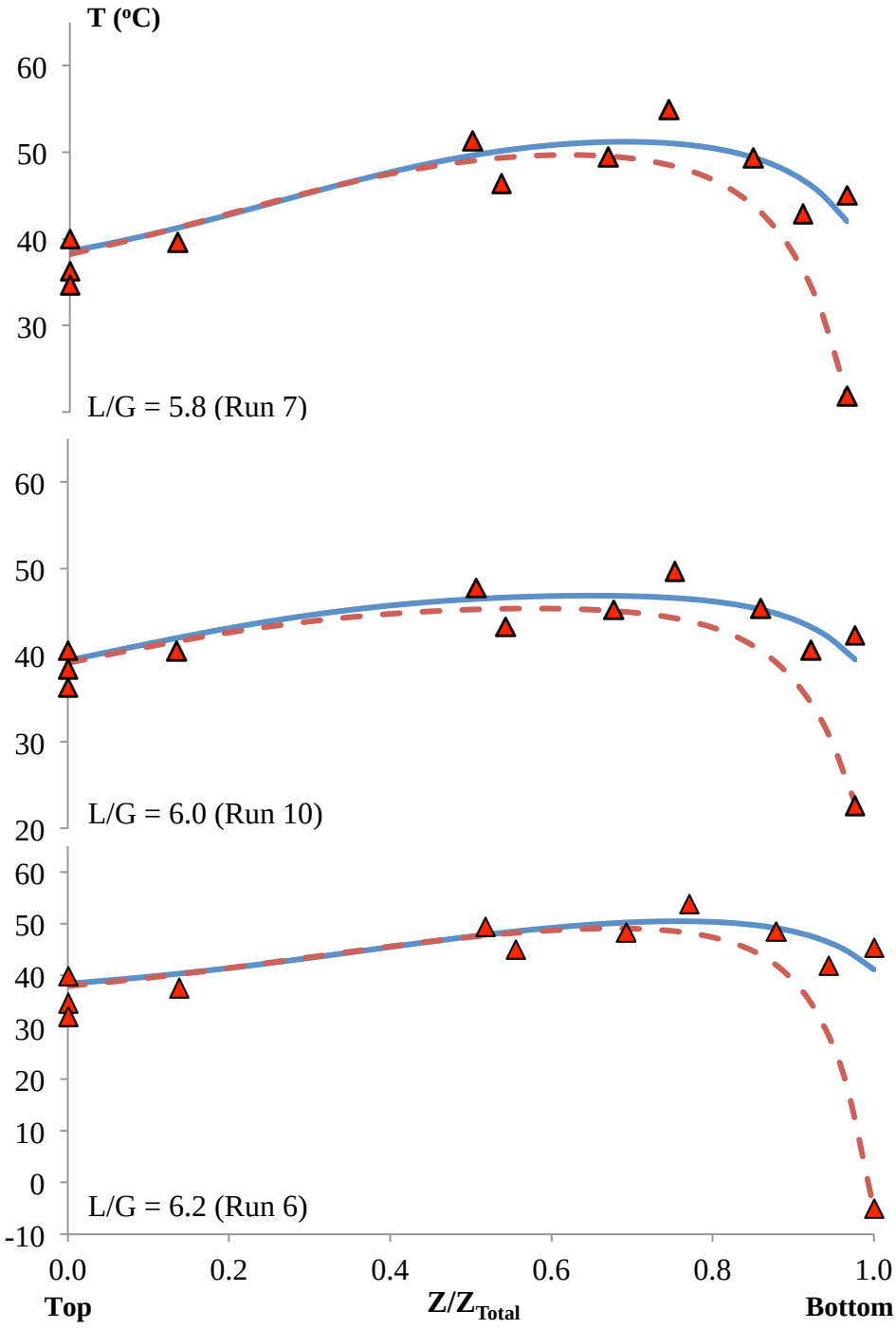


Figure 4.32: T profiles. PZ Nov. '08 Campaign. 7m PZ (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G mol/mol)

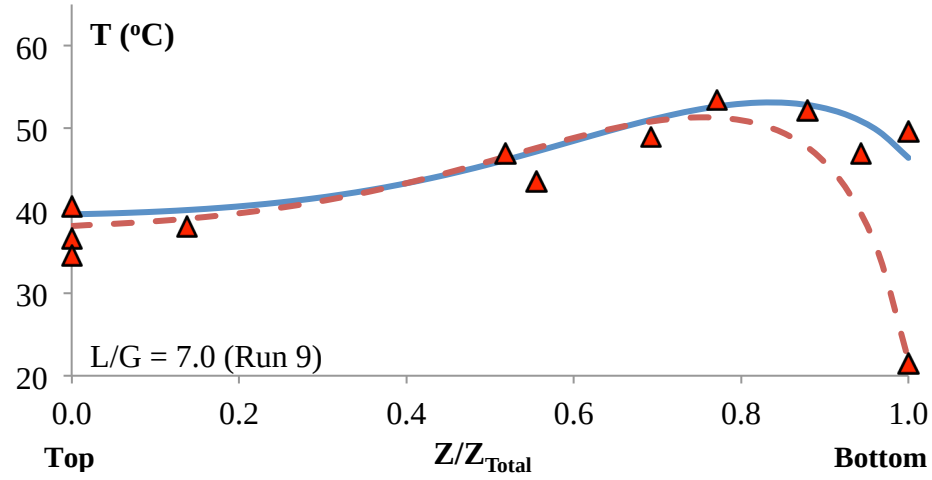


Figure 4.33: T profiles. PZ Nov. '08 Campaign. 7m PZ (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G mol/mol)

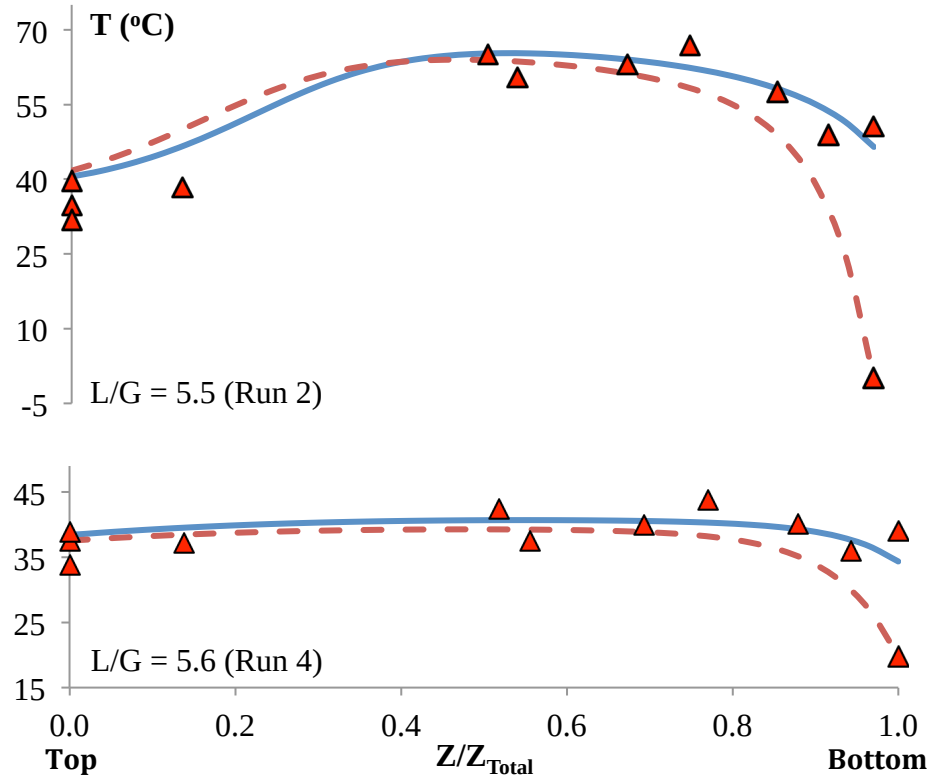


Figure 4.34: T profiles. PZ Nov. '08 Campaign - 8 m PZ. (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G mol/mol)

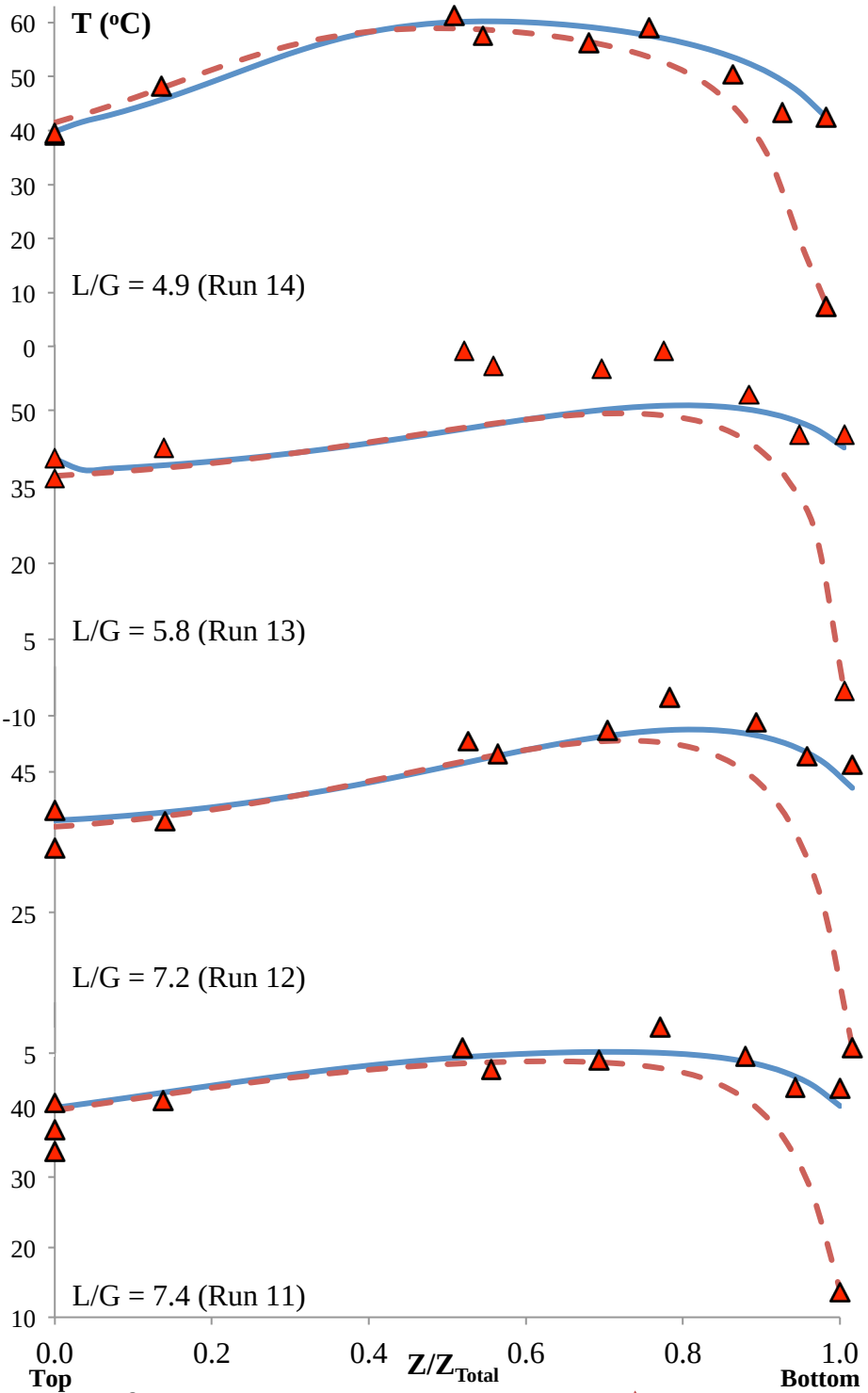


Figure 4.35: T profiles. PZ Nov. '08 Campaign. 4 m PZ. (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G mol/mol)

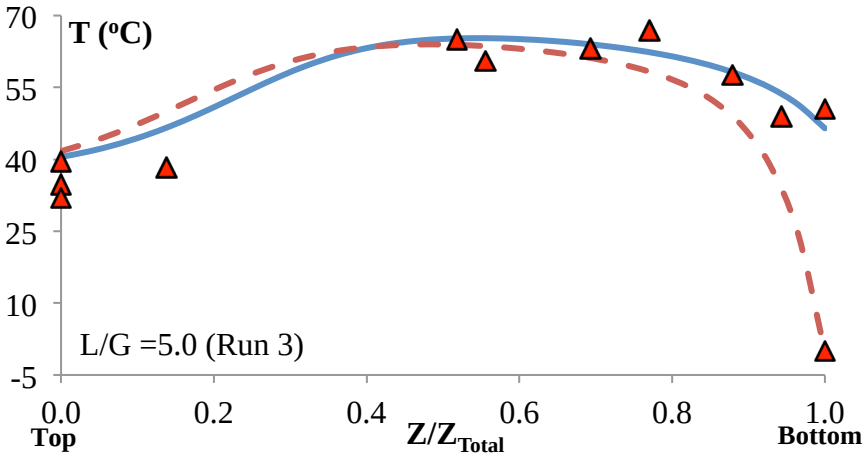


Figure 4.36: T profiles. PZ Nov. '08 Campaign. 9 m PZ. (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G mol/mol)

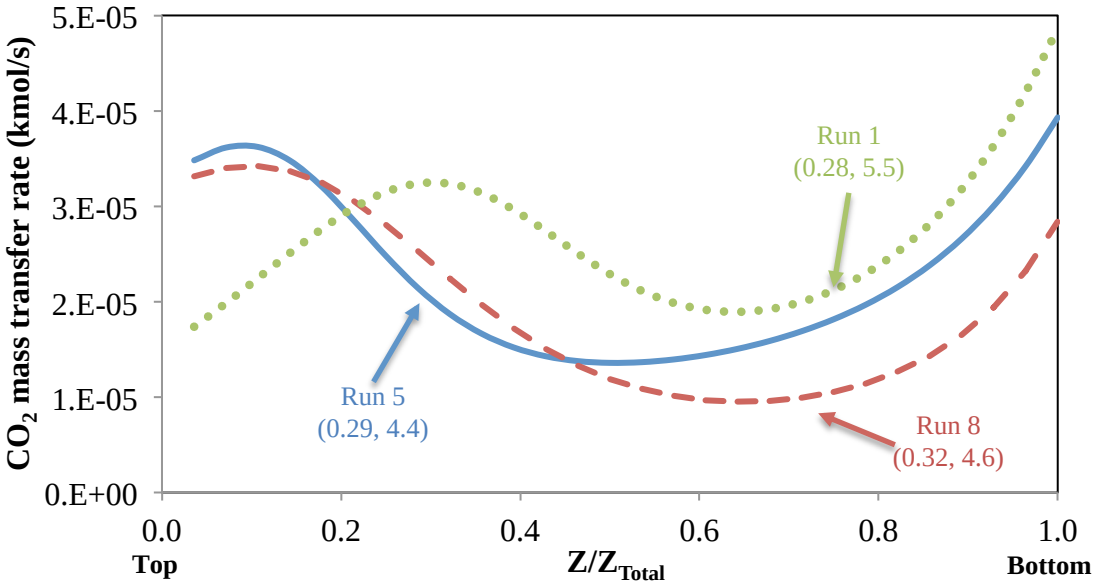


Figure 4.37: Predicted CO_2 mass transfer rate profiles. PZ Nov. '08 Campaign. 7m PZ (lean loading, L/G).

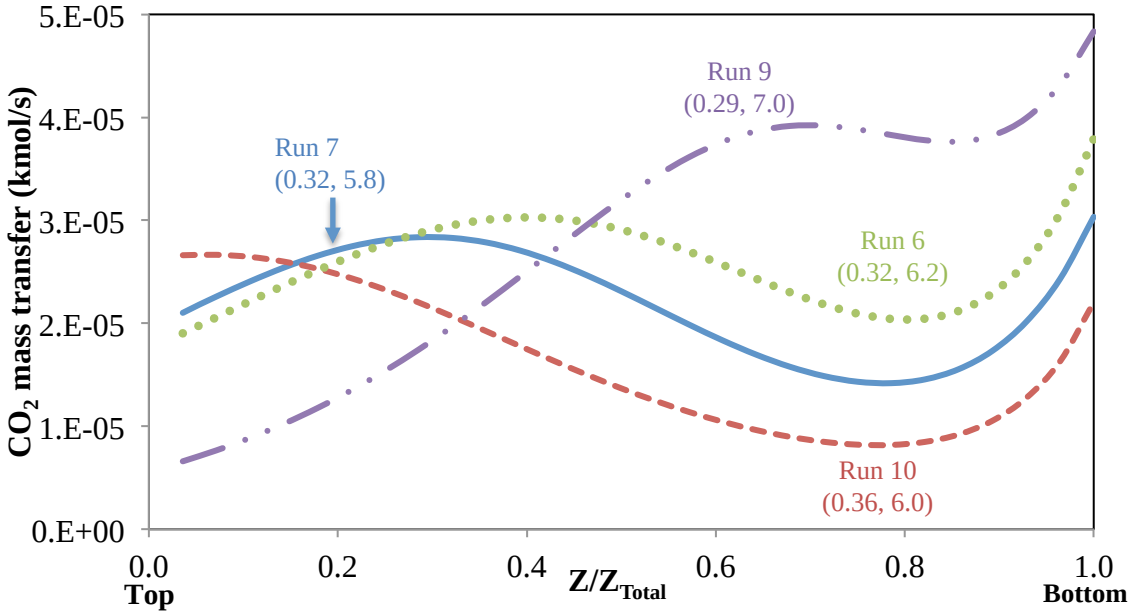


Figure 4.38: Predicted CO_2 mass transfer rate profiles. PZ Nov. '08 Campaign. 7m PZ (lean loading , L/G).

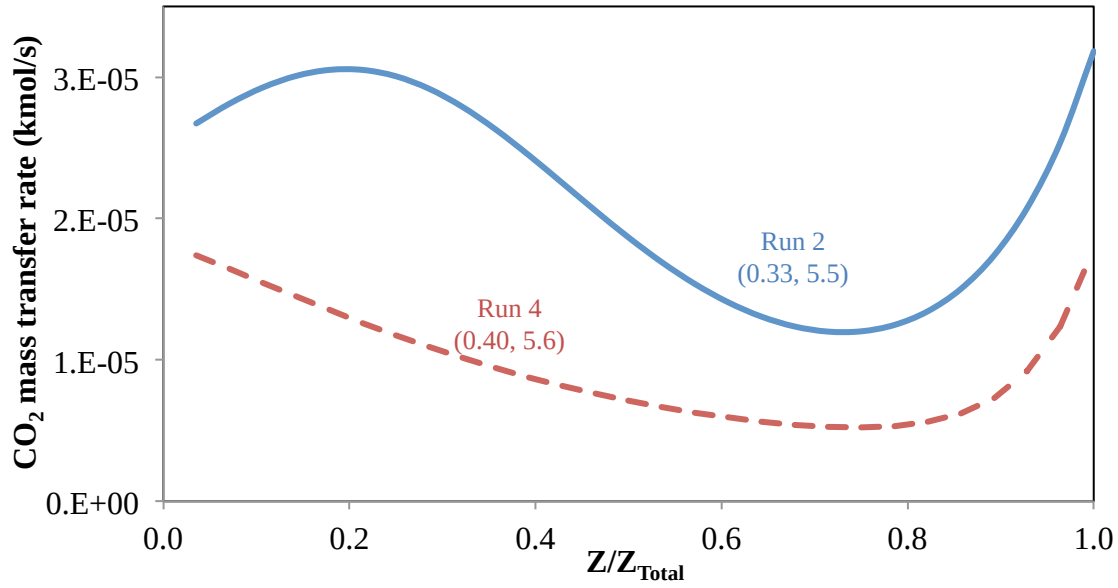


Figure 4.39: Predicted CO_2 mass transfer rate profiles. PZ Nov. '08 Campaign. 8m PZ (lean loading, L/G).

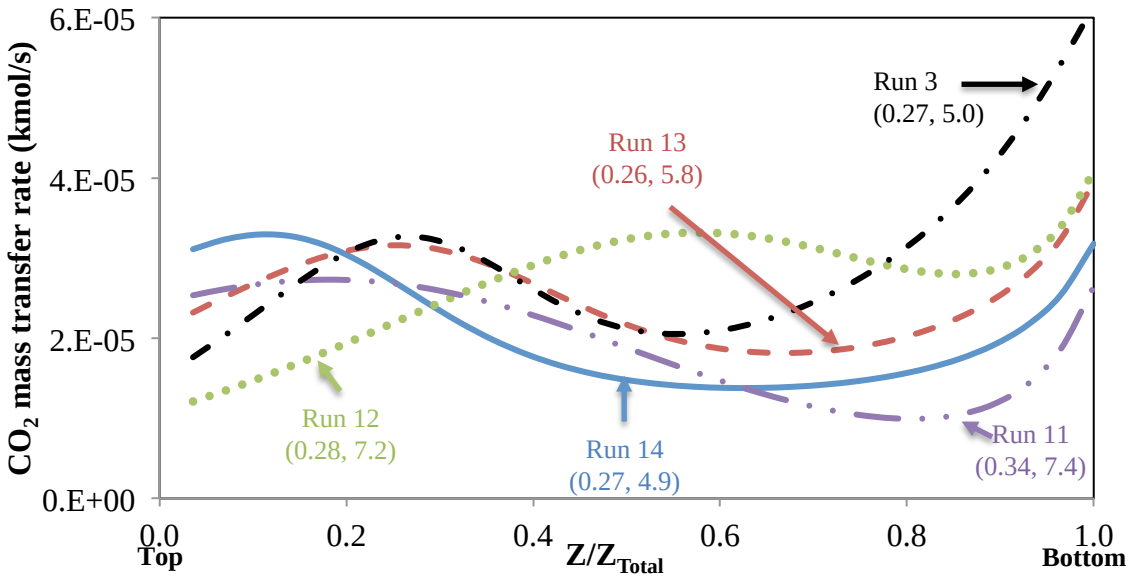


Figure 4.40: Predicted CO_2 mass transfer rate profiles. PZ Nov. '08 Campaign. 4m PZ , 9m PZ for Run 3 (lean loading, L/G).

The PZ November'09 campaign also varied amine concentration. Comparison between Run 3 and Run 14 with lean loading of 0.27 and L/G = 5 show a 50% increase in NTU by changing amine concentration from 4m to 9m. Similar increase in performance with higher amine concentration can be observed when comparing Run 12 and Run 9, with a loading around 0.28 and L/G around 7, NTU increases from 2.15 for 4m PZ to 2.76 for 7m PZ.

Temperature profiles in Figures 4.31 to 4.36 have been organized in ascending L/G order for each PZ concentration. It is possible to observe how the temperature bulge moves from near the top of the column at low L/G values (4.4 mol/mol) to towards the bottom at the high L/G (7.0 mol/mol).

Figures4.37 to 4.40 show how CO_2 mass transfer rates in the absorber are affected by the lean loading feed and the L/G. The highest absorber removal (NTU = 2.8) is achieved in Run 9. Its lean loading with a higher L/G (7 mol/mol) limits the temperature

increase across the column maintaining the CO_2 mass transfer rates through most of the column. However, it is possible to observe that the column is close to a lean side pinch (mass transfer rate = 0) and the highest removal rates occur at the bottom.

Runs 5, 8, 7, and 10 show an approach to a mass transfer pinch coinciding with the column temperature bulge. These cases are limited by the increase in temperature and benefit the most from the use of intercooling, as it will be later shown. Run 10 fed a high loading lean solvent that is close to pinching towards the bottom of the column. However, its performance at the top is slightly boosted by the lower temperatures due to the high L/G (mol/mol).

In Run 4 temperatures are kept low due to the limited absorption and high L/G (5.6 mol/mol). Mass transfer rates are limited due to the high loading of the lean solvent.

Run 9, with higher amine concentration (9m), has the second best performance of the campaign. It is aided by the lower lean loading and higher L/G (mol/mol). Its mass transfer profile shows that it is approaching a lean side pinch with most removal occurring at the bottom of the column.

Run 11 was the highest L/G tested during the campaign but removal was limited by the high lean loading (0.34) and lower amine concentration (4m). Removal in Run 13 may have been slightly boosted at the bottom of the column due to the low temperature feed gas (-10°C)

4.5.2. September 2010

This main objective of this campaign was to evaluate the increase in absorber CO_2 removal performance with intercooling. The absorber used Raschig-Jaeger Raschig Super Pak 250 (RSP-250). It is said to offer great separation efficiency and high loading capacity with low-pressure drop. It is considered a hybrid packing offering performance

between random and structured packing. Various L/G values were tested. The resulting reconciliation manipulated variables are presented in Table 4.8.

The interfacial area was increased only 2%. CO_2 content was adjusted 5.4% across all campaign runs. As with the November'09 campaign this generated a positive shift in lean and rich loadings. However, loadings were maintained within 10% of the absorber side titration results (Figure 4.9). It is possible to match the CO_2 gas side calculated removal (Figure 4.13) closing the mass balance.

PZ concentration adjustments were much less than the required for the November'09 campaign (maximum -2.3%). Similarly the CO_2 inlet mol fraction was adjusted less than ± 0.5% which is lower than the resulting adjustments for the previous campaign.

Temperature profiles are adequately matched with better results for the cases with no intercooling (Figure 4.41). For the intercooled cases temperature profiles are well represented (Figures 4.32 – 4.44) with a slight under prediction of the temperature bulge for the lower L/G (< 4.0) and overestimation for the higher L/G (5 and above).

Heat loss values are approximately 8 kW for all runs. This value is lower than the average for November'09 and agrees with the higher ambient temperatures experienced for this campaign (25°C ±5°). As with the previous PZ campaign the heat loss is largely attributed to the gas side. The parameters for the liquid side heat loss (A_L, B_L) are small.

Figures 4.45 to 4.48 show the predicted CO_2 mass transfer profile rates for the runs in the campaign. Run 9 with a low L/G (2.8 mol/mol) and no intercooling shows a high temperature bulge (75°C) at the top of the column. However, most of the removal occurs at the bottom of the column and the top tends towards a pinch. A slight increase in removal rate at the bottom of the column is probably due to the low feed gas temperature (25°C).

Table 4.26: Pilot Plant Reconciliation Result Deviations PZ September '10 Campaign

		Deviation[a]												
	Specified	Actual (%) – except for T												
Run		1	2	3	4	5	6	7	8	9	10	11	12	13
L/G (mol/mol)	b	0.0	0.0	0.0	0.1	0.0	-0.1	0.0	0.1	0.0	-0.2	-0.3	0.1	0.0
Lean ldg	20.0%	5.4	5.5	5.1	2.8	4.4	5.9	5.5	3.1	5.5	9.0	9.2	4.8	6.9
Rich ldg	20.0%	1.6	6.0	3.5	4.2	1.1	7.0	7.7	6.0	4.4	5.5	4.9	0.4	2.2
Δ ldg	N/A	-11.6	7.9	-2.7	10.3	-8.0	9.7	14.1	14.2	2.8	0.7	-0.9	-7.0	-5.0
PZ *(m)*	b	0.0	-0.1	0.2	1.3	0.3	-0.7	-0.1	1.2	-0.2	-2.2	-2.3	0.3	-0.7
y_{CO2} - in	3.0%[c]	-0.1	0.0	-0.1	-0.1	0.1	0.3	-0.1	-0.2	-0.1	0.4	0.4	0.0	-0.1
y_{CO2} - out	N/A	2.0	5.8	8.3	4.2	2.6	1.1	6.0	2.8	0.8	-4.3	-0.2	0.1	0.0
T_G – in (°C)	4.0	0.0	0.0	0.0	0.0	-0.1	0.0	0.1	0.0	-0.3	0.0	0.1	0.2	0.3
T_G – out (°C)	4.0	1.6	3.3	0.9	1.3	5.5	6.0	1.5	1.5	3.3	3.4	2.1	3.3	2.1
T_L – in (°C)	4.0	-0.8	-0.4	-0.9	-0.8	0.1	0.6	-0.1	-1.8	-1.7	-0.5	0.3	-2.5	-1.4
T_L – out (°C)	4.0	-0.2	0.5	-0.4	0.1	1.5	0.9	-0.7	-0.3	3.2	2.5	0.9	0.1	-0.9
NTU	1.0%	2.5	2.2	2.4	1.6	2.2	2.2	2.2	1.9	2.6	2.4	2.5	2.3	1.7
(Dev)[d]		(-1.8)	(-2.8)	(-2.9)	(-3.1)	(-1.1)	(-0.2)	(-2.6)	(-2.4)	(-1.3)	(0.7)	(0.6)	(-0.3)	(0.0)
Q_{Loss} (kW)[e]	---	7.9	7.8	8.0	7.9	7.9	7.9	8.1	8.1	7.9	7.6	8.0	7.7	8.0

[a] Positive value indicates that reconciled value is greater than measured .
[b] Specified using individual component streams.
[c] Specified using the lean CO_2 stream and the CO_2 multiplier (see Figure 4.15)
[d] This row contains the resulting NTU and the deviation in parenthesis.
[e] Calculated heat loss from reconciliation.

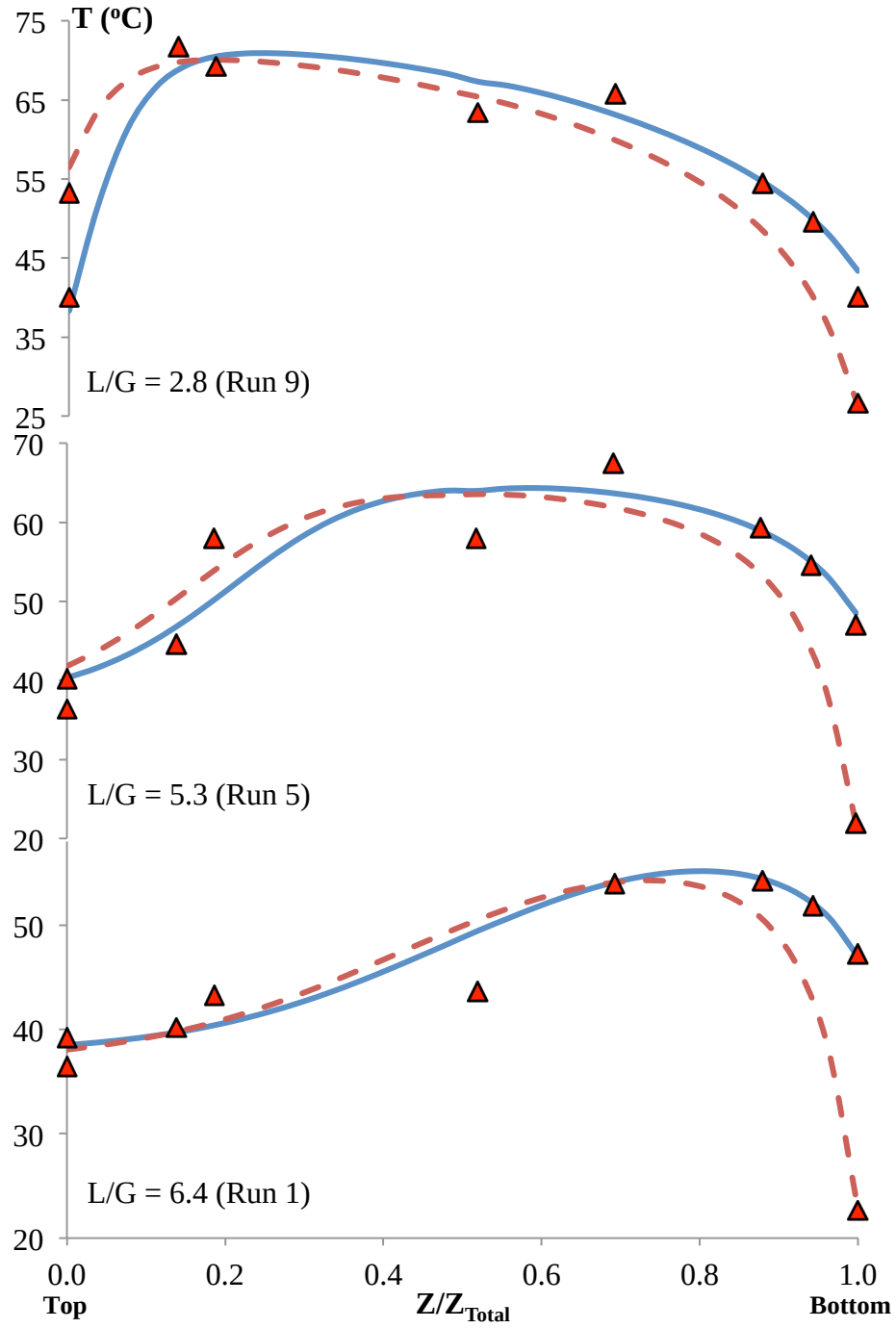


Figure 4.41: T profiles. PZ Sept. '10 Campaign. No intercooling: (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G mol/mol)

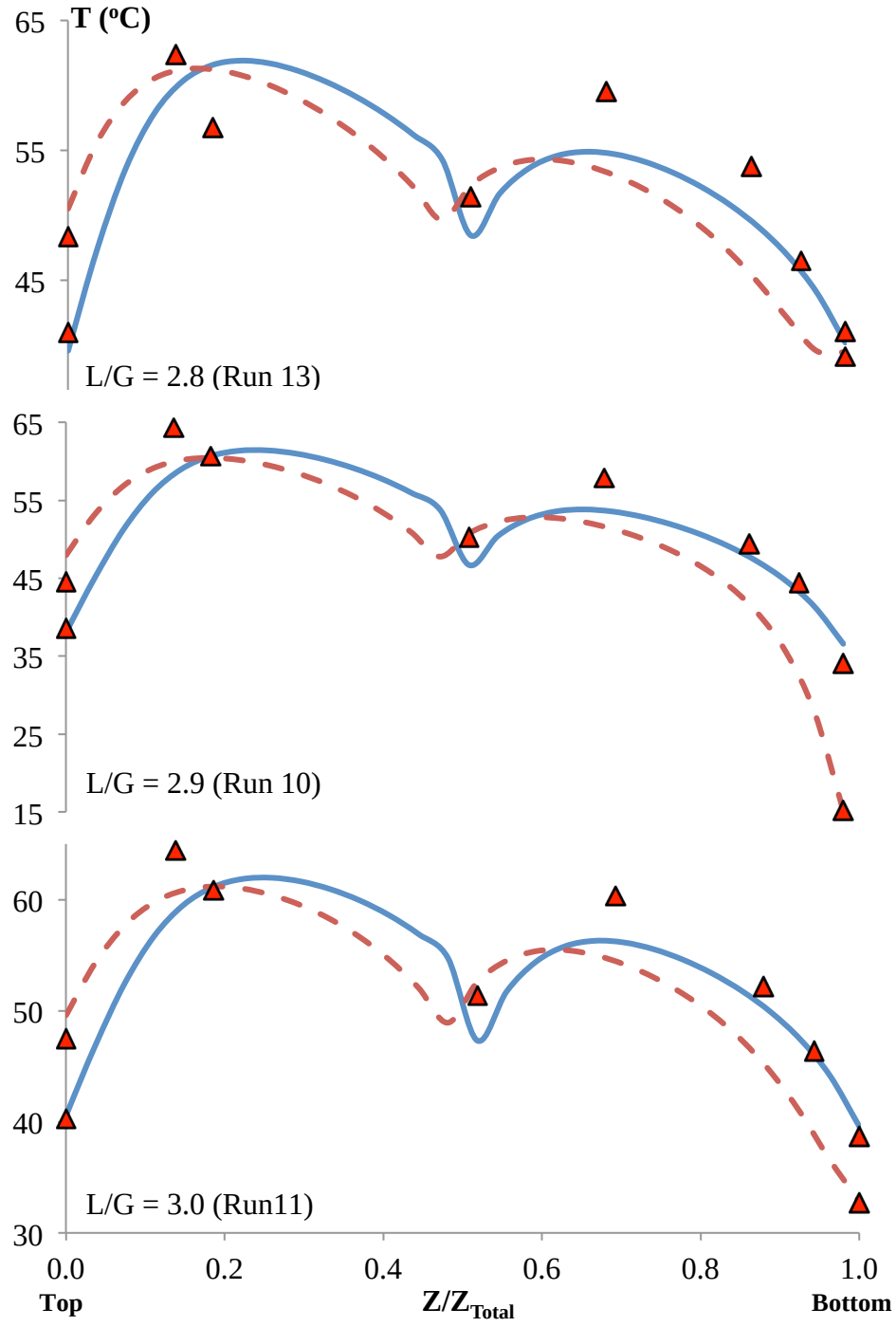


Figure 4.42: T profiles. PZ Sept. '10 Campaign. Intercooled: (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G mol/mol)

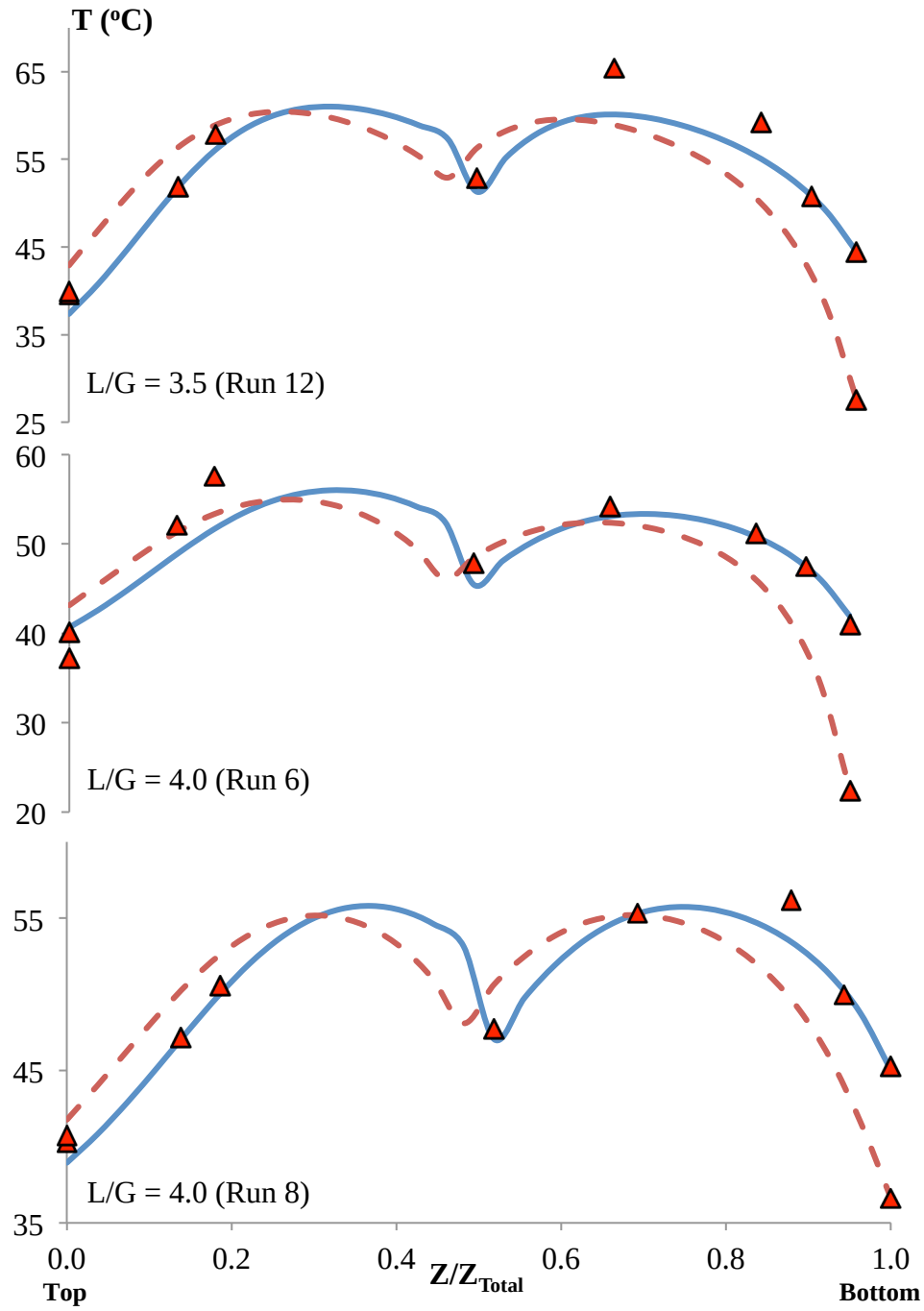


Figure 4.43: T profiles. PZ Sept. '10 Campaign. Intercooled: (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G mol/mol)

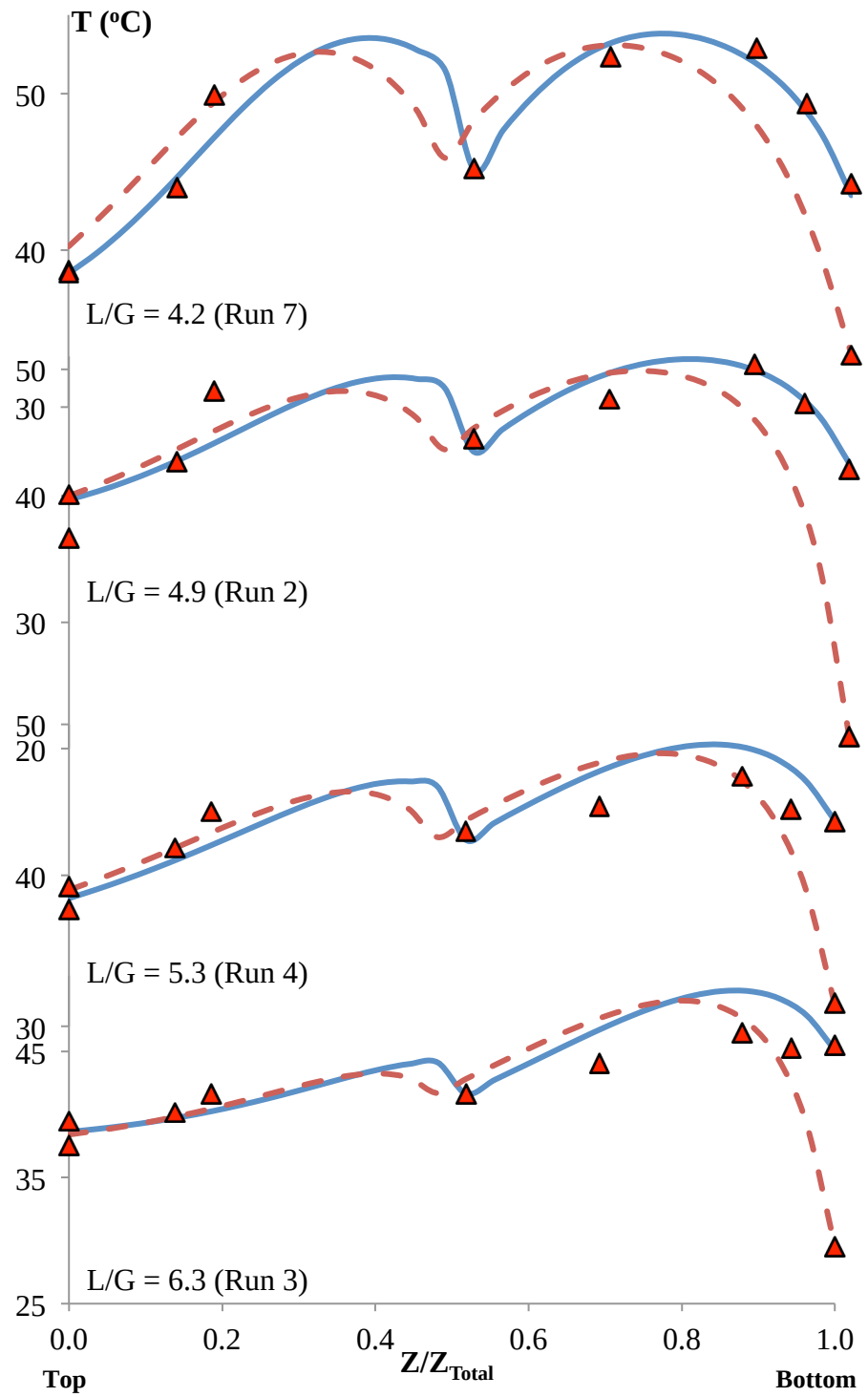


Figure 4.44: T profiles. PZ Sept. '10 Campaign. Intercooled: (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G mol/mol)

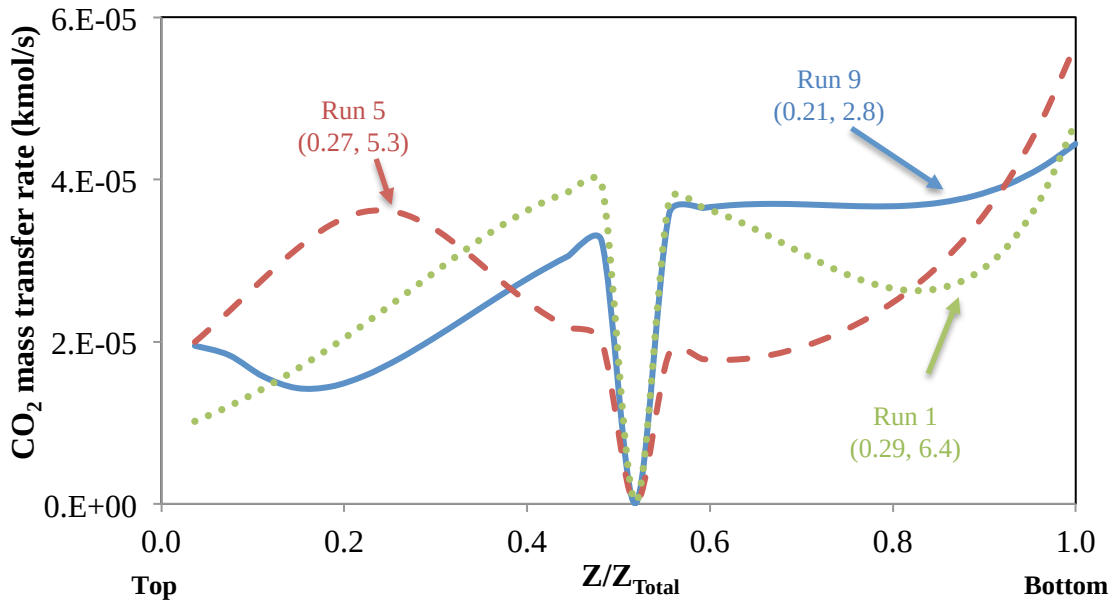


Figure 4.45: Predicted CO_2 mass transfer rate profiles. PZ Sept. '10 Campaign. No intercooling (lean loading, L/G).

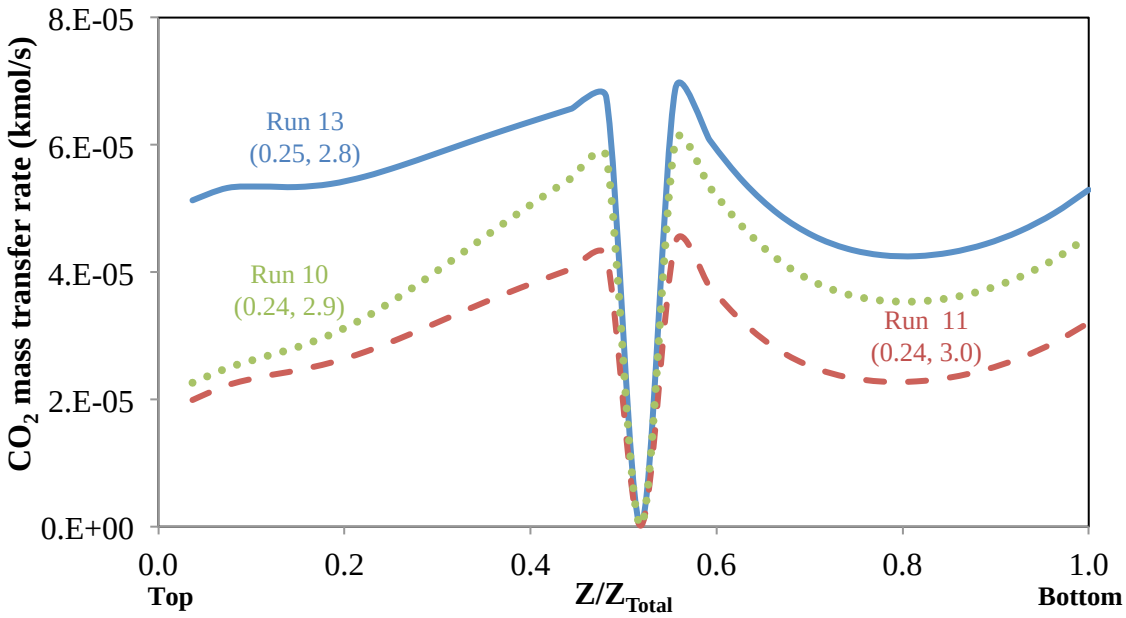


Figure 4.46: Predicted CO_2 mass transfer rate profiles. PZ Sept. '10 Campaign. Intercooled (lean loading, L/G). Runs 10, 11, 13

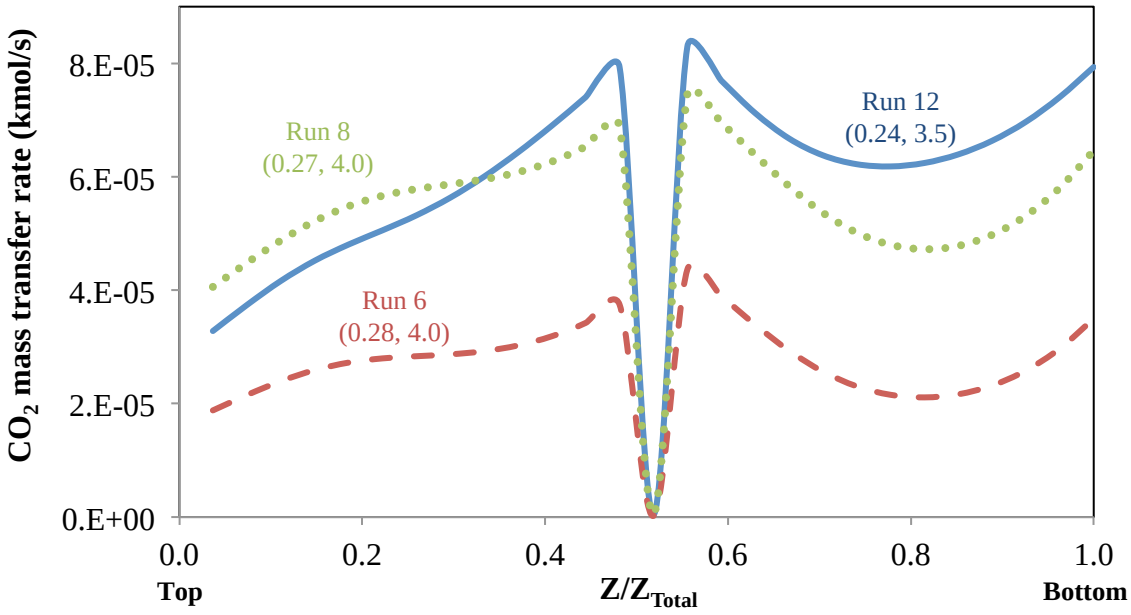


Figure 4.47: Predicted CO_2 mass transfer rate profiles. PZ Sept. '10 Campaign. Intercooled (lean loading, L/G). Runs 6, 8, 12.

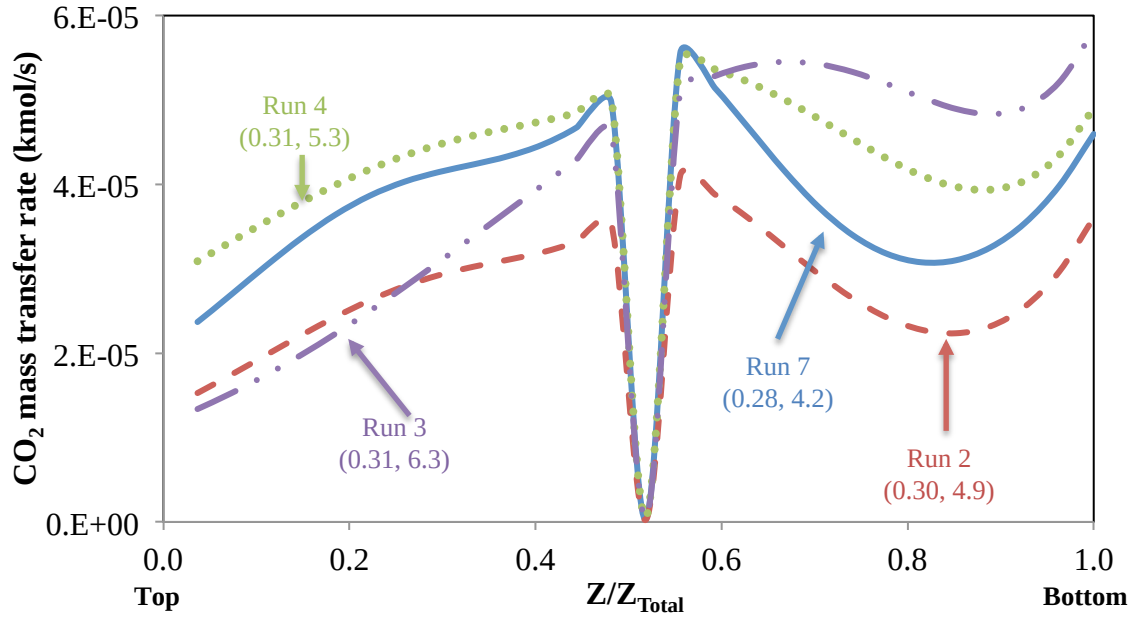


Figure 4.48: Predicted CO_2 mass transfer rate profiles. PZ Sept. '10 Campaign. Intercooled (lean loading in parenthesis). Runs 2,3,4,7.

Run 5 with a relative high L/G (5.3 mol/mol) has a temperature bulge slightly towards the bottom of the column coinciding with the lower CO_2 mass transfer rate. The higher CO_2 mass transfer rates are towards the extremes of the column. At the bottom the low temperature of the gas feed increases removal rate and the top section the lower column temperature does not restrict mass transfer rates.

Run 1 has the richest lean solvent feed (0.29) for the non-intercooled rate that combined with the high L/G (6.4 mol/mol) obtains considerable removal with a NTU = 2.5. The CO_2 mass transfer profile shows that most of the removal occurs towards the middle of the column approaching a lean end pinch.

In general, intercooled mass transfer profiles for low L/G show higher CO_2 mass transfer removal rates towards the top section of packing and moves towards around the intercooling section as the L/G is increased. The highest L/G have increased performance at the bottom of the column and tend to approach a lean end pinch. In these runs (2, 3, 4) the high L/G coupled with intercooling maintains low temperatures throughout the column causing mass transfer to benefit more from the higher bottom section driving forces.

During the PZ September ’10 campaign liquid samples were drawn from the middle of the column for loading determination. Figure 4.49 compares the pilot plant reported results to the 5deMayo model results. The data is overall within +20% of the reported data. However, considering the loading shift (approx. 6%) from the reconciliation, the pilot plant data and the Aspen Plus® model results are within 10%. This is a good indicator that the kinetics used in the model are adequately representing the rate behavior of the pilot plant absorber.

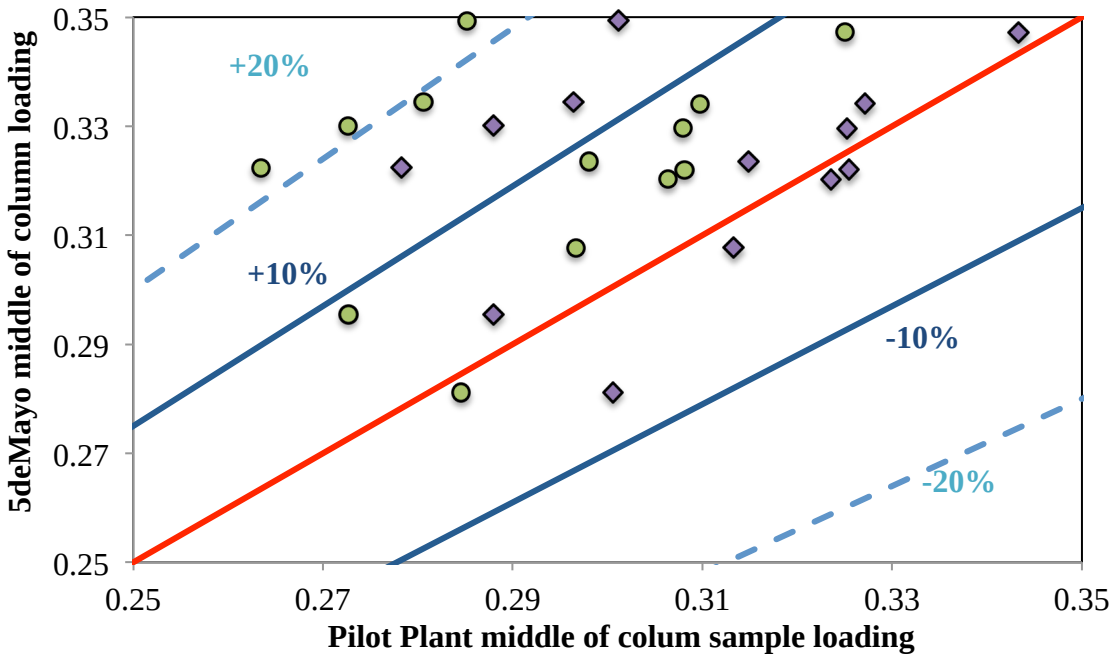


Figure 4.49: Parity plot for loading at the middle of the absorber for the PZ Sept'10 Campaign. (●) Reported pilot plant data (◆) Pilot Plant loading as shifted by the Aspen Plus ® Reconciliation.

4.5.3. December 2010

The main goal of this campaign was to test the two-stage flash process configuration to regenerate the solvent. This campaign used the same packing as the September '10 (RSP-250) and all runs used absorber intercooling. The manipulated variables are presented in Table 4.24.

The interfacial area for this campaign was reduced 28% and the CO_2 content shift was of 6% leading to lean loading adjustments below 11%. These adjustments are in line with the observed discrepancies for CO_2 removal between liquid and gas side, (Figure 4.14). However, calculated removal values between phases show deviations above 20% in some cases. Amine concentration was adjusted maximum 3% as well as

CO_2 gas mol fraction. This probably led to the observed decrease in interfacial area to match equipment performance.

Temperature profiles are adequately matched. Heat loss was estimated as the highest in all of the campaigns studied (around 20 kW). As with previous campaigns assigned mostly to the gas side. Average ambient temperature was around $10^oC \pm 5^o$, the lowest of all campaigns.

In Run 5 (Figure 4.50) the observed temperature profile is different from that expected for such low L/G. The small temperature bulge is observed towards the bottom of the column. However the outlet liquid temperature is extremely low compared to the surrounding values and is not matched by the model.

Temperature in the profiles of the column for this campaign are lower than most of the runs of the PZ September'10 Campaign. Overall removal is improved by intercooling for most runs (NTU above 2.0). The CO_2 mass transfer rate is increased around the intercooling section. Some runs show an increase in the removal rate for the bottom section and a lean end pinch approach (Runs 1, 3, 4, 6).

Run 8 with a richer lean solvent (0.31) has low CO_2 mass transfer rates throughout the column. However, there is no pinch across the column so performance is better than similar non-intercooled cases.

Table 4.27: Pilot Plant Reconciliation Result Deviations PZ December'10 Campaign

		Deviation[a]								
	Specified	Actual (%) – except for T								
Run		1	2	3	4	5	6	7	8	9
L/G (mol/mol)	b	-0.1	-5.0	-1.3	-2.3	-2.2	-1.7	1.6	0.8	0.4
Lean ldg	15%	6.9	5.7	9.7	7.8	5.3	7.1	5.2	6.8	4.1
Rich ldg	15%	10.5	10.1	6.0	5.5	4.3	2.0	0.3	5.2	1.4
Δ ldg	N/A	21.5	21.2	-2.0	0.8	1.5	-13.5	-17.5	0.3	-5.7
PZ *(m)*	b	-0.4	6.0	-0.9	1.8	3.1	1.6	-1.7	-1.6	0.7
y_{CO2} - in	3.0%[c]	0.3	1.1	1.1	0.6	-0.6	-1.1	0.6	0.7	-0.5
y_{CO2} - out	3.0%	0.7	-2.3	0.8	-0.2	-2.7	-2.5	1.4	1.3	-0.4
T_G – in (°C)	4.0	0.5	0.0	0.5	0.5	-1.2	-1.1	0.3	0.1	0.0
T_G – out (°C)	4.0	3.5	5.9	2.6	1.9	4.1	1.2	3.4	4.2	2.5
T_L – in (°C)	4.0	0.8	4.0	1.5	0.4	-2.4	-2.4	-0.2	2.0	-0.1
T_L – out (°C)	4.0	-6.0	NA	-2.2	-1.7	23.6	20.3	-3.6	0.0	NA
NTU	1.0%	2.2	2.2	1.2	1.0	2.5	2.3	2.0	1.5	1.75
(Dev)[d]		(-0.1)	(1.7)	(0.4)	(0.8)	(0.8)	(0.6)	(-0.4)	(-0.3)	(0.0)
Q_{Loss} (kW)[e]	---	19.6	19.7	19.6	19.1	18.9	19.1	19.5	19.8	19.5

[a] Positive value indicates that reconciled value is greater than measured .
[b] Specified using individual component streams.
[c] Specified using the lean CO_2 stream and the CO_2 multiplier (see Figure 4.15)
[d] This row contains the resulting NTU and the deviation in parenthesis.
[e] Calculated heat loss from reconciliation.

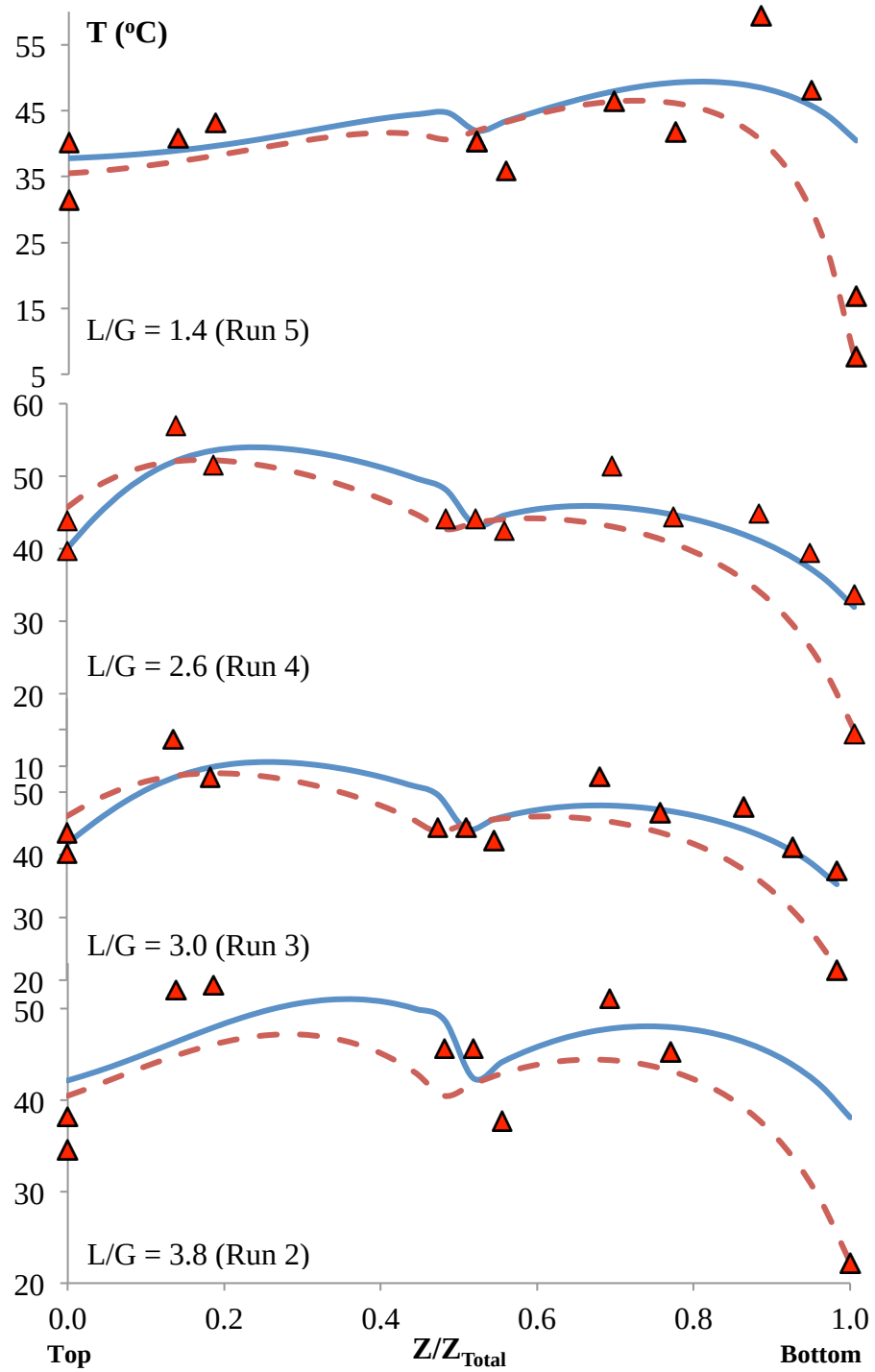


Figure 4.50: T profiles. PZ December'10 Campaign. (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G mol/mol) *L/G*= 1.4 - 3.8

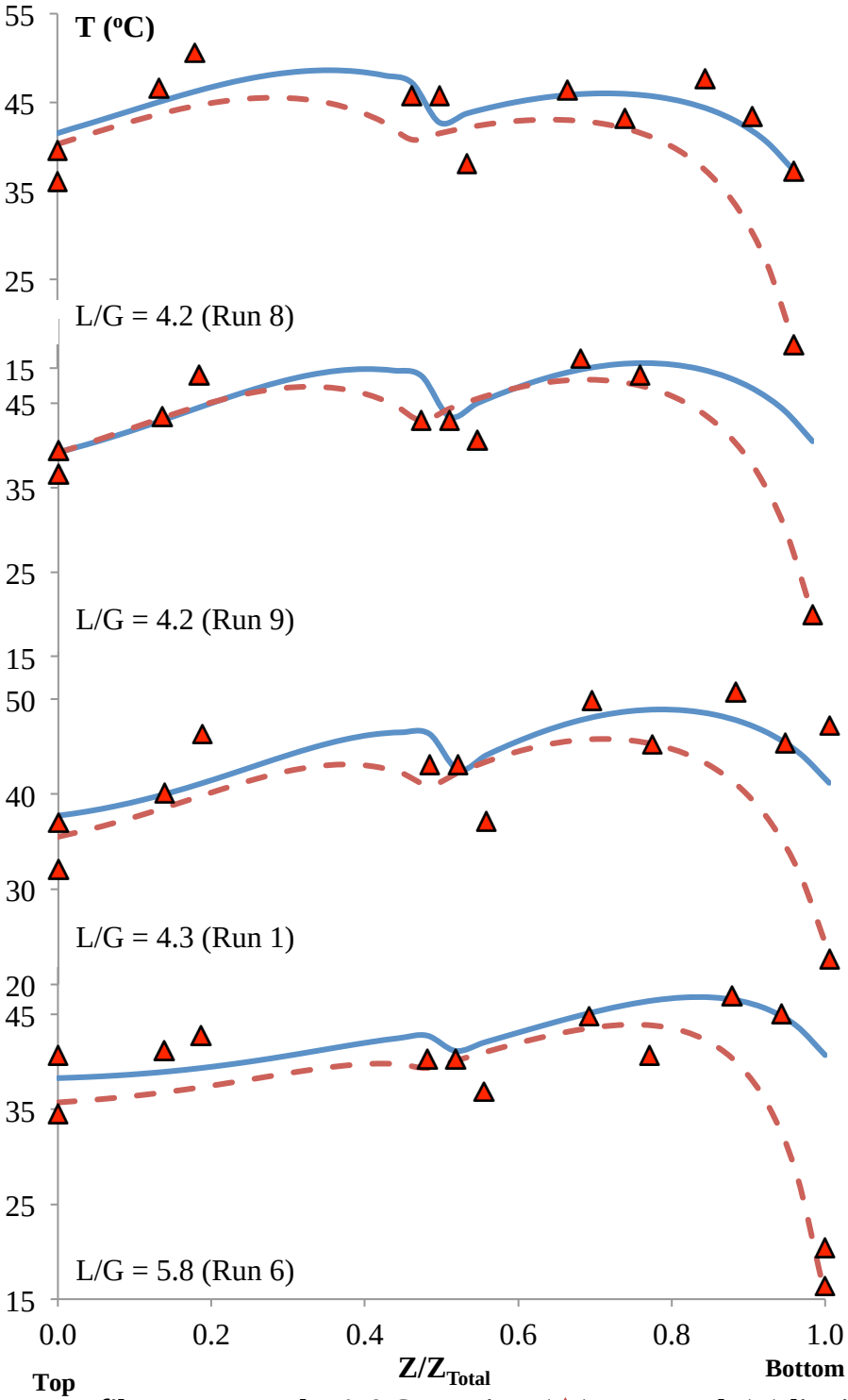


Figure 4.51: T profiles. PZ December'10 Campaign. (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G mol/mol). *L/G*= 4.2 - 5.8

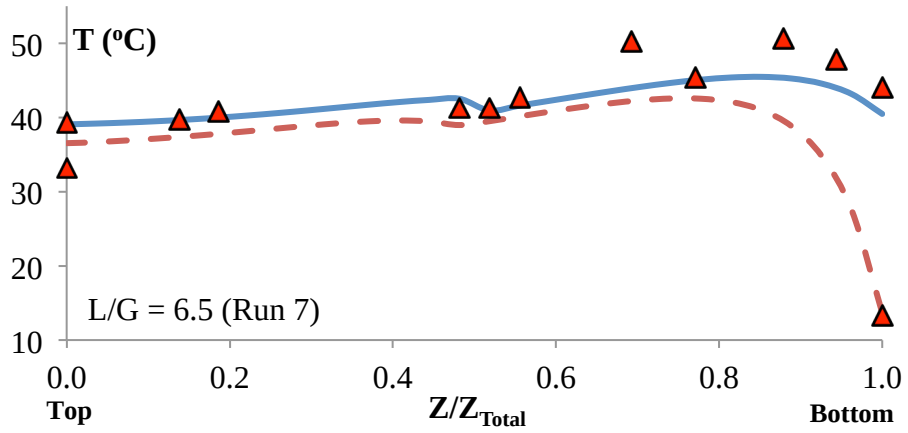


Figure 4.52: T profiles. PZ December'10 Campaign. (▲) measured, (–) liquid reconciled, (– –) gas reconciled (L/G = 6.5 mol/mol)

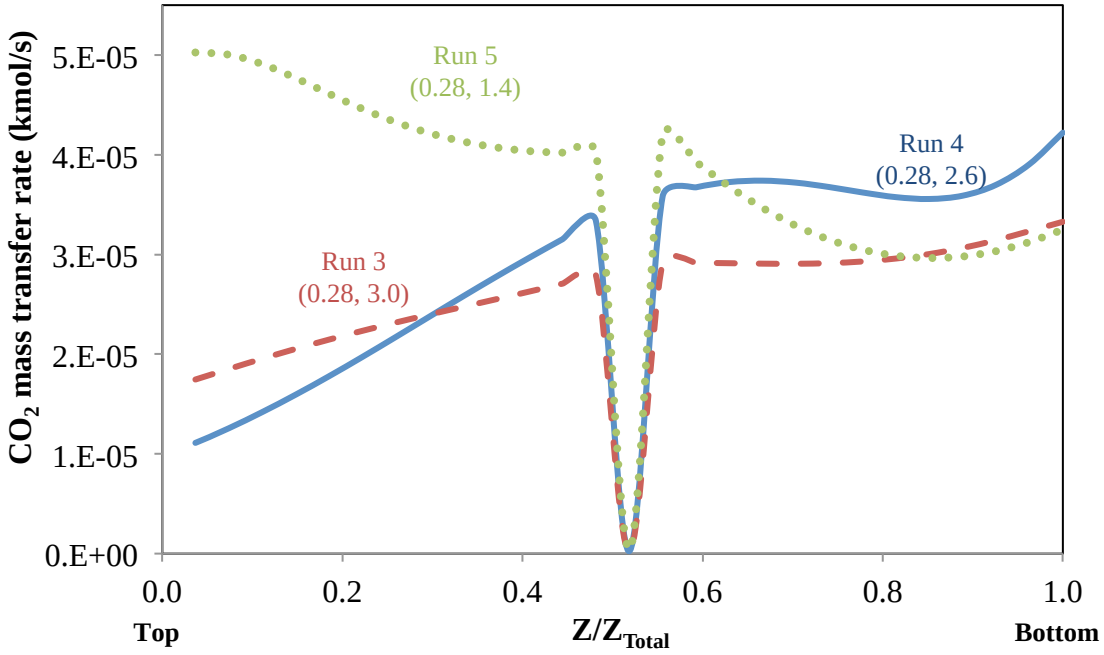


Figure 4.53: Predicted CO_2 mass transfer rate profiles. PZ Sept'10 Campaign. Intercooled (lean loading, L/G). Runs 3, 4, 5

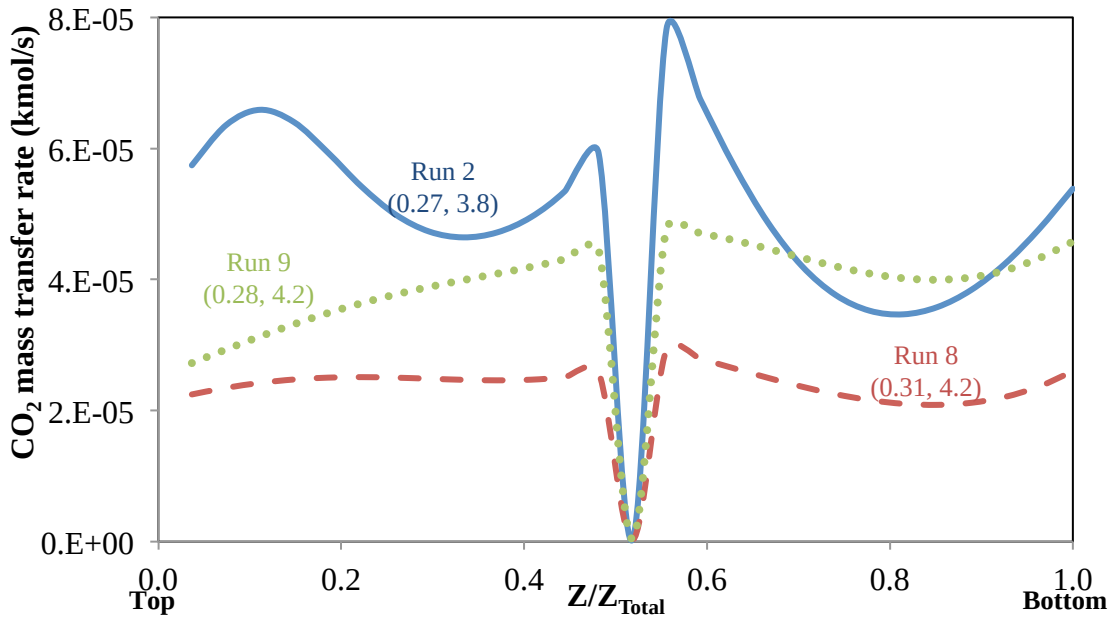


Figure 4.54: Predicted CO_2 mass transfer rate profiles. PZ Sept'10 Campaign. Intercooled (lean loading, L/G). Runs 2, 8, 9

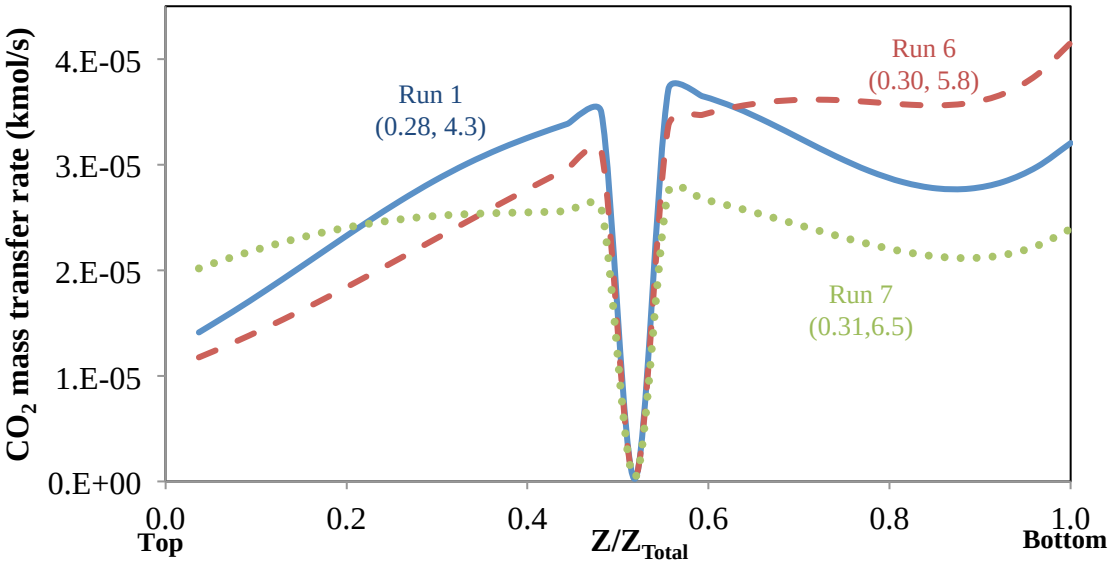


Figure 4.55: Predicted CO_2 mass transfer rate profiles. PZ Sept'10 Campaign. Intercooled (lean loading, L/G). Runs 1, 6, 7

Run 9 has the highest performance of the campaign (NTU=2.6). Column temperature is kept low due to intercooling and low gas feed temperature. The mass transfer rate profile is relatively even in both sections of packing although a slight lean side pinch is observed.

4.6. General Analysis and Observations

The model represents the pilot plant at the J.J. Pickle Research Center with MEA and PZ with the four selected packing types. Performance was matched within 10% of NTU for most runs. Temperature profiles are adequately represented in all campaigns.

The MEA run using Flexipac AQ Style20 required a 12% adjustment of the interfacial area calculated using Hanley and Chen (2011). The absorber performance was well matched using parameters for Flexipac 1Y. NTU for this case was 1.035 (65% removal), which is the lowest performance for this solvent.

The temperature profiles generated for the studied campaigns show the effect of the L/G on the location and magnitude of the temperature bulge. As the L/G is increased the temperature bulge moves from near the top of the column towards the bottom and its magnitude decreases. This is clearly observed for the runs without intercooling and is also noticeable for the intercooled runs.

Performance improvement due to intercooling is validated across the campaigns that evaluated this process option. In the MEA September'10 campaign comparison between Run 1 and Run 2 show that with equal L/G and lean loading (LG =5.9, lean ldg around 0.30) it is possible to increase performance up to 30% (for NTU and 7% removal). The improvement to performance by intercooling is also dependent on the L/G. Runs 3 and 4 with an L/G = 4.5 show a 35% increase in NTU with intercooling, which is higher than the result between Run 1 and 2. However, as proposed by Plaza

(2009) this improvement goes through a maximum, “the critical L/G”. Figure 4.41 show the NTU increase as related to L/G. This behavior is tied to the effect of temperature on mass transfer rates. When the temperature bulge coincides with a mass transfer pinch (zero mass transfer rate) intercooling breaks the pinch and considerably improves removal as it was seen in the intercooled CO_2 rate profiles. This was presented in Plaza (2009) and will be studied in the next chapter to optimize absorber design for these solvents.

The performance for PZ using Mellapak 2X and RSP-250 can be compared using Run2 from November’09 and Run 5 from September’10. These two campaigns have a molality of 8, a lean loading around 0.30 and an L/G close to 5.4. NTU increased from 1.22 to 2.24 using RSP-250.

Figure 4.43 shows the model predicted change in performance evaluated as the difference between the NTU for non-intercooled and an intercooled case. Two loading values were used along with variable L/G with the pilot plant packing height and RSP-250 to reach 90% with 9m MEA and 6.1 m RSP-250. Constant loading lines show a maximum for the increase in removal due to intercooling for the selected loadings moves towards the higher L/G as the lean loading is increased.

Comparison of the performance of the three PZ operating conditions was done using the pilot plant model to obtain the required L/G for absorber to obtain 90% removal varying lean solvent loading. Results show that for 90% removal and 350 ACFM, RSP-250 outperforms Mellapak 2X across all loadings although slightly at the lean loading where both packing types show similar behavior. The intercooled absorber with RSP-250 gives the best performance of all tested process conditions, requiring the lowest amount of solvent to reach 90% removal. Figure 4.57 also includes raw pilot plant data to show the adequate model prediction. However, it is clear that loading adjustments are

necessary when using the model (Figure 4.58). Applying the average % correction for each campaign to the pilot plant data places the points close to the predicted L/G line. The model predicts that the maximum lean loading to reach 90% removal with the available pilot plant packing is around 0.32 for PZ.

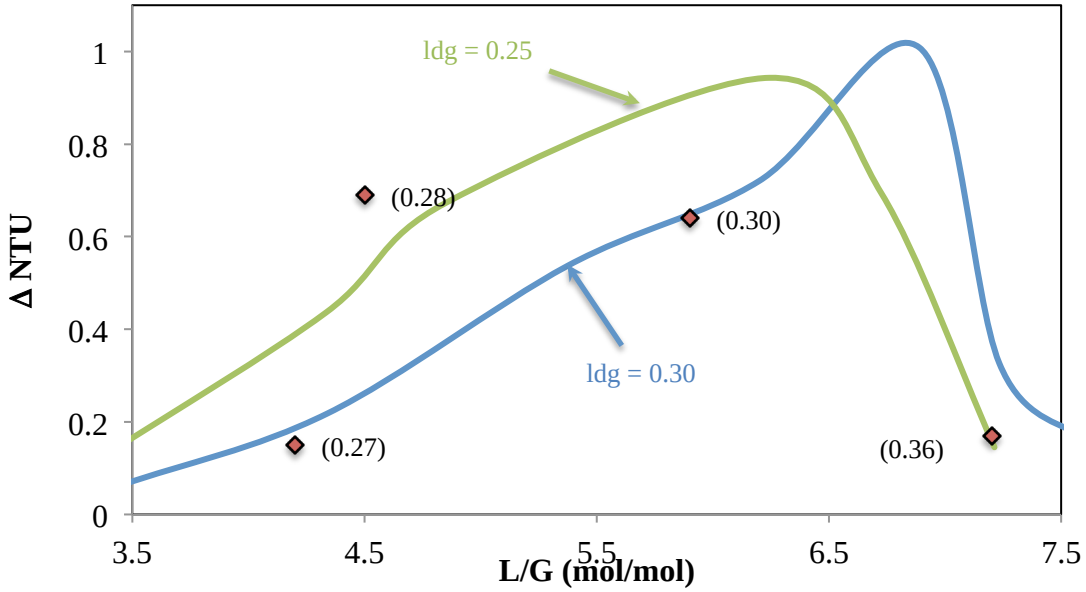


Figure 4.56: Effect of L/G in the change in performance due to intercooling for the MEA September'10 campaign. Points are pilot plant data (loading in parenthesis). 6.1m RSP-250, 9 m MEA, 350 ACFM. Lines are model predicted.

Comparison of the loading determinations across all campaigns using the density correlations and the sample titrations show discrepancies between measurements below 20%. Although this value is used as a process guide, it might be adequate to recalibrate the correlation to improve process control.

PZ parity plots for CO_2 removal show a possible loss of correspondence between the titrated data and the gas removal in the latter campaigns (Figure 4.11 and Figures 4.13 – 4.14). Similarly there is an increase in dispersion of loadings for the density meters and the reconciled values with the number of campaigns (Figure 4.7 and Figures 4.9 – 4.10).

These two observations added to larger modification for the interfacial area in the PZ December'10 campaign suggest a possible degradation effect of the solvent. This may have affected the PZ VLE requiring a larger CO_2 concentration shift, a larger change in loading and may have affected the titration results reported by the pilot plant.

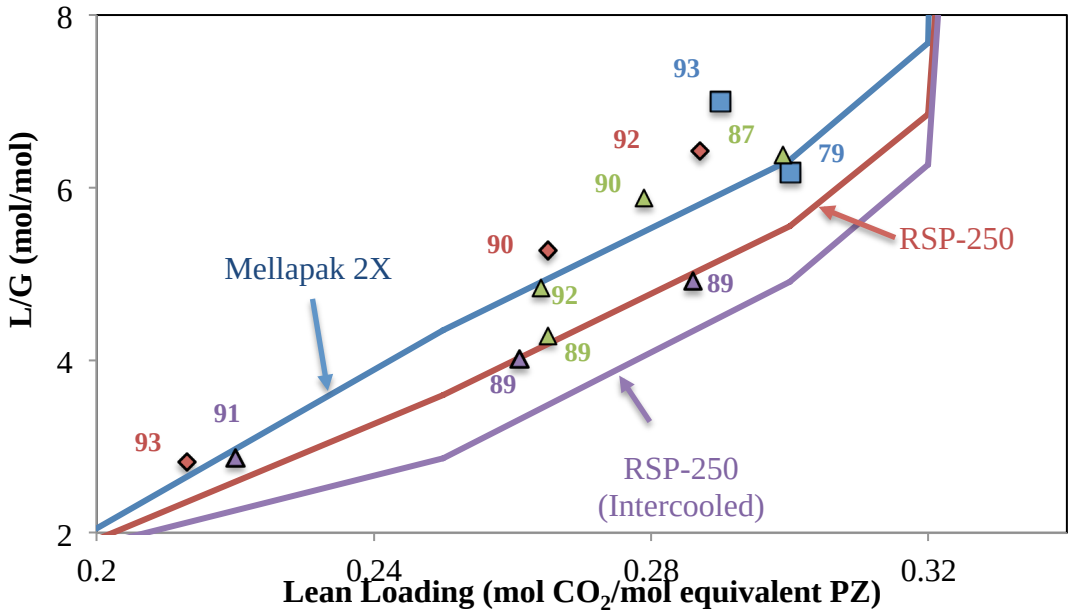


Figure 4.57: Required L/G for 90% removal with 8m PZ in Pilot Plant Absorber. Inlet gas: 350 ACFM, 21°C, 12%mol CO_2. Lines are model 5deMayo predictions. Points are pilot plant data, CO_2 removal (%): (■) Nov'09 (Mellapak 2X), (◆) Sept'10 (RSP250), (▲) Sept'10 (RSP250 - intercooled), (▲) Dec'10 (RSP250-intercooled)

Heat loss was reconciled mainly on the gas side. Liquid side heat loss parameters were very small if not reduce to zero. This could be attributable to the lower heat capacity of the gas. Furthermore, the PZ campaigns showed a higher heat loss than the MEA campaigns. There might be an effect of the heat capacity fit (Figure 3.40) in the 5deMayo model. The overall fit was lower than the measured values and missed the loading behavior. However, it seems unlikely that it could account for the order of

magnitude difference between solvent campaigns. Nevertheless, calculated heat loss is within values estimated by Dugas (2006) and Chen (2007).

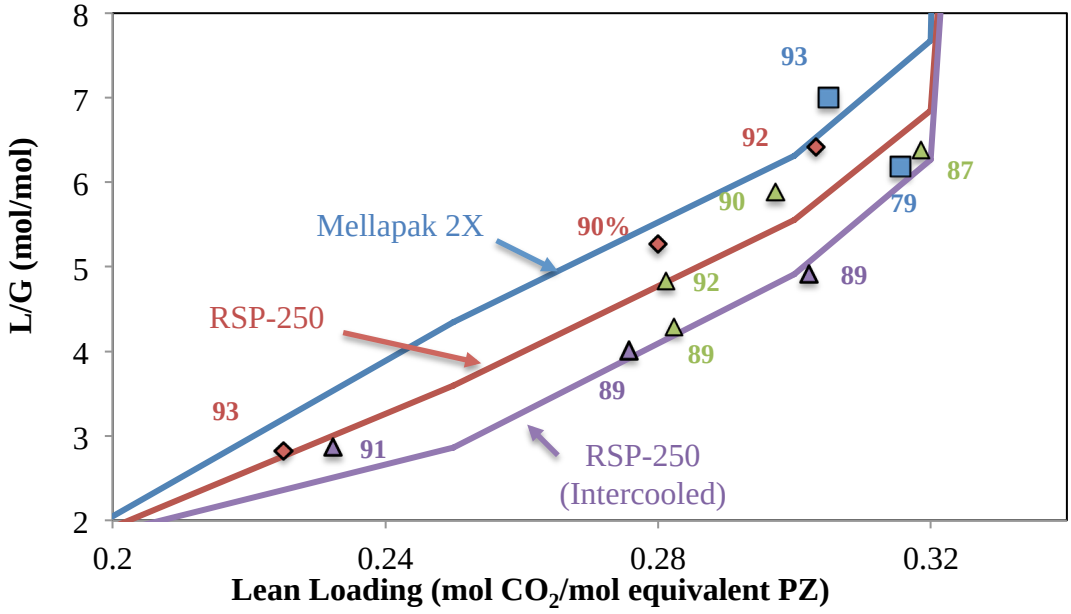


Figure 4.58: Required L/G for 90% removal in Pilot Plant Absorber. Inlet gas: 350 ACFM, 21ºC, 12%mol CO_2. Lines are model 5deMayo predictions. Shifted loading pilot plant points, CO_2 removal (%): (■) Nov'09 (Mellapak 2X-5.2%), (◆) Sept'10 (RSP250-5.6%), (▲) Sept'10 (RSP250 – intercooled-5.6%), (▲) Dec'10 (RSP250-intercooled-6.5%)

5

Chapter 5: Absorber Intercooling

Chapter 4 showed how intercooling is capable of increasing CO_2 absorber removal by lowering column temperature and breaking mass transfer rate pinches. However, there are column conditions that make its benefits marginal. This chapter uses the validated models to determine the optimum conditions for intercooling for the solvents studied in this work. Various process conditions are considered and additional benefits to this process modification are presented. The critical L/G (L/G_c) is also studied and a rule of thumb for its determination is presented.

5.1. Intercooling Application

Intercooling is a common strategy to increase the performance of absorption systems. Jackson and Sherwood (1941) showed that intercooling increased absorption up to 37% and even higher during the winter in refinery gas absorbers for absorption of C_4^+ from cracking coil gas. Linhoff (1930) reports the use of intercooling in a refinery with vapor recovery by an absorption oil. Sobel (1968) introduces the use of a computational method for absorbers, which routinely include a feature for modeling intercooling. A number of authors have shown that absorber intercooling can be effective with CO_2 capture by amines (Thompson and King, 1987; Freguia and Rochelle, 2003; Chang and Shih, 2005; Tobiesen and Dorao, 2008). Patents have also been filed with more complex

intercooling configurations to increase absorber performance (Shoeld, 1934; Geleff, 2003; Reddy et al., 2006).

Intercooling is especially useful for systems where the heat of absorption (i.e. heat of solution and/or reaction) results in an increase in temperature of the solvent affecting the vapor pressure of the dissolved species. Kvamsdal and Rochelle (2008) observed this behavior for the absorption of carbon dioxide from flue gas by aqueous monoethanolamine (MEA). They studied absorber parameters such as liquid/gas ratio (L/G), height of packing, and flue gas composition and its effect on the appearance of a temperature bulge in the absorber. Chen (2007) observed similar behavior for systems using piperazine (PZ) promoted potassium carbonate (K^+).

5.2. INTERCOOLING IN PIPERAZINE PROMOTED POTASSIUM CARBONATE

The developed model for 4.5m/4.5m K^+/PZ (Chapter 3) was used to evaluate the effect of solvent loading and intercooling on the performance of the absorber. The absorber was modeled with 15 m of packing divided into 30 segments. Table 5.1 shows the absorber design specifications.

Table 5.1: Absorber Design Conditions for K^+/PZ intercooling study

Variable	Value
Diameter (m)	9.8
Height (m)	15.0
Packing Characteristics	
Type	CMR
Vendor	MTL
Material	Metal
Dimension	NO-2P
Liquid hold up (%)	5

For this intercooling analysis random packing was selected. Koch-Glitsch, LP Cascade Mini Rings® 2-P (CMR™ NO-2P) are said to have high capacity with low pressure drop. According to the manufacturer its low aspect ratio increases effective mass transfer (www.koch-glitsch.com). Interfacial area and mass transfer coefficients were modeled using the Aspen Plus® implemented Onda correlations (Onda, 1968).

Intercooling for the K^+/PZ solvent was evaluated for two configurations: for the absorber coupled to a double matrix system and for a simple absorber. Flue gas characteristics used for both analyses are in included in Table 5.2.

Table 5.2: Flue Gas Conditions for K^+/PZ intercooling study

Variable	Value
Flow (kmol/s)	5.4879
Temperature (^{0}C)	40.0
Pressure (kPa)	111.33
Mol fraction	
H_2O	0.0670
CO_2	0.1270
N_2	0.7569
O_2	0.0491

5.2.1. The Double Matrix System.

Oyenekan (2007) proposed a stripper configuration for the regeneration of the solvent for the absorption of CO_2 that reduces the temperature change across the stripper by using a multipressure system (see Figure 5.1). The rich solution from the absorber is split into two streams: One goes to a higher-pressure system while the other goes to a lower pressure stripper. This configuration generates two return streams to the absorber (Lean and Semilean). Depending on the operating condition these streams will be at higher of lower loading and their flow will also vary.

Van Wagener (Rochelle et al., 2007), based on the work by Oyenekan, modeled the double matrix configuration using the 4.5 m K^+/4.5 m PZ solvent. Results from this analysis provided an optimum loading (moles of CO_2/moles of alkalinity) of 0.4012 for the lean stream and 0.4598 for the semilean stream. The flow split between the streams was 0.1850(mol semilean/mol lean). These values correspond to 0.5 kPa partial pressure of CO_2 in the lean stream at 40 ^oC. These data were used to set up an absorber optimization to maximize CO_2 removal using a fixed packing height and varying the position of the semilean feed and an additional intercooling point.

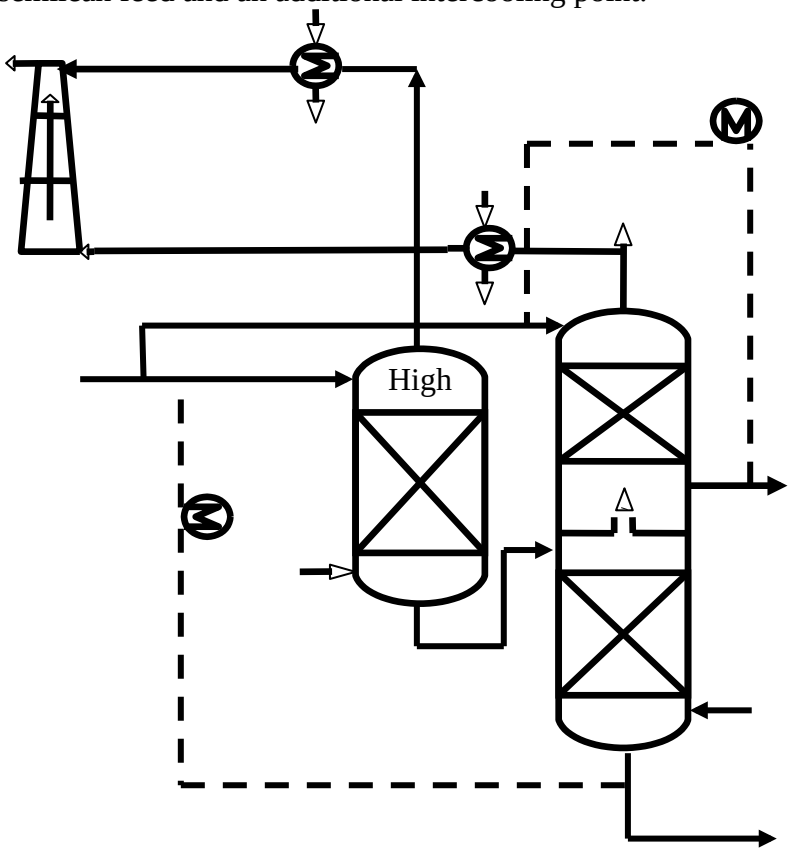


Figure 5.1: Double Matrix Stripper Configuration

Figure 5.2 demonstrates optimization of the case using a semilean feed without intercooling. The position of the semilean feed was adjusted to maximize CO_2 removal.

Removal is not very sensitive to the semilean feed position, but an optimum is found 30% from the top. Figure 5.3 gives the resulting liquid temperature and CO_2 rate profile.

Intercooling was defined to achieve 40°C liquid at the selected stage, which should be feasible with cooling water. Implementation in the model was done using a heat stream. When intercooling was used at the same point as the semilean feed, the optimum semilean feed position changed from the upper half of the column to the lower third. (See Figures 5.4 and 5.5)

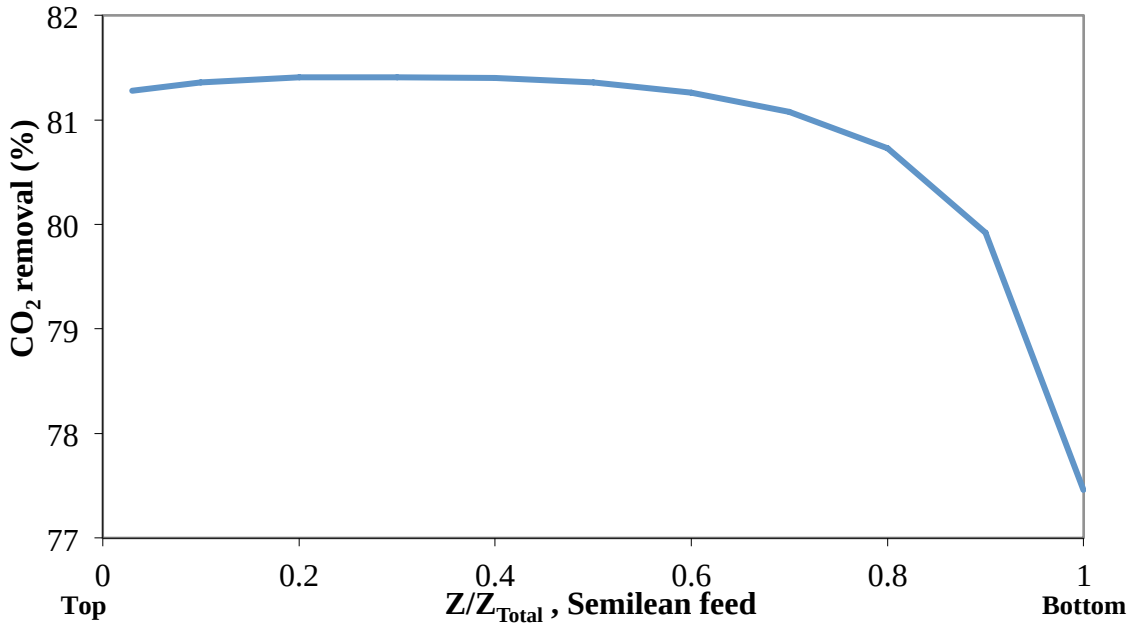


Figure 5.2: CO_2 removal with semilean feed with no intercooling for 4.5m/4.5 m K_2CO_3/ PZ. 0.5 kPa CO_2 lean solvent, 15 m packing

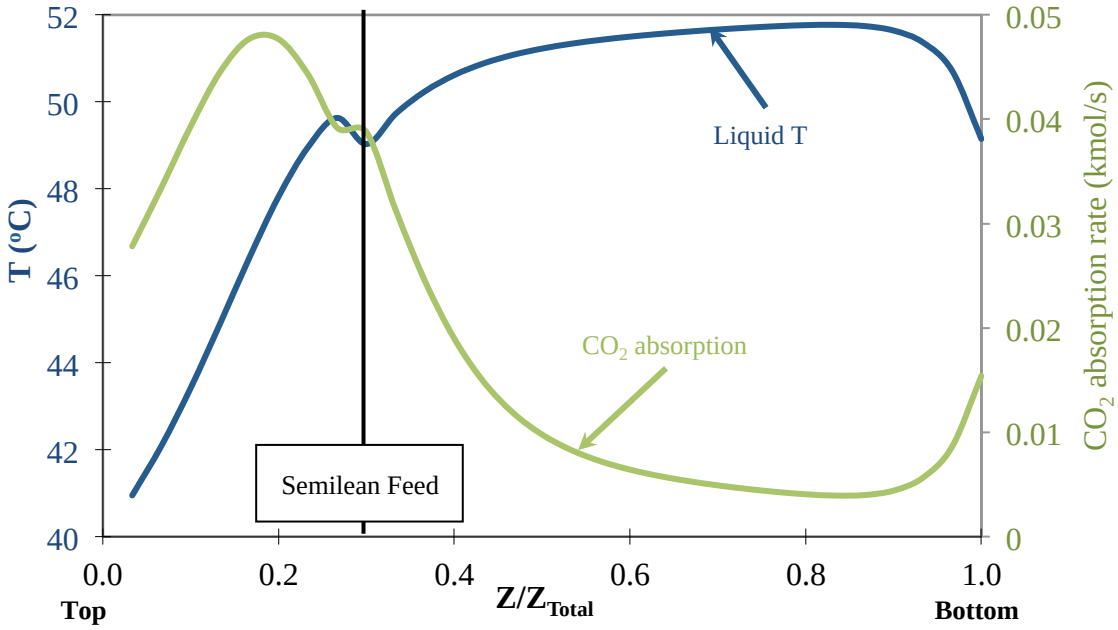


Figure 5.3: T and CO_2 absorption rate profiles for absorber with semilean feed at 0.30 column height no intercooling. Solvent 4.5m/4.5 m K_2CO_3/ PZ. 0.5 kPa CO_2 lean solvent. 15 m packing.

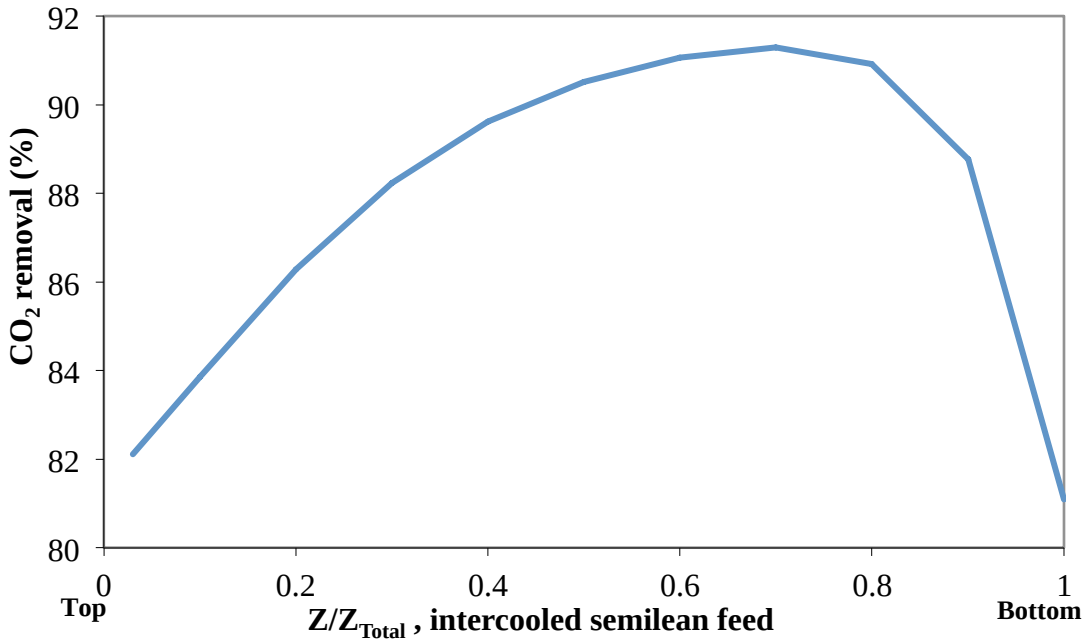


Figure 5.4: CO_2 removal with intercooled semilean feed for 4.5m/4.5 m K^+/PZ. 0.5 kPa CO_2 lean solvent. 15 m packing.

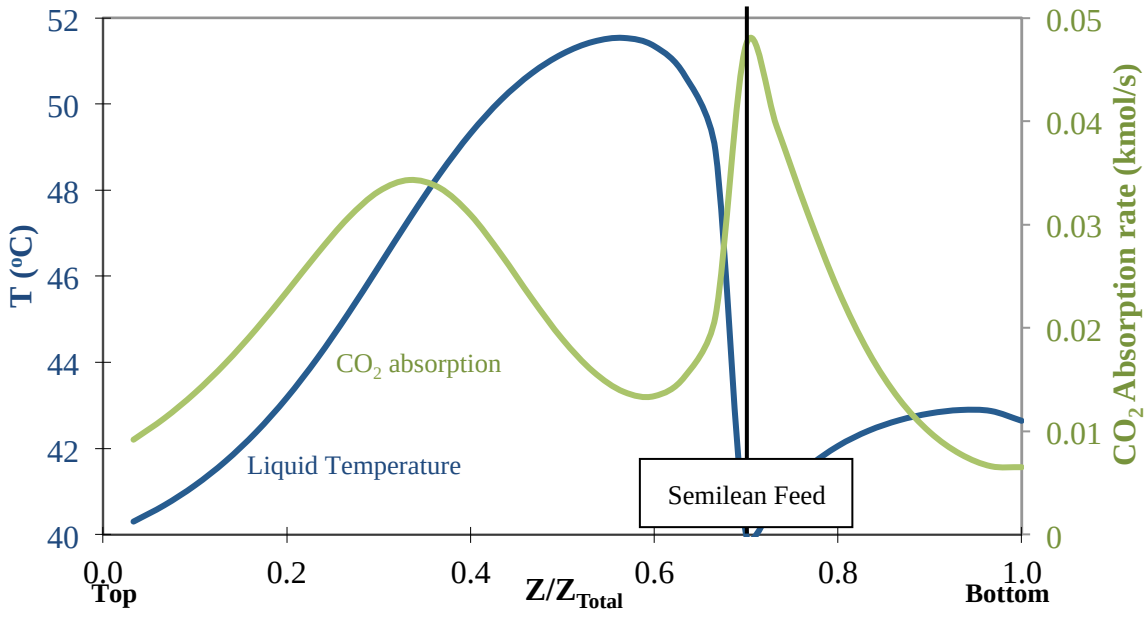


Figure 5.5: T and CO_2 rate profiles for absorber with semilean feed and intercooling at 0.70 column height. Solvent 4.5m/4.5 m K_2CO3/ PZ. 0.5 kPa CO2 lean solvent.

An additional intercooling stage was set up to furthermore increase absorber performance. Figure 5.6 shows the resulting profiles. The semilean feed with intercooling was simultaneously optimized along with the additional intercooling point. The optimum location for the semilean and the additional intercooling was found at 0.63 and 0.87 of the column height respectively.

Van Wagener reported conditions for the stripper using a higher loading lean solvent corresponding to 0.7 kPa CO_2 partial pressure. The lean solvent loading was 0.4208, the semilean loading was 0.4743 and the split was 0.1453. An analysis analogous to the 0.5 kPa loading was conducted, obtaining similar results. A summary of the results for the various operating conditions modeled is presented in Table 5.3. A single intercooling at the point of the semilean feed is very effective. Additional intercooling provides only marginal improvement.

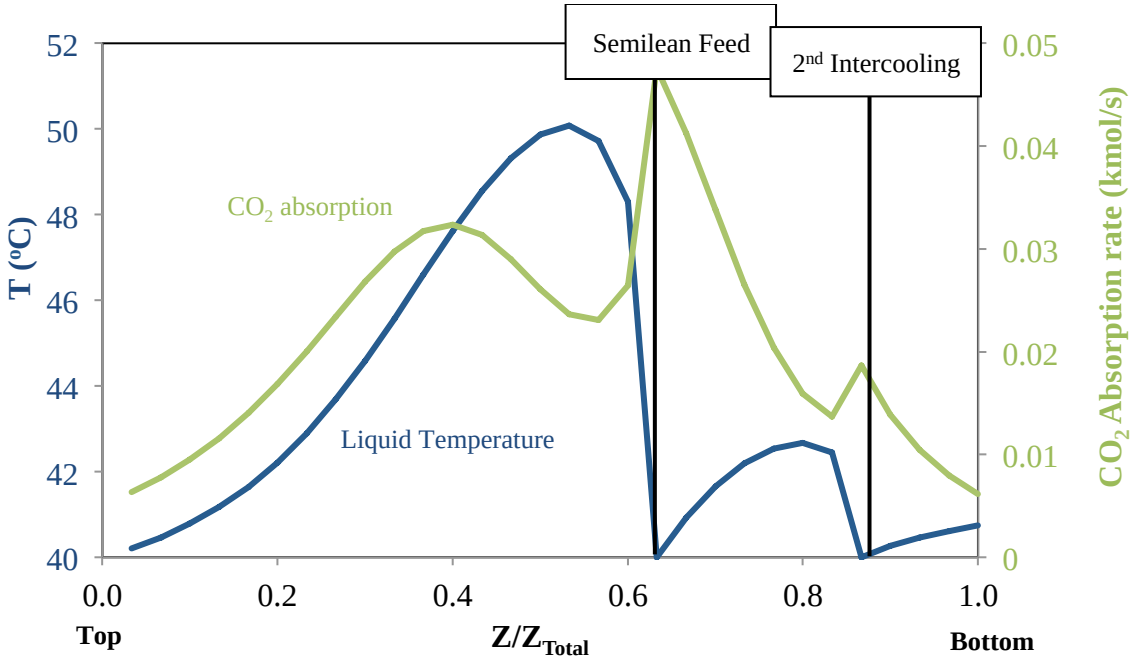


Figure 5.6: T and CO_2 rate profiles for absorber with intercooled semilean feed at 0.63 and intercooling at 0.87 column height. Solvent 4.5m/4.5 m K_2CO_3/ PZ. 0.5 kPa CO2 lean solvent.

Table 5.3: CO_2 removal results for K_2CO_3/PZ absorber configurations

CO_2 Pressure in lean Solvent @ 40ºC	**0.5 kPa**	**0.7 kPa**
Intercooling	**CO_2 Removal (%)**	
None	81.4	71.6
Single with semilean feed	91.3	82.9
Double	92.8	84.2

5.2.2. Intercooling Effect on Solvent Capacity and Rich Loading.

The lean solvent loading was varied to determine its effect on solvent capacity and rich loading for a simple absorber system with a single feed while maintaining removal at 90%. Solvent capacity is defined as the kmol of CO_2 removed per kg lean solvent feed.

Intercooling was set up in the model in the same matter to the analysis in the previous section. It was placed in the middle of the column and at the optimum point

(minimum amount of solvent for the level of lean loading). The flue gas and absorber specifications are the same as in Tables 5.1 and 5.2 but packing height was increased to 20 m. Figure 5.7 shows the effect of lean loading in solvent capacity Figure 5.8 presents the corresponding change in rich loading

In both figures the optimum curve and the Z/Z_{Total} = 0.5 curve overlap along the studied range. Thus, the position of the intercooling stage is not critical with 20 m of packing.

Figures 5.8 and 5.9 show that intercooling at high loading lean feed is not very beneficial because there is a limited temperature increase (7 ^{o}C) in the absorber. The higher solvent flow needed at high lean loading buffers any temperature increase due to reaction. The heat is absorbed by the increased solvent flow. Thus, mass transfer is not limited by the increase of temperature.

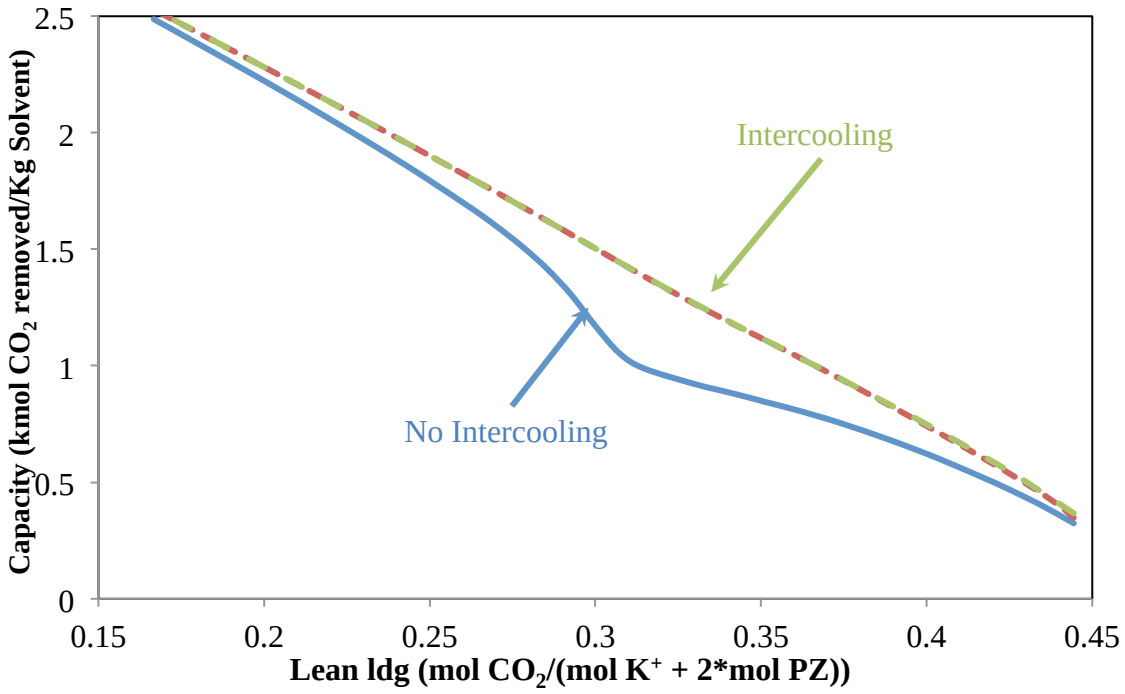


Figure 5.7: Change in solvent capacity vs. lean loading. 4.5 m K^+/4.5 m PZ. 90% CO_2 removal with 20 m of CMR#2 packing.

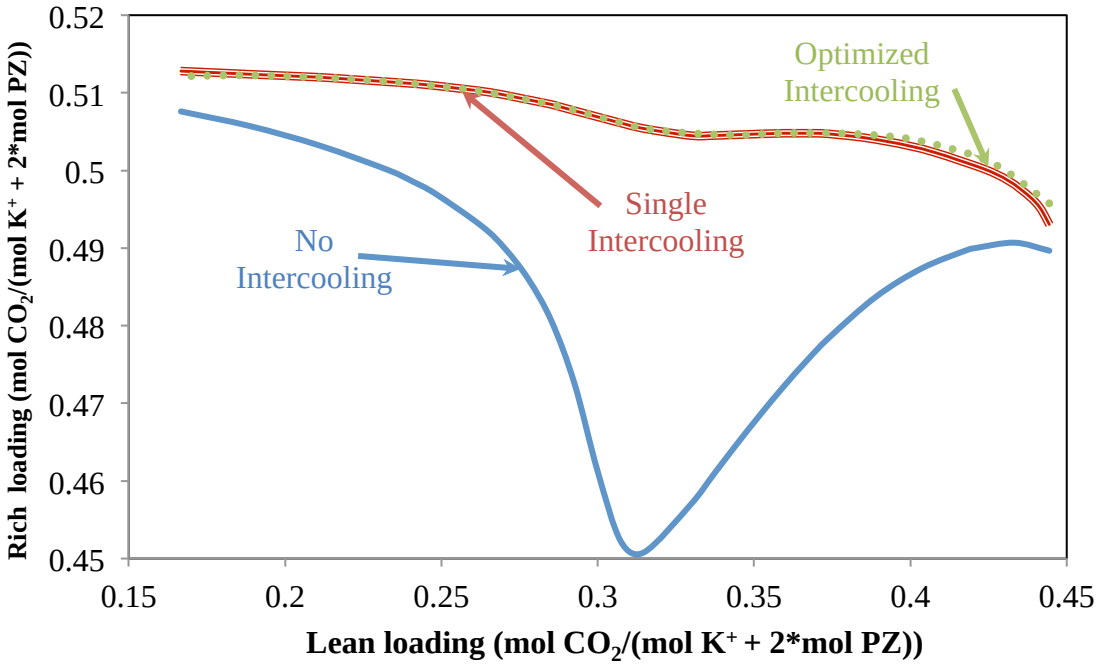


Figure 5.8: Variation of rich loading with lean loading. 4.5 m K^+/4.5 m PZ. 90% CO_2 removal with 20 m of CMR#2 packing.

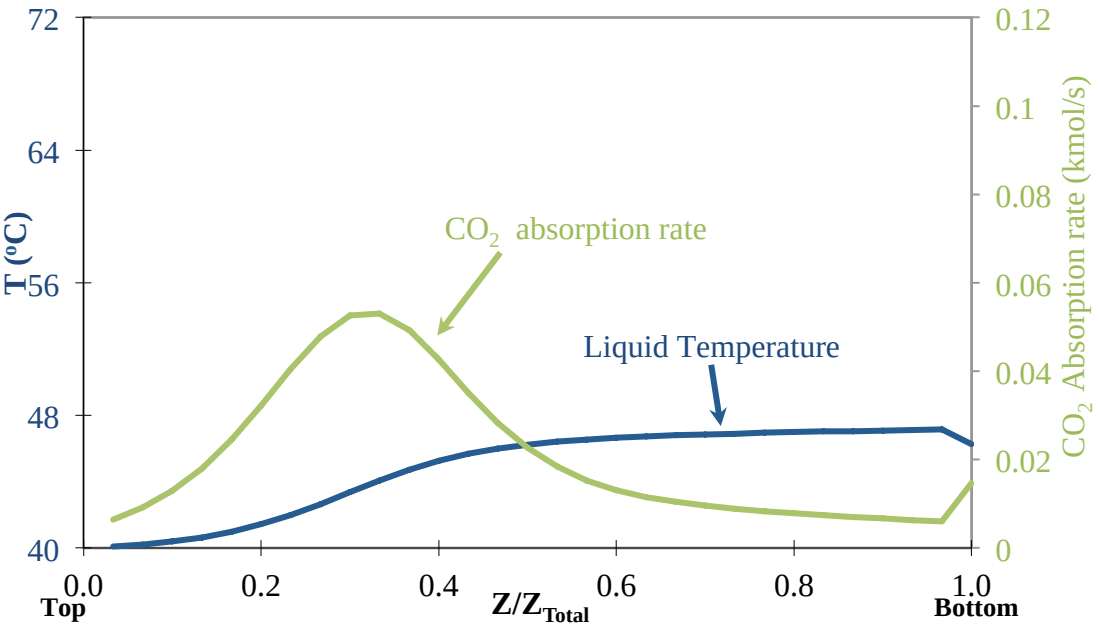


Figure 5.9: T and CO_2 rate profiles for absorber. 4.5 m K^+/4.5 m PZ, lean ldg= 0.44

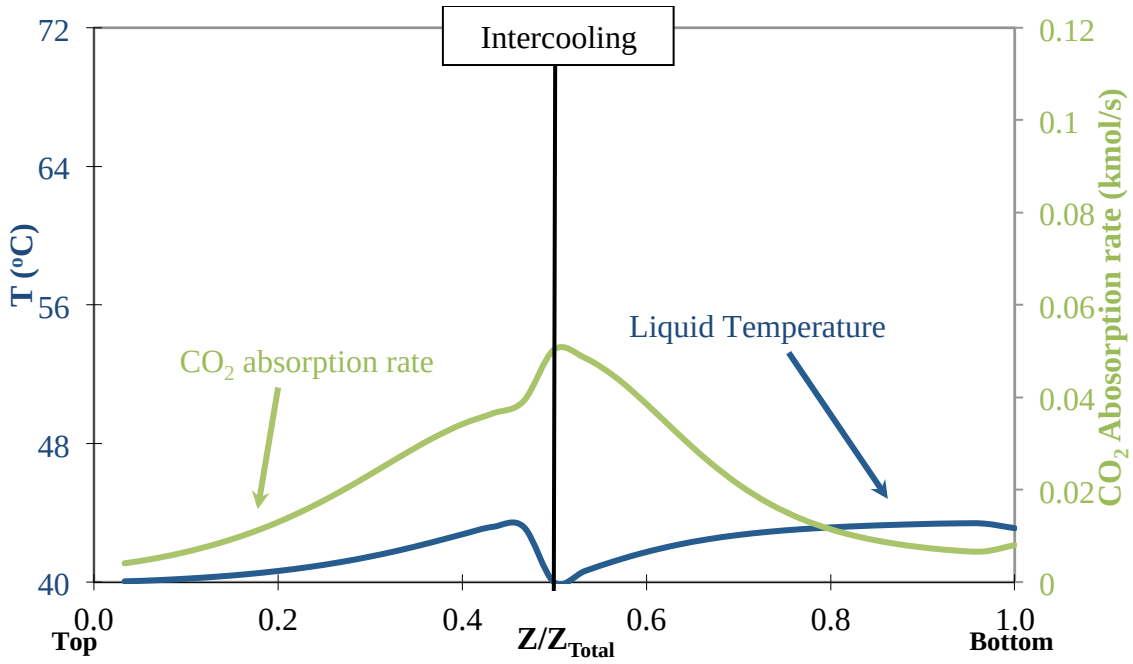


Figure 5.10: T and CO_2 rate profiles for absorber. 4.5 m K^+/4.5 m PZ, lean ldg= 0.44. Intercooling at Z/Z_{Total} = 0.5

On the other hand, lean loading solvent feeds (Figures 5.11 and 5.12) show a large increase in solvent temperature towards the top of the column. The low CO_2 content in the solvent offers an initial high driving force that allows for increased reaction rates at the top of the column causing a noticeable temperature increase. The lower solvent rates are not capable of absorbing all the generated heat and there is a top of column temperature bulge (around 70 ^oC). As temperature increases the equilibrium becomes a limiting factor, yet most of the CO_2 has already been absorbed so the bottom of the column does not react much. Figure 5.12 shows that the use of intercooling does not provide a considerable benefit in performance. The CO_2 rate profile obtained without intercooling (Figure 5.11) differs little from the former.

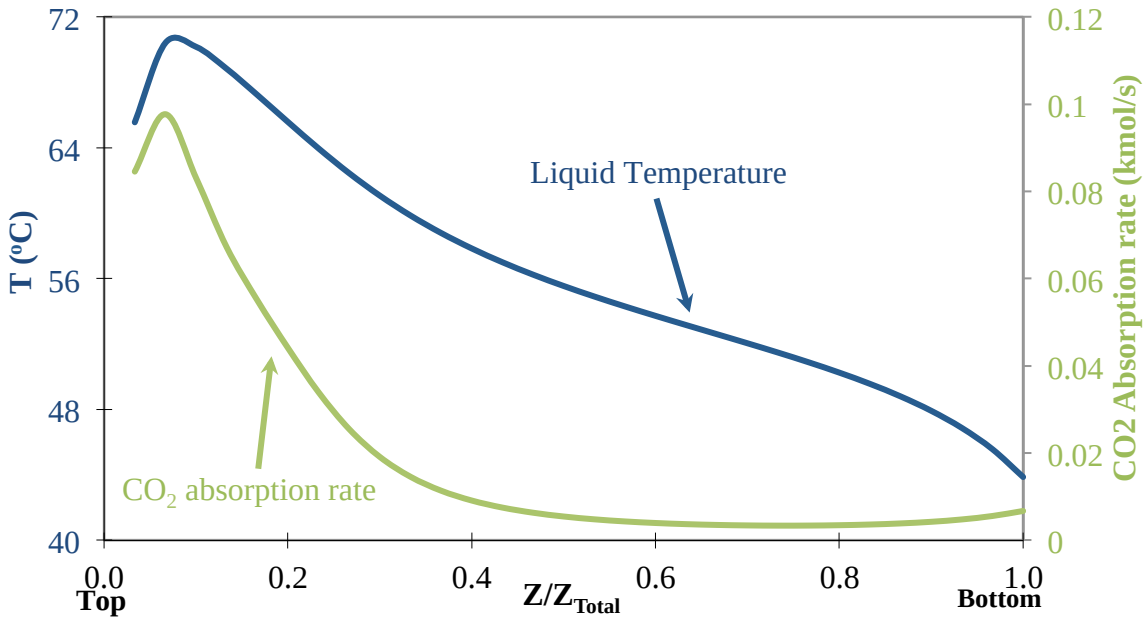


Figure 5.11: T and CO_2 rate profiles for absorber. 4.5 m K^+/4.5 m PZ, lean ldg= 0.21.

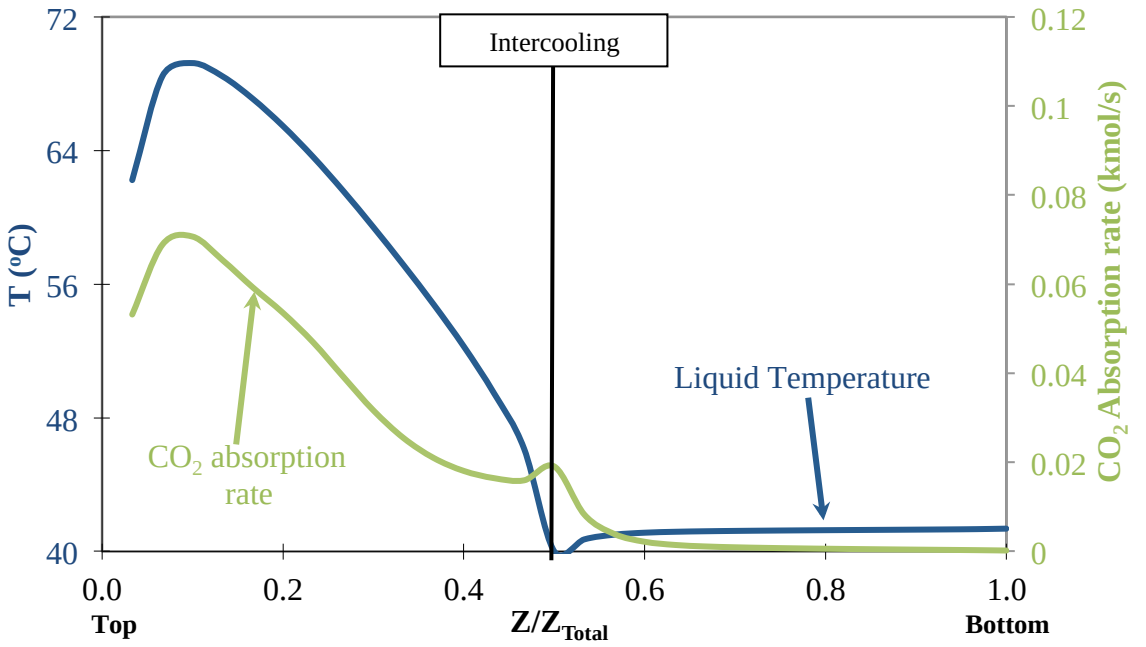


Figure 5.12: T and CO_2 rate profiles for absorber. 4.5 m K^+/4.5 m PZ, lean ldg= 0.21. Intercooling at Z/Z_{Total} = 0.5

However, absorbers operating at conditions within the loading bracket 0.27 to 0.40 mol CO_2/mol alkalinity benefit from intercooling. The temperature bulge is located near the center of the column and limits mass transfer rates. By adding intercooling it is possible to boost this phenomenon thanks to a lowering of the temperature of the bulge and of the column in general. (See Figures 5.13 and 5.14) This bracket is defined as the critical L/G region.

Looking at the behavior of the rich loading with respect to lean loading (Figure 5.8), intercooling proves especially beneficial between in the critical L/G (0.27–0.40 mol CO_2/mol alkalinity) for stripper performance. The higher loading from the absorber allows for lower energy consumption in the stripper.

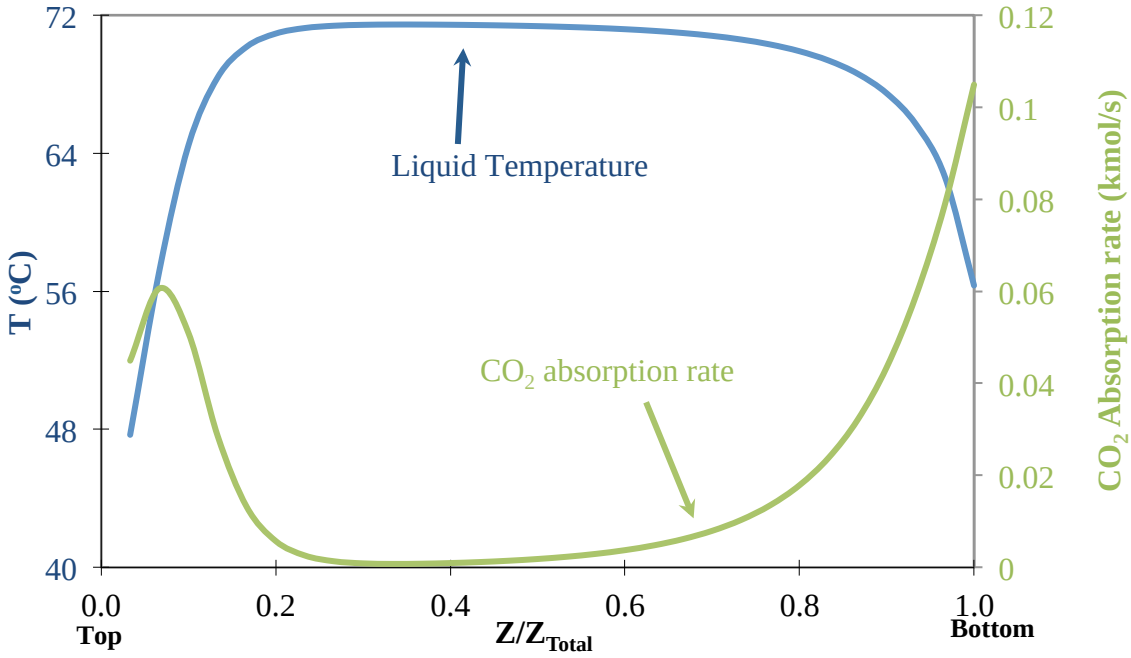


Figure 5.13: T and CO_2 rate profiles for absorber. 4.5 m K^+/4.5 m PZ, lean ldg= 0.315.

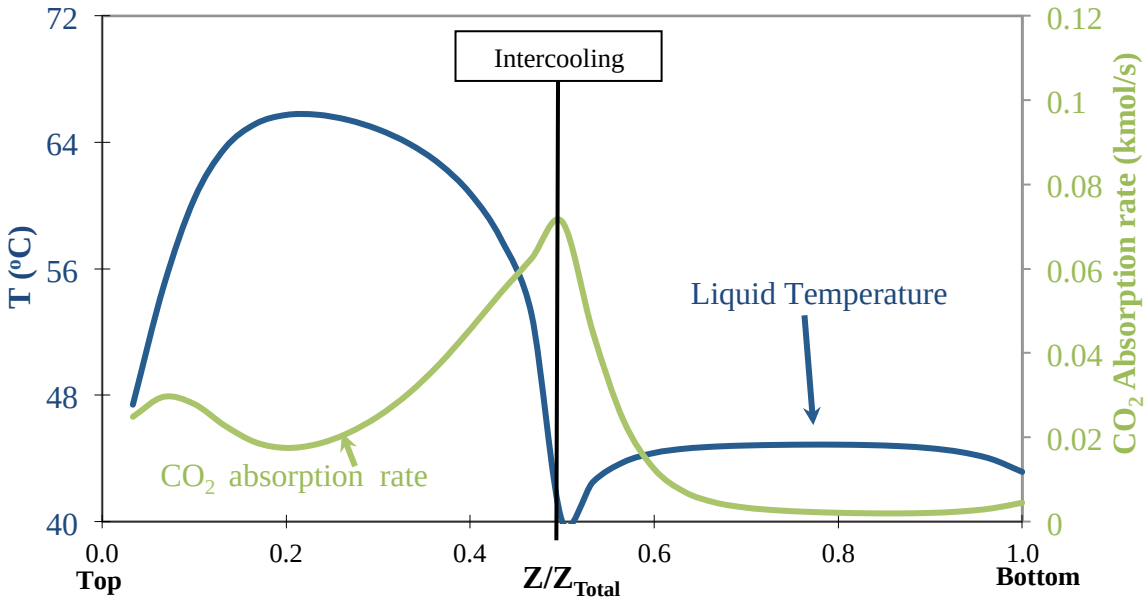


Figure 5.14: T and CO_2 rate profiles for absorber. 4.5 m K^+/4.5 m PZ, lean ldg= 0.315. Intercooling at Z/Z_{Total} = 0.5

5.3. INTERCOOLING IN MEA AND CONCENTRATED PZ

Chapter 4 showed the benefits of intercooling in absorption using aqueous MEA and PZ in the pilot plant. In this section an analysis of intercooling at various conditions for 8m PZ and 9m MEA will be presented. Simulations were done using gas conditions presented by Rochelle et al. (2005) for a 500 MW plant. Gas conditions are included in Table 5.4. Lean loading, removal, solvent flow rates and packing volume were varied across possible operating conditions maintaining the gas composition constant. Results are normalized to generate figures that can be used as initial estimates for system study and design. Inlet temperature is 40°C for the lean solvent is constant for all run. Feed gas conditions are presented in Table 5.4. Pressure drop is neglected since it is not in the scope of this work to test pressure models. Flooding is kept around 80%. Intercooling is modeled using the *pumparound* option in Aspen Plus® as it was described in Chapter 4.

The intercooling section is placed at the middle of the column with loop liquid flow equal to the lean solvent feed and cooled to 40°C.

Both solvents were analyzed to determine the minimum liquid flow rate by using a large amount of packing to assure pinching. This minimum flow rate at a fixed lean loading and CO_2 removal was determined for an absorber with and without intercooling. Then, absorber performance was evaluated using a liquid flow rate 10% and 20% more of the minimum flow rate with and without intercooling. The final section shows the effect of intercooling on amine emissions. Tables 5.5 and 5.6 summarize the conditions and results analyzed.

Table 5.4: Flue Gas Conditions for MEA and PZ intercooling study

Variable	Value
Temperature (°C)	40
Pressure (kPa)	101.125
Mol fraction	
H_2O	0.074
CO_2	0.126
N_2	0.751
O_2	0.049

Table 5.5: Summary of Studied Conditions and Results for 9m MEA – RSP-250. Intercooling at the middle to 40°C

No.	L/L_{min}	IC	Loading		*L/G (mol/mol)*	V_p per CO_2 removed	Bulge		Outlet T (°C)	
			Lean	Rich		m^3/kmol/s	T (°C)	Z/Z_T	Liquid	Gas
1	1	N	0.120	0.526	2.01	1892	72.5	0.1	67.0	42.3
2	1	N	0.200	0.521	2.54	2064	74.2	0.1	67.0	42.9
3	1	N	0.222	0.519	2.75	2136	74.8	0.1	66.8	43.3
4	1	N	0.250	0.514	3.09	2257	75.7	0.1	66.4	44.1
5	1	N	0.263	0.511	3.29	2336	76.1	0.1	66.1	44.7
6	1	N	0.300	0.469	4.81	2836	77.8	0.2	60.5	51.6
7	1	N	0.310	0.448	5.89	3061	78.2	0.2	53.9	55.0

No.	L/L_{min}	IC	Loading		L/G (mol/mol)	V_p per CO_2 removed	Bulge		Outlet T (°C)	
			Lean	Rich		m^3/kmol/s	T (°C)	Z/Z_T	Liquid	Gas
8	1	N	0.327	0.445	6.90	3230	78.4	0.2	46.0	56.8
9	1	N	0.350	0.458	7.55	3313	77.4	0.3	41.5	56.7
10	1	N	0.366	0.469	7.85	3306	75.2	0.4	40.9	56.2
11	1	N	0.386	0.484	8.27	3226	69.5	0.6	40.6	55.2
12	1	N	0.400	0.494	8.67	3084	64.5	0.8	40.5	54.3
13	1	N	0.420	0.504	9.74	3066	57.8	0.9	40.3	52.5
14	1	N	0.450	0.507	14.24	3616	49.8	1.0	40.1	48.5
15	1.1	N	0.150	0.491	2.40	1075	73.3	0.1	66.7	44.8
16	1.1	N	0.200	0.492	2.79	1273	74.4	0.1	66.3	45.5
17	1.1	N	0.222	0.492	3.03	1378	74.9	0.1	66.0	46.0
18	1.1	N	0.250	0.490	3.40	1567	75.7	0.1	65.4	46.9
19	1.1	N	0.262	0.488	3.62	1695	76.1	0.1	64.8	47.6
20	1.1	N	0.300	0.454	5.29	1974	77.6	0.2	57.4	53.8
21	1.1	N	0.310	0.436	6.48	1506	77.2	0.3	48.3	56.8
22	1.1	N	0.327	0.434	7.58	880	64.8	0.8	41.1	56.9
23	1.1	N	0.350	0.448	8.31	985	61.1	0.8	40.8	55.6
24	1.1	N	0.400	0.485	9.53	1751	57.9	0.8	40.5	53.3
25	1.1	N	0.420	0.496	10.71	2308	55.0	0.9	40.3	51.6
26	1.1	N	0.450	0.502	15.66	3143	48.9	0.9	40.1	47.9
27	1.2	N	0.150	0.462	2.61					
28	1.2	N	0.200	0.468	3.05					
29	1.2	N	0.250	0.470	3.70					
30	1.2	N	0.300	0.441	5.77					
31	1.2	N	0.310	0.425	7.07					
32	1.2	N	0.330	0.425	8.27					
33	1.2	N	0.350	0.440	9.06					
34	1.2	N	0.400	0.478	10.40					
35	1.2	N	0.450	0.498	17.09					
36	1	YN	0.150	0.537	2.11	4746				
37	1	Y	0.250	0.537	2.84	7752				
38	1	Y	0.300	0.532	3.51	8531	74.6	0.1	64.0	43.7
39	1	Y	0.327	0.525	4.11	10276	75.1	0.1	61.1	45.5
40	1	Y	0.350	0.520	4.87	10845	73.9	0.1	54.8	47.2
41	1	Y	0.386	0.517	6.21	11985	69.9	0.2	44.1	49.3
42	1	Y	0.400	0.519	6.83	10779	65.7	0.3	42.0	49.1
43	1	Y	0.420	0.524	7.77	12755	59.5	0.4	41.4	48.2
44	1	Y	0.450	0.536	9.52	20469	52.4	0.4	40.9	45.5
45	1.1	Y	0.150	0.502	2.32	1189	70.9	0.1	63.9	43.7
46	1.1	Y	0.250	0.511	3.13	1862	72.8	0.1	62.2	44.5
47	1.1	Y	0.300	0.511	3.86	2719	74.0	0.1	59.4	45.8
48	1.1	Y	0.327	0.507	4.52	3570	74.0	0.1	54.6	47.6

No.	L/L_{min}	IC	Loading		L/G (mol/mol)	V_p per CO_2 removed	Bulge		Outlet T (°C)	
			Lean	Rich		m^3/kmol/s	T (°C)	Z/Z_T	Liquid	Gas
49	1.1	Y	0.350	0.502	5.35	4578	73.5	0.1	47.5	49.7
50	1.1	Y	0.386	0.505	6.83	2501	61.3	0.4	40.7	50.2
51	1.1	Y	0.400	0.508	7.51	2730	57.3	0.4	40.6	49.4
52	1.1	Y	0.450	0.528	10.47	8193	47.7	0.9	40.4	46.3
53	1.2	Y	0.150	0.473	2.53	832				
54	1.2	Y	0.250	0.489	3.41	1293				
55	1.2	Y	0.300	0.494	4.21	1784				
56	1.2	Y	0.350	0.489	5.84	1715				
57	1.2	Y	0.386	0.495	7.45	1872				
58	1.2	Y	0.400	0.499	8.19	2165				
59	1.2	Y	0.450	0.521	11.42	5430				

Table 5.6: Summary of Studied Conditions and Results for 8m PZ – RSP-250. Intercooling at the middle to 40°C

No.	L_{min}	IC?	Loading		L/G	V_p per CO_2 removed	Bulge		Outlet T (°C)	
			Lean	Rich		m^3/kmol/s	T (°C)	Z/Z_T	G	L
1	1	N	0.150	0.429	1.63	2347	72.0	0.03	67.0	41.3
2	1	N	0.160	0.428	1.70	2383	72.0	0.03	66.8	41.5
3	1	N	0.200	0.420	2.07	2527	73.0	0.05	65.9	42.8
4	1	N	0.230	0.404	2.62	2737	74.7	0.11	64.0	45.8
5	1	N	0.240	0.369	3.51	2972	74.6	0.16	61.4	49.1
6	1	N	0.250	0.355	4.34	3132	74.6	0.24	52.4	55.0
7	1	N	0.270	0.357	5.24	3377	71.8	0.29	44.5	55.8
8	1	N	0.300	0.374	6.10	3411	62.7	0.55	41.3	53.8
9	1	N	0.330	0.391	7.40	3475	55.6	0.76	40.4	51.1
10	1	N	0.340	0.397	8.02	3356	53.7	0.82	40.2	50.1
11	1	N	0.350	0.401	8.82	3701	51.6	0.94	40.0	49.1
12	1.1	N	0.150	0.403	1.80	499	71.8	0.17	66.2	45.0
13	1.1	N	0.200	0.400	2.28	742	72.8	0.17	64.8	46.1
14	1.1	N	0.230	0.388	2.89	1094	74.2	0.19	62.4	48.6
15	1.1	N	0.240	0.358	3.86	757	73.7	0.33	55.9	54.0
16	1.1	N	0.250	0.345	4.77	484	69.2	0.54	47.5	56.6
17	1.1	N	0.270	0.349	5.77	401	59.5	0.80	42.7	55.1
18	1.1	N	0.300	0.368	6.71	537	55.9	0.86	41.1	52.8

No.	L_{min}	IC?	Loading		L/G	V_p per CO_2 removed	Bulge		Outlet T (^oC)	
			Lean	Rich		$m^3/kmol/s$	T (^oC)	Z/Z_T	G	L
19	1.1	N	0.330	0.386	8.14	868	52.0	0.90	40.3	50.2
20	1.1	N	0.340	0.391	8.82	1106	50.8	0.92	40.1	49.3
21	1.1	N	0.350	0.397	9.71	2580	49.5	0.96	40.0	48.3
22	1.2	N	0.150	0.382	1.96	376	71.6	0.23	65.5	47.8
23	1.2	N	0.200	0.383	2.49	550	72.5	0.24	63.7	48.7
24	1.2	N	0.230	0.374	3.15	796	73.8	0.26	60.7	50.8
25	1.2	N	0.240	0.348	4.22	545	72.0	0.40	52.6	55.6
26	1.2	N	0.250	0.337	5.20	350	62.3	0.78	44.7	56.6
27	1.2	N	0.270	0.342	6.29	353	56.3	0.89	42.2	54.2
28	1.2	N	0.300	0.362	7.32	465	53.4	0.92	41.0	51.8
29	1.2	N	0.330	0.381	8.88	761	50.4	0.94	40.3	49.4
30	1.2	N	0.340	0.387	9.63	972	49.4	0.95	40.1	48.5
31	1.2	N	0.350	0.393	10.59	2322	48.3	0.98	40.0	47.7
32	1	Y	0.150	0.430	1.62	4365	69	0.06	66.2	41.2
33	1	Y	0.200	0.425	2.02	2324	71	0.06	64.7	42.1
34	1	Y	0.240	0.418	2.56	2466	71	0.08	63.2	42.4
35	1	Y	0.250	0.411	2.82	3418	71	0.10	60.6	43.8
36	1	Y	0.270	0.397	3.57	2927	63.9	0.14	48.3	47.1
37	1	Y	0.300	0.401	4.51	3850	68.2	0.18	50.4	48.8
38	1	Y	0.320	0.405	5.50	3082	56.2	0.37	41.1	48.2
39	1	Y	0.340	0.409	6.62	3268	51.6	0.43	40.2	47.1
40	1	Y	0.350	0.377	16.54	4466	45	1	40	45
41	1.1	Y	0.150	0.405	1.78	452	69	0.16	63.2	60.2
42	1.1	Y	0.200	0.405	2.22	640	68.9	0.17	60.8	57.8
43	1.1	Y	0.220	0.403	2.49	758	69	0.15	59.2	45.8
44	1.1	Y	0.240	0.401	2.81	1051	69.8	0.14	59.4	45.4
45	1.1	Y	0.250	0.397	3.10	1145	69.1	0.16	54.5	47.4
46	1.1	Y	0.270	0.386	3.93	871	64.1	0.28	45.7	49.9
47	1.1	Y	0.300	0.391	4.97	905	55.5	0.41	41.4	49.6
48	1.1	Y	0.320	0.395	6.05	937				
49	1.1	Y	0.340	0.402	7.28	1446	50.5	0.89	40.2	48.1
50	1.1	Y	0.350	0.375	18.20	5647	44.6	1.00	44.6	40.0

In Tables 5.5 and 5.6 IC refers to intercooling and the column with the L_{min} heading refers to the solvent rate in reference to the minimum solvent rate. Rows with 1

is the minimum solvent rate and the volume of packing used is large enough to approach infinite packing height.

5.3.1. Intercooling in 9m MEA

5.3.1.1. Intercooling and L_{min}/G

Figure 5.15 shows the resulting L_{min}/G and rich loading for 90% removal with changing lean loading. Column height was over specified to 18 m to assure that the resulting lean liquid flow rate is approximately the minimum solvent rate (L_{min}) and a pinch is expected at some point in the column. The minimum solvent flow rate is the flow rate required when the column has infinite packing volume (Anthony et al., 1987) Based on the results for the MEA Sept'10 Campaign the interfacial area (a_{intf}) was multiplied by 0.73 in all the runs using RSP-250 and 9m MEA.

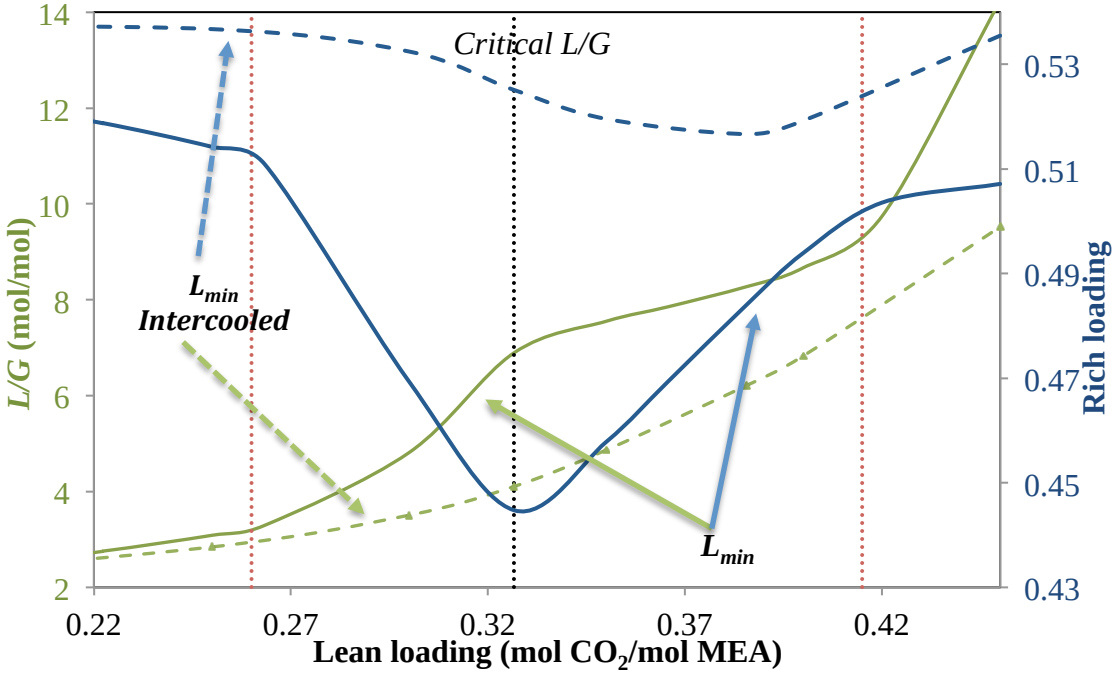


Figure 5.15: Variation of rich loading and *L/G* with change in lean loading for 9m MEA and 90% removal. Variable RSP-250. Diameter set for 80% flooding. Liquid intercooled to 40°C in the middle of the column

Behavior of rich loading and L_{min}/G for the non-intercooled cases shows four column performance regions. Characterization of these regions can be done using a figure that shows the liquid and gas temperature profiles and the flux rate as a function of the relative position in the column (Z/Z_{Total}). In the first region (lean loading <0.26) L_{min}/G has a negligible increase and the rich solvent leaves the absorber less saturated. The temperature bulge is located at the top of the column and there is a mass transfer pinch towards the bottom of the column as shown in Figure 5.16. In the second region (0.26<lean loading<0.33) there is a considerable increase in solvent flow rate leading to a noticeable decrease in rich loading. The temperature profile shows a higher magnitude and broader temperature bulge than in the previous region and mass transfer pinches towards the top of the column (Figure 5.17). At the limit between the second and third region is an inflection point similar to the one observed in Figure 5.8. This point corresponds to the critical $(L/G)_c$. At this point the temperature bulge is in the middle of the column and encompasses almost half of the packing height. The mass transfer pinch is near or at the middle of the column (Figure 5.18). In the third region from the critical L/G to a loading of 0.42 the solvent flow rate increases with a lower slope than in the second region leading to an increase in the rich loading. The temperature bulge shifts towards the bottom of the column and is reduced in magnitude. The increased solvent flow rate absorbs additional heat lowering the column temperature. The pinch is still near the middle of the column but it has moved towards the bottom (Figure 5.19). The last region has a considerable increase in solvent flow rate yet the rich loading is almost constant. The solvent has limited capacity so any increase in loading requires a considerable increase in solvent flow. The lean solvent has a high loading so the column has a lean end pinch and the slight increase in temperature at the bottom along with the approach to equilibrium generates a rich end pinch. Removal occurs around the middle

of the column. The mass transfer rate is limited compared to the other regions (Figure 5.20).

Outlet temperatures for liquid and gas also vary within the 4 regions. The temperature difference between liquid and gas stream at the top of the column is larger at the top for low L_{min}/G and decreases to almost negligible for high L_{min}/G. At the bottom of the column the temperature difference seems to go through a maximum as the liquid temperature is the highest at the critical L_{min}/G (57°C) and approaches gas feed temperature (40°C) at the extremes of Figure 5.15 where there is either very small amounts of gas and the generated heat leaves with the gas at the top or at the bottom where the liquid flow is much larger than the gas and most of the heat leaves with the solvent. Figure 5.21 shows the location (Z/Z_{Total}) and magnitude of the temperature bulge as a function of loading. The bulge moves from the top of the column to the bottom as the lean loading and the liquid flow rate are increased to achieve 90% removal. The magnitude of the temperature bulge increases with lean loading until the critical L/G where it starts to decrease approaching inlet liquid values.

Similar to what is observed in Figure 5.15 there is an inflection for each of the constant removal lines. Figure 5.22 shows the variation of rich loading with respect to the change for lean loading for 90%-60% removal. Solid lines show constant L_{min}/G.

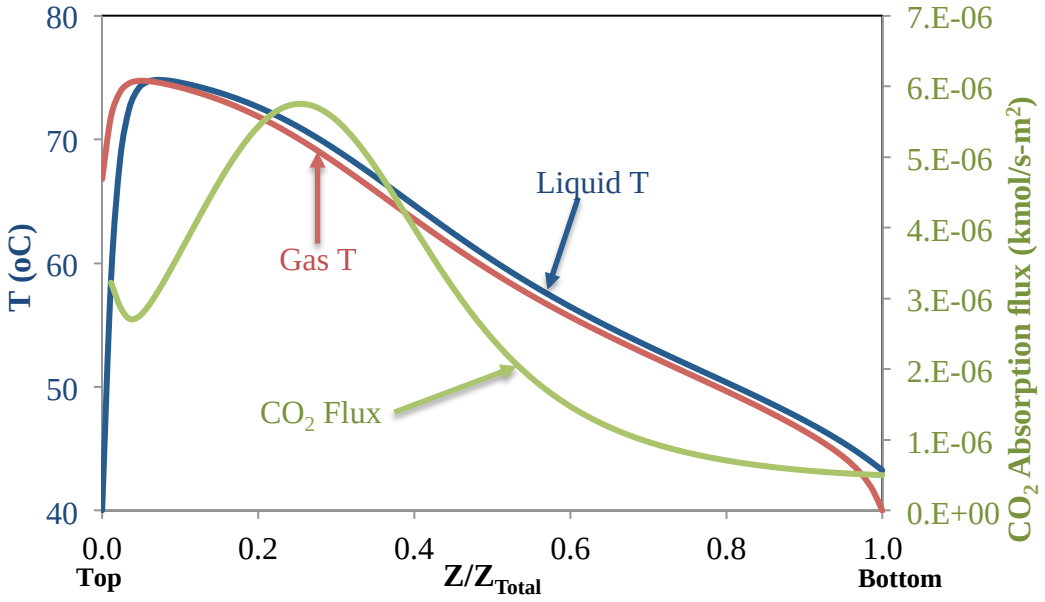


Figure 5.16: Temperature and CO_2 mass transfer flux profiles. Lean ldg = 0.22 L_{min}/G = 2.75. 2090 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. 9 m MEA

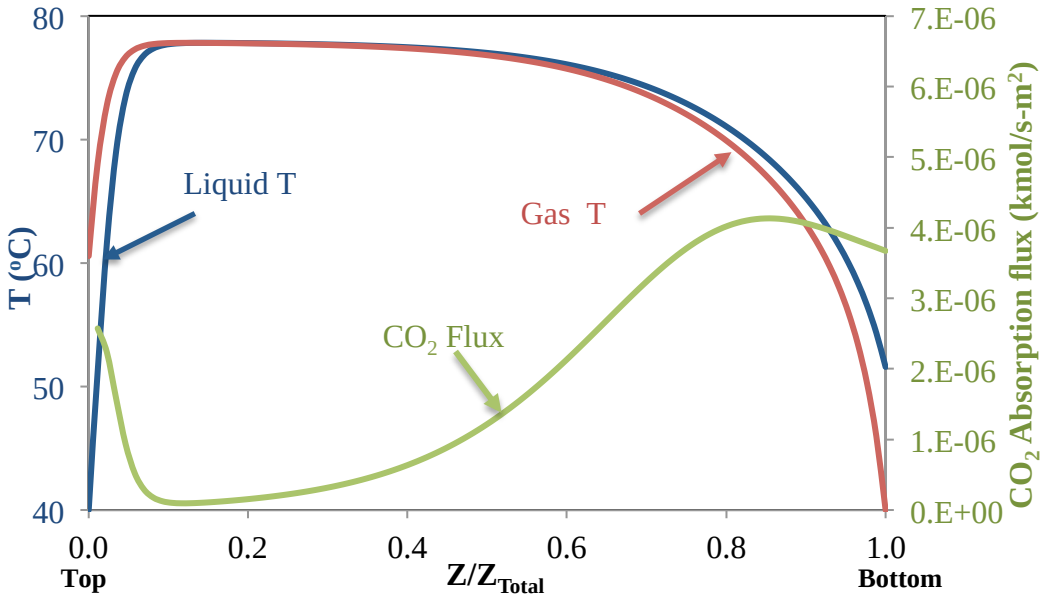


Figure 5.17: Temperature and CO_2 mass transfer flux profiles. Lean ldg = 0.30 L_{min}/G = 4.8. 2774 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. 9 m MEA.

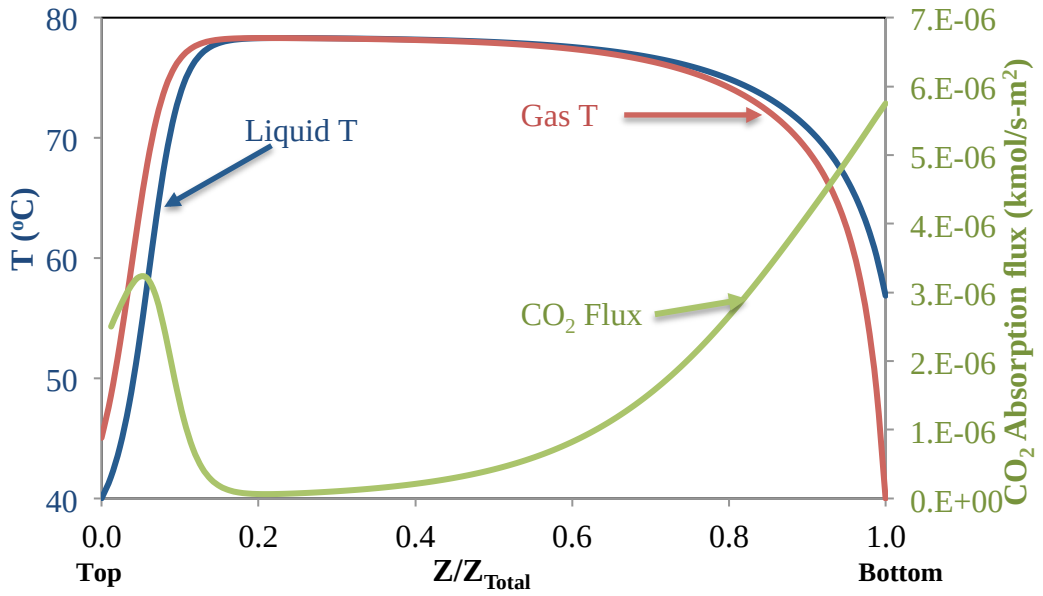


Figure 5.18: Temperature and CO_2 mass transfer flux profiles. Lean ldg = 0.33 (Critical point) L_{min}/G = 6.9 3159 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. 9 m MEA.

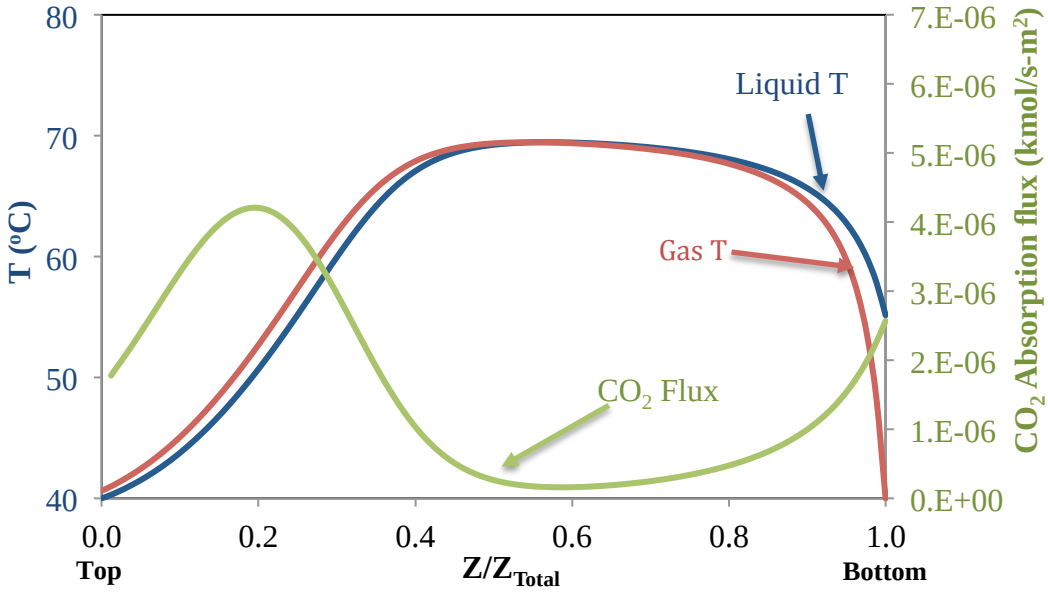


Figure 5.19: Temperature and CO_2 mass transfer flux profiles. Lean ldg = 0.386 L_{min}/G = 8.3. 3155 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. 9 m MEA

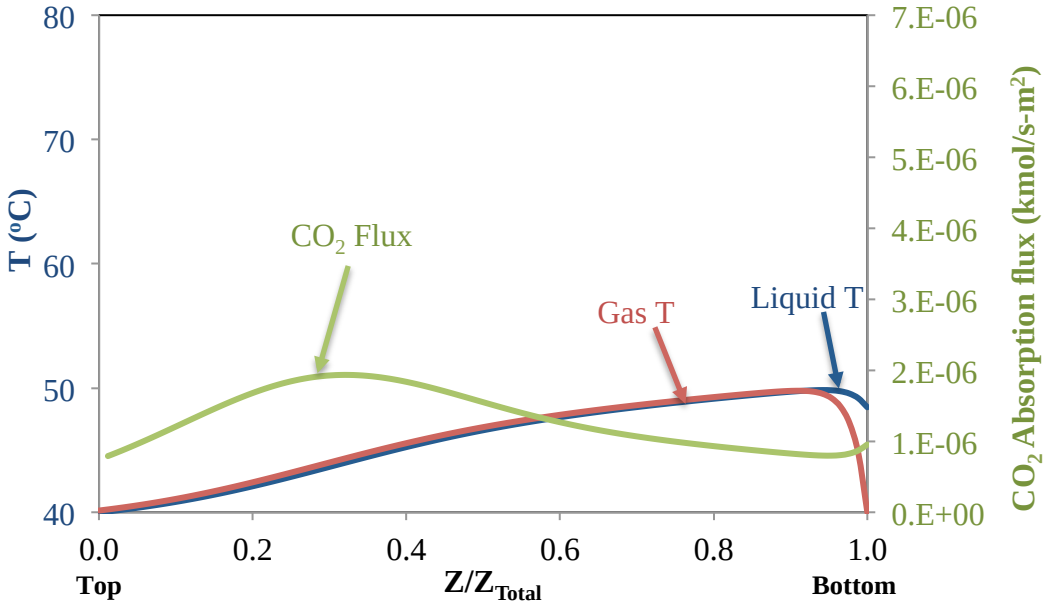


Figure 5.20: Temperature and CO_2 mass transfer flux profiles. Lean ldg = 0.45 L_{min}/G = 14.2 3537 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. 9 m MEA

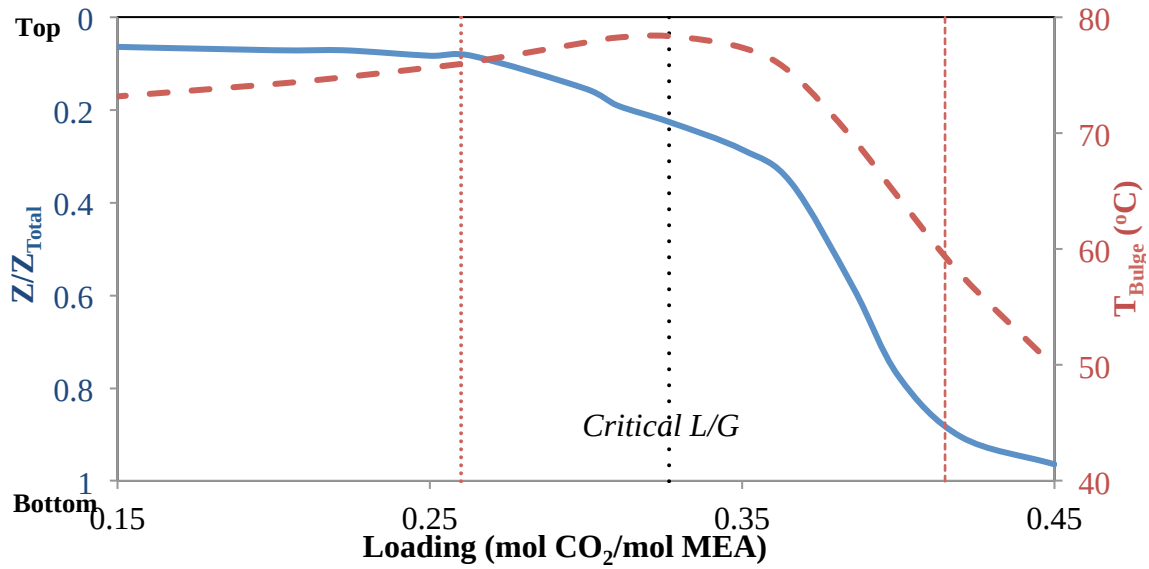


Figure 5.21: Location and Magnitude of the Temperature bulge as a function of loading L_{min}/G. RSP-250. Diameter set for 80% flooding. 90% removal. 9 m MEA

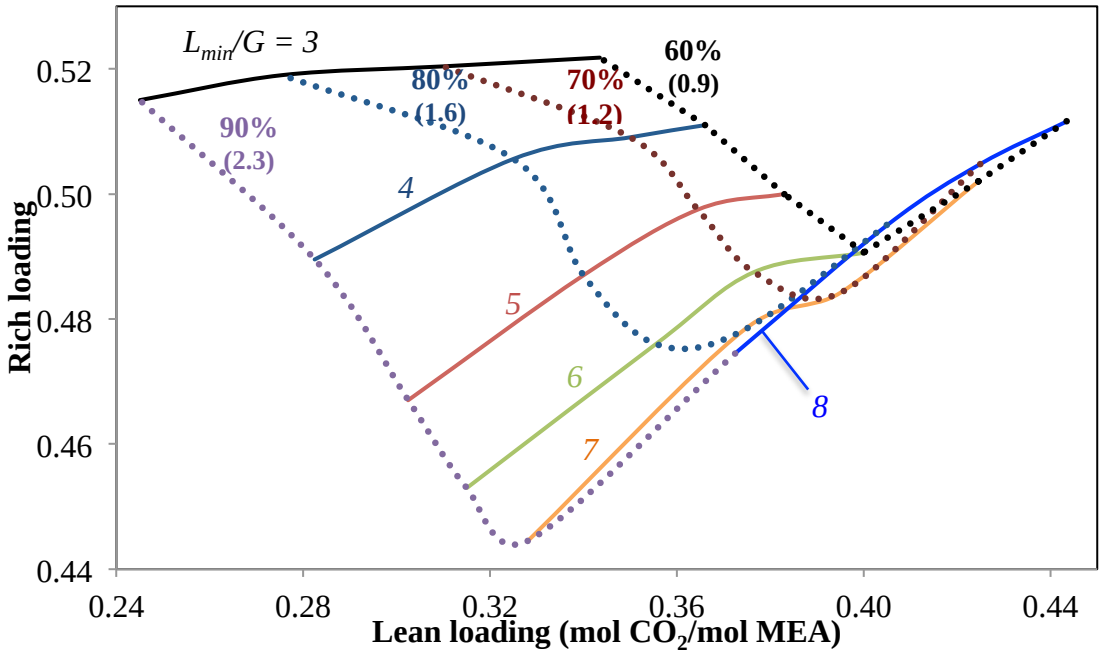


Figure 5.22: Variation of Rich loading with change in lean loading for 9m MEA. Solid lines at constant L_{min}/G. Dashed lines constant removal (NTU in Parenthesis). ∞ Packing: RSP-250. Diameter set for 80% flooding.

Dashed lines in Figure 5.15 represent L_{min}/G. with intercooling in the middle of the column to 40°C. The four regions observed for the absorber without intercooling are no longer visible for L/G and slightly noticeable in the rich loading curve. The difference between L_{min}/G with and without intercooling is maximized as the lean loading approaches the critical L/G region. This implies that less solvent is required to achieve the required removal.

Around the critical $(L/G)_c$ region, removal is increased the most by implementing intercooling. Temperature limits mass transfer and the temperature bulge coincides with a mass transfer pinch. Intercooling breaks the mass transfer pinch, reduces column temperature and improves packing use increasing removal as it is shown in Figure 5.23

for the critical L_{min}/G. The L_{min}/G for this case goes from 6.9 for the non-intercooled case to 4.5 for the intercooled absorber.

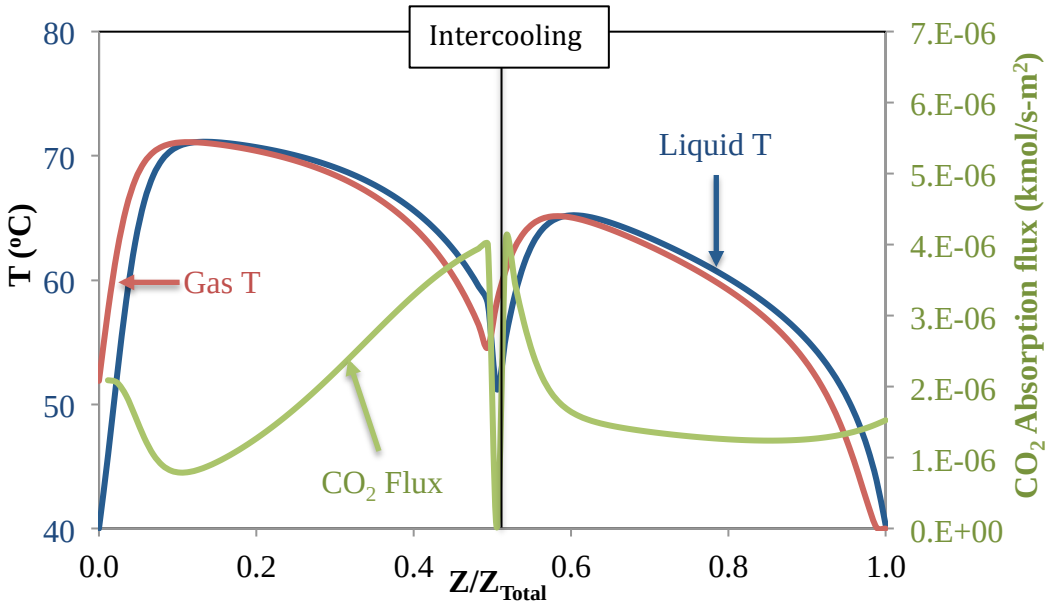


Figure 5.23: Temperature and CO_2 mass transfer flux profiles. 9m MEA Lean ldg = 0.33 L_{min}/G = 4.5 2805 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. Intercooled at Z/Z_{Total} = 0.5

Intercooling allows the column to have a closer approach to equilibrium leading to a richer solvent to the stripper. The rich loading in Figure 5.15 shows a slight minimum but is mostly constant throughout the loading range. This is another advantage to intercooling since the absorber can operate at higher rich and lean loadings, which translates into lower energy requirements for solvent regeneration.

5.3.1.2. Intercooled Absorber Performance for 1.1 and 1.2 L_{min}

Operation at the minimum solvent flow rate would require an infinite packing volume so it is necessary to determine an optimum flow rate that would reduce solvent requirements without requiring excessive amounts of packing. Frailie (2011) showed that 10% L_{min} is a good criterion for determining absorber solvent rate. This number comes

from looking at the solvent rate and specified packing volume. Figure 5.24 shows the solvent rate required for 90% removal corresponding to a specified packing volume. The curves in this plot approach asymptotes on both axes. The x-axis asymptote represents the minimum L/G while the y-axis asymptote corresponds to the minimum packing height, infinite solvent rate. Increasing the minimum solvent flowrate by 10% decreases packing volume to accomplish the specified removal without major solvent increase.

The four regions are still visible for the non-intercooled cases and less visible for the intercooled runs when analyzing the change of rich loading and *L/G* with variation of lean loading for *1.1* L_{min}. Intercooling decreases solvent flow rate requirements and yields higher rich loadings that should improve stripper energy performance.

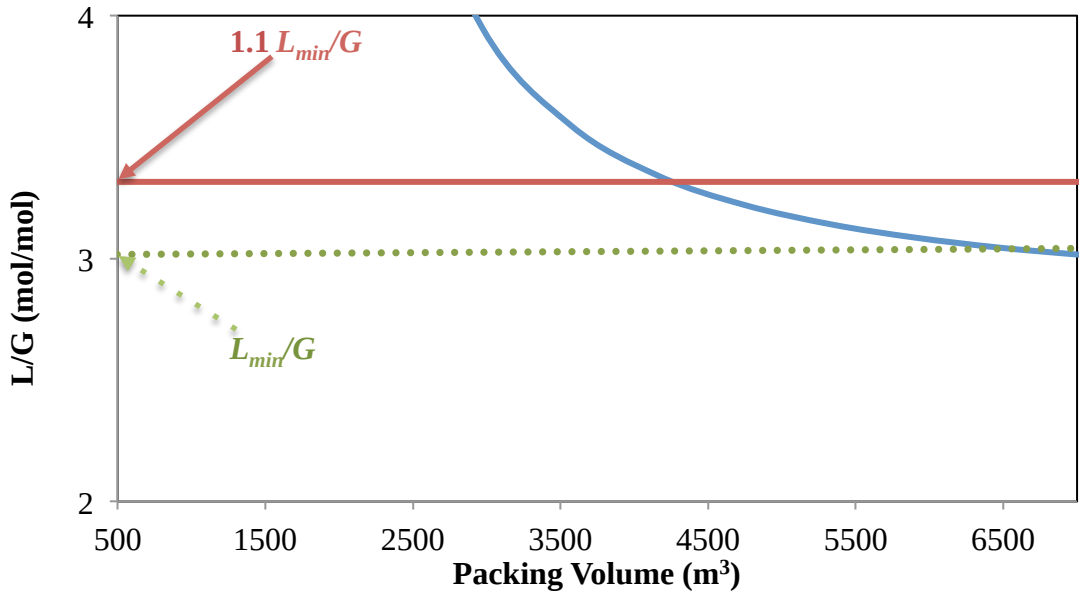


Figure 5.24: Variation of *L/G* with respect to packing volume. RSP-250. 90% Removal. Constant gas flow rate. Diameter for 80% flooding 9 m MEA. Loading = 0.25 mol CO_2/mol MEA

It is still possible to observe a change in the behavior of mass transfer and temperature profiles between the regions for the intercooled cases. For the first region,

there is still a large bulge at the top of the column with a lower magnitude (70°C). There is an approach to a mass transfer pinch towards the bottom of the column with most of the removal occurring at the top (Figure 5.26).

Similar to the minimum flow rate case as the lean loading is increased the bulge temperature increases slightly, the mass transfer flux rates drops across the column and the mass transfer pinch at the bottom of the column is accentuated. (Figure 5.27)

At the critical L/G (without intercooling, ldg=0.33) temperatures are still higher at the top but a second bulge is more noticeable in the bottom section of the column. Flux rates are lower in magnitude and remain larger in the top section (Figure 5.28)

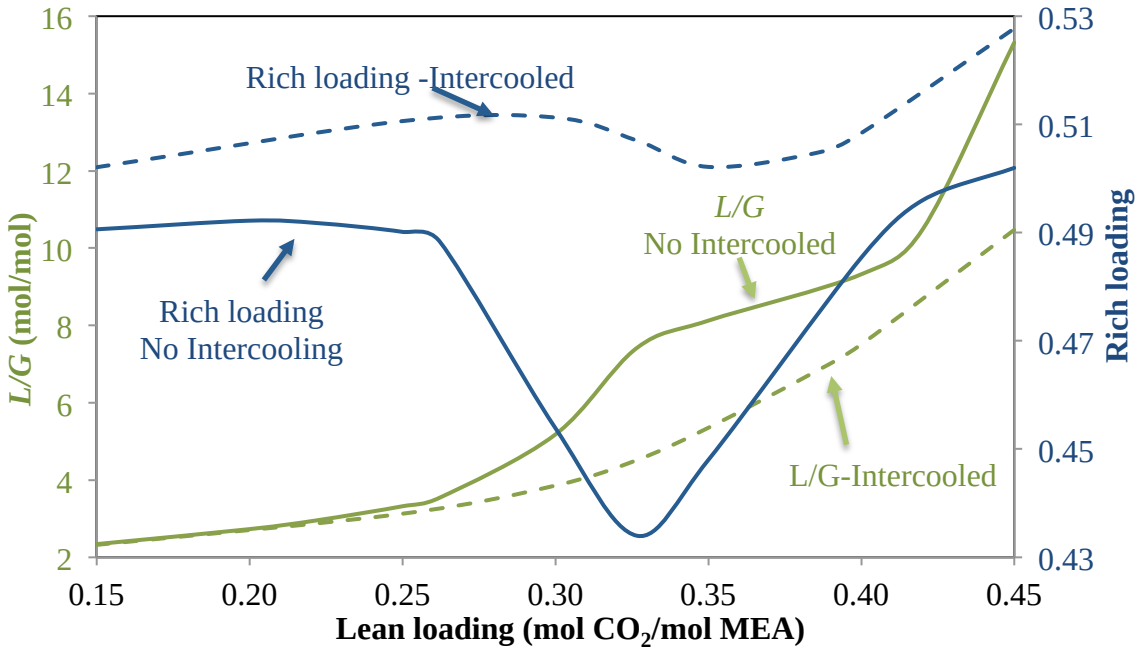


Figure 5.25: Variation of rich loading and *L/G* with change in lean loading for 9m MEA and 90% removal. RSP-250. Diameter set for 80% flooding. Liquid intercooled to 40°C in the middle of the column. 1.1 L_{min}

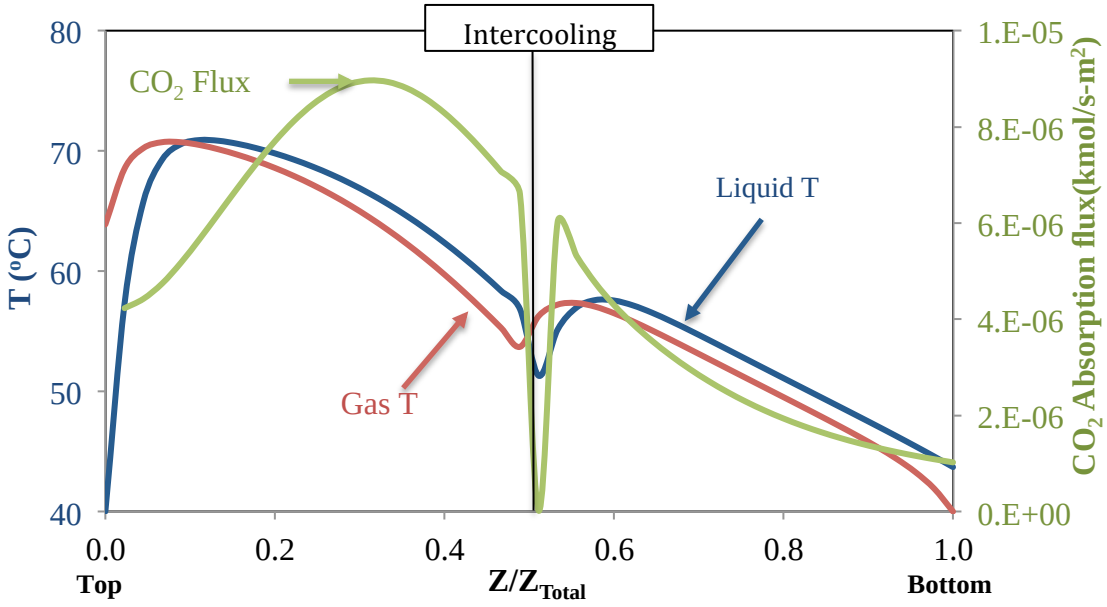


Figure 5.26: Temperature and CO_2 mass transfer flux profiles. 9 MEA. Lean ldg = 0.15 1.1 L_{min}/G=2.3. 1189 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. Intercooled at Z/Z_{Total} = 0.5.

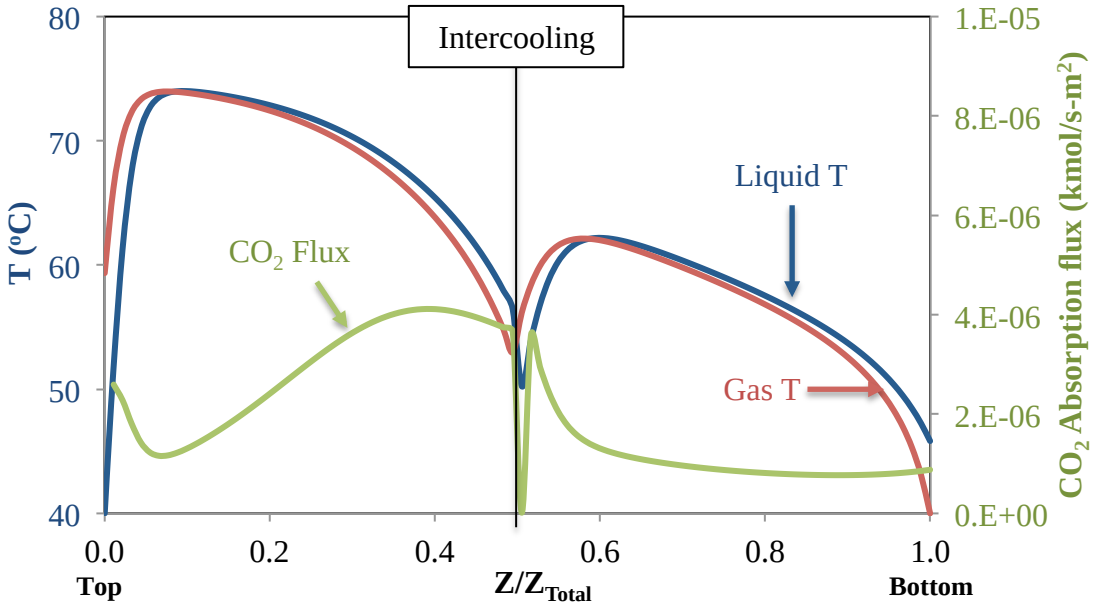


Figure 5.27: Temperature and CO_2 mass transfer flux profiles. 9 MEA. Lean ldg = 0.30 1.1 L_{min}/G= 3.9. 1863 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. Intercooled at Z/Z_{Total} = 0.5

Figure 5.29 shows the temperature and mass transfer profiles for the third region of Figure 5.15. The CO_2 mass transfer flux is higher at the top of the column and closer to pinch at lower bottom. There is a temperature decrease across the column and the bulge moved towards the bottom of the column due to an increase in the heat generated leaving with the solvent. This portion of the rich vs. lean loading curve shows a minimum.

The last region with high loading has a minimum temperature increase due to the high solvent flow rate. Mass transfer flux rates are very low, evenly distributed across the column and with a pinch at the bottom (Figure 5.31).

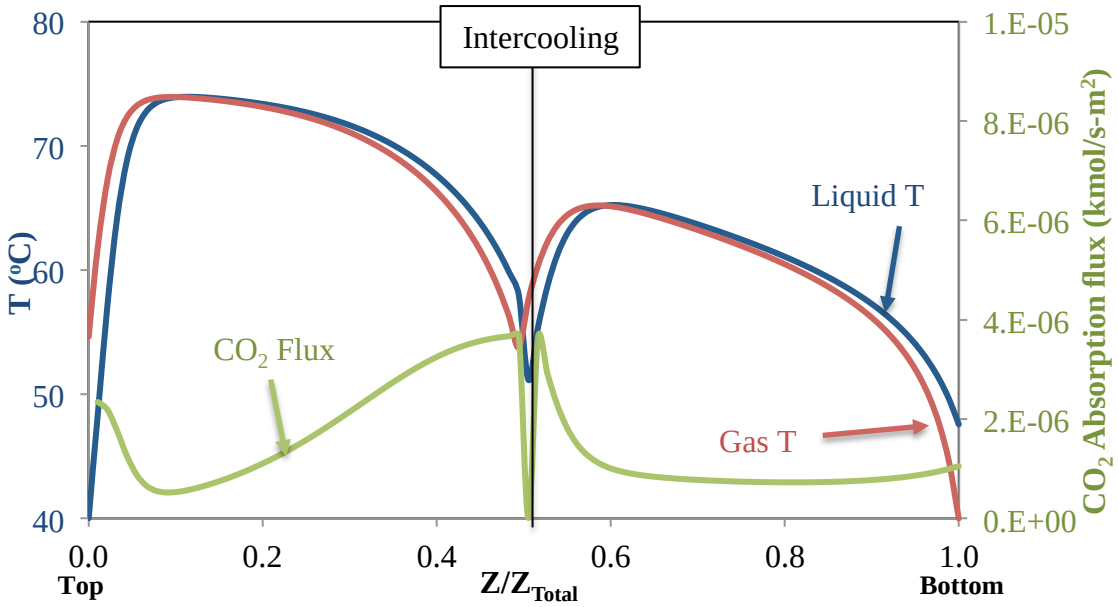


Figure 5.28: Temperature and CO_2 mass transfer flux profiles. 9 m MEA. Lean ldg = 0.33 1.1 L_{min}/G. 3570 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. Intercooled at Z/Z_{Total} = 0.5

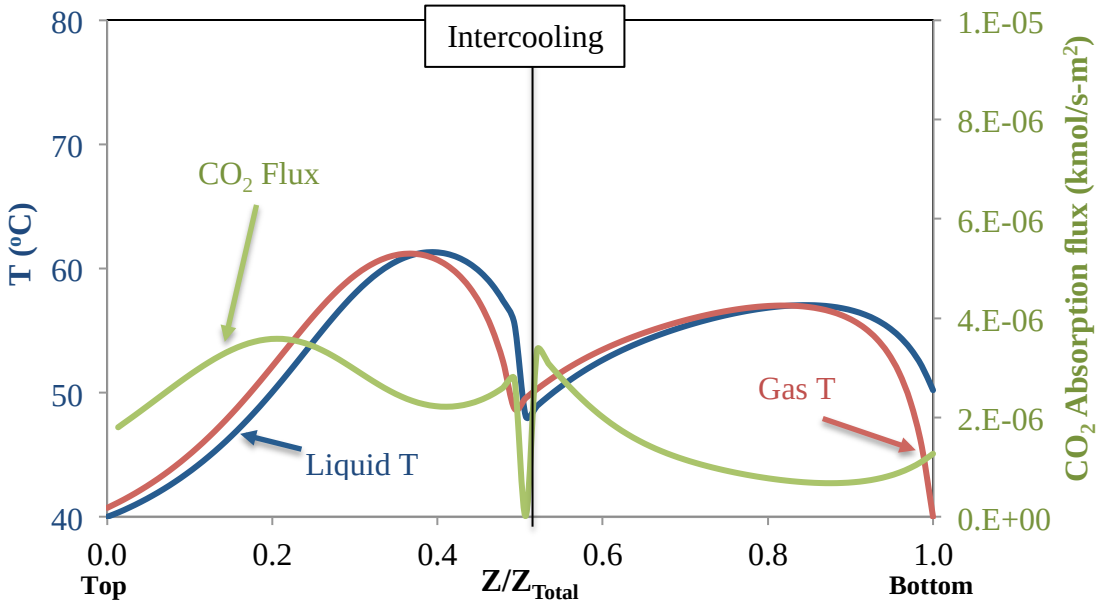


Figure 5.29: Temperature and CO_2 mass transfer flux profiles. 9m MEA Lean ldg = 0.386 1.1 L_{min}/G=6.8 2498 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. Intercooled at Z/Z_{Total} = 0.5

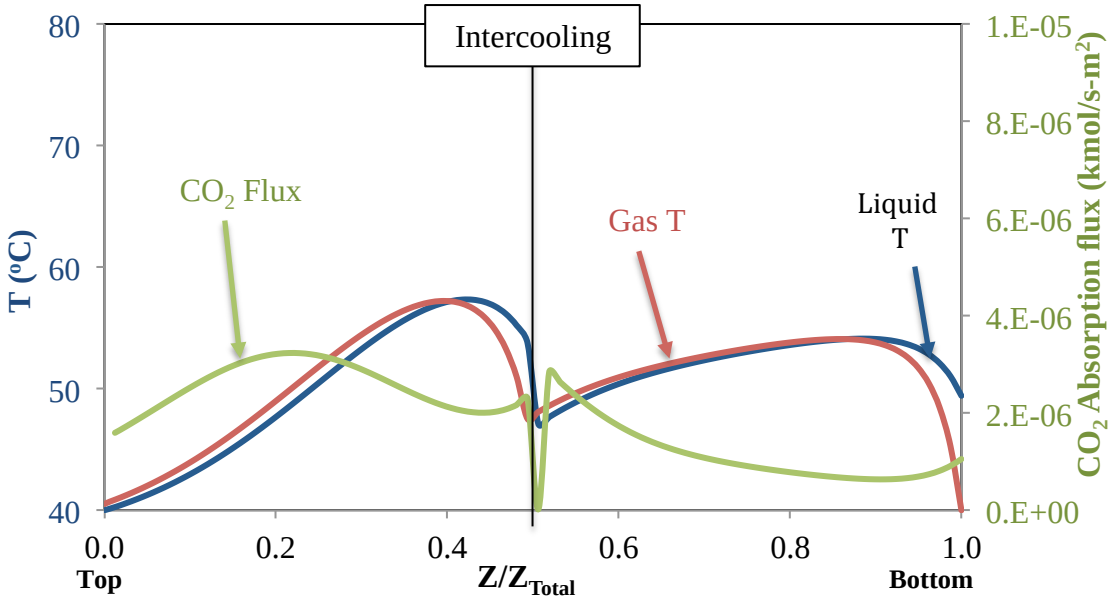


Figure 5.30: Temperature and CO_2 mass transfer flux profiles. 9m MEA. Lean ldg = 0.40 1.1 L_{min}/G. 2730 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. Intercooled at Z/Z_{Total} = 0.5

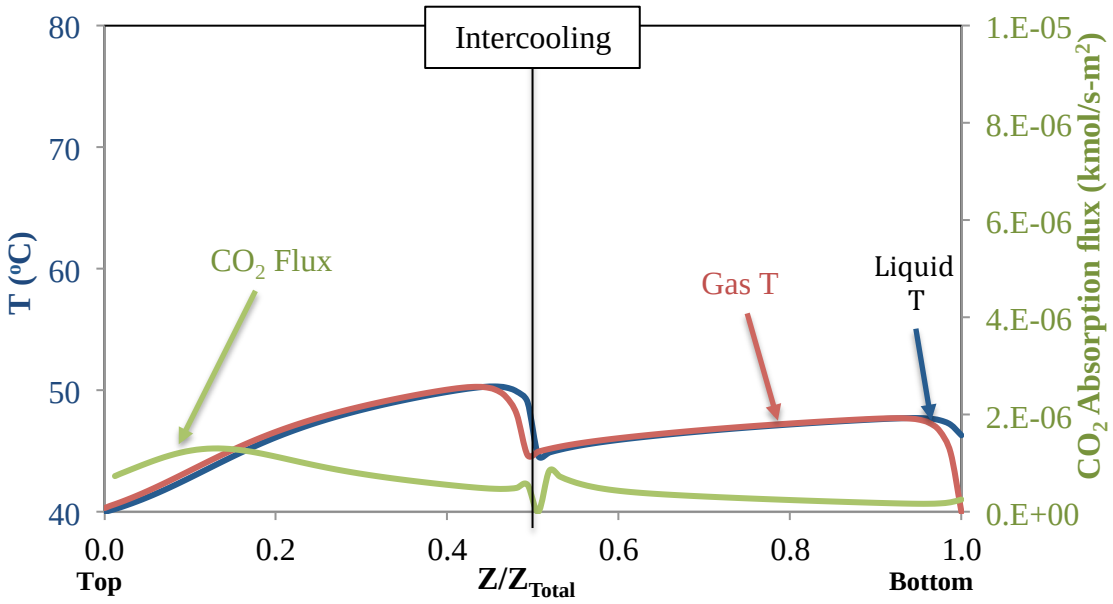


Figure 5.31: Temperature and CO_2 mass transfer flux profiles. 9m MEA. Lean ldg = 0.45 1.1 L_{min}/G=10.5 8193 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. Intercooled at Z/Z_{Total} = 0.5

A similar behavior to 90% removal for the rich loading can be observed for lower removal requirements as is displayed in Figure 5.22 that plots the change in rich loading with lean loading for constant removal and L/G.

Figure 5.32 shows the change in required packing area per mol of CO_2 per hour removed as a function of *L/G* at 90% removal with an absorber with and without intercooling to 40°C in the middle of the column. It is a plot of an inverse of flux that can be used to determine required packing volume for a specified removal and *L/G*. The area in the y-axis corresponds to the nominal interfacial area (a_n) that is equal to the product of the volume of packing (V_p) and the specific packing area (a_p). The volume of packing is calculated specifying 80% flooding for the diameter of the column and changing the height of packing to match the desired removal. Equation 4.3 was used to determine packing height segmentation. If the final packing height did not correspond to the

calculated number of segments, the column was re-segmented and the packing height recalculated. The moles of CO_2 removed were calculated from the specified gas rate and the required removal.

This figure shows that although intercooling allows operation at higher solvent loadings and lower solvent rates, it requires an increase in packing volume. It also shows a bimodal shape for the performance of both intercooled and non-intercoooled absorbers with an expected minimum at lean loading and a second minimum using higher solvent rate and richer lean feed. This second minimum corresponds to the critical L/G.

The additional packing required to maintain high rich loadings is clear from looking at Figure 5.33. At around the critical L/G, without intercooling, the rich loading presents its largest decrease with increasing lean loading so more intercooled area assures a closer approach to equilibrium, a higher rich loading.

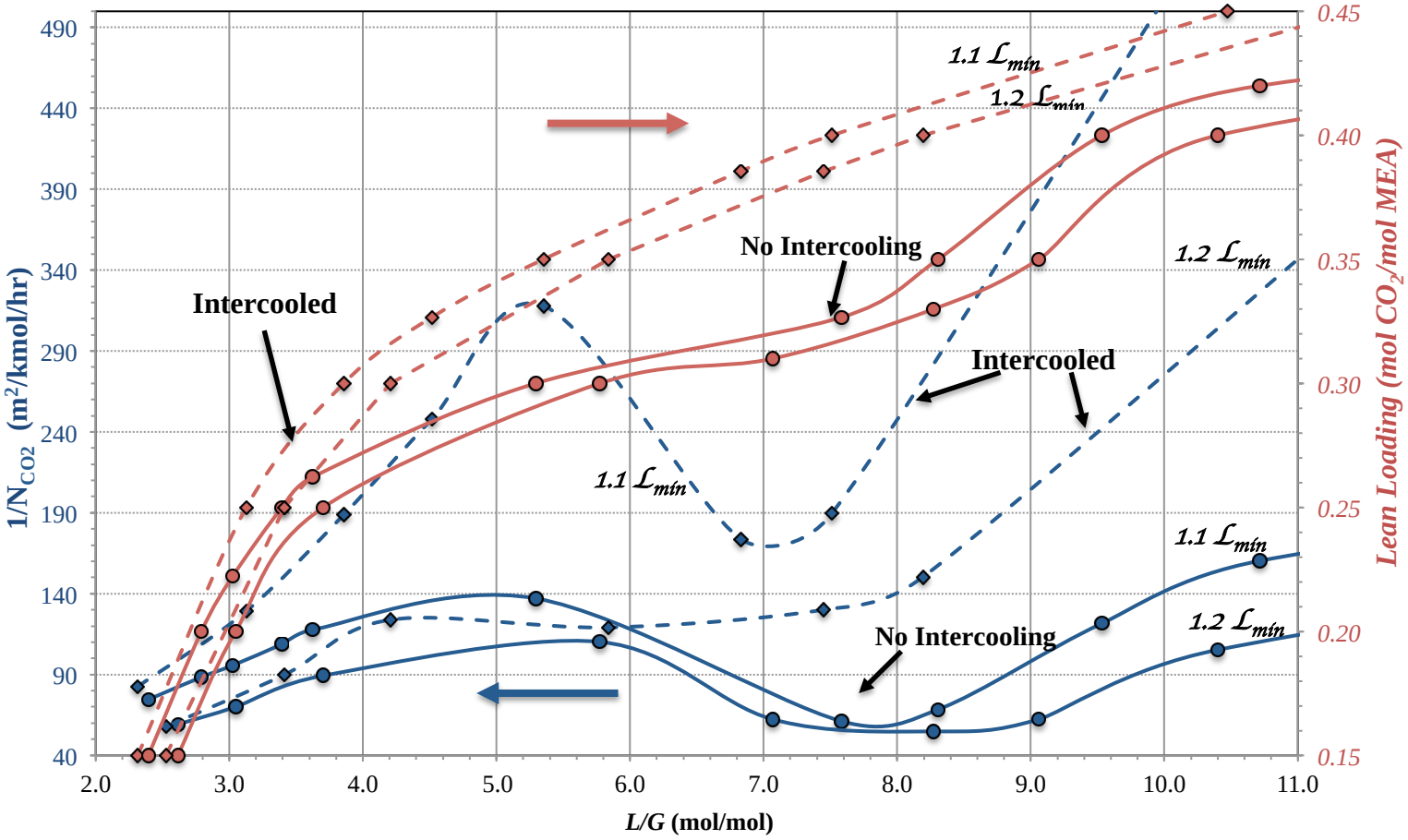


Figure 5.32 Required packing area per mol of CO_2 per hour removed as a function of *L/G* and lean loading for 90% Removal. Absorber intercooled to 40°C at Z/Z_{Total} = 0.5. x L_{min}/G. RSP-250. Diameter set for 80% flooding, 9m MEA . Gas and solvent feed at 40°C. Inlet y_{CO2} = 12.6%

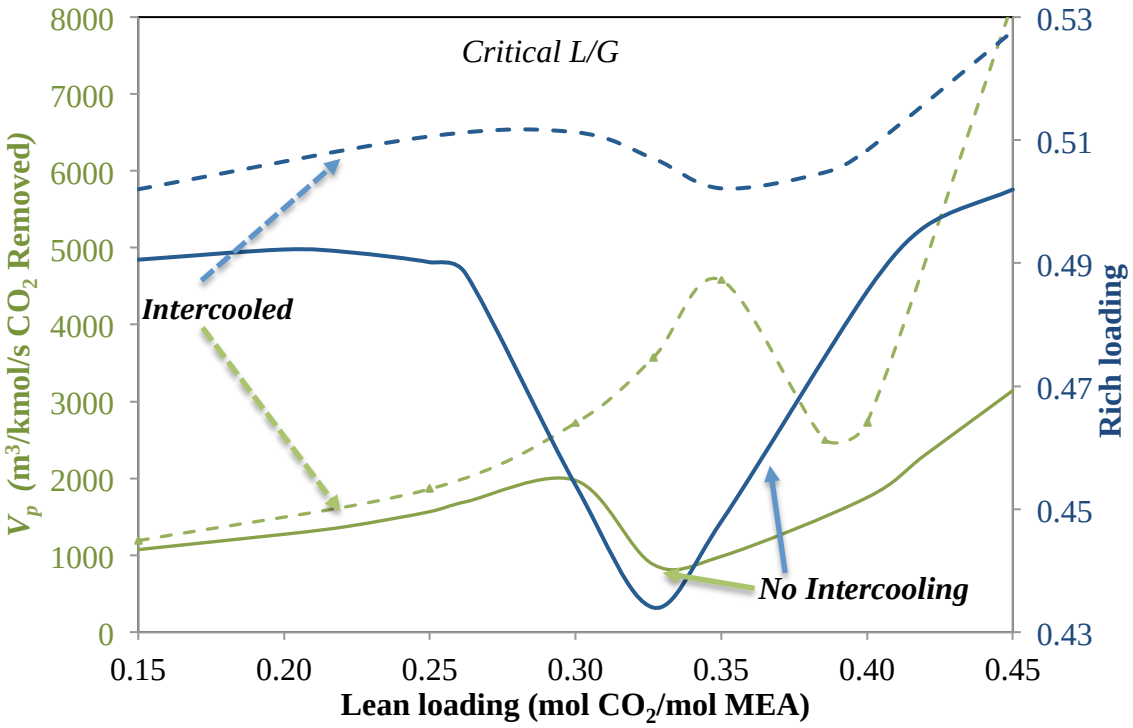


Figure 5.33: Variation of rich loading and required packing volume of RSP-250 with change in lean loading for 9m MEA and 90% removal. Diameter set for 80% flooding. Liquid intercooled to 40°C in the middle of the column. 1.1 L_{min}

However, using 20% of L_{min} the bimodality observed for required packing tends to disappear and there is a considerable reduction in packing area that at L/G < 6.0 is lower than the values for no intercooling and 1.1 L_{min}. This increase in solvent rate may prove to be optimal since it still allows operation at high loadings with small increase in solvent rate and considerably lower packing requirements.

Figure 5.25 can be used to determine the 1.1 L_{min}/G required for a desired removal and pre-determined lean loading. These values can then be used to find the required nominal area per mol/hr of CO_2 removed. Using the required removal, gas composition and flow rate along with specific are of packing it is possible to estimate the required volume for the selected type of structured packing.

Using 1.1 L_{min} for a lean loading of 0.33 and the gas flow rate estimated by Rochelle et al. (2005) for a 500 MW plant, the required volume of RSP-250 for 90% CO_2 removal is 2299 m^3. The *L/G* for this case is 7.6. Maintaining the same lean loading (0.33) the *L/G* for the intercooled case from Figure 5.32 is 4.52. The required volume of RSP-250 for an absorber with intercooled liquid to 40°C in the middle of the column can be calculated using the same figure and an *L/G* of 4.52. The value of the y-axis for these conditions is 248 m^2/kmol/hr CO_2 removed. Calculations to determine the volume of packing are as follows:

$$V_p = \left(248 \frac{m^2}{\frac{kmol\ CO_2}{hr}}\right)\left(250 \frac{m^2}{m^3}\right)^{-1} (0.126)\left(82959 \frac{kmol\ CO_2}{hr}\right)(0.90)$$

$$V_p = 9332\ m^3$$

This volume of packing is 4 times the required for the non-intercooled case. However, the solvent flow rate is almost half and the rich loading is almost 20% more than the non intercooled case Running two more non-intercooled cases at 1.1 L_{min} and comparing with the results from the estimated values using Figure 5.32 corroborates that intercooling has a largest impact at the critical L/G region. Increasing the solvent flow rate to 1.2 L_{min} reduces in half the packing area (4516 m^3) for a loading of 0.33.

Table 5.7: Required absorber RSP-250 packing volume for 90% removal for a 500 MW coal power plant. 1.1 L_{min}. 80% flooding. 9m MEA

Lean Loading	No-Intercooling			Intercooled to 40°C		
	L/G (mol/mol)	Rich ldg	V_p (m^3)	L/G (mol/mol)	Rich ldg	V_p (m^3)
0.22	3.03	0.492	3600	2.95	0.508	5645
0.33	7.59	0.433	2299	4.52	0.507	9332
0.42	10.71	0.496	6031	8.85	0.515	13171

Figure 5.32 can also be used to estimate the packing volume for an absorber using structured packing different to RSP-250. Table 5.8 compares the resulting volume from the absorber model with Mellapak 2X for the cases in Table 5.7 and the values obtained with the specific packing area for Mellapak 2X (205 m^2/m^3) and Figure 5.32. The interfacial factor used for Mellapak 2X was the same used for RSP-250 (0.73). The predicted values are within ±25% of the modeled results. This corresponds to the closely to the adjustment applied to the interfacial area of Mellapak 2X. However, the deviation for the richer solvent loading is higher (40%). This might be related to the higher solvent rates used at the richer solvent feeds.

Table 5.8: Required absorber packing volume for 90% removal for a 500 MW coal power plant. 1.1 L_{min}. 80% flooding. 9m MEA for Mellapak 2X Intercooling at Z/Z_{Total}=0.5 to 40°C.

Lean Loading	Aspen Plus ® Model			Figure 5.28			V_p deviation %
	L/G (mol/mol)	Rich ldg	V_p (m^3)	L/G (mol/mol)	Rich ldg	V_p (m^3)	
0.22	2.8	0.520	5646	2.95	0.508	6884	21.9
0.33	4.8	0.503	9221	4.52	0.507	11381	23.4
0.42	8.5	0.515	11440	8.85	0.515	16062	40.4

5.3.2. Intercooling in 8m PZ

Piperazine has higher rates and capacity than MEA but these are limited by the amount of CO_2 that enters the liquid phase. Intercooling by lowering column temperatures may help take advantage of the characteristics of this solvent. The following sections present results to an analysis similar to the previous section for MEA. First the behavior of L_{min}/G with loading was explored and then a look at 1.1 L_{min}/G to determine process performance is included. Figure 5.34 shows the change in L/G as a

function of volume of packing. The asymptotes represent the minimum packing volume and the minimum *L/G*.

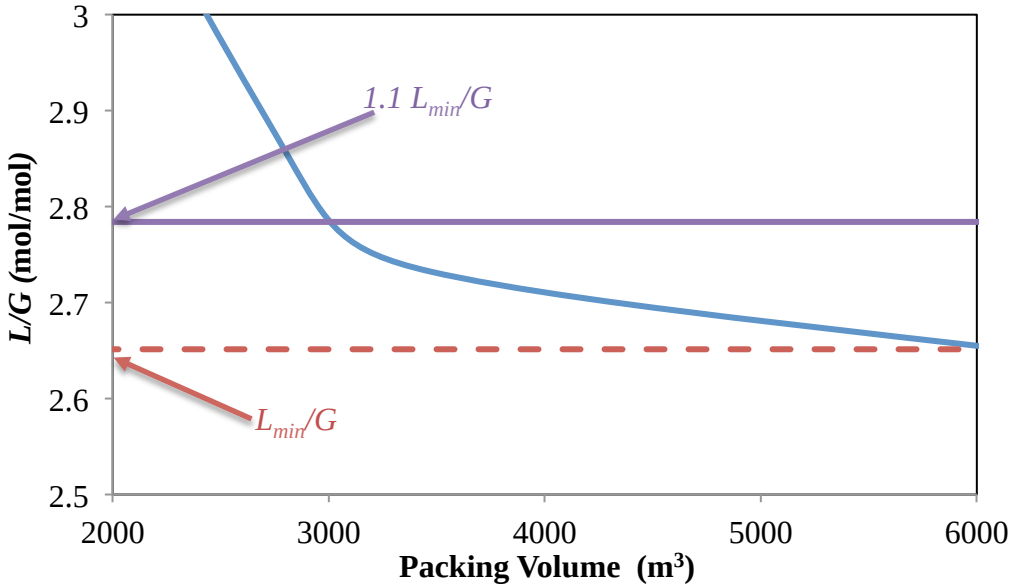


Figure 5.34: Variation of *L/G* with respect to packing volume. RSP-250. 90% Removal. Constant gas flow rate. Diameter for 80% flooding 8 m PZ. Loading = 0.30 mol CO_2/mol equivalent PZ

5.3.2.1. Intercooling and L_{min}/G

For this analysis the lean loading was varied and its effect on rich loading and L_{min}/G was observed (Figure 5.35). Simulation runs used the same methodology of the MEA analysis. The interfacial area (a_{intf}) was multiplied by 1.02 in all the runs using RSP-250 with PZ per the results of the PZ Sept'10 Campaign.

In Figure 5.35 the four regions previously observed for MEA are also identified. The fourth performance region is not clearly observed since the operational loading window of PZ is limited compared to MEA due solvent solubility.

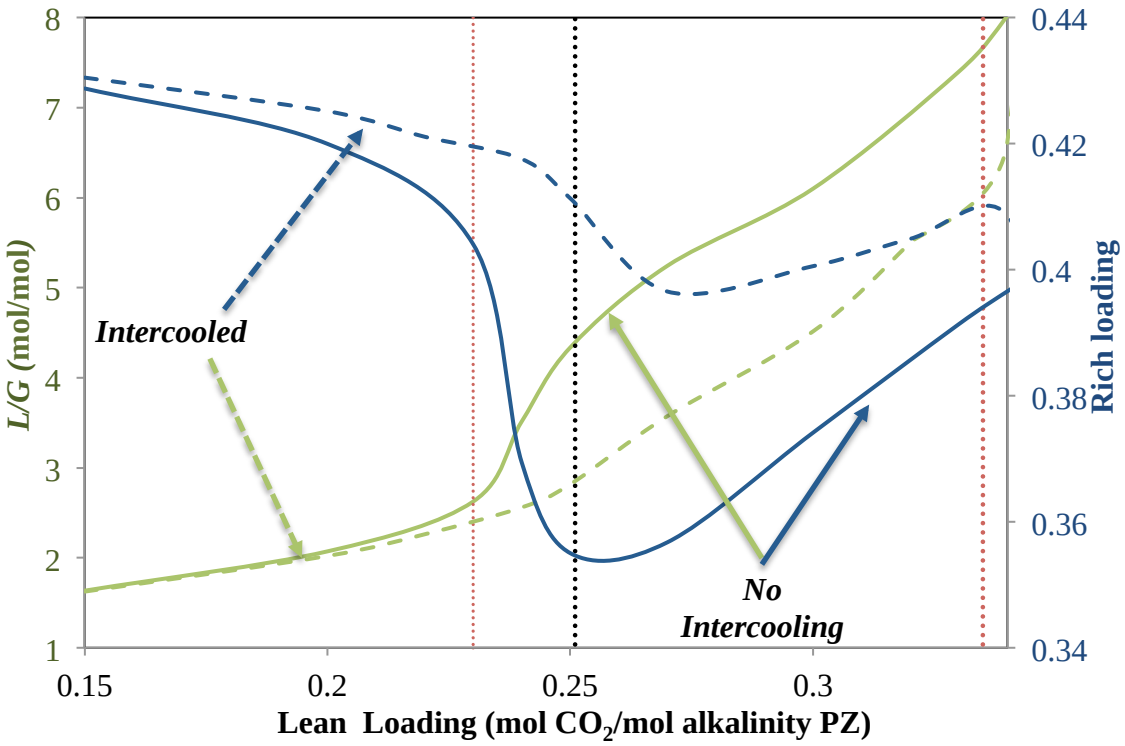


Figure 5.35: Variation of rich loading and *L/G* with change in lean loading for 8m PZ and 90% removal. Infinite packing RSP-250. Diameter set for 80% flooding. Liquid intercooled to 40°C in the middle of the column.

Temperature and mass transfer profiles have similar behavior to those for MEA. The first two regions including the critical *L/G* have very similar temperature behavior but are differentiated by the mass transfer flux profiles. In the first region the temperature bulge is located towards the top of the column and a mass transfer pinch at the bottom. As with MEA most of the heat leaves with the gas so its temperature is considerably higher than the liquid temperature (Figure 5.36). Driving forces are high due to the lean loading so most of the mass transfer occurs at the top.

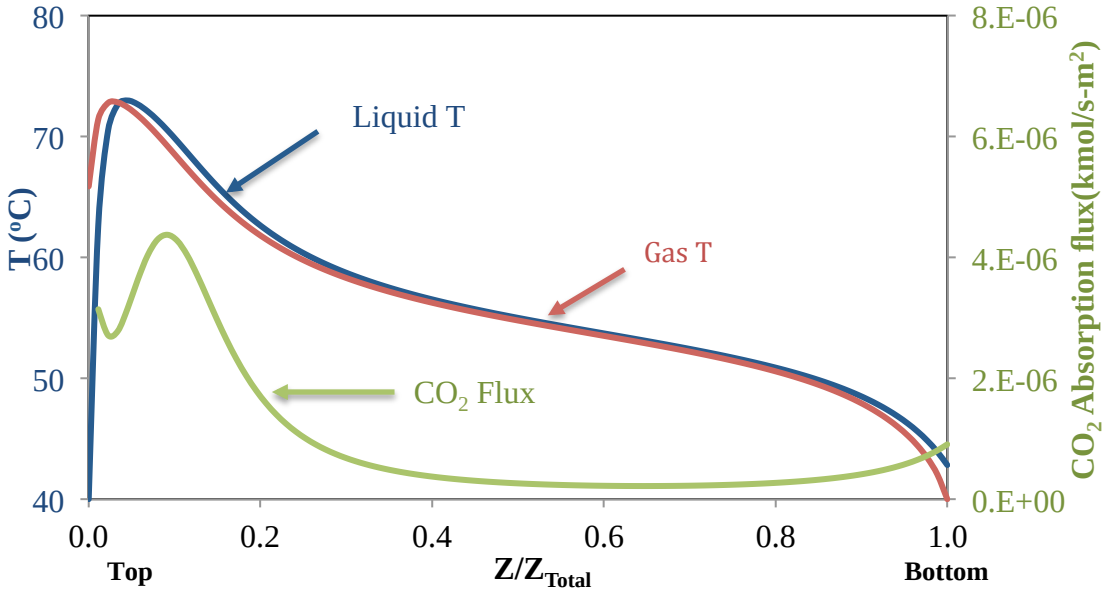


Figure 5.36: Temperature and CO_2 mass transfer flux profiles. Lean ldg = 0.20 L_{min}/G = 2.1 2507 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal . 8m PZ

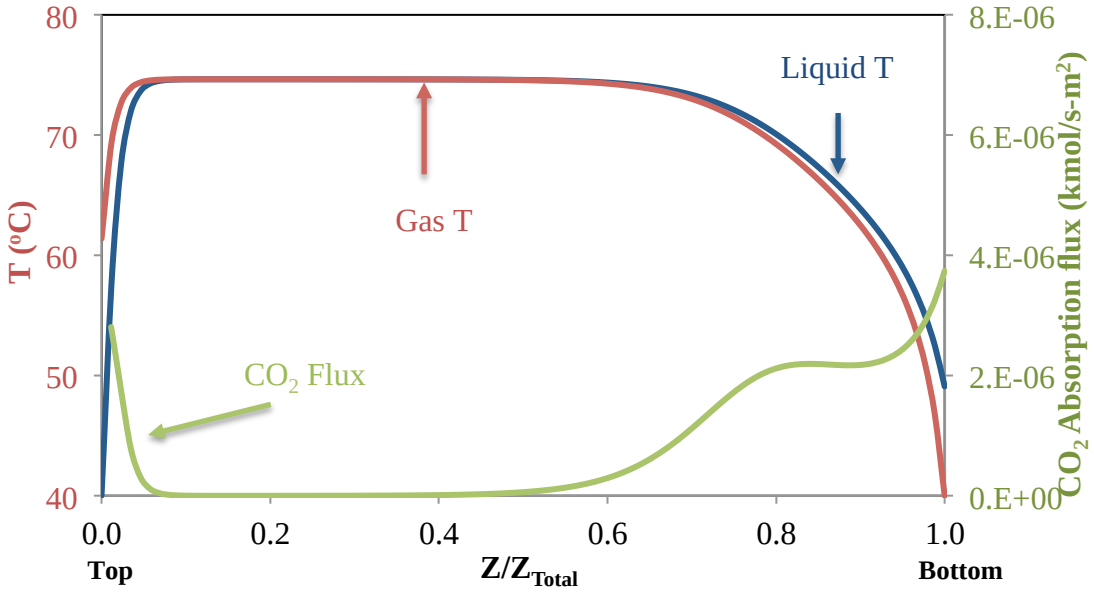


Figure 5.37: Temperature and CO_2 mass transfer flux profiles. Lean ldg = 0.24 L_{min}/G = 3.5 2949 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal 8m PZ.

At the limit of the second region a similar discontinuity to the one observed for MEA and K^+/PZ can be noted although not as marked as in the other two solvents. This is the critical *L/G* point that defines the critical *L/G* region. Here the temperature bulge is broad and located towards the middle of the column. As with MEA and K^+/PZ, in this region intercooling is a more valuable alternative to increase absorber removal (Figure 5.38).

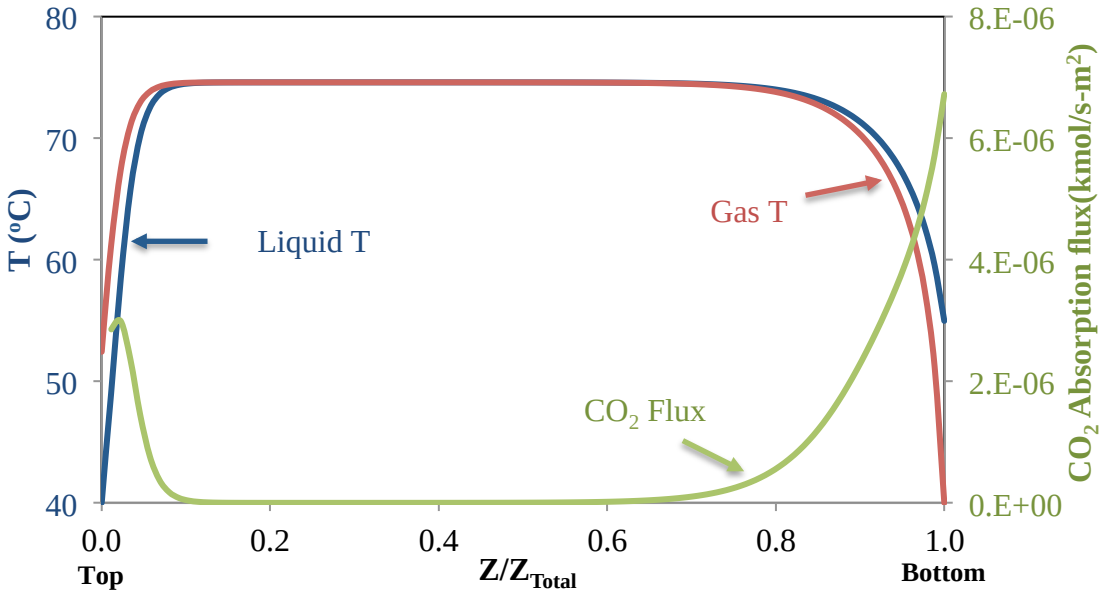


Figure 5.38: Temperature and CO_2 mass transfer flux profiles. Lean ldg = 0.25 L_{min}/G = 4.3 3108 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. 8m PZ.

After the critical *L/G* the temperature bulge continues to move towards the bottom of the column corresponding to more heat being carried with the additional liquid. Figure 5.38

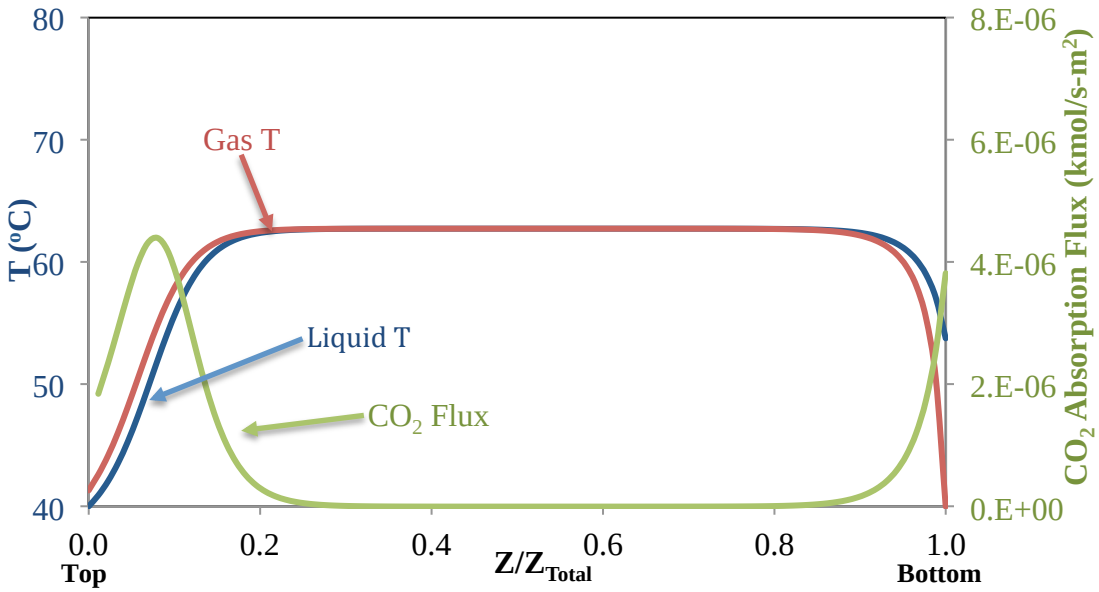


Figure 5.39: Temperature and CO_2 mass transfer flux profiles. Lean ldg = 0.30 L_{min}/G = 6.1 3384 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. 8m PZ.

The 4th zone is not obvious from Figure 5.29 but looking at the profiles for a loading of 0.35 it is clear that the column operates under a different regime from the previous zone (Figure 5.39). The top of the column is completely pinched and all of the removal occurs in the lower half of the absorber.

Figure 5.41 shows the change of rich loading at constant removal and L_{min}/G for 90%-70% removal. Lower removal curves show similar behavior to 90% but with a higher value for the minimum rich loading. Removal values verge towards a single line on the rich side.

The temperature bulge changes in location and magnitude moving towards the bottom of the column with the increase of lean loading. The maximum magnitude of the bulge is lower than MEA and decreases with loading (Figure 5.42)

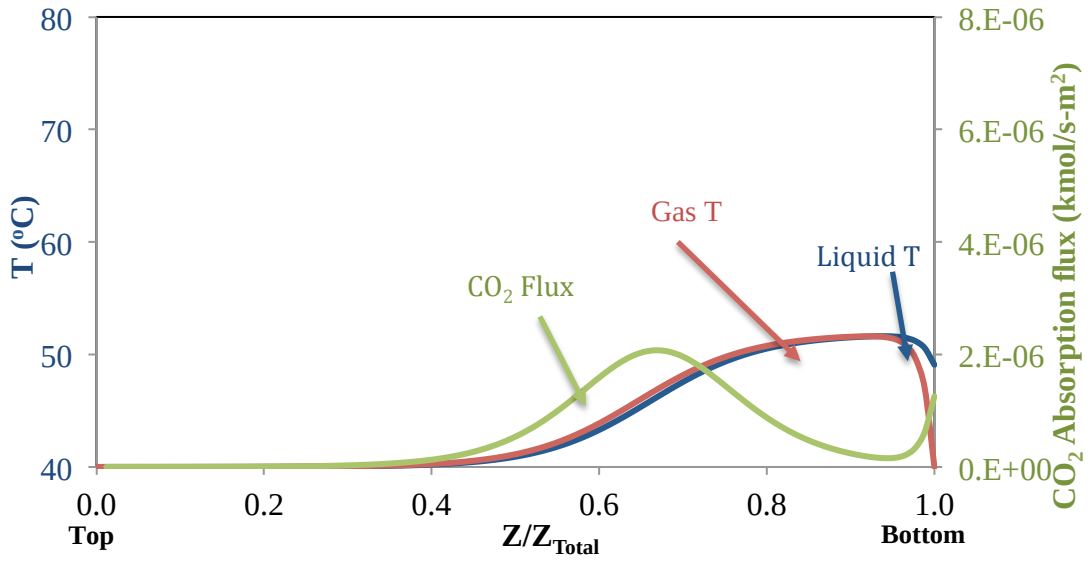


Figure 5.40: T and CO_2 mass transfer flux profiles. Lean ldg = 0.35 L_{min}/G = 8.8. 3672 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. 8m PZ.

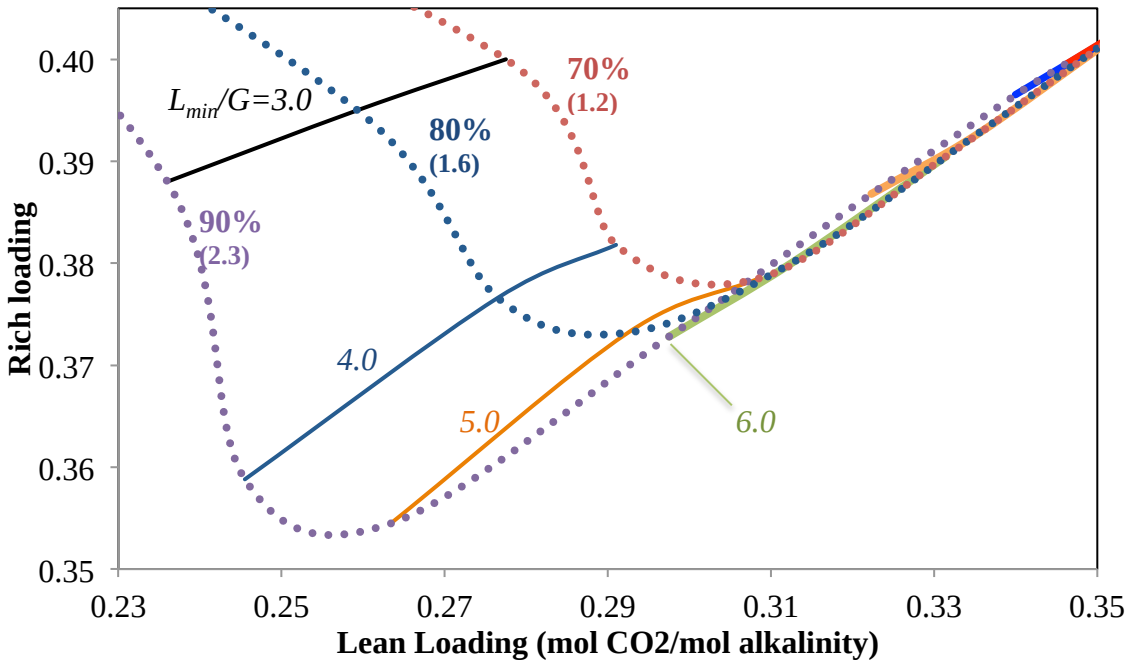


Figure 5.41 Variation of Rich loading with change in lean loading for 8m PZ. Solid lines at constant *L/G*. Dashed lines constant removal (NTU in Parenthesis). ∞ Packing: RSP-250 Diameter set for 80% flooding.

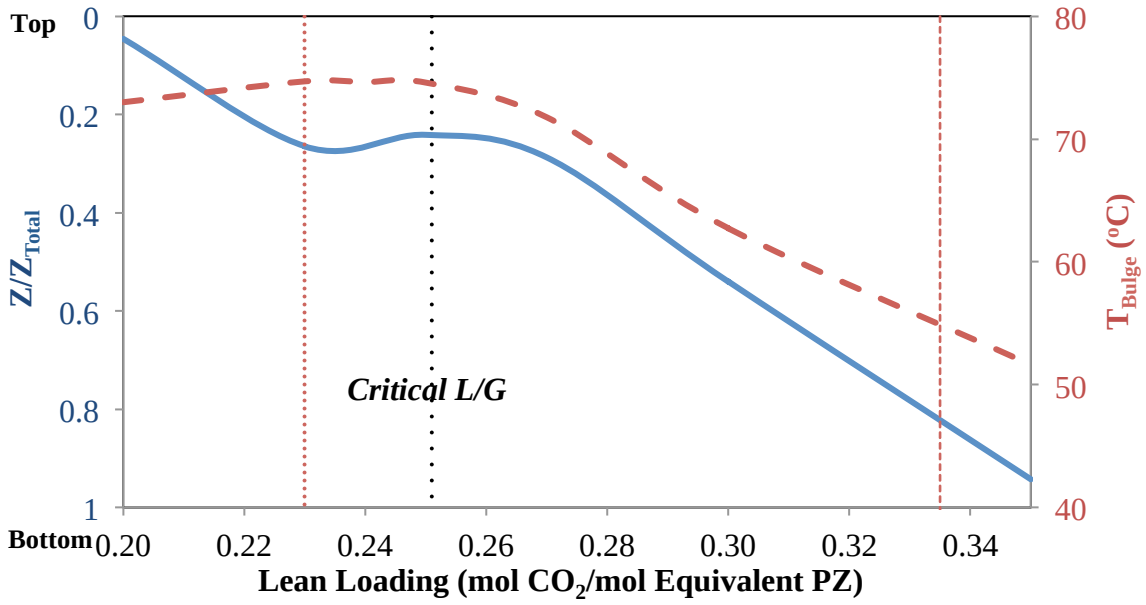


Figure 5.42: Location and Magnitude of the Temperature bulge as a function of loading L_{min}/G. RSP-250. Diameter set for 80% flooding. 90% removal. 8 m PZ.

5.3.2.2. Intercooled Absorber Performance for 1.1 and 1.2 L_{min}

Using the same criterion of 10% additional solvent to the minimum flow rate implemented with MEA, intercooling with PZ was studied. Figure 5.35 shows the four performance regions studied earlier. Similar to MEA, intercooled profiles show a shift in the temperature bulge (lower than the non-intercooled cases) from the top to the bottom of the column as lean loading increases. The mass transfer of CO_2 is better distributed across absorber packing height. Its maximum values move from above the intercooling section to the bottom of the column as lean loading is increased. Figures 5.44 to 5.49 show the resulting temperature and mass transfer flux profiles.

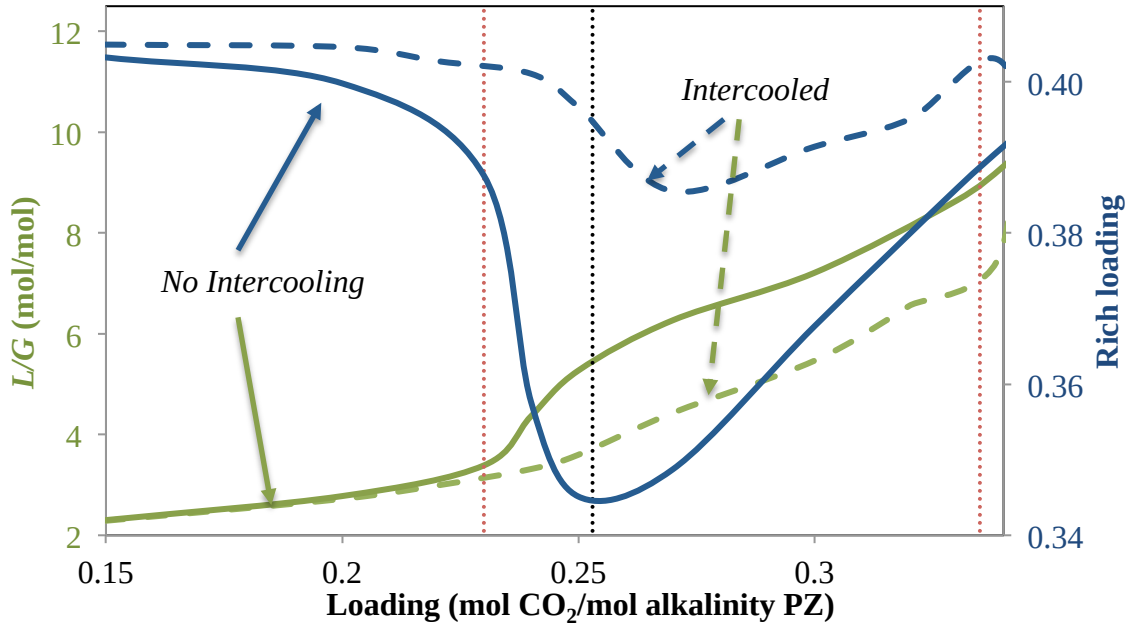


Figure 5.43: Variation of rich loading and *L/G* with change in lean loading for 8m PZ and 90% removal. Variable packing volume. RSP-250. Diameter set for 80% flooding. Liquid intercooled to 40°C in the middle of the column. 1.1 L_{min}.

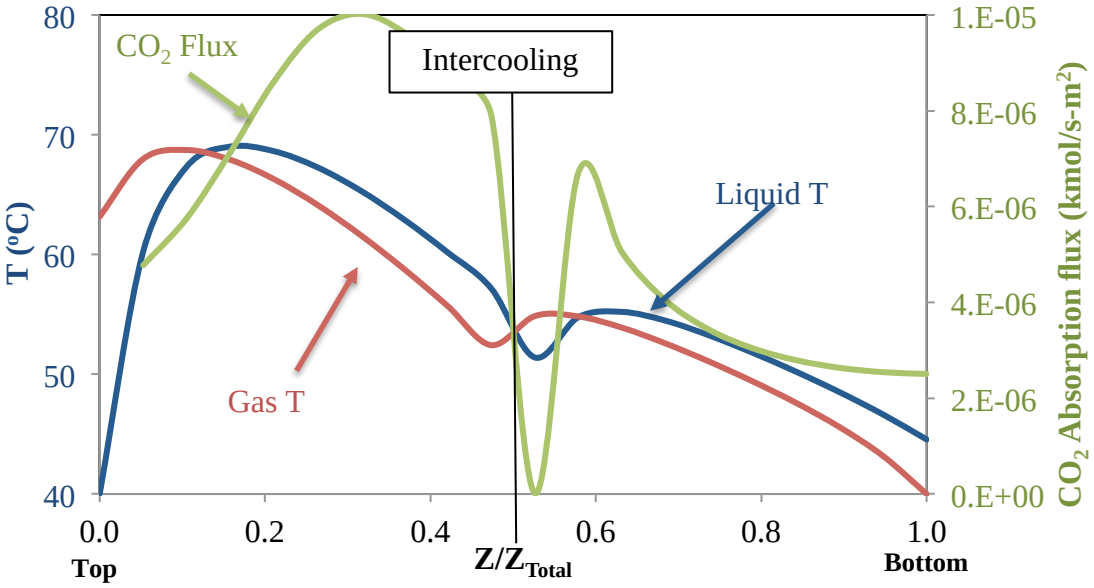


Figure 5.44: Temperature and CO_2 mass transfer flux profiles. 8m PZ Lean ldg = 0.15 1.1 L_{min}/G=1.8 . 452 m^3 RSP-250/kmol/s. Diameter set for 80% flooding. 90% removal. Intercooled at Z/Z_{Total} = 0.5.

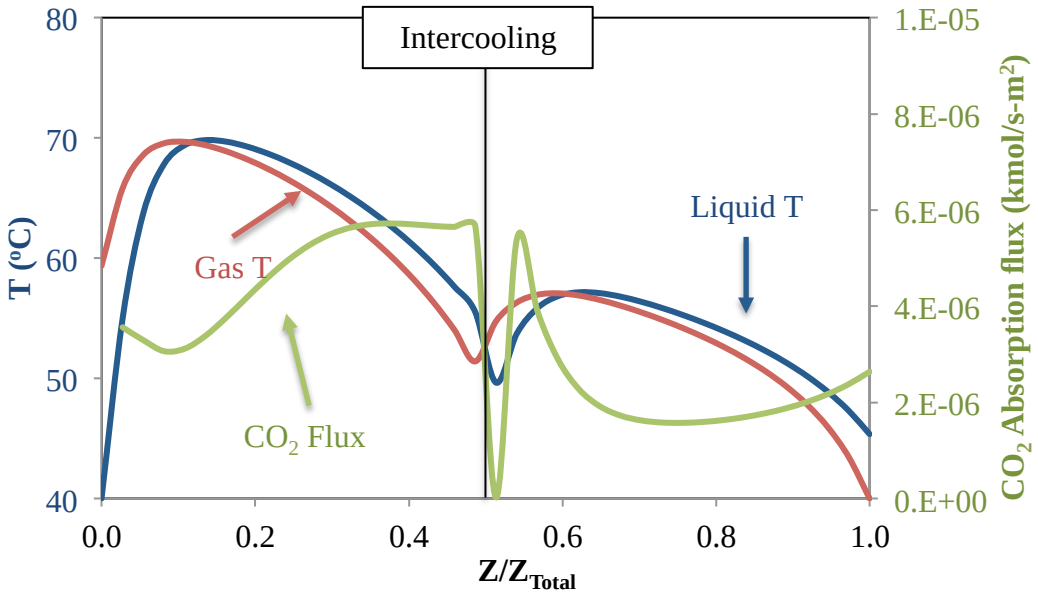


Figure 5.45: Temperature and CO_2 mass transfer flux profiles. 8m PZ Lean ldg = 0.24 , 1.1 L_{min}/G=2.8 . 1051 m^3 RSP-250/kmol/s. Diameter set for 80% flooding. 90% removal. Intercooled at Z/Z_{Total} = 0.5

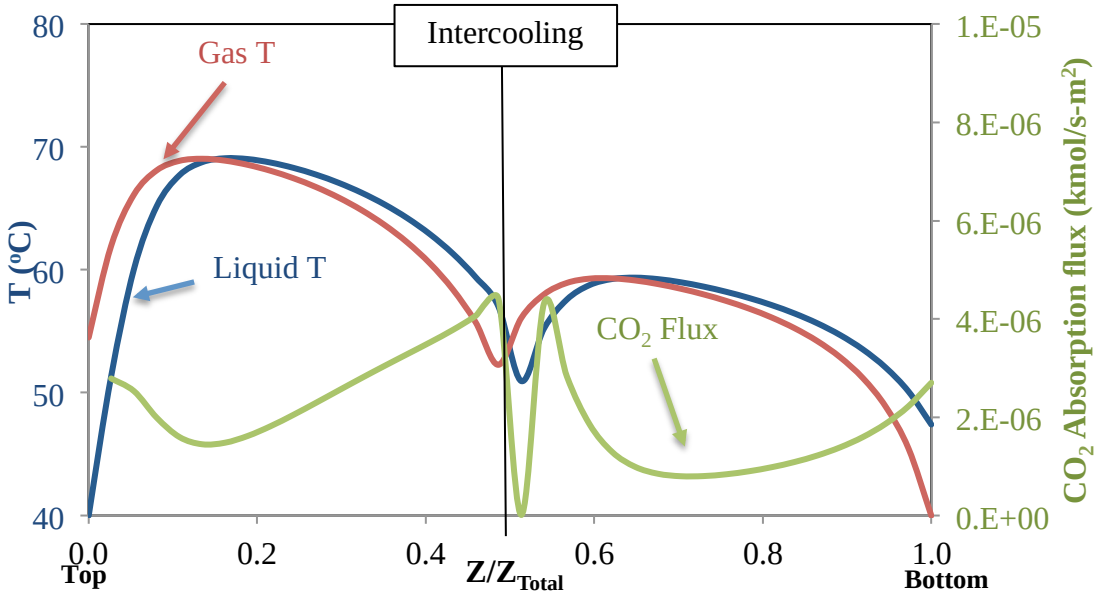


Figure 5.46: Temperature and CO_2 mass transfer flux profiles. 8m PZ Lean ldg = 0.25 , 1.1 L_{min}/G=3.1. 1145 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. Intercooled at Z/Z_{Total} = 0.5

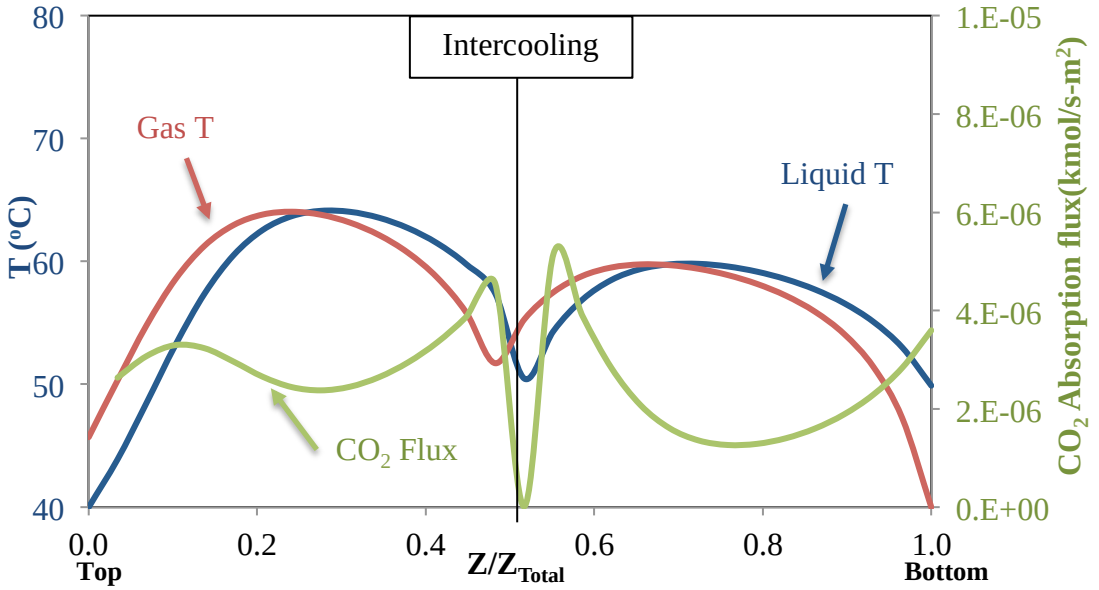


Figure 5.47: Temperature and CO_2 mass transfer flux profiles. 8m PZ Lean ldg = 0.27, 1.1 L_{min}/G=3.9 871 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. Intercooled at Z/Z_{Total} = 0.5

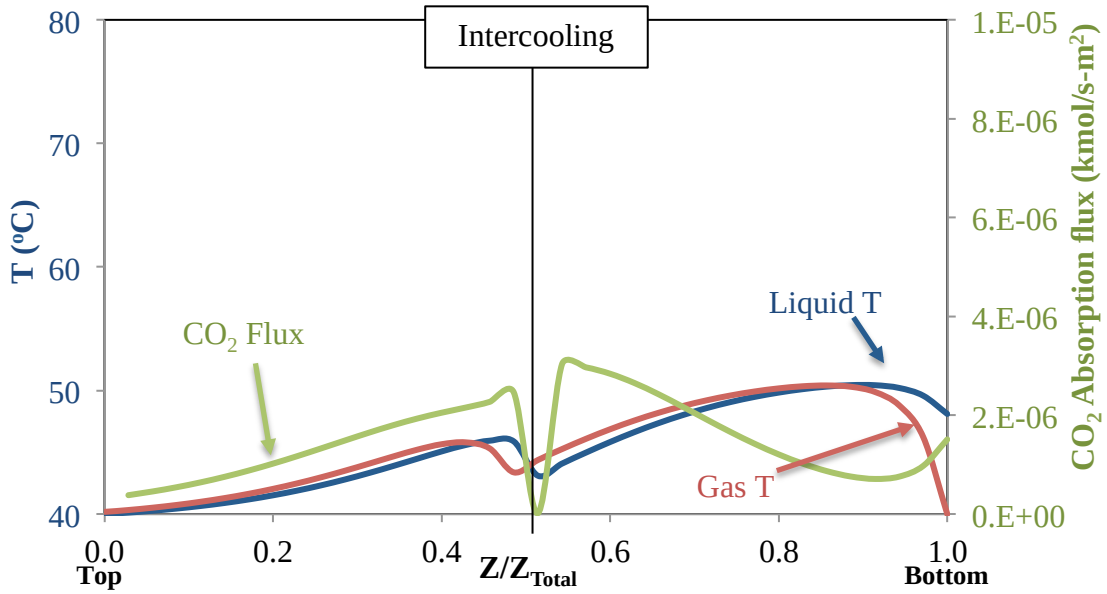


Figure 5.48: Temperature and CO_2 mass transfer flux profiles. Lean ldg = 0.34 1.1 L_{min}/G=7.3 , 1446 m^3 RSP-250/kmol/s CO_2 removed. Diameter set for 80% flooding. 90% removal. Intercooled at Z/Z_{Total} = 0.5

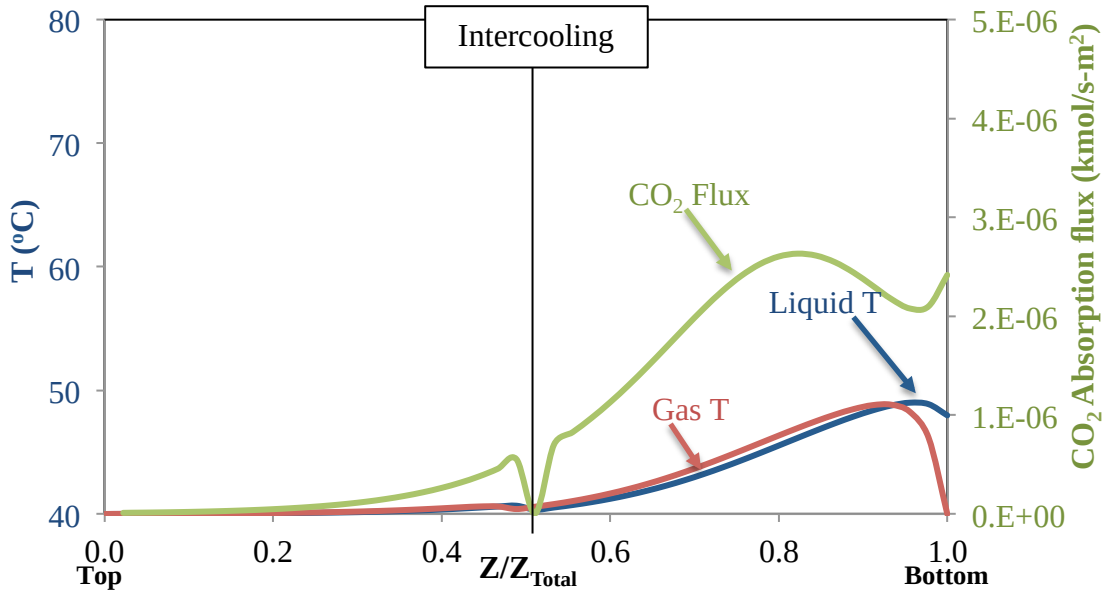


Figure 5.49: Temperature and CO_2 mass transfer flux profiles. Lean ldg = 0.35 1.1 L_{min}/G=18.20 , 5647 m^3 RSP-250/kmol/s CO_2 removed . Diameter set for 80% flooding. 90% removal. Intercooled at Z/Z_{Total} = 0.5

The intercooled cases using PZ show higher mass transfer flux than their MEA counterparts (above $8x10^{-6}$ kmol/s-m^2 for some cases) and overall lower column temperatures.

Figure 5.50 shows the required packing area per kmol/hr of CO_2 removed as a function of 1.1 L_{min}/G. for an intercooled absorber and 1.1 and 1.2 L_{min}/G for an absorber without intercooling for PZ. Alike Figure 5.32, Figure 5.50 relates the lean loading to the L/G and the required area for 90% removal. Contrary to MEA, intercooling reduces required packing volume as well as increases solvent lean and rich loadings. This might be related to the greater rates of reaction and capacity for PZ. By eliminating the temperature restrictions using intercooling the solvent benefits are maximized. The non-intercooled system using 1.2 L_{min} reduces the packing requirements and rich solvent loading. At L/G below 6 mol/mol this might be a better operating condition for the non

intercooled absorber since the decrease in loading is low compared to the reduction in packing volume (over 30% at 1.2 L_{min}/G =3.15).

Using this plot it is possible to estimate the packing volume for an absorber using PZ with an intercooling stage to 40°C at Z/Z_{Total} = 0.5 specifying the lean loading, required removal and the specific structured packing area.

Table 5.9 shows a comparison between required RSP-250 packing volume for an intercooled and non-intercooled absorber with 8m PZ and the gas flow rate estimated by Rochelle et al. (2005) for a 500 MW plant and 90% CO_2 removal. Results demonstrate the benefits of intercooling reducing the required packing to reach the desired removal.

Table 5.9: Required absorber RSP-250 packing volume for 90% removal for a 500 MW coal power plant. 1.1 L_{min}. 80% flooding. 8m PZ

Lean Loading	No-Intercooling			Intercooled to 40°C		
	L/G (mol/mol)	Rich ldg	V_p (m^3)	L/G (mol/mol)	Rich ldg	V_p (m^3)
0.20	2.3	0.40	8095	2.2	0.405	1440
0.25	4.8	0.345	5290	3.1	0.397	3647
0.34	8.8	0.391	12072	7.3	0.402	4610

Figure 5.50 just as with MEA shows two possible packing volume minima. However the solubility limitations for 8m PZ rule out the leaner minimum thus it is possible to define an optimum around a lean loading of 0.25

Figure 5.50 was used to estimate the packing volume for an absorber with Mellapak 2X. These results were compared with the values obtained using the absorber model for Mellapak 2X and the interfacial area factor from the PZ November'09 campaign (Table 5.10). Figure extracted values are 65% of the model results. This larger difference might be related to the fact that Mellapak 2X was not validated with intercooling.

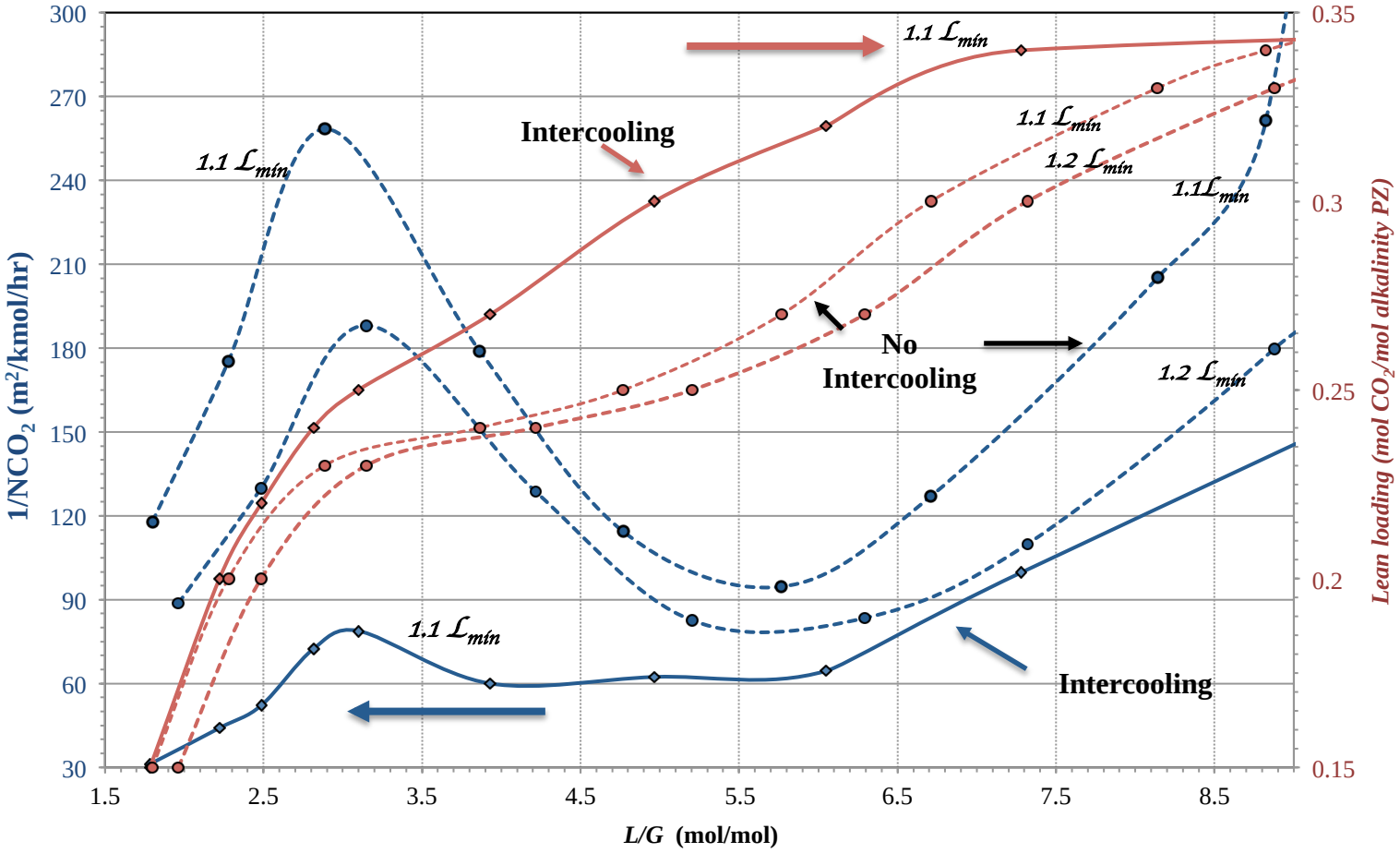


Figure 5.50: Required packing area per mol of CO_2 per hour removed as a function of *L/G* and lean loading for 90% removal. Absorber intercooled to 40°C at Z/Z_{Total} = 0.5. *x* L_{min}/G. RSP-250. Diameter set for 80% flooding. 8m PZ

Table 5.10: Required absorber packing volume for 90% removal for a 500 MW coal power plant. 1.1 L_{min}. 80% flooding. 8m PZ for Mellapak 2X Intercooling at Z/Z_{Total}=0.5 to 40°C.

Lean Loading	Aspen Plus ® Model			Figure 5.28			V_p deviation %
	L/G (mol/mol)	Rich ldg	V_p (m^3)	L/G (mol/mol)	Rich ldg	V_p (m^3)	
0.25	2.74	0.415	10548	3.1	0.397	3618	65%

5.4. AMINE VOLATILITY AND INTERCOOLING

An important issue in carbon capture using aqueous amines is the control of any additional emissions related to the process. In the absorber the main concern is the control of emissions leaving the unit along with the lean gas. Possible pollutants are amine degradation products and amine itself. Current technology development proposes the use of a water wash stage to reduce any possible amine emissions to the atmosphere to acceptable levels.

Intercooling can provide an additional benefit in controlling amine emissions due to its effects on column temperature profiles. High temperature towards the top of the column may increase amine emissions while a cooler column may help control emissions and reduce the requirements of the downstream water wash.

Mol fraction of MEA and PZ was plotted for the intercooled and non-intercooled cases against the change in loading (Figures 5.51 to 5.54). In the volatility plot it is possible to observe a discontinuity around the critical L/G for the mole fraction of amine in the vapor phase as well as the temperature. Amine volatility follows the position of the temperature bulge. In the rich loading region the top of the column is cold because all of the heat is leaving with the solvent. The steep change in the middle is due to the shift from a bottom section temperature bulge to a top section bulge. The intercooled cases display a lower mole fraction of amine in the exiting gas for the leaner loadings than the

non-intercooled cases. This changes at around the critical *L/G* and is equal at higher rich loadings since it is at the 40°C equilibrium value, This relates to the values of the temperature at the top of the column.

In Figures 5.42 and 5.43 the lean loading amine content is higher than the corresponding equilibrium value since the gas mass transfer has limited rates and the evaporated amine coming from lower parts of the column is not capable of reaching equilibrium with the colder incoming solvent.

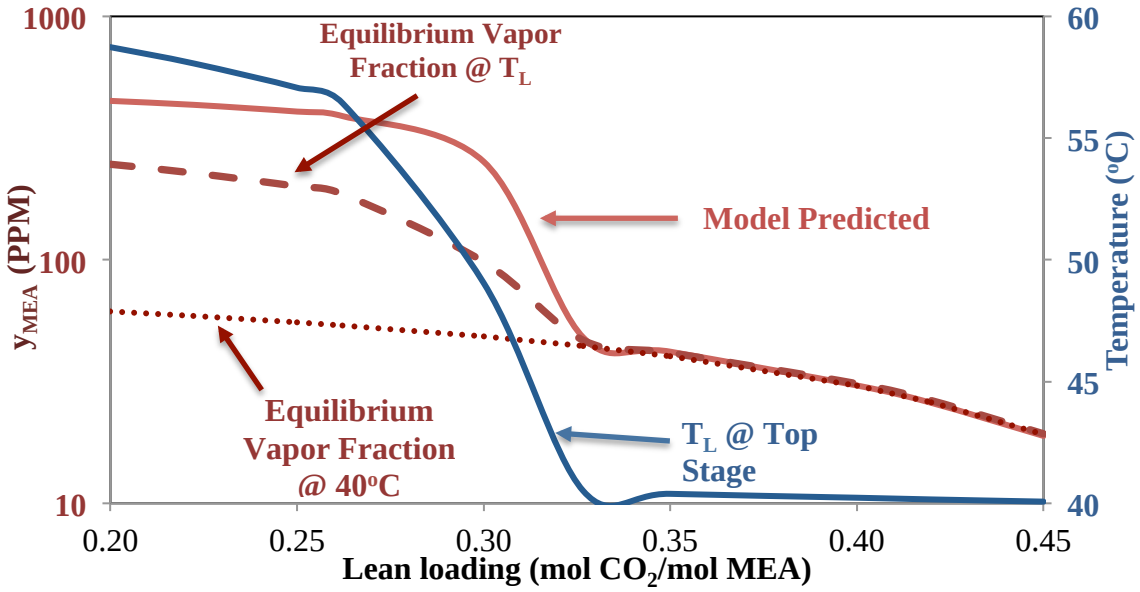


Figure 5.51: MEA in the exit vapor from absorber, 9m MEA and 90% removal. Variable RSP-250. 1.1 L_{min} Diameter set for 80% flooding. No Intercooling.

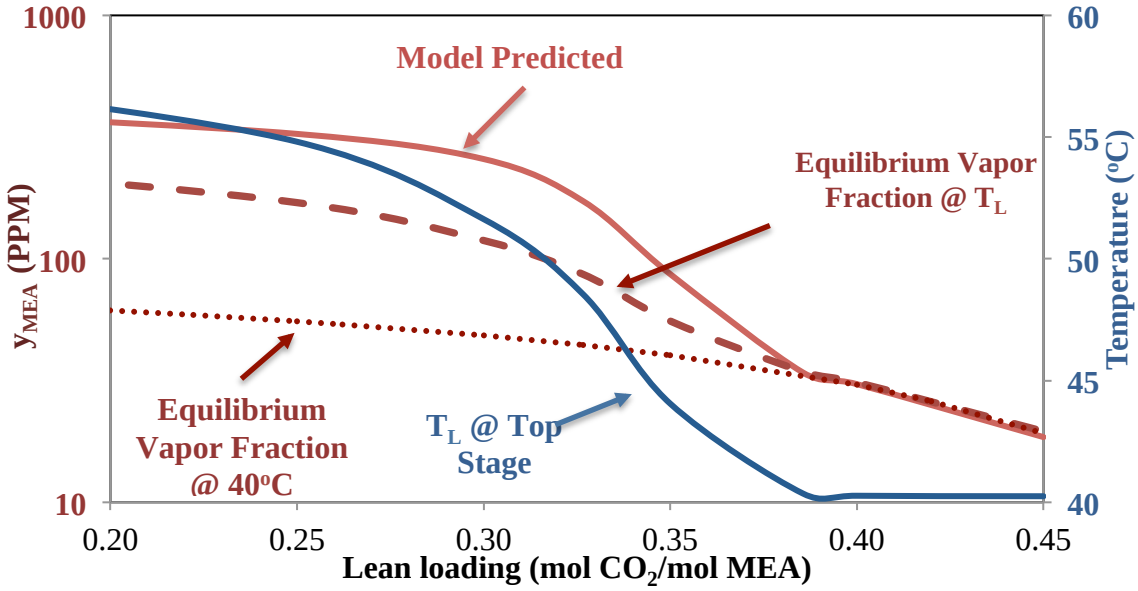


Figure 5.52 :MEA in the exit vapor from absorber, 9m MEA and 90% removal. Variable RSP-250. 1.1 L_{min} Diameter set for 80% flooding. Intercooling to 40°C at Z/Z

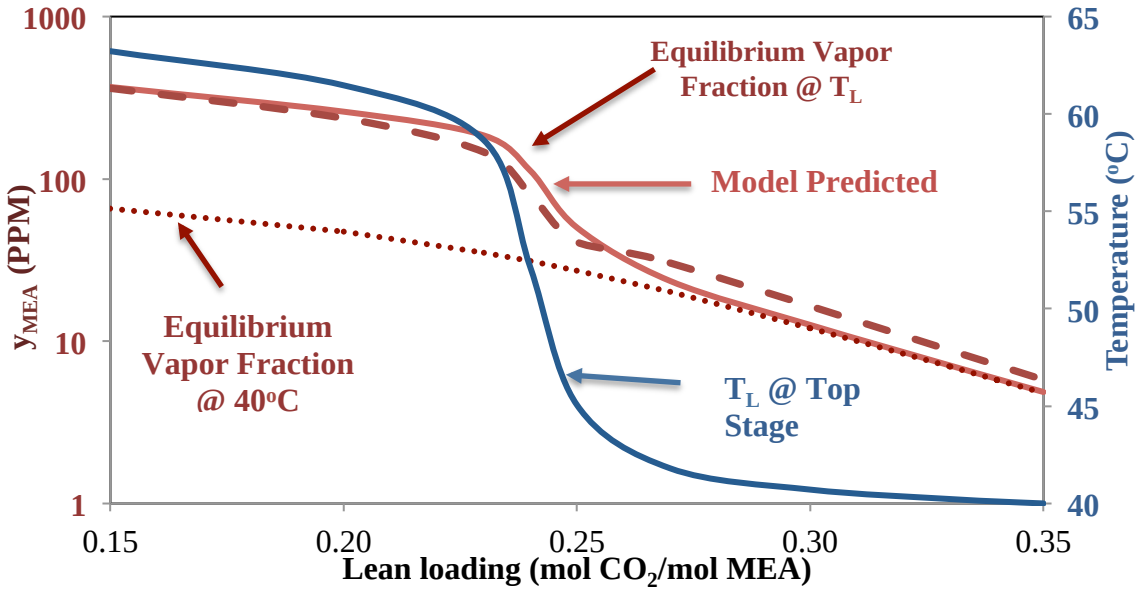


Figure 5.53: PZ in the exit vapor from absorber, 8m PZ and 90% removal. Variable RSP-250. 1.1 L_{min} Diameter set for 80% flooding. No Intercooling.

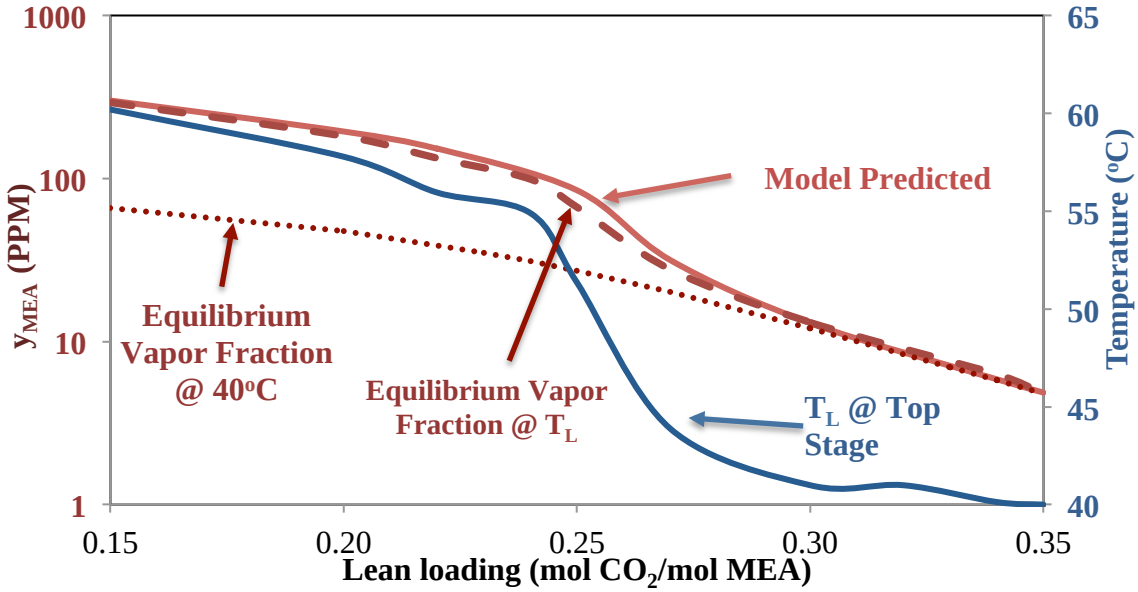


Figure 5.54: PZ in the exit vapor from absorber, 8m PZ and 90% removal. Variable RSP-250. 1.1 L_{min} Diameter set for 80% flooding. Intercooling to 40°C at Z/Z

5.5. Determination of the Critical Liquid-Gas Ratio $(L/G)_c$

The benefit of using intercooling is maximized when the selected operating conditions cause the temperature bulge to coincide with a mass transfer pinch. This relates to the capacity of the gas and the liquid to carry heat out of the column. At the critical L/G the heat generated by the absorption of CO_2 is removed evenly between the liquid and the gas producing a temperature bulge towards the middle of the column.

In order to determine the critical L/G it is useful to set up global mass and energy balances and balances between the top of the column and the location of the temperature bulge. The global energy balance can be written as follows:

$$L_{in}H_{in}^{L} + G_{in}H_{in}^{G} = L_{out}H_{out}^{L} + G_{out}H_{out}^{G} \quad (5.1)$$

Where: L and G are the liquid and gas flow rates respectively (moles/s)

H is the enthalpy of the stream.

^{L,G} are superscripts for gas and liquid properties

$_{in,\ out}$ label inlet and outlet streams around the absorber.

Likewise, an energy balance around the top of the absorber and the location of the temperature bulge results in the following equality:

$$L_{in}H_{in}^{L} + G_{b}H_{b}^{G} = L_{b}H_{b}^{L} + G_{out}H_{out}^{G} \qquad (5.2)$$

Where the sub index *b* stands for conditions at the temperature bulge location.

The following approximations and assumptions will be made to determine $(L/G)_c$ based on engineering criteria and observations from the presented modeling cases:

- In the global energy balance enthalpies are determined using the inlet liquid temperature (T_{in}^{L}) as reference temperature (T_o). This eliminates the inlet liquid term from equation 5.1.
- The energy contribution due to CO_2 absorption and vaporization of water are included in the outlet gas enthalpy as follows:

$$G_{out}H_{out}^{G} = G_{out}Cp_{out}^{G}(T_{out}^{G} - T_o) + \left(n_{out}^{CO_2(g)} - n_{in}^{CO_2(g)}\right)h_{abs}|_{T_o} + \left(n_{out}^{H_2O(g)} - n_{in}^{H_2O(g)}\right)h_{vap}|_{T_o} \qquad (5.3)$$

where: *n* is the flow rate (moles/s) of ^{CO2} or ^{H2O} respectively in the gas;

T_o is the reference temperature;

$h_{abs}|_{To}$ is the CO_2 heat of absorption at T_o;

$h_{vap}|_{To}$ is the heat of vaporization of water at T_o;

$Cp_{out}^{\ G}$ is the heat capacity of the gas at outlet conditions.

- Replacing Equation 5.3 in the global energy balance and including the expressions for outlet liquid and inlet gas enthalpy results in:

$$\begin{aligned} &G_{out}Cp_{out}^{G}\left(T_{out}^{G} - T_{in}^{L}\right) + \left(n_{out}^{CO_2(g)} - n_{in}^{CO_2(g)}\right)h_{abs}|_{T_{in}^{L}} + \left(n_{out}^{H_2O(g)} - n_{in}^{H_2O(g)}\right)h_{vap}|_{T_{in}^{L}} \\ &+ L_{out}Cp_{out}^{L}\left(T_{out}^{L} - T_{in}^{L}\right) = G_{in}Cp_{in}^{G}\left(T_{in}^{G} - T_{in}^{L}\right) \end{aligned} \qquad (5.4)$$

Where: Cp_{out}^{L}, is the heat capacity of the liquid at outlet conditions.

Cp_{in}^{G} is the heat capacity of the gas at inlet conditions.

- Observations from the rigorous simulations have shown that the maximum temperature at the bulge is approximately equal to the temperature that the gas would have if all of the heat of absorption were carried out with the gas. The global energy balance (Equation 5.4) for a system in which all the heat leaves with the gas out the top of the absorber ($T_{out}^{L} = T_{in}^{L}$) is reduced to:

$$G_{out}Cp_{out}^{G}(T_{out}^{G} - T_o) + \left(n_{out}^{CO_2(g)} - n_{in}^{CO_2(g)}\right)h_{abs}|_{T_o} + \left(n_{out}^{H_2O(g)} - n_{in}^{H_2O(g)}\right)h_{vap}\Big|_{T_o} = 0 \qquad (5.5)$$

The right side of Equation 5.5 is set to zero because for all of the cases the inlet liquid temperature was set equal to the gas inlet temperature (40°C)

- By defining a desired removal (R) the outlet gas water content is calculated. Assuming that the outlet gas leaves in equilibrium with the inlet solvent it is possible to determine the outlet temperature using equations of state or steam tables. This value is used as an approximation to the bulge temperature (T_b):

$$\left(\frac{n_{out}^{H_2O}}{G_{out}}\right)\frac{P_{Total}}{x_{in}^{H_2O}} = \frac{y_{out}^{H_2O}}{x_{in}^{H_2O}}P_{Total} = P_{sat}^{H_2O} = f(T) \qquad (5.6)$$

Here the outlet gas flow (G_{out}) is calculated using the inlet gas composition and taking into account the outlet water content and the removed carbon dioxide. x_{in}^{H2O} is the mole fraction of water in the inlet solvent.

- If in Equation 5.2 the reference temperature (T_o) is set to the bulge temperature (T_b) the resulting energy balance is:

$$L_{in}Cp_{in}^{L}(T_{in}^{L} - T_b) + \left(n_{out}^{CO_2(g)} - n_b^{CO_2(g)}\right)h_{abs}|_{T_b} + \left(n_{out}^{H_2O(g)} - n_b^{H_2O(g)}\right)h_{vap}\Big|_{T_b} = G_{out}Cp_{out}^{G}(T_{out}^{G} - T_b) \qquad (5.7)$$

- The change in liquid flow rate across the column is neglected and the gas flow rate is defined as:

$$G = G^i + n^{CO_2(g)} + n^{H_2O(g)} \quad (5.8)$$

Here G^i, represents the inert species present in the gas stream: oxygen and nitrogen

- Introducing Equation 5.8 into Equation 5.7 and after some manipulation:

$$\left(\frac{L}{G^i}\right)_c = \left(1 + (1-R)Y_{in}^{CO_2} + Y_{out}^{H_2O}\right)\left(\frac{Cp_{out}^G}{Cp_{in}^L}\right) + \frac{\left(Y_b^{CO_2} - (1-R)Y_{in}^{CO_2}\right)h_{abs}\big|_{T_b} + \left(Y_{out}^{H_2O} - Y_b^{H_2O}\right)h_{vap}\big|_{T_b}}{Cp_{in}^L(T_{in}^L - T_b)} \quad (5.9)$$

Where: $(L/G^i)_c$ is the critical ratio of liquid to inert gas species.

Y^{H2O}, Y^{CO2} are the fractions of water and carbon dioxide respectively to inert species in the gas stream (n^{CO2}/G^i, n^{H2O}/G^i)

- The outlet gas water content can be calculated using an equilibrium relation:

$$y_{out}^{H_2O} = x_{in}^{H_2O} P_{sat}^{H_2O}(T_{in}^L) \quad (5.10)$$

Where: y_{out}^{H2O}, x_{in}^{H2O} are mole fractions of water in the gas and liquid

P_{sat}^{H2O} is the vapor pressure of water at the inlet liquid temperature (T_{in}^L)

- The CO_2 content at the temperature bulge is approximated based on results for simulations at various removals. Figure 5.44 gives total CO_2 removal as a function of the removal obtained at the bulge. Results were fit to the following equation depending only on the desired final removal:

$$R_b^{CO_2} = AR_{Total}^{CO_2} + Bh_{vap}\big|_{T_b} + C \quad (5.11)$$

Where: R_b^{CO2} is the removal of CO_2 at the temperature bulge

$h_{vap}|_{Tb}$ is the heat of absorption of CO_2

The constants in Equation 5.11 are included in Table 5.8. These were obtained from regressing the data in Figure 5.44

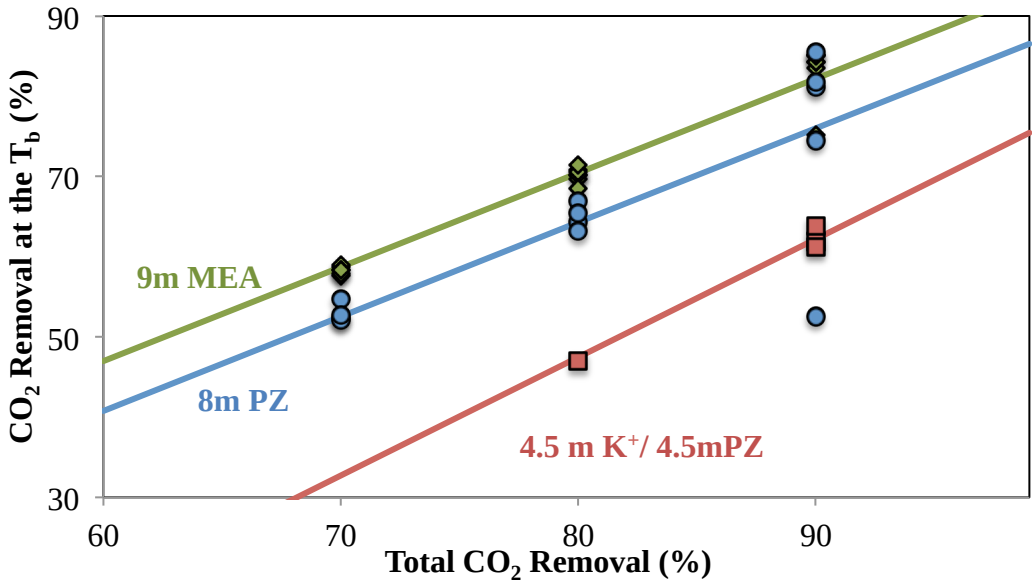


Figure 5.55: CO_2 removal at the bulge as a function of total removal. Points are simulation results. Lines are from regressed Equation 5.11

Table 5.11: Parameters for Equation 5.11

Solvent	A	B (mol/kcal)	C
9 m MEA 8 m PZ	0.01881	0.0117	-0.6110
4.5m/4.5m K^+/PZ	1.4728	0	-0.7038

Equation 5.9 is independent of the height of packing and was generated from simple energy and heat balances. Table 5.9 compares model results for 90% removal with calculated value using Equation 5.9

Table 5.12: Comparison of predictions for $(L/G)_c$ using Equation 5.9 and Aspen Plus® models for 90% Removal

Solvent	$(L/G)_c$		T_b (°C)		y_{out}^{H2O}		T_{out}^{G} (°C)	
	Aspen	Approx.	Aspen	Approx.	Aspen	Approx.	Aspen	Approx.
MEA	6.9	4.9	78	69		0.06	48	40
PZ	4.3	4.3	74.6	66.9		0.06	52	40
K^+/PZ	4.1	4.0	72	68	0.10	0.06	46	40

The proposed approximation gives a relatively good estimate of the critical L/G (less than 10% deviation) for PZ and K^+/PZ. The approximated MEA critical L/G is almost 30% off the Aspen Plus® result. As for the temperature bulge, it is under predicted up to 9 °C for MEA.

Table 5.9 shows an increase in the critical L/G that might be related to the increase in heat of absorption. This trend is maintained by the developed approximation but falls short on the magnitude of this change.

6

Chapter 6: Conclusions and Recommendations

Detailed development of three models for the absorption of carbon dioxide using aqueous amine solvents has ben covered in previous chapters. Two of them (5deMayo and Phoenix) were successfully validated and fine-tuned using pilot plant results obtained at the J.J Pickle Research Campus of The University of Texas at Austin. Intercooling as a process improvement strategy was explored and experimentally tested with positive results showing that its use is capable of increasing absorber CO_2 removal. This chapter presents the conclusions obtained through this work as well as recommendations on the direction to expand this study and on the industrial application of this technology.

6.1. Model Development

Three models were developed in this work. The model for 4.5m/4.5 m K^+/PZ was developed following the methodology in Chen (2007). The two other models, Phoenix and 5deMayo used a new model development methodology divided into the implementation of three major components: physical properties, thermodynamics and kinetics.

6.1.1. Physical Properties

Viscosity for both Phoenix and 5deMayo has been adequately represented using FORTRAN user subroutines. The implementation of physical properties using these subroutines offers the advantage of independence among speciation, VLE and density and viscosity. Unlike internal Aspen Plus® correlations these subroutines do not require additional regressions when the VLE is modified by the inclusion of additional data offering more flexibility and reducing development time.

For density Phoenix uses a FORTRAN subroutine, which provides similar advantages. 5deMayo used the Clarke model within Aspen Plus® for liquid molar volume for electrolyte solutions with satisfactory results.

6.1.2. Thermodynamics

One of the major changes from previous model development is the reduction of the reaction sets used to represent the amine system. This is done by combining and eliminating reactions based on the removal of species with lower concentration (H^+, OH^- ,CO_3^{-2}). Their contribution to the description of the amine chemistry is minimal but its elimination considerably facilitates model development and analysis.

For Phoenix the following reaction set was implemented:

$$2\,MEA + CO_2 \leftrightarrow MEA^+ + MEACOO^- \tag{6.1}$$

$$MEA + CO_2 + H_2O \leftrightarrow MEA^+ + HCO_3^- \tag{6.2}$$

5deMayo reduced piperazine reactions resulting in the following

$$2\,PZ + CO_2 \leftrightarrow PZH^+ + PZCOO^- \tag{6.3}$$

$$2\,PZCOO^- + CO_2 \leftrightarrow PZ(COO^-)_2 + H^+PZCOO^- \tag{6.4}$$

$$PZCOO^- + CO_2 + H_2O \leftrightarrow HCO_3^- + H^+PZCOO^- \quad (6.5)$$

$$PZ + H^+PZCOO^- \leftrightarrow PZH^+ + PZCOO^- \quad (6.6)$$

CO_2 solubility in both models was well fit up to the high temperature stripper operating range. In the case of Phoenix this solved over prediction issues that had been identified by Van Wagener (2011)for stripper conditions. 5deMayo is the first model to use concentrated PZ data for its development.

Amine volatility was modeled but its fit in both models requires improvement. Phoenix over predicts 60ºC data for MEA and slightly under predicts 40ºC data. 5deMayo matches 60ºC for PZ and over predicts 40ºC. These results suggest review of the model fit but also showed possible inaccuracies of the collected experimental data.

Speciation was fit well in both models using the eNRTL parameters. Bicarbonate formation in the models is improved by using pK_a values for the protonation of MEA and PZ. Even though the hydrogen ion has been removed, using the complete reaction set to give initial values for the regression for aqueous Gibbs energy and aqueous enthalpy of formation and aqueous heat capacity for the protonated species (PZH^+, MEA^+) yields a better speciation fit for HCO_3^-. However, in the case of 5deMayo the pK_a was not fit well so the related eNRTL parameters compensated this flaw. This might have influenced the amine vapor pressure fit and the heat capacity behavior.

The new model development methodology defines amines as Henry's components. This provides additional handles to better-fit amine volatility and activity coefficients. This yielded a major improvement in Phoenix with respect to previous MEA models related to the inadequate temperature and loading behavior of the activity

coefficient of MEA as it had been identified by Dugas (2009) in the Hilliard (2008) model.

The activity of CO_2 was fit after the main VLE regression using eNRTL parameters of CO_2 with amine species. This allowed a good fit of the activity coefficient without affecting the VLE. The fitted activity coefficients were extracted from physical solubility data using the N_2O analogy.

6.1.3. Kinetics

Phoenix and 5deMayo are activity based models for which the new model development methodology uses a WWC model implemented in Aspen Plus® RateSep™ to extract kinetic data.

The forward bicarbonate reaction constants for both models were approximated using the rate constants of tertiary amines corresponding to the base dissociation constant for MEA and $PZCOO^-$. For this purpose the following equation was used:

$$k^f_{HCO_3^-} = 3.278\, e^{-1.11\, pK_b} \tag{6.7}$$

The energy of activation was approximated as the reported value for MDEA (49 kJ/gmol).

Carbamate reaction constants were obtained using the WWC model and the previously obtained bicarbonate constants. Kinetic values were regressed in the model to match reported laboratory flux data for each solvent from the WWC. For MEA this narrowed down to a single reaction set and a forward reference rate constant and energy of activation. For PZ it required establishing a ratio between rates of reaction. For both systems this was significantly simplified due to the implemented reduced reaction set.

The resulting activity based kinetic expressions for Phoenix are as follows:

$$r^{f}_{MEACOO^-} = 8.56\text{x}10^{10} e^{\frac{-3964}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} a^{2}_{MEA} a_{CO_2} \quad (6.8)$$

$$r^{r}_{MEACOO^-} = 24800 e^{\frac{-59600}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} a_{MEACOO^-} a_{MEA^+} \quad (6.9)$$

$$r^{f}_{HCO_3^-} = 22991\ e^{-\frac{49000}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} a_{MEA} a_{CO_2} \quad (6.10)$$

$$r^{r}_{HCO_3^-} = 18.35\ e^{-\frac{96230}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} \frac{a_{HCO_3^-} a_{MEA^+}}{a_{H_2O}} \quad (6.11)$$

The resulting energies of activation for MEA reflect the predominance of the forward carbamate reaction for the absorption of CO_2

For 5deMayo the activity based kinetic expressions are:

$$r^{f}_{PZCOO^-} = 3\times10^{9}\ e^{-\frac{32042}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} a_{PZ} a_{PZCOO^-} a_{CO_2} \quad (6.12)$$

$$r^{r}_{PZCOO^-} = 1843\ e^{-\frac{102455}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} \frac{a_{PZH^+} a^{2}_{PZCOO^-}}{a_{PZ}} \quad (6.13)$$

$$r^{f}_{PZ(COO^-)_2} = 2.85\times10^{9}\ e^{-\frac{32042}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} a^{2}_{PZCOO^-} a_{CO_2} \quad (6.14)$$

$$r^{r}_{PZ(COO^-)_2} = 1854\ e^{-\frac{940594}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} a_{H^+PZCOO^-} a_{PZ(COO^-)_2} \quad (6.15)$$

$$r^{f}_{HCO_3^-} = 21951.76\ e^{-\frac{49000}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} a_{PZCOO^-} a_{CO_2} \quad (6.16)$$

$$r^{r}_{HCO_3^-} = 947.54\ e^{-\frac{84471}{8.314}\left(\frac{1}{T}-\frac{1}{298.15}\right)} \frac{a_{H^+PZCOO^-} a_{HCO_3^-}}{a_{H_2O}} \quad (6.17)$$

6.1.3.1. Implementation of Diffusion Coefficients

Aspen Plus® uses Maxwell-Stefan equations with an approximation to calculate mass transfer rates across liquid and vapor phases. This generates an incorrect dependence of $k_g^{'}$ on the diffusion coefficients of reactants and products. The proposed

solution outlined in section 3.2.3.3. did not resolve this issue. Its solution may require a deeper work around within the Aspen Plus® framework.

6.2. k'_g BEHAVIOR

Phoenix and 5deMayo were used to study mass transfer performance under various conditions. This was assessed using k'_g behavior. For both solvents kg' decreases with an increase in loading. Temperature effects are minimal.

6.2.1. Diffusion of reactants and products

The effect of diffusion of reactants and products was evaluated using the sensitivity of k_g' to k^o_L that varies according with the temperature and loading of the solution. For MEA, at low temperature, diffusion of reactants and products has limited effect on the overall mass transfer (<20%) with a slight increase at rich loadings (ldg>0.40). However, at higher temperatures (80°C) diffusion of reactants and products rapidly becomes a limiting factor. At low loadings there is more available free amine so even though it is being depleted faster it is not a limiting factor. Nevertheless, as loading increases there is less amine available at the interface and CO_2 absorption rates become hindered by the diffusion of reactants and products to the point of being the controlling mechanism (ldg>0.45) (Figure 3.28)

Diffusion of reactants and products proves to be more significant with 8m PZ. At 40°C and low loading (ldg<0.30) the effect can be neglected yet it rapidly increases to over 20% at a loading of 0.45. At higher temperatures (80°C) its effect is significant across the operating range becoming a major limiting factor even at low loadings (Figure 3.45)

6.2.2. Bicarbonate formation

Phoenix was used to examine the effect of bicarbonate formation in CO_2 absorption. This was assessed by comparing carbamate and bicarbonate forward rate of reaction. Results show that the bicarbonate reaction has a very limited effect at low temperatures and only begins to become significant past a loading of 0.5. However, at rich loading (>0.46) and temperature (80°C) it rapidly becomes significant with rates as high as 20% of the carbamate formation at the interface.

6.3. PILOT PLANT STUDIES

A major component of this work was the analysis of data from 5 pilot plant campaigns, accounting for 48 pilot plant runs at the J.J. Pickle Research Center of The University of Texas at Austin. These campaigns allowed evaluating the performance of 8m PZ and 9m MEA as well as intercooling as a process innovation. They also provided data to validate and fine tune Phoenix and 5deMayo.

6.3.1. Measurements

CO_2 Online loading measurements in the pilot plant require further revision. Even though they are used as operational reference, it is important to consider its better implementation. Deviations between the titration results and the online measurements are as high as 0.09 units for MEA and 0.04 for PZ. However this might also be related to solvent degradation. The MEA solvent has been used for at least 15 runs. The PZ solvent shows an increase in discrepancies and error scatter between the initial Noevmber'09 campaign and December'10 campaign. Similar behavior is observed when looking at mass balance results. Deviation between gas side CO_2 removal and liquid side calculated values are between ±10% for the PZ November'09 campaign and increases to

close to +20% for the PZ September'10 Campaign and up to over 20% for the December'10 Campaign.

Results for the PZ December'10 campaign show that the loading correlation does not apply well for the new density meters located in the two stage flash regeneration.

Temperature measurements were increased between campaigns providing more detailed information of the absorber performance. However, the lack of insulation and impossibility to measure temperatures closer to the center of the column provides an undesirable uncertainty in the measurements.

6.3.2. Pilot Plant Models

Phoenix and 5deMayo models of the pilot plant absorber column adequately represent the performance of the unit. Interfacial area reconciliation was necessary but stayed within 20% of the calculated interfacial area. The first MEA campaign required a reduction of the interfacial area of 12% for Flexipak AQ Style 20. The second campaign (RSP-250) required larger adjustment of 27%. For PZ the first campaign increased the interfacial area of Mellapak 2X 17%. For RSP-250 the first campaign (Sept'10) adjusted the interfacial area +2%. In the December'10 campaign the area was reduced 28%. Both solvents display a reduction in the interfacial area with time that may be related to factors including run temperatures, solvent flow rates and concentrations. However, this consistent reduction may also be related to solvent degradation occurring in each campaign affecting solvent kinetics.

Model discretization is very important to adequately represent absorber performance. Interface boundary layer discretization was optimized to reduce computing time but assure convergence. This leads to the use of more segments towards the liquid-

vapor interface, where the composition changes rapidly and less segments closer to the bulk of the solution where the profiles become almost flat.

Discretization of the column packing can be implemented to account for the flow non-idealities of a real unit. Using mixed segments of an adequate size may provide enough inaccuracies to represent such phenomenon as back mixing in real columns. This work used a simple rule to account for this, which proved useful for the studied campaigns:

$$N_m = \frac{h_s}{h_p} \tag{6.18}$$

Where: N_m is the number of segments to represent a packing section.

h_s is the height of packing for a section

h_p is the characteristic element height of the installed packing.

Intercooling modeling schemes range from the simple heat stream to the more elaborate two columns to represent each section connected by a heat exchanger. Results showed that the heat stream adds additional mass and heat transfer that is not available in the real unit. For the more complex options mass transfer is very similar and varies more on the convergence complexity, making the two column approach perhaps too elaborate, generating additional computing time and convergence difficulties. The single packing section pumparound used in this work provided adequate results without a major sacrifice of convergence, robustness, and computing time.

6.3.3. Data Reconciliation

The Phoenix and 5deMayo absorber models were successful at representing pilot plant data. Temperature profiles were mostly matched along with the temperature bulge location. Adjustment of lean loading and amine concentrations were required and ranged

from around a percent to 10% for lean loading and as high as 18% for amine concentration in 5m PZ runs.

CO_2 inlet and outlet concentrations were also adjusted as well as liquid and gas inlet temperatures to close mass and energy balances.

Heat loss was introduced as a manipulated variable applied to the whole column and calculated considering ambient temperature. Results attributed most of the heat loss to the gas side. Values for all campaigns were around 8 kW except for the PZ December'10 Campaign that had an average gas side heat loss of 20 kW. This was also a December campaign registering ambient temperatures around 10°C

Intercooled temperature profiles were effectively modeled. Temperature bulge location and magnitude were matched for the most part thus it is possible to conclude that intercooling was adequately modeled.

Loading measurements taken from the middle of the column for the PZ September'10 Campaign were around 10% of the reconciled model results. This offers a positive evaluation on the implemented 5deMayo rate model.

Reconciled pilot plant campaign results allow concluding that absorber removal is increased for all conditions using intercooling. However, the increase in performance goes through an expected maximum (the critical L/G). For the pilot plant PZ campaigns intercooled RSP-250 offers the lowest required solvent flow to reach 90% removal.

6.4. ABSORBER INTERCOOLING

Absorber intercooling is most effective at the critical *L/G*. At the critical *L/G* the temperature bulge without intercooling is in the middle of the column. In this case it will allow for higher absorption by reducing the magnitude of the bulge temperature. However for systems with low *L/G* where the bulge is located towards the top of the

column and away from the mass transfer pinch the use of intercooling will offer limited improvement. In the same way, the use of intercooling for systems with high *L/G*, such that the temperature bulge is small in magnitude and its location matches the pinch at the bottom of the column, intercooling will also offer very limited enhancement to the performance of the system. In both cases intercooling may prove to be impractical especially considering the increase of capital costs required for its implementation.

The volume of packing to get 90% removal with L/L_{min} =1.1 at the critical *L/G* is reduced by 30% for 8m PZ. At the rich solvent side a simple absorber using PZ goes from prohibitive packing volumes to a feasible result with intercooling.

The critical *L/G* is 4.3 for 8m PZ, 6.9 for 9m MEAand 4.1 for 4.5m K^+/ 4.5m PZ

MEA showed different packing volume behavior than PZ. For a solvent flow rate of 1.1 L_{min} packing volume is increased with intercooling at constant *L/G*. This increase translates in higher solvent loadings that suggest lower stripping energy requirements. Higher solvent rates (1.2 L_{min}) may provide conditions for intercooling that offer the similar packing reduction as with PZ. This difference between intercooling with PZ and MEA lies in the properties of the solvent. The temperature bulge limits the faster kinetics and capacity of PZ. Removal of the bulge allows maximizing the use of these benefits. MEA has a higher heat of absorption but lower reaction rates. Intercooling reduces mass transfer limitations and reaction rates, so higher richer loadings are possible but more packing volume is required.

Four column performance regions can be identified using a plot of rich loading and L_{min}/G as a function of lean loading for constant removal. In the low loading, the temperature bulge is located at the top of the column with a mass transfer pinch towards the bottom. The following region is characterized by a marked decrease in rich loading as solvent flow rates are increased to maintain removal. The magnitude of the

temperature bulge is the highest in this region. As lean loading is increased the temperature bulge and mass transfer pinch move towards the middle of the column until they coincide. This is the critical L/G. It is an inflection on the rich loading curve. Even though there it is a single point, the region around it presents similar characteristics with the pinch closer to the top for the lean side of the inflexion and to the bottom to the right of the critical L/G. It is in this region of operation that intercooling provides the biggest performance benefit. At constant removal and packing height, liquid flow rate is considerably reduced when compared to the non-intercooled case. If solvent flow rate is kept constant and packing volume is varied, this region provides the highest decrease in packing volume.

To the right of the critical L/G region the temperature bulge moves towards the bottom of the column and the mass transfer pinch continues towards the top.

Magnitude of the temperature bulge changes through all four regions being the highest on the lean side, decreasing to similar values around the critical L/G and reaching its lowest values on the rich lean loading side.

The temperature effects of the mentioned regions are more noticeable with MEA since it has the highest heat of absorption of studied solvents. These also lead to higher magnitude temperature bulges. However, the balance in PZ between capacity, heat of absorption and faster kinetics makes intercooling specially beneficial for PZ over the other studied solvents. Volume of packing at the critical L/G can be reduced as much as 70% for the PZ intercooled case. For MEA there is a 20% decrease in packing volume. At the rich solvent side a simple absorber using PZ goes from prohibitive packing volumes to a feasible result with intercooling.

Intercooling is capable of increasing removal capabilities as high as 10 % for the double matrix system K^+/PZ. Adequate positioning of the intercooled semilean feed and

the additional intercooling assure the best possible performance of the column in this configuration. Solvent capacity with intercooling for K^+/PZ in a simple absorber is increased as high as 45% in the critical L/G.

Intercooling provides an additional benefit in controlling amine emissions due to its effects on column temperature profiles. High temperature towards the top of the column may increase amine emissions while the cooler column reduces emissions and possibly the requirements of the downstream water wash.

Two tools for absorber design have been developed. Using Figures 5.28 for 9m MEA and 5.41 for 8m PZ is possible to estimate the structured packing volume for an absorber with a liquid intercooling stage to 40°C in the middle of the column. This tool was tested with two packing types so further verification is required.

An approximation for the critical L/G has been proposed. It predicted the critical L/G adequately for PZ and K^+/PZ however, it under predicts considerably the $(L/G)_c$ for the higher heat of absorption solvent (MEA). Additional work is needed to establish an adequate dependence on the heat of absorption.

6.5. RECOMMENDATIONS

6.5.1. Model Development

Additional physical properties should be evaluated and included in process development to improve model rigorousness. Properties such as surface tension and conductivity are required in the design of heat exchangers for conductivity, and are used in mass transfer coefficient correlations.

5deMayo requires correction of the protonated PZ pK_a regression and revision of the heat capacity fit.

Issues with the implementation of diffusion of reactants and products need to be addressed to assure proper representation of this phenomenon in the developed models.

Additional and better quality amine volatility data needs to be obtained to improve model fit of this variable. Data is also needed to better assess the activity coefficient of the amines in the Amine-CO_2-H_2O system especially for 8m PZ.

Physical solubility of N_2O in concentrated PZ should be measured to improve model representation of the activity coefficient of CO_2 in concentrated PZ. Current model extrapolates from very limited data.

There is a need for a closer look at the assumptions used for the bicarbonate reaction. Although results seem adequate, the assumptions used for its implementation should be validated.

6.5.2. Pilot Plant Studies

The online loading measurement using the density meters should be improved and recalibrated to assure better process control improving the quality of the gathered data.

There is a need to further validate the packing height segmentation rule used in this work. This rule may prove to be a valuable tool in better representing real flow behavior in absorber models.

Temperature measurements need to be improved in the pilot plant absorber. This implies better RTD placement and more importantly column insulation to reduce any possible heat loss that may affect temperature readings.

Inlet gas temperatures fluctuate considerably depending on ambient conditions. Intercooling to lower temperature may further increase performance and might be an option when operating in cooler locations or where chilled water is readily available. In some cases the extreme cold temperatures (5°C) generate unexpected cooling that may

cause more water condensation and affect overall column performance. Similarly since the column uses a blend of CO_2 with air, changes in humidity cause variations in feed gas conditions and more noise to absorber studies. Considerations should be taken to improve the conditioning of absorber feed gas.

Middle of column liquid samples provide valuable insight into column mass transfer rates. More and better sampling from various points within the column should be implemented with the objective of better assessing the rate model.

6.5.3. Absorber Modeling, Intercooling and Configurations

This work proved the benefits of intercooling as a process improvement. Further work should continue to evaluate this option with other solvents and process conditions. Intercooling to lower temperatures may further increase performance and might be an option when operating in cooler locations or where chilled water is readily available.

Since the highest flux is achieved around the intercooled section, study of process configurations that may increase interfacial area while intercooling should be considered.

The intercooled cases implemented in this study removed all liquid from the middle of the column to be cooled. Using double or triple the amount of liquid flowing around the intercooling loop may provide additional cooling and obtain lower temperature values around this section of the column.

The hot gas temperature limits the effect of intercooling in reducing middle of the column temperature. Further study in ways to reduce gas temperature along with liquid temperatures may provide additional configurations that may increase column CO_2 removal.

Absorber pressure drop and hold up models were not evaluated in this work but are required in developing a more rigorous absorber model since pressure drop may become an important variable in the design optimization process.

A water wash model needs to be developed as part of completing modeling of the absorption section of the capture unit.

Absorber optimization requires a balance between operational costs and capital costs. The first are tied to stripper performance so tools need to be developed to assess the impact of absorber conditions on the stripper. Capital cost requires implementing a tool that considers packing, solvent and material costs.

A

Appendix A: Reviewed MEA Absorber Models.

This section contains the results of an extensive literature review of the available models for absorption of CO_2 using aqueous MEA. Since MEA is the industry standard it has been the most study solvent with models ranging in capability and complexity from the equilibrium models with enhancement factors to the more rigorous rate based modeling.

Table A1: Kinetic expressions used in the 5deMayo model.

Reference	Conditions	Kinetics	VLE	Framework	Description
Alatiqi, Sabri et al. (1994)	16% 20% CO_2 3.6 – 4.1 m MEA	Enhancement factors based on Danckwerts (1966)	Licht and Weiland (1989)	BASIC	Model developed for gas treating units of the Kuwait National Petroleum Company and The Petrochemical Industries Company.
Al-Baghli, Pruess et al. (2001)	7% CO_2, 4 m MEA	Danckwerts (1979), Glasscock, Critchfield et al. (1991)	Kent and Eisenberg (1976)		Rate-based model for a trays absorber using enhancement factors
Kucka, Muller, et al. (2003)	2.1–4.4 m MEA 2.5%-15.1% CO_2	(Kucka et al., 2002, 2003)	Austgen, Rochelle et al. (1989)	Aspen Custom Modeler	Validation against pilot plant data. Includes analysis of film discretization
Freguia and Rochelle (2003)	8 m. MEA Heat Stable Salts (HSS) 0.1mol/mol MEA	From a FORTRAN subroutine, data by Flour Daniel, Inc. and Dang (2000)	Modified to include data from Jou, Mather et al. (1995)	Aspen Plus RateFrac®	Sensitivity analysis of process variables for low steam requirements. Absorber intercooling is explored
Geuzebroek et al. (2004)	Gas-fired flue gas (3.3 mol% CO_2)	Not Available	Not Available	Aspen Plus	Aspen Plus model is coupled with Exercom for exergy analysis
Kopcke (2004)	7-24 mMEA 12 mol % CO_2	Enhancement factors to account for kinetics	Austgen, Rochelle et al. (1989)	Aspen Plus RateFrac®	Analysis with multiple MEA concentrations, intercooling and plant integration

Reference	Conditions	Kinetics	VLE	Framework	Description
Alie, Backham et al. (2005)	3%, 14%, 25% mol CO_2 7 m MEA	Efficiencies	Austgen (1989)	Aspen Plus RateFrac®	Proposes the decomposition of absorber/stripper system for ease of modeling.
Chang and Shih (2005)	4 – 7 m MEA	Enhancement factors	Austgen (1989)	Ratefrac with subroutines for boundary layer point modeling (Pacheco and Rochelle, 1998)	Model is used to explore intercooling and split flow configurations
Liu, Yu et al. (2006)	2 % CO_2	Enhancement factor as defined by Wellek, Brunson et al. (1978) using Hikita, Asai et al. (1979)		FLUENT 6.1	Model validation with Pilot Plant data by Tontiwachwuthikul et al. (1992) and Pintola, Tontiwachwuthikul et al. (1993)12/19/11 3:31 PM
Asprion (2006a)	Not Available	Not Available	Not Available	BASF's CHEMASIM	Study of optimum film discretization in reactive systems.
Abu-Zahra et al. (2007)	13 % CO_2 0.5—11 m MEA	Versteeg et al.(1996); Jamal et al.(2006)12/19/11 3:31 PM	Austgen (1989)	Aspen Plus Radfrac	Optimization and study of effect of lean loading and temperature and stripper pressure in energy requirements

Reference	Conditions	Kinetics	VLE	Framework	Description
Tobiesen et al. (2007)	7 m MEA	Equilibrium & kinetics matching the Castor report B.	Model described by Hoff et al. (2004), using own data, Jou, Mather et al. (1995) and Ma'mun et al. (2005)	FORTRAN 90	Model uses penetration theory. A comparison with pilot plant data is presented.
Oi (2007)	10.5 m MEA. Flue gas from a gas-fired power plant.	Murphree efficiency (0.25)	Kent and Eisenberg (1976)	Aspen HYSYS	Analysis of process variables their impact on energy consumption
Mofarahi et al. (2008)	7 m MEA Flue gas from a gas fired power plant	Murphree efficiency (0.35)	Kent and Eisenberg (1976)	MATLAB	Study of various solvents and conditions for CO_2removal at the Sarkhun gas refinery power station
Dugas et al. (2008)	7 m MEA 11-13 % CO_2	Versteeg et al. (1996) with activity coefficient corrections	Freguia (2002)	Aspen Plus RateSep	Model developed to study the results of the CASTOR pilot plant campaign #2
Plaza et al. (2008)	9 m MEA 12 mol% CO_2	Fitted from selected raw data from Aboudheir (2002)	Hilliard (2008)	Aspen Plus RateSep	Model accuracy was assessed by comparison with pilot plant data. Lean loading optimization for a novel stripper configuration

Reference	Conditions	Kinetics	VLE	Framework	Description
Kvamsdal et al. (2009)	8 m MEA	Freguia and Rochelle (2003) with adjusted parameters to match pilot plant data in Kvamsdal and Rochelle (2008)	Equilibrium speciation model by Hoff et al. (2004)	gPROMS	Dynamic model to study start up and load reduction.

B

Appendix B: Laminar Jet Model.

In addition to the Aspen Plus® model developed to extract MEA and PZ kinetic data from the WWC. A laminar jet model was developed to obtain kinetic expressions for 7m MEA from the work conducted by Aboudheir (2002). It was implemented using the thermodynamic framework developed by Hilliard (2007). Kinetics were represented using the same reaction set used in Phoenix (See section 3.2.3.2). Bicarbonate reaction constants were determined using the same methodology implemented for the Phoenix model. The resulting model was used in Plaza et al.(2010) to match pilot plant data from a run conducted in October 2007 at the Pickle Research Center Pilot Plant. This section presents details of the laminar jet model developed in Aspen Plus®

The laminar jet absorber consists of a liquid jet generated using nozzle that flows intact through a media filled with gas to be captured by a slightly larger capillary receiver. (Danckwerts, 1970; Aboudheir, 2002). Rate measurements are collected by changing the length of the jet, the temperature and liquid flow. The flexibility of this method to measure kinetics lies in the possibility to change the interfacial area and the k^o_L fairly easy by increasing liquid flow rate and length of the jet.

The laminar jet Aspen Plus® model uses FORTRAN subroutines to calculate the interfacial area and the physical mass transfer coefficient. (k^o_L). The former is calculated as a simple cylinder with the height and the diameter of the jet:

$$A_{intf} = \pi d h \tag{B.1}$$

where: A_{intf} is the interfacial area

h is the height of the jet

d is the diameter of the jet

The physical mass transfer coefficient is calculated using an equation derived analytically (Aboudheir, 2002):

$$k_L^o = \frac{4}{\pi d}\sqrt{D_i \frac{L}{h}} \tag{B.2}$$

where: D_i is the liquid diffusion of specie i

L is the liquid flow rate.

The gas mass transfer coefficient (k_G) was fixed at 10^{-6} kgmol*m^{-2}*Pa^{-1}s^{-1}. It was defined to minimize gas phase resistance.

The dimensions of the laminar jet used in Plaza et al. (2010) required additional scaling since the dimensions of the jet were smaller than the allowed in Aspen Plus®. The new Aspen Plus® version 7.3 allows the user to specify packing height, column diameter and flow rates that correspond to laboratory scale equipment so there is no longer a need to scale up the reported experimental values.

The methodology to extract kinetic constants from the laminar jet model is the same as the one used in this work for the WWC. The PFD of the laminar jet simulation is

included in Figure B.1. As in the WWC model gas and solvent specifications are manipulated with individual component streams that go through a mixer. The two flash tanks (EQ1 and EQ2) are used to generate the equilibrium constants. The heat exchanger upstream of the column controls the liquid temperature.

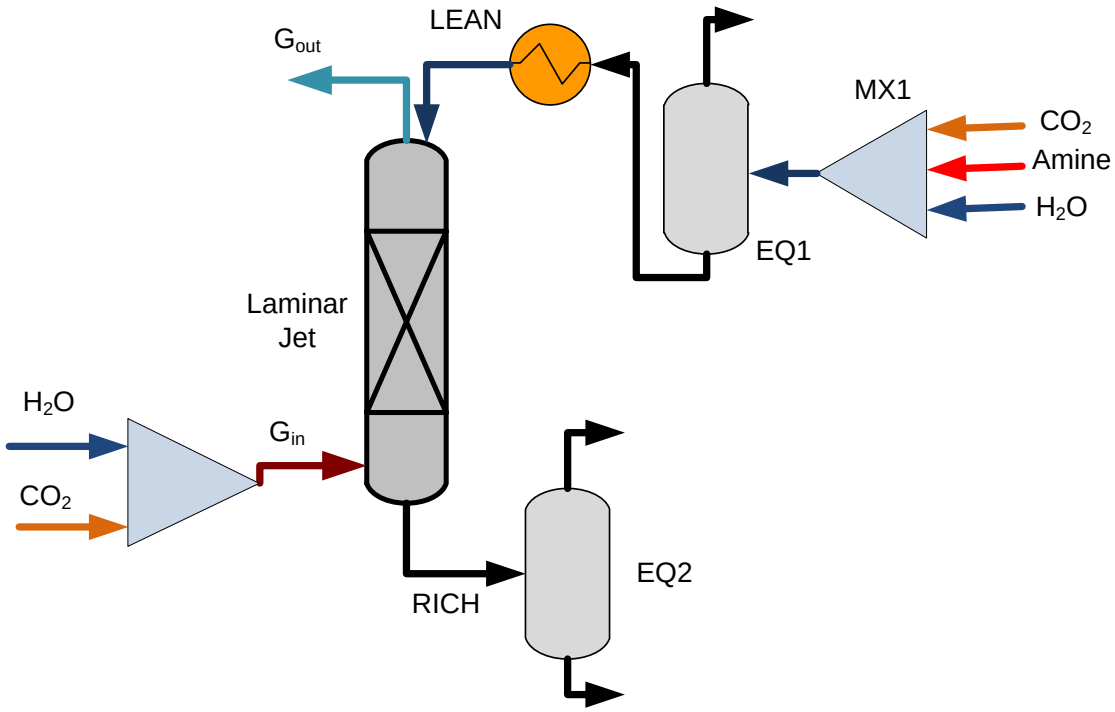


Figure B.1: PFD for the Laminar Jet Aspen Plus® simulation.

C

Appendix C: FORTRAN User Subroutines for the Aspen Plus® Models.

This section includes the subroutines developed for the models presented in this work. They follow examples included in Aspen Plus®. The subroutines for 5deMayo are part of the license for this model and are not included in this section.

C.1. DENSITY SUBROUTINE FOR MEA

```
C Log keyword added
C
C$ #1 BY: PING LI 14-MAY-2004 USER ROUTINE FOR MIXTURE PROPERTIES USING
C$                            MIXING RULE
C ==========================cvs revision history=======================
      SUBROUTINE VL2U2 (T, P, X, N, IDX, XMW, SG, VLSTD, VL2U2A,
     *                  VI, DVI, DPVI, KSW, KOP, NDS, KDIAG,
     *                  VMX, DVMX, DPVMX, KER )
C***********************************************************************
C     Template for VL2U2 routine for mixture liquid molar volume
C     and its temperature, pressure derivatives
C
C     VMX is the calculated liquid mixture molar volume (output)
C
C     DVMX is the temperature derivative of VMX (output)
C
C     DPVMX is the pressure derivative of VMX (output)
C
C     All input and output in this user routine are in SI Units
C      with Gas constant = 8314.33
C
C***********************************************************************
C     ARGUMENT LIST VARIABLES:
C
C        VARIABLE I/O TYPE-SPEC DIMENSION     DESCRIPTION AND RANGE
C
C         T        I   REAL*8            OPERATING TEMPERATURE
C         P        I   REAL*8            OPERATING PRESSURE
C         Z        I   REAL*8    N       COMPONENT MOLE FRACTION VECTOR
C         N        I   INTEGER           NUMBER OF COMPONENTS IN MIXTURE
C         IDX      I   INTEGER   N       VECTOR OF COMPONENT POINTERS
C         XMW      I   REAL*8    NCC     MOLECULAR WEIGHT FOR EACH COMPONENT
C         SG       I   REAL*8    NCC     SPECIFIC GRAVITY FOR EACH COMPONENT
C         VLSTD    I   REAL*8    NCC     STD. LIQUID VOLUME FOR EACH COMPONENT
C         VL2U2A   I   REAL*8    5,NCC   USER DEFIND PARAMETER FOR THIS MODEL
C         VI       I   REAL*8    N       PURE LIQUID MOLAR VOLUME
C         DVI      I   REAL*8    N       TEMPERATURE DERIVATIVE OF VI
C         DPVI     I   REAL*8    N       PRESSURE DERIVATIVE OF VI
C         KSW      I   INTEGER   3       CALCULATION CODE
C                                        KSW(1) FOR PROPERTY
C                                        KSW(2) FOR TEMP. DERIVATIVE
C                                        KSW(3) FOR PRES. DERIVATIVE
C                                        VALUE = 1: CALCULATE
C                                        VALUE = 0: DO NOT CALCULATE
C         KOP      I   INTEGER   10      MODEL OPTION CODE
C         NDS      I   INTEGER           DATA SET NUMBER
C         KDIAG    I   INTEGER           MESSAGE PRINTING CODE
C                                        IF .GE. 2: PRINT ERROR MESSAGES
C                                        IF .GE. 3: PRINT WARNING MSGS.
C         VMX      O   REAL*8            LIQUID MOLAR VOLUME [CUM/KGMOL]
C         DVMX     O   REAL*8            TEMPERATURE DERIVATIVE OF VMX [CUM/KGMOL-K]
C         DPVMX    O   REAL*8            PRESSURE DERIVATIVE OF VMX [CUM/KGMOL-PA]
C         KER      O   INTEGER           ERROR RETURN CODE
C
C***********************************************************************
      IMPLICIT NONE
#include "dms_global.cmn"
C
C     DECLARE VARIABLES USED IN DIMENSIONING
C
      INTEGER N
C
C     DECLARE ARGUMENTS
C
      INTEGER IDX(N), KSW(3), KOP, NDS, KDIAG, KER
      INTEGER DMS_KCCIDC
      INTEGER IWATER, IMEACOO, ICO2, IMEAH, IHCO3
      INTEGER ICO3, IMEA
      REAL*8 X(N), T, P, XMW(N), SG(1), VLSTD(1), VL2U2A(5,1),
     .       VI(N), DVI(N), DPVI(N), VMX, DVMX, DPVMX
      REAL*8 WATER, MEACOO, CO2, MEAH, HCO3
      REAL*8 MEA,CO3
      REAL*8 A, B, C, D, E, F, MWMEA, VH2O, XCO2T, XMEAT
```

```
      REAL*8 VCO2, VST, VAM
      REAL*8 XPZT, AA, BB, ML, XH2O, LDG, RHOX, WWATER
      REAL*8 PPUTL_AVEMW, MWT1
C
C     DECLARE LOCAL VARIABLES
C
      INTEGER IPROG(2)
C
C      DATA STATEMENTS
C
      DATA IPROG /4HVL2U, 4H2   /
C
C     BEGIN EXECUTABLE CODE
C
C
C     Molar volume calculated BASED ON WEILAND ET AL. (J. Chem. Eng. Data 1998, 43, 378-382)
C
C     JORGE M. PLAZA 01/26/11
C
C
      INTEGER I
      REAL*8 SUM, DSUM, DPSUM
      SUM = 0D0
      DSUM = 0D0
      DPSUM = 0D0
      A = -0.000000535162D0
      B = -0.000451417D0
      C = 1.19451D0
      D = 0D0
      E = 0D0
      VCO2 = 0.04747D0
      VST = -1.8218D0
      MWMEA = 61.08D0
C
C     CALCULATE AVERAGE MW
C
      MWT1 = PPUTL_AVEMW (N, IDX, X)
C
C    INDEX VALUES FOR COMPONENTS IN SIMULATION
C
      IWATER = DMS_KCCIDC('H2O')
      IMEACOO = DMS_KCCIDC('MEACOO-')
      ICO2 = DMS_KCCIDC('CO2')
      IMEAH = DMS_KCCIDC('MEA+')
      IHCO3  = DMS_KCCIDC('HCO3-')
      IMEA = DMS_KCCIDC('MEA')
      ICO3 = DMS_KCCIDC('CO3--')
C
      WWATER = XMW(IWATER)

C     ASSIGNMENT OF INDEX NUMBERS FOR SPECIES PRESENT
C
      DO 50 I = 1, N
      IF (IDX(I). EQ. IWATER) IWATER = I
      IF (IDX(I). EQ. IMEACOO) IMEACOO = I
      IF (IDX(I). EQ. ICO2) ICO2 = I
      IF (IDX(I). EQ. IMEAH) IMEAH = I
      IF (IDX(I). EQ. IHCO3) IHCO3 = I
      IF (IDX(I). EQ. ICO3) ICO3 = I
      IF (IDX(I). EQ. IMEA) IMEA = I
   50 CONTINUE
C
C     MOLAR VOLUME OF WATER
C
      VH2O = VI(IWATER)*1000D0
C
C     LOADING CALCULATIONS
C
      XH2O = X(IWATER)+X(IHCO3)
      XCO2T = X(IMEACOO)+X(IHCO3)+X(ICO2)
      XMEAT = X(IMEACOO)+X(IMEAH)+X(IMEA)
C
C
C     MOLAR VOLUME
C
```

```
      VAM = MWMEA/(A*(T**2D0)+ B*T + C)
      VMX = XMEAT*VAM + XH2O*VH2O + XCO2T*VCO2 + XMEAT*XH2O*VST
C
C   VMX
      IF (KSW(1) .EQ. 1) VMX = VMX/1000D0
C   dVMX/dT
      IF (KSW(2) .EQ. 1) DVMX = DSUM
C   dVMX/dP
      IF (KSW(3) .EQ. 1) DPVMX = DPSUM
C
  200 CONTINUE
      RETURN
      END
```

C.2. VISCOSITY SUBROUTINE FOR MEA

```
C Log keyword added
C
C$ #1 BY: PING LI 14-MAY-2004 USER ROUTINE FOR MIXTURE PROPERTIES USING
C$                            MIXING RULE
C ==========================cvs revision history=======================
      SUBROUTINE MUL2U2 (T, P, X, N, IDX, XMW, SG, VLSTD, MULU2A,
     *                   MUI, DMUI, DPMUI, KSW, KOP, NDS, KDIAG,
     *                   MUMX, DMUMX, DPMUMX, KER )
C***********************************************************************
C     Template for MUL2U2 routine for mixture liquid viscosity
C     and its temperature, pressure derivatives
C
C     MUMX is the calculated liquid mixture viscosity (output)
C
C     DMUMX is the temperature derivative of MUMX (output)
C
C     DPMUMX is the pressure derivative of MUMX (output)
C
C     All input and output in this user routine are in SI Units
C      with Gas constant = 8314.33
C
C***********************************************************************
C     ARGUMENT LIST VARIABLES:
C
C        VARIABLE I/O TYPE-SPEC DIMENSION     DESCRIPTION AND RANGE
C
C         T        I   REAL*8            OPERATING TEMPERATURE
C         P        I   REAL*8            OPERATING PRESSURE
C         Z        I   REAL*8    N       COMPONENT MOLE FRACTION VECTOR
C         N        I   INTEGER           NUMBER OF COMPONENTS IN MIXTURE
C         IDX      I   INTEGER   N       VECTOR OF COMPONENT POINTERS
C         XMW      I   REAL*8    NCC     MOLECULAR WEUGHT FOR EACH COMPONENT
C         SG       I   REAL*8    NCC     SPECIFIC GRAVITY FOR EACH COMPONENT
C         VLSTD    I   REAL*8    NCC     STD. LIQUID VOLUME FOR EACH COMPONENT
C         MULU2A   I   REAL*8    5,NCC   USER DEFIND PARAMETER FOR THIS MODEL
C         MUI      I   REAL*8    N       PURE LIQUID VISCOSITY
C         DMUI     I   REAL*8    N       TEMPERATURE DERIVATIVE OF MUI
C         DPMUI    I   REAL*8    N       PRESSURE DERIVATIVE OF MUI
C         KSW      I   INTEGER   3       CALCULATION CODE
C                                        KSW(1) FOR PROPERTY
C                                        KSW(2) FOR TEMP. DERIVATIVE
C                                        KSW(3) FOR PRES. DERIVATIVE
C                                        VALUE = 1: CALCULATE
C                                        VALUE = 0: NO CALCULATION
C         KOP      I   INTEGER           OPTION CODE
C         NDS      I   INTEGER           DATA SET NUMBER
C         KDIAG    I   INTEGER           MESSAGE PRINTING CODE
C                                        IF .GE. 2: PRINT ERROR MESSAGES
C                                        IF .GE. 3: PRINT WARNING MSGS.
C         MUMX     O   REAL*8            LIQUID VISCOSITY
C         DMUMX    O   REAL*8            TEMPERATURE DERIVATIVE OF MUMX
C         DPMUMX   O   REAL*8            PRESSURE DERIVATIVE OF MUMX
C         KER      O   INTEGER           ERROR RETURN CODE
C
C***********************************************************************
      IMPLICIT NONE
#include "dms_global.cmn"
C
```

```
C     DECLARE VARIABLES USED IN DIMENSIONING
C
      INTEGER N
C
C     DECLARE ARGUMENTS
C
      INTEGER IDX(N), KSW(3), KOP, NDS, KDIAG, KER
      INTEGER DMS_KCCIDC
      INTEGER IWATER, IMEACOO, ICO2, IMEAH, IHCO3
      INTEGER ICO3, IMEA
      REAL*8  X(N), T, P, XMW(N), SG(1), VLSTD(1), MULU2A(5,1),
     .        MUI(N), DMUI(N), DPMUI(N), MUMX, DMUMX, DPMUMX
      REAL*8  WATER, MEACOO, CO2, MEAH, HCO3
        REAL*8  CO3, MEA
      REAL*8  LDG, A, B, C, D, E, F, G, MUW, XCO2T
      REAL*8  XMEAT, XWMEA, MWMEA, MWH2O, MWCO2, MWT, XH2O
        REAL*8  MUBLEND
C
C     DECLARE LOCAL VARIABLES
C
      INTEGER IPROG(2)
C
C      DATA STATEMENTS
C
      DATA IPROG /4HMUL2, 4HU2  /
C
C     BEGIN EXECUTABLE CODE
C
C     Viscosity is calculated from the regressed data using Weiland et al.1998 FOR MEA
C
C     JORGE M. PLAZA 01/27/11
C
C     VALUES OBTAINED FROM THE VISCOSITY REGRESSION
      INTEGER I
      REAL*8 SUM, DSUM, DPSUM
      SUM = 0D0
      DSUM = 0D0
      DPSUM = 0D0
      A = 0D0
      B = 0D0
      C = 21.186D0
      D = 2373D0
      E = 0.01015D0
        F = 0.0093D0
        G = -2.2589D0
C
C     INDEX VALUES FOR COMPONENTS IN SIMULATION
C
      IWATER = DMS_KCCIDC('H2O')
      IMEACOO = DMS_KCCIDC('MEACOO-')
      ICO2 = DMS_KCCIDC('CO2')
      IMEAH = DMS_KCCIDC('MEA+')
      IHCO3  = DMS_KCCIDC('HCO3-')
      IMEA = DMS_KCCIDC('MEA')
        ICO3 = DMS_KCCIDC('CO3--')
C
C
C     ASSIGNMENT OF INDEX NUMBERS FOR SPECIES PRESENT
C
      DO 50 I = 1, N
      IF (IDX(I). EQ. IWATER) IWATER = I
      IF (IDX(I). EQ. IMEACOO) IMEACOO = I
      IF (IDX(I). EQ. ICO2) ICO2 = I
      IF (IDX(I). EQ. IMEAH) IMEAH = I
      IF (IDX(I). EQ. IHCO3) IHCO3 = I
      IF (IDX(I). EQ. IMEA) IMEA = I
        IF (IDX(I). EQ. ICO3) ICO3 = I
   50 CONTINUE
C
C     VISCOSITY OF WATER
C
      MUW = MUI(IWATER)
C
C     LOADING CALCULATIONS
C
```

```
      XCO2T = X(IMEACOO)+ X(IHCO3)+ X(ICO2)
      XMEAT = X(IMEACOO)+ X(IMEAH)+ X(IMEA)
      LDG = XCO2T/XMEAT
C
C     AMINE MASS FRACTION CALCULATION
C
      MWMEA = 61.08D0
      MWCO2 = 44.01D0
      MWH2O = 18D0
      XH2O = X(IWATER)+ X(IHCO3)
      MWT =  XMEAT*MWMEA + XH2O*MWH2O
      XWMEA = ((XMEAT*MWMEA)/MWT)*100
C
C
C     VISCOSITY CALCULATION
C
      MUBLEND = (A*XWMEA + B)* T + (C*XWMEA +D)
        MUBLEND = MUBLEND *(LDG*(E*XWMEA + F*T + G)+1)*XWMEA
        MUBLEND = MUBLEND/T**2D0
      MUBLEND = DEXP(MUBLEND)
        MUMX = MUBLEND * MUW
C
C   MUMX
      IF (KSW(1) .EQ. 1) THEN
      MUMX = MUMX
      END IF
C   dMUMX/dT
      IF (KSW(2) .EQ. 1) DMUMX = DSUM
C   dMUMX/dP
      IF (KSW(3) .EQ. 1) DPMUMX = DPSUM
C
  200 CONTINUE
      RETURN
      END
```

C.3. DIFFUSION SUBROUTINE FOR MEA

```
C Log keyword added
C
C$ #1 BY: SUPHAT WATANASIRI 09-SET-2007 USER ROUTINE FOR LIQUID BINARY
C                                       DIFFUSION COEFFICIENTS
C
C ==========================cvs revision history=======================
      SUBROUTINE DL0U ( T, P, X, N, IDX, IRW, IIW, KCALC, KOP,
     *                  NDS, KDIAG, QBIN, KER )
C***********************************************************************
C     Template for DL0U routine for binary liquid diffusion coefficients
C     STUB ROUTINE
C
C     T = temperature
C     P = pressure (system)
C     X(N) = mole fraction
C     N = number of components present in X
C     IDX(N) = index of component present
C     IRW = real work area index
C     IIW = integer work area index
C     KCALC = calculation code (0=do not calculate, 1 = calculate)
C     KOP(10) = model option code
C     NDS = data set number
C     KDIAG = diagnostic message level
C     QBIN(N,N) = results. Binary diffusion coeffcients.
C     QBIN(i,j) is binary diffusion coefficient of component i in component j
C     KER = error return code (0 = no error)
C     All input and output in this user routine are in SI Units
C     with Gas constant = 8314.33
C***********************************************************************
C
      IMPLICIT NONE
C
C     DECLARE VARIABLES USED IN DIMENSIONING
C
      INTEGER N
#include "dms_global.cmn"
#include "dms_errout.cmn"
```

```
#include "ppexec_user.cmn"
#include "dms_maxwrt.cmn"
#include "dms_plex.cmn"
C
C     DECLARE ARGUMENTS
C
      INTEGER IDX(N), IRW, IIW, KCALC, NDS, KDIAG, KER, KOP
        INTEGER IWATER, IMEACOO, ICO2, IHCO3
        INTEGER IMEAH, IMEA, ICO3, IN2, IO2
        INTEGER DMS_KCCIDC, DMS_IFCMNC
      REAL*8 X(N), QBIN(N,N), T, P
        REAL*8 WATER, MEACOO, CO2, HCO3, MEAH, MEA
        REAL*8 CO3, VBMEA, MUWO
        REAL*8 LDG, XCO2T, XMEAT, EFACT1
        REAL*8 MWMEA, MWCO2, MWH2O, XH2O, MWT, XWMEA
        REAL*8 XWAMINE, B(1), DFACTCO2, DFACTMEA
        REAL*8 IOND, CO2D, MEAD, XMOLT, CO2DW
        REAL*8 MA, MB, MC, MD, ME, MUMX, MUMX1
        REAL*8 A, E, BB, THET, C, MU0, MUW, R, HG
        Real*8 VISC, LVISC, B1, C1, DZERO, MU0MEA, T0
        EQUIVALENCE (B(1), IB(1))
        Integer DFACT1, EFACT
        integer nbopst(6), name(2)
        CHARACTER*256 BUFFER(1)
C
C     DECLARE LOCAL VARIABLES
C
      INTEGER IPROG(2), I, J, K
C
C      DATA STATEMENTS
C
      DATA IPROG /4HDL0U, 4H    /
C
C     BEGIN EXECUTABLE CODE
C       DIFFUSIVITIES CALCULATED BY (...) METHOD
C       VALUES OBTAINED FROM THE DIFFUSIVITY REGRESSION
        KER = 0
        IF (KCALC .EQ. 0) RETURN
c
C       INDEX VALUES FOR COMPONENTS IN SIMULATION
C
      IWATER = DMS_KCCIDC('H2O')
      IMEACOO = DMS_KCCIDC('MEACOO-')
      ICO2 = DMS_KCCIDC('CO2')
        IHCO3  = DMS_KCCIDC('HCO3-')
      IMEAH = DMS_KCCIDC('MEAH+')
      IMEA = DMS_KCCIDC('MEA')
        ICO3 = DMS_KCCIDC('CO3--')
        IN2 = DMS_KCCIDC('N2')
        IO2 = DMS_KCCIDC('O2')
C
C
C       ASSIGNMENT OF INDEX NUMBERS FOR SPECIES PRESENT
C
        DO 50 I = 1, N
        IF (IDX(I). EQ. IWATER) IWATER = I
        IF (IDX(I). EQ. IMEACOO) IMEACOO = I
        IF (IDX(I). EQ. ICO2) ICO2 = I
        IF (IDX(I). EQ. IHCO3) IHCO3 = I
        IF (IDX(I). EQ. IMEAH) IMEAH = I
        IF (IDX(I). EQ. IMEA) IMEA = I
        IF (IDX(I). EQ. ICO3) ICO3 = I
        IF (IDX(I). EQ. IN2) IN2 = I
        IF (IDX(I). EQ. IO2) IO2 = I
   50 CONTINUE
C
C       LOADING CALCULATION
        XCO2T = X(IMEACOO)+X(ICO2)+X(IHCO3)
        XMEAT = X(IMEACOO)+X(IMEAH)+X(IMEA)
        LDG = XCO2T/(XMEAT)
C
C
C       AMINE MASS FRACTION CALCULATION
C
        MWMEA = 61.8D0
```

```
         MWCO2 = 44.01D0
         MWH2O = 18D0
         XH2O = X(IWATER)
         MWT = XCO2T*MWCO2 + XMEAT*MWMEA + XH2O*MWH2O

c        Viscosity of solution from Aspen
         call PPUTL_GOPSET ( NBOPST , NAME )
c
         CALL PPMON_VISCL (T, P, X, N, IDX, NBOPST, KDIAG, VISC, KER)
         LVISC = VISC
         MUMX = LVISC
C
C     Viscosity of water according to Likhachev E.R. Technical Physics, Vol. 48 N0.4 2003 pp. 514-515
C     Viscosity in Pa-s
      E = 4.753D0
      MU0 = 0.000024055D0
      THET = 139.7D0
         A = 0.000442D0
         BB = 0.0009565D0
         C = 0.0124D0
         R = 0.008314D0
         P = P / 100000D0
         HG = A * P +((E - BB * P)/(R * (T - THET - C * P)))
         MUW = (MU0 * EXP(HG))
C
C     DFACT1, EFACT STORE THE POSITION OF REGRESSED PARAMETERS FOR DIFF. CORRELATION
C        THEY REFER TO THE VALUES SPECIFIED IN PROPERTY-PARAMETER-USRDEF

      DFACT1 = DMS_IFCMNC('DFACT1')
      EFACT = DMS_IFCMNC('EFACT')
C        DFACT2 = DMS_IFCMNC('DFACT2')
C     ASSIGNS THE DFACT1 POSITION TO CO2 AND DFACT2 POSITION TO MEA
         DFACTCO2 = B(DFACT1+IDX(ICO2))
      DFACTMEA = B(DFACT1+IDX(IMEA))
      EFACT1 = B(EFACT+IDX(ICO2))
C
C     DIFFUSIVITY OF CO2 IN WATER
      CO2DW = 0.00000235D0 * EXP(-2119D0 / T)
C
C     DIFFUSIVITY OF CO2 IN SOLUTION BASED ON VERSTEEG, 1988
      CO2D = CO2DW * (MUW / MUMX)**(0.8D0)*((T/313.15)**(EFACT1))
      CO2D = CO2D * DFACTCO2

      CO2D = ((DFACTCO2)**2)/DFACTMEA * (MUW/MUMX)**0.8
      CO2D = CO2D*(T/313.15)**(EFACT1)
C

C     Diffusivity of Amine in water Hayduk and Laudie, 1974. AIChe Journal Vol.20 No. 3
C     DZERO in cm^2/s
C


C
C     Diffusivity of amine in solution - Aboudheir
C
      MEAD = (1/((MUMX/MUW)**0.8D0))*((T/313.15)**(EFACT1))
      MEAD = MEAD * DFACTMEA
C
C
C
C     ASSIGNING VALUES IN THE DIFFUSIVITY MATRIX
C
C
      DO 200 I = 1, N
        DO 100 J = 1, N
          IF (I.EQ.J) THEN
            QBIN(I,J) = 0D0

          ELSE
            QBIN(I,J) = MEAD

            IF (I.EQ.ICO2)QBIN(I,J) = CO2D
            IF (J.EQ.ICO2)QBIN(I,J) = CO2D
            IF (I.EQ.IN2)QBIN(I,J) = CO2D
            IF (J.EQ.IN2)QBIN(I,J) = CO2D
          END IF
```

```
  100   CONTINUE
  200 CONTINUE

C
C       WRITE VARIABLES TO HISTORY FILE
C
C        THE WRITE TO UNIT USER_NHSTRY WRITES TO THE HISTORY FILE
        WRITE (BUFFER, *) 'Executed fortran subroutine'
        CALL DMS_WRTALN(USER_NHSTRY, BUFFER(1))
        WRITE (BUFFER, *) 'Pressure ', P
        CALL DMS_WRTALN(USER_NHSTRY, BUFFER(1))
        WRITE (BUFFER, *) 'Temperature ', T
        CALL DMS_WRTALN(USER_NHSTRY, BUFFER(1))
        WRITE (BUFFER, *) 'LVISC ', LVISC
        CALL DMS_WRTALN(USER_NHSTRY, BUFFER(1))
        WRITE (BUFFER, *) ' '
        CALL DMS_WRTALN(USER_NHSTRY, BUFFER(1))


C 999      RETURN
      END
```

C.4. LIQUID AND GAS MASS TRANSFER COEFFICIENT (K_L^o, K_G) SUBROUTINE FOR THE WWC

```
      SUBROUTINE USRMTRFC (KSTG,   NCOMPS, IDX,    NBOPST, KPDIAG,
     1                     XCOMPB, FRATEL, YCOMPB, FRATEV, PRESS,
     2                     TLIQ,   TVAP,   AVMWLI, AVMWVA, VISCML,
     3                     DENMXL, SIGMAL, VISCMV, DENMXV, AREAIF,
     4                     PREK,   EXPKD,  COLTYP, USRCOR, TWRARA,
     5                     COLDIA, HTPACK, PACSIZ, SPAREA, CSIGMA,
     6                     PFACT,  PKPRMS, VOIDFR, IPAKAR, IPTYPE,
     7                     IVENDR, IPMAT,  IPSIZE, WEIRHT, DCAREA,
     8                     ARAACT, FLOPTH, NPASS,  WEIRL,  IFMETH,
     9                     SYSFAC, HOLEAR, ITTYPE, TRASPC, PITCH,
     A                     IPHASE, NINT,   INT,    NREAL,  REAL)
      IMPLICIT NONE
      INTEGER KSTG, NCOMPS, IDX(NCOMPS), NBOPST(6), KPDIAG,
     +        COLTYP, USRCOR, IPAKAR, IPTYPE, IVENDR, IPMAT,  IPSIZE,
     +        NPASS, IFMETH, ITTYPE, NINT, INT(NINT), IPHASE, NREAL
      REAL*8  XCOMPB(NCOMPS), FRATEL, YCOMPB(NCOMPS), FRATEV,
     +        PRESS, TLIQ, TVAP, AVMWLI, AVMWVA, VISCML, DENMXL,
     +        SIGMAL, VISCMV, DENMXV, AREAIF, PREK, EXPKD,
     +        TWRARA, COLDIA, HTPACK, PACSIZ, SPAREA, CSIGMA,
     +        PFACT,  PKPRMS(20), VOIDFR, WEIRHT, DCAREA, ARAACT,
     +        FLOPTH, WEIRL, SYSFAC, HOLEAR, TRASPC, PITCH,
     +        REAL(NREAL)
C***********************************************************************
C  LICENSED MATERIAL.  PROPERTY OF ASPEN TECHNOLOGY, INC.  TO BE       *
C  TREATED AS ASPEN TECH PROPRIETARY INFORMATION UNDER THE TERMS       *
C  OF THE ASPEN PLUS SUBSCRIPTION AGREEMENT.                           *
C***********************************************************************
C-----------------------------------------------------------------------
C         COPYRIGHT (C) 2004
C          ASPEN TECHNOLOGY, INC.
C          CAMBRIDGE, MA
C-----------------------------------------------------------------------
C
C     DESCRIPTION: User provided RateSep routine to calculate the
C                  liquid (IPHASE=0) and vapor (IPHASE=1) binary mass
C                  transfer coefficient parameters (PREK, EXPKD).
C
C     VARIABLES IN ARGUMENT LIST
C
C     VARIABLE I/O  TYPE   DIMENSION   DESCRIPTION AND RANGE
C     -------- ---  ----   ---------   --------------------------------
C     KSTG      I    I         -       SEGMENT NUMBER
C     NCOMPS    I    I         -       NUMBER OF COMPONENTS
C     IDX       I    I       NCOMPS    COMPONENT INDEX VECTOR
C     NBOPST    I    I         6       PHYSICAL PROPERTY OPTION
C                                      SET BEAD POINTER
C     KPDIAG    I    I         -       PHYSICAL PROPERTY
```

```
C                                          DIAGOSTIC CODE
C     XCOMPB    I    R       NCOMPS      BULK LIQUID MOLE FRACTION
C     FRATEL    I    R         -         FLOW OF LIQUID (KMOL/SEC)
C     YCOMPB    I    R       NCOMPS      BULK VAPOR MOLE FRACTION
C     FRATEV    I    R         -         FLOW OF VAPOR (KMOL/SEC)
C     PRESS     I    R         -         PRESSURE (N/SQ.M)
C     TLIQ      I    R         -         LIQUID TEMPERATURE (K)
C     TVAP      I    R         -         VAPOR TEMPERATURE (K)
C     AVMWLI    I    R         -         AVERAGE MOLECULAR WEIGHT
C                                        OF LIQUID MIXTURE
C                                        (KG/KMOL)
C     AVMWVA    I    R         -         AVERAGE MOLECULAR WEIGHT
C                                        OF VAPOR MIXTURE (KG/KMOL)
C     VISCML    I    R         -         VISCOSITY OF LIQUID
C                                        (N-SEC/SQ.M)
C     DENMXL    I    R         -         DENSITY OF LIQUID MIXTURE
C                                        (KMOL/CU.M)
C     SIGMAL    I    R         -         SURFACE TENSION OF LIQUID
C                                        (N/M)
C     VISCMV    I    R         -         VISCOSITY OF VAPOR MIXTURE
C                                        (N-SEC/SQ.M)
C     DENMXV    I    R         -         DENSITY OF VAPOR MIXTURE
C                                        (KMOL/CU.M)
C     AREAIF    I    R         -         INTERFACIAL AREA
C                                        (SEE NOTE-1 BELOW)
C     PREK      O    R         -         BINARY MASS TRANSFER =
C     EXPRKD    O    R         -            PREK*DIFFUSIVITY**EXPKD
C                                        (SEE NOTE-2 BELOW)
C     COLTYP    I    I         -         TYPE OF COLUMN
C                                        1 = PACKED
C                                        2 = TRAY
C     USRCOR    I    I         -         CALCULATION METHOD (I.E.
C                                        CHOICE OF USER CORRELATION)
C                                          1  = USER1
C                                          2  = USER2
C                                          3  = USER3
C                                          4  = USER4
C     TWRARA    I    R         -         CROSS-SECTIONAL AREA OF
C                                        TOWER (SQ.M)
C     COLDIA    I    R         -         COLUMN DIAMETER (M)
C     HTPACK    I    R         -         HEIGHT OF PACKING IN THE
C                                        SEGMENT (M)
C     PACSIZ    I    R         -         SIZE OF PACKING (M)
C     SPAREA    I    R         -         SPECIFIC SURFACE AREA OF
C                                        PACKING (SQ.M/CU.M)
C     CSIGMA    I    R         -         CRITICAL SURFACE TENSION
C                                        OF PACKING MATERIAL (N/M)
C     PFACT     I    R         -         PACKING FACTOR (1/M)
C     PKPRMS    I    R        20         PACKING PARAMETERS
C                                        PKPRMS(1) = STICHLMAIR CONSTANT C1
C                                        PKPRMS(2) = STICHLMAIR CONSTANT C2
C                                        PKPRMS(3) = STICHLMAIR CONSTANT C3
C                                        PKPRMS(4) = CL IN BILLET 93
C                                        PKPRMS(5) = CV IN BILLET 93
C                                        PKPRMS(6) = B IN BRF 85
C                                        PKPRMS(7) = S IN BRF 85
C                                        PKPRMS(8) = H IN BRF 85
C                                        PKPRMS(9) = Fse IN BRF 92
C                                        PKPRMS(10) = CE IN BRF 92
C                                        PKPRMS(11) = THETA IN BRF 92
C     VOIDFR    I    R         -         VOID FRACTION OF PACKING
C     IPAKAR    I    I         -         PACKING ARRANGEMENT
C                                          1 = RANDOM
C                                          2 = STRUCTURED
C     IPTYPE    I    I         -         PACKING TYPE
C                                        See IPTYPE in packsr.f
C     IVENDR    I    I         -         PACKING VENDOR CODE
C     IPMAT     I    I         -         PACKING MATERIAL CODE
C     IPSIZE    I    I         -         PACKING SIZE CODE
C     WEIRHT    I    R         -         AVERAGE WEIR HEIGHT (M)
C     DCAREA    I    R         -         TOTAL AREA OF DOWNCOMER
C                                        ON TRAY (SQ.M)
C     ARAACT    I    R         -         TOTAL ACTIVE AREA AVAILABLE
C                                        ON TRAY (SQ.M)
C     FLOPTH    I    R         -         AVERAGE FLOWPATH LENGTH (M)
```

```
C     NPASS     I    I         -       NUMBER OF TRAY PASSES
C     WEIRL     I    R         -       AVERAGE WEIRH LENGTH (M)
C     IFMETH    I    I         -       FLOODING CALCULATION
C                                      METHOD; REQUIRED FOR SIEVE
C                                      TRAY
C     SYSFAC    I    R         -       SYSTEM FACTOR; REQUIRED FOR
C                                      SIEVE TRAY
C     HOLEAR    I    R         -       HOLE AREA/ACTIVE AREA; REQUIRED
C                                      FOR SIEVE TRAY
C     ITTYPE    I    I         -       TRAY TYPE
C                                        1 - BUBBLE CAPS
C                                        2 - SIEVE
C                                        3 - GLITSCH BALLAST
C                                        4 - KOCH FLEXITRAY
C                                        5 - NUTTER FLOAT VALVE
C     TRASPC    I    R         -        TRAY SPACING (M)
C     PITCH     I    R         -       SIEVE TRAY HOLE PITCH (M)
C     IPHASE    I    I         -       PHASE QUALIFIER
C                                        0 = LIQUID
C                                        1 = VAPOR
C     NINT      I    I         -       Size of INT
C     INT      I/O   I       NINT      User correlation INT array
C     NREAL     I    I         -       Size of REAL
C     REAL     I/O   I       NREAL     User correlation REAL array
C
C    NOTE-1:
C         SPECIFIC INTERFACIAL AREA "AREAIF" HAS THE FOLLOWING UNITS.
C           FOR PACKED COLUMNS, THE UNITS IS "SQ.M/CU.M OF PACKING"
C           FOR TRAY COLUMNS, THE UNITS IS "SQ.M/SQ.M ACTIVE TRAY AREA"
C
C    NOTE-2:
C         BINMTP = PREK * DIFFUSIVITY**EXPKD
C         BINARY MASS TRANSFER COEFFCIENTS "BINMTP" HAVE UNITS (KMOL/SEC)
C         DIFFUSIVITY HAVE UNITS (SQ.M/SEC)
C         BINMTP HAS MOLAR DENSITY AND INTERFACIAL AREA INCLUDED
C
C**********************************************************************
C     Declare local variables used in the user correlations
C
      REAL*8 RS_BennettHL
      REAL*8 RS_BennettA
      REAL*8 RS_BennettC
      REAL*8 ScLB,   ScVB,   rhoLms, rhoVms, ReLPrm,
     +       dTemp,  uL,     uV,     Fs,     QL,
     +       C,      alphae, hL,     ShLB,   ReV,
     +       vel,    hydia,  qsoln,  w,      dtempa
C
C     Instead of computing BINMTP from diffusivity as in RATEFRAC
C     compute PREK and EXPKD for RateSep
C
      IF (COLTYP .EQ. 1) THEN
C
C**** PACKED COLUMN
c
c     This is the begining of the Dugas Modification
c
         IF (USRCOR .EQ. 9) THEN
C
            IF (IPHASE.EQ.0) THEN
C
C              Liquid phase
C
               qsoln = FRATEL / DENMXL / 100
C              The factor of 100 is needed since the simulation has 10x diamter (100x flow).
C
               w = 0.03958407
C              w is the circumfrence of the column in meters.  Diamter of WWC is 0.0126m
C
               dTemp = 3**0.3333 * 2**0.5 / 3.1416**0.5
               dTemp = dTemp * qsoln**.3333*0.091**.5*w**.6667/0.003852
               dTemp = dTemp * (9.81*DENMXL/VISCML*AVMWLI)**.1667
C              The proceeding equation is a simplification of the equaitons in Cullinane's thesis,
c              pages 57-60.  The simplification for theta is used to allow the form Aspen requires.
c              The constants 0.091, 0.003852, and 9.81 refer to the height of the WWC, the area of the
c              WWC and acceleration due to gravity.
```

```
c
C              CONVERT K FROM M/S TO KMOL/S
               dTemp = dTemp * TWRARA * HTPACK * AREAIF * DENMXL
C              This is the conversion used in the Onda mass transfer routine

               PREK  = dtemp * 1.0
               EXPKD = 0.5D0
C
            ELSE
C
C              Vapor phase
C
C              From Pacheco's correlation:  R*T*kg*d/DCO2=1.075(Re*Sc*d/h)^0.85
C              Simplified, this gives RTkg=1.075*DCO2^.15*d^.7*(v/h)^.85
C
                vel = FRATEV / TWRARA / DENMXV
                hydia  = 0.0044D0
C               This corresponds to the estimated hydraulic diameter of the WWC, 0.44cm.
c
                dTemp = 1.075D0 * hydia ** 0.7D0
                dTemp = dTemp *(vel / (0.091D0)) ** 0.85D0
C               The constant, 0.091, corresponds to the height of the WWC.  Aspen has a argument for
the
C               height of a stage but nothing for the # of stages.  Therefore the total height was
hardwired.
C
                dTemp = dTemp * DENMXV * AREAIF * TWRARA * HTPACK
C               This time the number of stages is not need bc this mass tranfer coeffient is the moles
reacted by stage
C               Note: this correlation results in a MT value (in mol/s) 100 times greater than the
calculated excel value due to 10x diameter.
C
                PREK = dtemp
                EXPKD = 0.15D0
C
             END IF
C            END OF IF (IPHASE)
C
         END IF
C        END OF IF (USRCOR)
c
C     This is the end of the Dugas Modification
C
         IF (USRCOR .EQ. 1) THEN
C           user subroutine example for packed column: Onda 68
C
C           Onda, K., Takeuchi, H. and Okumoto, Y., "Mass Transfer
C             Coefficients between Gas and Liquid Phases in Packed
C             Columns", J. Chem. Eng. Jap., 1, (1968) P56
C
            IF (IPHASE.EQ.0) THEN
C
C              Liquid phase
C
               rhoLms = DENMXL * AVMWLI
               uL = FRATEL / TWRARA / DENMXL
               ReLPrm = rhoLms * uL / VISCML / AREAIF
               dTemp = (rhoLms/9.81D0/VISCML)**(0.33333333D0)
               dTemp = 0.0051D0 * (ReLPrm**(0.66666667D0))
     +                *((SPAREA*PACSIZ)**(0.4D0)) / dTemp
C
C              CONVERT K FROM M/S TO KMOL/S
               dTemp = dTemp * TWRARA * HTPACK * AREAIF * DENMXL
C
C              COMPOSITION INDEPENDENT PART OF SCHMIDT NUMBER
               ScLB = VISCML / rhoLms
C
               PREK  = dTemp / DSQRT(ScLB)
               EXPKD = 0.5D0
C
            ELSE
C
C              Vapor phase
C
                rhoVms = DENMXV * AVMWVA
```

```
                  uV = FRATEV / TWRARA / DENMXV
                  ReV = rhoVms * uV / VISCMV / SPAREA
                  dTemp = SPAREA*PACSIZ
                  dTemp = dTemp * dTemp
                     IF (PACSIZ .GE. 0.015D0) THEN
                         dTemp = 5.23D0 / dTemp
                     ELSE
                         dTemp = 2.0D0 / dTemp
                     END IF
                  dTemp = dTemp * (ReV**(0.7D0)) * SPAREA
C
C                 CONVERT K FROM M/S TO KMOL/S
                  dTemp = dTemp * TWRARA * HTPACK * AREAIF * DENMXV
C
C                 COMPOSITION INDEPENDENT PART OF SCHMIDT NUMBER
                  ScVB = VISCMV / rhoVms
C
                  PREK = dTemp * ScVB ** 0.33333333D0
                  EXPKD = 0.66666667D0
               END IF
C              END OF IF (IPHASE)
C
            END IF
C           END OF IF (USRCOR)
C
         ELSE IF (COLTYP .EQ. 2) THEN
C
C**** TRAY COLUMN
C
            IF (USRCOR .EQ. 1) THEN
C              user subroutine example for tray column: AIChE 58
C
C              AIChE, Bubble Tray Design Manual: Prediction of Fractionation
C                Efficiency, New York, 1958
C
C              For bubble cap, valve, and sieve trays
C
               IF (IPHASE.EQ.0) THEN
C
C                 Liquid phase
C
                  rhoVms = DENMXV * AVMWVA
                  rhoLms = DENMXL * AVMWLI
                  uV = FRATEV /DENMXV /ARAACT
                  Fs = uV * DSQRT(rhoVms)
                  C = 0.5D0 + 0.438D0 * DEXP(-137.8 * WEIRHT)
                  QL = FRATEL/DENMXL
                  ALPHAE = DEXP(-12.55D0*(uV*DSQRT(RHOVMS/DABS(RHOLMS-
     1                     RHOVMS)))**0.91D0)
                  hL =ALPHAE*(WEIRHT + C*(QL/WEIRL/ALPHAE)**0.67D0)
                  dTemp = 19700.0D0 *(0.4D0*Fs+0.17D0) * hL
     +                    * ARAACT * DENMXL
C
                  PREK = dTemp
                  EXPKD = 0.5D0
C
               ELSE
C
C                 Vapor phase
C
                  rhoVms = DENMXV * AVMWVA
                  uV = FRATEV /DENMXV /ARAACT
                  Fs = uV * DSQRT(rhoVms)
                  QL = FRATEL/DENMXL
                  dTemp = 0.776 + 4.57*WEIRHT - 0.238*Fs
     +                  + 104.8*QL/WEIRL
                  dTemp = dTemp * uV * ARAACT * DENMXV
C
C                 COMPOSITION INDEPENDENT PART OF SCHMIDT NUMBER
                  ScVB = VISCMV / rhoVms
C
                  PREK = dTemp /DSQRT(ScVB)
                  EXPKD = 0.5D0
               END IF
C              END OF IF (IPHASE)
```

```
C
         END IF
C        END OF IF (USRCOR)
C
        END IF
C     END OF IF (COLTYP)
C
      RETURN
      END
```

C.5. INTERFACIAL AREA (A_{INTF}) SUBROUTINE FOR THE WWC

```
      SUBROUTINE AREA    (KSTG,   NCOMPS, IDX,    NBOPST, KPDIAG,
     1                    XCOMPB, FRATEL, YCOMPB, FRATEV, PRESS,
     2                    TLIQ,   TVAP,   AVMWLI, AVMWVA, VISCML,
     3                    DENMXL, SIGMAL, VISCMV, DENMXV, AREAIF,
     4                    COLTYP, USRCOR, TWRARA, COLDIA, HTPACK,
     5                    PACSIZ, SPAREA, CSIGMA, PFACT,  PKPRMS,
     6                    VOIDFR, IPAKAR, IPTYPE, IVENDR, IPMAT,
     7                    IPSIZE, WEIRHT, DCAREA, ARAACT, FLOPTH,
     8                    NPASS,  WEIRL,  IFMETH, SYSFAC, HOLEAR,
     9                    ITTYPE, TRASPC, PITCH,  NINT,   INT,
     A                    NREAL,  REAL)
      IMPLICIT NONE
      INTEGER KSTG, NCOMPS, IDX(NCOMPS), NBOPST(6), KPDIAG,
     +        COLTYP, USRCOR, IPAKAR, IPTYPE, IVENDR, IPMAT,  IPSIZE,
     +        NPASS, IFMETH, ITTYPE, NINT, INT(NINT), NREAL
      REAL*8  XCOMPB(NCOMPS), FRATEL, YCOMPB(NCOMPS), FRATEV,
     +        PRESS, TLIQ, TVAP, AVMWLI, AVMWVA, VISCML, DENMXL,
     +        SIGMAL, VISCMV, DENMXV, AREAIF, TWRARA, COLDIA,
     +        HTPACK, PACSIZ, SPAREA, CSIGMA, PFACT, PKPRMS(20),
     +        VOIDFR, WEIRHT, DCAREA, ARAACT, FLOPTH, WEIRL,
     +        SYSFAC, HOLEAR, TRASPC, PITCH, REAL(NREAL)
C***********************************************************************
C  LICENSED MATERIAL.  PROPERTY OF ASPEN TECHNOLOGY, INC.  TO BE       *
C  TREATED AS ASPEN TECH PROPRIETARY INFORMATION UNDER THE TERMS       *
C  OF THE ASPEN PLUS SUBSCRIPTION AGREEMENT.                           *
C***********************************************************************
C-----------------------------------------------------------------------
C         COPYRIGHT (C) 2004
C          ASPEN TECHNOLOGY, INC.
C          CAMBRIDGE, MA
C-----------------------------------------------------------------------
C
C     DESCRIPTION: User provided RateSep routine to calculate the
C                  specific interface area AREAIF (see NOTE-1).
C
C     VARIABLES IN ARGUMENT LIST
C
C     VARIABLE I/O  TYPE   DIMENSION   DESCRIPTION AND RANGE
C     -------- ---  ----   ---------   ---------------------------------
C     KSTG      I    I         -       SEGMENT NUMBER
C     NCOMPS    I    I         -       NUMBER OF COMPONENTS
C     IDX       I    I       NCOMPS    COMPONENT INDEX VECTOR
C     NBOPST    I    I         6       PHYSICAL PROPERTY OPTION
C                                      SET BEAD POINTER
C     KPDIAG    I    I         -       PHYSICAL PROPERTY
C                                      DIAGOSTIC CODE
C     XCOMPB    I    R       NCOMPS    BULK LIQUID MOLE FRACTION
C     FRATEL    I    R         -       FLOW OF LIQUID (KMOL/SEC)
C     YCOMPB    I    R       NCOMPS    BULK VAPOR MOLE FRACTION
C     FRATEV    I    R         -       FLOW OF VAPOR (KMOL/SEC)
C     PRESS     I    R         -       PRESSURE (N/SQ.M)
C     TLIQ      I    R         -       LIQUID TEMPERATURE (K)
C     TVAP      I    R         -       VAPOR TEMPERATURE (K)
C     AVMWLI    I    R         -       AVERAGE MOLECULAR WEIGHT
C                                      OF LIQUID MIXTURE
C                                      (KG/KMOL)
C     AVMWVA    I    R         -       AVERAGE MOLECULAR WEIGHT
C                                      OF VAPOR MIXTURE (KG/KMOL)
C     VISCML    I    R         -       VISCOSITY OF LIQUID
C                                      (N-SEC/SQ.M)
C     DENMXL    I    R         -       DENSITY OF LIQUID MIXTURE
C                                      (KMOL/CU.M)
```

```
C     SIGMAL    I    R          -       SURFACE TENSION OF LIQUID
C                                       (N/M)
C     VISCMV    I    R          -       VISCOSITY OF VAPOR MIXTURE
C                                       (N-SEC/SQ.M)
C     DENMXV    I    R          -       DENSITY OF VAPOR MIXTURE
C                                       (KMOL/CU.M)
C     AREAIF    O    R          -       INTERFACIAL AREA
C                                       (SEE NOTE-1 BELOW)
C     COLTYP    I    I          -       TYPE OF COLUMN
C                                       1 = PACKED
C                                       2 = TRAY
C     USRCOR    I    I          -       CALCULATION METHOD (I.E.
C                                       CHOICE OF USER CORRELATION)
C                                         1 = USER1
C                                         2 = USER2
C                                         3 = USER3
C                                         4 = USER4
C     TWRARA    I    R          -       CROSS-SECTIONAL AREA OF
C                                       TOWER (SQ.M)
C     COLDIA    I    R          -       COLUMN DIAMETER (M)
C     HTPACK    I    R          -       HEIGHT OF PACKING IN THE
C                                       SEGMENT (M)
C     PACSIZ    I    R          -       SIZE OF PACKING (M)
C     SPAREA    I    R          -       SPECIFIC SURFACE AREA OF
C                                       PACKING (SQ.M/CU.M)
C     CSIGMA    I    R          -       CRITICAL SURFACE TENSION
C                                       OF PACKING MATERIAL (N/M)
C     PFACT     I    R          -       PACKING FACTOR (1/M)
C     PKPRMS    I    R         20       PACKING PARAMETERS
C                                       PKPRMS(1) = STICHLMAIR CONSTANT C1
C                                       PKPRMS(2) = STICHLMAIR CONSTANT C2
C                                       PKPRMS(3) = STICHLMAIR CONSTANT C3
C                                       PKPRMS(4) = CL IN BILLET 93
C                                       PKPRMS(5) = CV IN BILLET 93
C                                       PKPRMS(6) = B IN BRF 85
C                                       PKPRMS(7) = S IN BRF 85
C                                       PKPRMS(8) = H IN BRF 85
C                                       PKPRMS(9) = Fse IN BRF 92
C                                       PKPRMS(10) = CE IN BRF 92
C                                       PKPRMS(11) = THETA IN BRF 92
C     VOIDFR    I    R          -       VOID FRACTION OF PACKING
C     IPAKAR    I    I          -       PACKING ARRANGEMENT
C                                         1 = RANDOM
C                                         2 = STRUCTURED
C     IPTYPE    I    I          -       PACKING TYPE
C                                       See IPTYPE in packsr.f
C     IVENDR    I    I          -       PACKING VENDOR CODE
C     IPMAT     I    I          -       PACKING MATERIAL CODE
C     IPSIZE    I    I          -       PACKING SIZE CODE
C     WEIRHT    I    R          -       AVERAGE WEIR HEIGHT (M)
C     DCAREA    I    R          -       TOTAL AREA OF DOWNCOMER
C                                       ON TRAY (SQ.M)
C     ARAACT    I    R          -       TOTAL ACTIVE AREA AVAILABLE
C                                       ON TRAY (SQ.M)
C     FLOPTH    I    R          -       AVERAGE FLOWPATH LENGTH (M)
C     NPASS     I    I          -       NUMBER OF TRAY PASSES
C     WEIRL     I    R          -       AVERAGE WEIRH LENGTH (M)
C     IFMETH    I    I          -       FLOODING CALCULATION
C                                       METHOD; REQUIRED FOR SIEVE
C                                       TRAY
C     SYSFAC    I    R          -       SYSTEM FACTOR; REQUIRED FOR
C                                       SIEVE TRAY
C     HOLEAR    I    R          -       HOLE AREA/ACTIVE AREA; REQUIRED
C                                       FOR SIEVE TRAY
C     ITTYPE    I    I          -       TRAY TYPE
C                                         1 - BUBBLE CAPS
C                                         2 - SIEVE
C                                         3 - GLITSCH BALLAST
C                                         4 - KOCH FLEXITRAY
C                                         5 - NUTTER FLOAT VALVE
C     TRASPC    I    R          -       TRAY SPACING (M)
C     PITCH     I    R          -       SIEVE TRAY HOLE PITCH (M)
C     NINT      I    I          -       Size of INT
C     INT      I/O   I       NINT       User correlation INT array
C     NREAL     I    I          -       Size of REAL
```

```
C     REAL     I/O   I       NREAL     User correlation REAL array
C
C     NOTE-1:
C           SPECIFIC INTERFACIAL AREA "AREAIF" HAS THE FOLLOWING UNITS.
C            FOR PACKED COLUMNS, THE UNITS IS "SQ.M/CU.M OF PACKING"
C            FOR TRAY COLUMNS, THE UNITS IS "SQ.M/SQ.M ACTIVE TRAY AREA"
C
C***********************************************************************
C     Declare local variables used in the user correlations
C
      REAL*8 WeL,   dTemp,  uV,    rhoVms,
     +       uL,    rhoLms, ReL,   FrL,    uL2,
     +       ReV,    d,     Wprime,
     +       AREAE, At, hp, Ft, Fse, ap,
     +       S, cosg, pi, theta
C
C     Compute specific interface area as described above
C     Check COLTYP/USRCOR if providing multiple area correlations
C
      IF (COLTYP .EQ. 1) THEN
C
C**** PACKED COLUMN
C
         IF (USRCOR .EQ. 1) THEN
C           user subroutine example for packed column: Onda 68
C
C           Onda, K., Takeuchi, H. and Okumoto, Y., "Mass Transfer
C             Coefficients between Gas and Liquid Phases in Packed
C             Columns", J. Chem. Eng. Jap., 1, (1968) p. 56
C
            rhoLms = DENMXL * AVMWLI
            uL = FRATEL / TWRARA / DENMXL
            uL2 = uL * uL
            ReL = rhoLms * uL / VISCML / SPAREA
            FrL = SPAREA * uL2 / 9.81D0
C           WHERE 9.81D0 IS GRAVITY CONSTANT IN M/S**2
            WeL    = rhoLms * uL2 / SIGMAL / SPAREA
            dTemp = -1.45D0*((CSIGMA/SIGMAL)**0.75D0)
     +                    *(ReL**0.1D0)*(FrL**(-0.05D0))
     +                    *(WeL**0.2D0)
            dTemp = 1.D0 - DEXP(dTemp)

            AREAIF = SPAREA*dTemp

C           Uses specific area of the packing for both random and structured

         ELSEIF (USRCOR .EQ. 2) THEN

            AREAIF = SPAREA  !(sq.m/cu.m)

C           Uses the Rocha-Bravo-Fair (1992) Model as defined in Aspen Plus

         ELSEIF (USRCOR .EQ. 3) THEN

            IF (SIGMAL .GE. 0.055) THEN
                cosg = 5.211*(10**(-16.835 * SIGMAL))
            ELSE
                cosg = 0.9
            END IF

            pi      = 3.141592654
            theta   = PKPRMS(11)*pi/180

            rhoLms  = DENMXL * AVMWLI
            uL      = FRATEL / TWRARA / DENMXL
            uL2     = uL * uL
            S       = PKPRMS(7)

            WeL     = uL2 * rhoLms * S / SIGMAL
            FrL     = uL2 / (S * 9.81D0)
            ReL     = uL * S * rhoLms / VISCML
            Ft      = (29.12*((WeL*FrL)**0.15)*(S**0.359))/(ReL**0.2)
     +                       /(VOIDFR**0.6)/(dsin(theta)**0.3)
     +                       /(1-(0.93*cosg))
```

```
            Fse     = PKPRMS(9) !Surface enhancement factor
            ap      = SPAREA  !Specific area of packing
C           At      = TWRARA  (cross sectional area of column)
C           hp      = HTPACK  (height of packing)

            AREAIF  = Ft*Fse*ap

            IF (IPSIZE .eq. 606 .AND. IPTYPE .eq. 701) THEN
                AREAIF = AREAIF*1.147643+172.01
            ELSE
                AREAIF = AREAIF
            END IF

C           AREAIF = dsin(PKPRMS(11)*Pi)
C           WRITE (*,*) denmxl, avmwli, fratel, twrara, S

         Elseif (USRCOR .EQ. 9) THEN
         AREAIF = 325.444D0
c        Actual wetted are of the WWC is 38.52cm2.
c        Column diamter is listed as 0.128655m (a factor of 10 bigger than the area which matches gas
flow area) and height as 9.1cm.
c        This gives a surface area of 325.444m2/m3 for the Aspen Simulations.

         END IF
C        END OF IF (USRCOR)
C
      ELSE IF (COLTYP .EQ. 2) THEN
C
C**** TRAY COLUMN
C
         IF (USRCOR .EQ. 1) THEN
C           user subroutine example for tray column: Scheffe-Weiland 87
C
C           Scheffe, R.D. and Weiland, R.H., "Mass Transfer
C           Characteristics of Valve Trays." Ind. Eng. Chem. Res.
C           26, (1987) p. 228
C
C           The original paper only mentioned valve tray.
C           It is also used for bubble-cap tray and sieve tray.
C
C           CHARACTERISTIC LENGTH IS ALWAYS 1 METER.
            d = 1.0D0
            rhoLms = DENMXL * AVMWLI
            rhoVms = DENMXV * AVMWVA
            uL = FRATEL / TWRARA / DENMXL
            uV = FRATEV / TWRARA / DENMXV
            ReL = rhoLms * uL * d / VISCML
            ReV = rhoVms * uV * d / VISCMV
            Wprime = WEIRHT / d
            AREAIF = 0.27D0 * ReV**0.375D0 * ReL**0.247D0
            AREAIF = AREAIF * Wprime**0.515
         END IF
C        END OF IF (USRCOR)
C
        END IF
C     END OF IF (COLTYP)
C
      RETURN
        END
```

C.6. ABSORBER LIQUID AND GAS MASS TRANSFER COEFFICIENT (K_L^o, K_G) SUBROUTINE

```
      SUBROUTINE USRMTRF3(KSTG,   NCOMPS, IDX,    NBOPST, KPDIAG,
     1                    XCOMPB, FRATEL, YCOMPB, FRATEV, PRESS,
     2                    TLIQ,   TVAP,   AVMWLI, AVMWVA, VISCML,
     3                    DENMXL, SIGMAL, VISCMV, DENMXV, AREAIF,
     4                    PREK,   EXPKD,  COLTYP, USRCOR, TWRARA,
     5                    COLDIA, HTPACK, PACSIZ, SPAREA, CSIGMA,
     6                    PFACT,  PKPRMS, VOIDFR, IPAKAR, IPTYPE,
     7                    IVENDR, IPMAT,  IPSIZE, WEIRHT, DCAREA,
     8                    ARAACT, FLOPTH, NPASS,  WEIRL,  IFMETH,
     9                    SYSFAC, HOLEAR, ITTYPE, TRASPC, PITCH,
     A                    IPHASE, NINT,   INT,    NREAL,  REAL)
```

```
      IMPLICIT NONE
      INTEGER KSTG, NCOMPS, IDX(NCOMPS), NBOPST(6), KPDIAG,
     +        COLTYP, USRCOR, IPAKAR, IPTYPE, IVENDR, IPMAT,  IPSIZE,
     +        NPASS, IFMETH, ITTYPE, NINT, INT(NINT), IPHASE, NREAL
      REAL*8  XCOMPB(NCOMPS), FRATEL, YCOMPB(NCOMPS), FRATEV,
     +        PRESS, TLIQ, TVAP, AVMWLI, AVMWVA, VISCML, DENMXL,
     +        SIGMAL, VISCMV, DENMXV, AREAIF, PREK, EXPKD,
     +        TWRARA, COLDIA, HTPACK, PACSIZ, SPAREA, CSIGMA,
     +        PFACT,  PKPRMS(20), VOIDFR, WEIRHT, DCAREA, ARAACT,
     +        FLOPTH, WEIRL, SYSFAC, HOLEAR, TRASPC, PITCH,
     +        REAL(NREAL)
C***********************************************************************
C  LICENSED MATERIAL.  PROPERTY OF ASPEN TECHNOLOGY, INC.  TO BE       *
C  TREATED AS ASPEN TECH PROPRIETARY INFORMATION UNDER THE TERMS       *
C  OF THE ASPEN PLUS SUBSCRIPTION AGREEMENT.                           *
C***********************************************************************
C-----------------------------------------------------------------------
C         COPYRIGHT (C) 2004
C          ASPEN TECHNOLOGY, INC.
C          CAMBRIDGE, MA
C-----------------------------------------------------------------------
C
C     DESCRIPTION: User provided RateSep routine to calculate the
C                  liquid (IPHASE=0) and vapor (IPHASE=1) binary mass
C                  transfer coefficient parameters (PREK, EXPKD).
C
C     VARIABLES IN ARGUMENT LIST
C
C     VARIABLE I/O  TYPE   DIMENSION   DESCRIPTION AND RANGE
C     -------- ---  ----   ---------   ---------------------------------
C     KSTG      I    I         -       SEGMENT NUMBER
C     NCOMPS    I    I         -       NUMBER OF COMPONENTS
C     IDX       I    I       NCOMPS    COMPONENT INDEX VECTOR
C     NBOPST    I    I         6       PHYSICAL PROPERTY OPTION
C                                      SET BEAD POINTER
C     KPDIAG    I    I         -       PHYSICAL PROPERTY
C                                      DIAGOSTIC CODE
C     XCOMPB    I    R       NCOMPS    BULK LIQUID MOLE FRACTION
C     FRATEL    I    R         -       FLOW OF LIQUID (KMOL/SEC)
C     YCOMPB    I    R       NCOMPS    BULK VAPOR MOLE FRACTION
C     FRATEV    I    R         -       FLOW OF VAPOR (KMOL/SEC)
C     PRESS     I    R         -       PRESSURE (N/SQ.M)
C     TLIQ      I    R         -       LIQUID TEMPERATURE (K)
C     TVAP      I    R         -       VAPOR TEMPERATURE (K)
C     AVMWLI    I    R         -       AVERAGE MOLECULAR WEIGHT
C                                      OF LIQUID MIXTURE
C                                      (KG/KMOL)
C     AVMWVA    I    R         -       AVERAGE MOLECULAR WEIGHT
C                                      OF VAPOR MIXTURE (KG/KMOL)
C     VISCML    I    R         -       VISCOSITY OF LIQUID
C                                      (N-SEC/SQ.M)
C     DENMXL    I    R         -       DENSITY OF LIQUID MIXTURE
C                                      (KMOL/CU.M)
C     SIGMAL    I    R         -       SURFACE TENSION OF LIQUID
C                                      (N/M)
C     VISCMV    I    R         -       VISCOSITY OF VAPOR MIXTURE
C                                      (N-SEC/SQ.M)
C     DENMXV    I    R         -       DENSITY OF VAPOR MIXTURE
C                                      (KMOL/CU.M)
C     AREAIF    I    R         -       INTERFACIAL AREA
C                                      (SEE NOTE-1 BELOW)
C     PREK      O    R         -       BINARY MASS TRANSFER =
C     EXPRKD    O    R         -          PREK*DIFFUSIVITY**EXPKD
C                                      (SEE NOTE-2 BELOW)
C     COLTYP    I    I         -       TYPE OF COLUMN
C                                      1 = PACKED
C                                      2 = TRAY
C     USRCOR    I    I         -       CALCULATION METHOD (I.E.
C                                      CHOICE OF USER CORRELATION)
C                                        1  = USER1
C                                        2  = USER2
C                                        3  = USER3
C                                        4  = USER4
C     TWRARA    I    R         -       CROSS-SECTIONAL AREA OF
C                                      TOWER (SQ.M)
```

```
C     COLDIA    I    R          -       COLUMN DIAMETER (M)
C     HTPACK    I    R          -       HEIGHT OF PACKING IN THE
C                                       SEGMENT (M)
C     PACSIZ    I    R          -       SIZE OF PACKING (M)
C     SPAREA    I    R          -       SPECIFIC SURFACE AREA OF
C                                       PACKING (SQ.M/CU.M)
C     CSIGMA    I    R          -       CRITICAL SURFACE TENSION
C                                       OF PACKING MATERIAL (N/M)
C     PFACT     I    R          -       PACKING FACTOR (1/M)
C     PKPRMS    I    R         20       PACKING PARAMETERS
C                                       PKPRMS(1) = STICHLMAIR CONSTANT C1
C                                       PKPRMS(2) = STICHLMAIR CONSTANT C2
C                                       PKPRMS(3) = STICHLMAIR CONSTANT C3
C                                       PKPRMS(4) = CL IN BILLET 93
C                                       PKPRMS(5) = CV IN BILLET 93
C                                       PKPRMS(6) = B IN BRF 85
C                                       PKPRMS(7) = S IN BRF 85
C                                       PKPRMS(8) = H IN BRF 85
C                                       PKPRMS(9) = Fse IN BRF 92
C                                       PKPRMS(10) = CE IN BRF 92
C                                       PKPRMS(11) = THETA IN BRF 92
C     VOIDFR    I    R          -       VOID FRACTION OF PACKING
C     IPAKAR    I    I          -       PACKING ARRANGEMENT
C                                         1 = RANDOM
C                                         2 = STRUCTURED
C     IPTYPE    I    I          -       PACKING TYPE
C                                       See IPTYPE in packsr.f
C     IVENDR    I    I          -       PACKING VENDOR CODE
C     IPMAT     I    I          -       PACKING MATERIAL CODE
C     IPSIZE    I    I          -       PACKING SIZE CODE
C     WEIRHT    I    R          -       AVERAGE WEIR HEIGHT (M)
C     DCAREA    I    R          -       TOTAL AREA OF DOWNCOMER
C                                       ON TRAY (SQ.M)
C     ARAACT    I    R          -       TOTAL ACTIVE AREA AVAILABLE
C                                       ON TRAY (SQ.M)
C     FLOPTH    I    R          -       AVERAGE FLOWPATH LENGTH (M)
C     NPASS     I    I          -       NUMBER OF TRAY PASSES
C     WEIRL     I    R          -       AVERAGE WEIRH LENGTH (M)
C     IFMETH    I    I          -       FLOODING CALCULATION
C                                       METHOD; REQUIRED FOR SIEVE
C                                       TRAY
C     SYSFAC    I    R          -       SYSTEM FACTOR; REQUIRED FOR
C                                       SIEVE TRAY
C     HOLEAR    I    R          -       HOLE AREA/ACTIVE AREA; REQUIRED
C                                       FOR SIEVE TRAY
C     ITTYPE    I    I          -       TRAY TYPE
C                                         1 - BUBBLE CAPS
C                                         2 - SIEVE
C                                         3 - GLITSCH BALLAST
C                                         4 - KOCH FLEXITRAY
C                                         5 - NUTTER FLOAT VALVE
C     TRASPC    I    R          -       TRAY SPACING (M)
C     PITCH     I    R          -       SIEVE TRAY HOLE PITCH (M)
C     IPHASE    I    I          -       PHASE QUALIFIER
C                                         0 = LIQUID
C                                         1 = VAPOR
C     NINT      I    I          -       Size of INT
C     INT      I/O   I       NINT       User correlation INT array
C     NREAL     I    I          -       Size of REAL
C     REAL     I/O   I       NREAL      User correlation REAL array
C
C    NOTE-1:
C         SPECIFIC INTERFACIAL AREA "AREAIF" HAS THE FOLLOWING UNITS.
C           FOR PACKED COLUMNS, THE UNITS IS "SQ.M/CU.M OF PACKING"
C           FOR TRAY COLUMNS, THE UNITS IS "SQ.M/SQ.M ACTIVE TRAY AREA"
C
C    NOTE-2:
C         BINMTP = PREK * DIFFUSIVITY**EXPKD
C         BINARY MASS TRANSFER COEFFCIENTS "BINMTP" HAVE UNITS (KMOL/SEC)
C         DIFFUSIVITY HAVE UNITS (SQ.M/SEC)
C         BINMTP HAS MOLAR DENSITY AND INTERFACIAL AREA INCLUDED
C
C*********************************************************************
C     Declare local variables used in the user correlations
C
```

```
      REAL*8 RS_BennettHL
      REAL*8 RS_BennettA
      REAL*8 RS_BennettC
      REAL*8 ScLB,   ScVB,   rhoLms, rhoVms, ReLPrm, P, GM,
     +       uL,     uV,     Fs,     QL, tL, THETA, dEQ, C,
     +       uSV,   alphae, hL,     ShLB,   ReV, uSL, S, B, H, CE,
     +       dTemp, Gz, Ka, Fth, uLE, uVE, DELTA, hld
C
C     Instead of computing BINMTP from diffusivity as in RATEFRAC
C     compute PREK and EXPKD for RateSep
C
      IF (COLTYP .EQ. 1) THEN
C
C**** PACKED COLUMN
C
         IF (USRCOR .EQ. 1) THEN
C           user subroutine example for packed column: Onda 68
C
C           Onda, K., Takeuchi, H. and Okumoto, Y., "Mass Transfer
C             Coefficients between Gas and Liquid Phases in Packed
C             Columns", J. Chem. Eng. Jap., 1, (1968) P56
C
            IF (IPHASE.EQ.0) THEN
C
C              Liquid phase
C
               rhoLms = DENMXL * AVMWLI
               uL = FRATEL / TWRARA / DENMXL
               ReLPrm = rhoLms * uL / VISCML / AREAIF
               dTemp = (rhoLms/9.81D0/VISCML)**(0.33333333D0)
               dTemp = 0.0051D0 * (ReLPrm**(0.66666667D0))
     +                 *((SPAREA*PACSIZ)**(0.4D0)) / dTemp
C
C              CONVERT K FROM M/S TO KMOL/S
               dTemp = dTemp * TWRARA * HTPACK * AREAIF * DENMXL
C
C              COMPOSITION INDEPENDENT PART OF SCHMIDT NUMBER
               ScLB = VISCML / rhoLms
C
               PREK  = dTemp / DSQRT(ScLB)
               EXPKD = 0.5D0
C
            ELSE
C
C              Vapor phase
C
                rhoVms = DENMXV * AVMWVA
                uV = FRATEV / TWRARA / DENMXV
                ReV = rhoVms * uV / VISCMV / SPAREA
                dTemp = SPAREA*PACSIZ
                dTemp = dTemp * dTemp
                   IF (PACSIZ .GE. 0.015D0) THEN
                       dTemp = 5.23D0 / dTemp
                   ELSE
                       dTemp = 2.0D0 / dTemp
                   END IF
                dTemp = dTemp * (ReV**(0.7D0)) * SPAREA
C
C               CONVERT K FROM M/S TO KMOL/S
                dTemp = dTemp * TWRARA * HTPACK * AREAIF * DENMXV
C
C               COMPOSITION INDEPENDENT PART OF SCHMIDT NUMBER
                ScVB = VISCMV / rhoVms
C
                PREK = dTemp * ScVB ** 0.33333333D0
                EXPKD = 0.66666667D0
             END IF
C            END OF IF (IPHASE)
C
         ELSE IF (USRCOR .EQ. 2) THEN
C        This code is for mass transfer coefficients using Wang Chao from April 11,2011.
C        The gas coefficient uses a modified version BRF-96
C        Modifying factor kL = REAL(1) kG = REAL(2)
C        HOLDUP = REAL(3)
C        THETA = REAL(4)
```

```
C        S = REAL(5)
C
         dTemp = REAL(1)
         CE = REAL(2)
         hld = REAL(3)
         THETA = REAL(4)
         S = REAL(5)


            IF (IPHASE.EQ.0) THEN
C
C           Liquid phase
C
C           Access to user variable array
                      uL = FRATEL / (TWRARA * DENMXL)
                uLE = uL/(VOIDFR*(1-hld)/sind(THETA))
                   DELTA = (3 * VISCML * uL/(DENMXL * hld * AVMWLI * 9.81D0))
                   DELTA = (DELTA**(0.5D0))/sind(THETA)
                   Gz = S * uLE * DELTA * sind(THETA)/HTPACK
                   Ka = AVMWLI * DENMXL * SIGMAL**(3.0D0)
                      Ka = Ka/(9.81D0 * VISCML**(4D0))
                PREK = 829.85D0 * (Gz**(0.5D0))/(Ka**(0.09D0))/S
C
C           Conversion of K from m/s to kmol/s
                PREK = dTemp * PREK * DENMXL *AREAIF * HTPACK * TWRARA
                EXPKD = 0.5D0
            ELSE
C
C           Vapor phase
C
                      uV = FRATEV / TWRARA / DENMXV
                      uVE = uV/(VOIDFR * (1-hld) * sind(THETA))
                PREK = 0.0317D0 * ((uVE/SPAREA)**(0.88D0)) * SPAREA
                      PREK = PREK * ((DENMXV * AVMWVA/VISCMV)**(0.55D0))
C
C           Conversion of K from m/s to kmol/s
                PREK = CE * PREK * DENMXV *AREAIF * HTPACK * TWRARA
                EXPKD = 0.67D0
            END IF
C           END OF IF (PHASE)
C
         END IF
C        END OF IF (USRCOR)
C
      ELSE IF (COLTYP .EQ. 2) THEN
C
C**** TRAY COLUMN
C
         IF (USRCOR .EQ. 1) THEN
C           user subroutine example for tray column: AIChE 58
C
C           AIChE, Bubble Tray Design Manual: Prediction of Fractionation
C             Efficiency, New York, 1958
C
C           For bubble cap, valve, and sieve trays
C
            IF (IPHASE.EQ.0) THEN
C
C              Liquid phase
C
               rhoVms = DENMXV * AVMWVA
               rhoLms = DENMXL * AVMWLI
               uV = FRATEV /DENMXV /ARAACT
               Fs = uV * DSQRT(rhoVms)
               C = 0.5D0 + 0.438D0 * DEXP(-137.8 * WEIRHT)
               QL = FRATEL/DENMXL
               ALPHAE = DEXP(-12.55D0*(uV*DSQRT(RHOVMS/DABS(RHOLMS-
     1                    RHOVMS)))**0.91D0)
               hL =ALPHAE*(WEIRHT + C*(QL/WEIRL/ALPHAE)**0.67D0)
               dTemp = 19700.0D0 *(0.4D0*Fs+0.17D0) * hL
     +                 * ARAACT * DENMXL
C
               PREK = dTemp
               EXPKD = 0.5D0
C
```

```
            ELSE
C
C              Vapor phase
C
               rhoVms = DENMXV * AVMWVA
               uV = FRATEV /DENMXV /ARAACT
               Fs = uV * DSQRT(rhoVms)
               QL = FRATEL/DENMXL
               dTemp = 0.776 + 4.57*WEIRHT - 0.238*Fs
     +                + 104.8*QL/WEIRL
               dTemp = dTemp * uV * ARAACT * DENMXV
C
C              COMPOSITION INDEPENDENT PART OF SCHMIDT NUMBER
               ScVB = VISCMV / rhoVms
C
               PREK = dTemp /DSQRT(ScVB)
               EXPKD = 0.5D0
             END IF
C            END OF IF (IPHASE)
C
          END IF
C         END OF IF (USRCOR)
C
        END IF
C     END OF IF (COLTYP)
C
      RETURN
      END
```

C.7. ABSORBER INTERFACIAL AREA (A_{INTF}) SUBROUTINE.

```
      SUBROUTINE awang(KSTG,   NCOMPS, IDX,    NBOPST, KPDIAG,
     1                  XCOMPB, FRATEL, YCOMPB, FRATEV, PRESS,
     2                  TLIQ,   TVAP,   AVMWLI, AVMWVA, VISCML,
     3                  DENMXL, SIGMAL, VISCMV, DENMXV, AREAIF,
     4                  COLTYP, USRCOR, TWRARA, COLDIA, HTPACK,
     5                  PACSIZ, SPAREA, CSIGMA, PFACT,  PKPRMS,
     6                  VOIDFR, IPAKAR, IPTYPE, IVENDR, IPMAT,
     7                  IPSIZE, WEIRHT, DCAREA, ARAACT, FLOPTH,
     8                  NPASS,  WEIRL,  IFMETH, SYSFAC, HOLEAR,
     9                  ITTYPE, TRASPC, PITCH,  NINT,   INT,
     A                  NREAL,  REAL)
      IMPLICIT NONE
      INTEGER KSTG, NCOMPS, IDX(NCOMPS), NBOPST(6), KPDIAG,
     +        COLTYP, USRCOR, IPAKAR, IPTYPE, IVENDR, IPMAT,  IPSIZE,
     +        NPASS, IFMETH, ITTYPE, NINT, INT(NINT), NREAL
      REAL*8  XCOMPB(NCOMPS), FRATEL, YCOMPB(NCOMPS), FRATEV,
     +        PRESS, TLIQ, TVAP, AVMWLI, AVMWVA, VISCML, DENMXL,
     +        SIGMAL, VISCMV, DENMXV, AREAIF, TWRARA, COLDIA,
     +        HTPACK, PACSIZ, SPAREA, CSIGMA, PFACT, PKPRMS(20),
     +        VOIDFR, WEIRHT, DCAREA, ARAACT, FLOPTH, WEIRL,
     +        SYSFAC, HOLEAR, TRASPC, PITCH, REAL(NREAL)
C***********************************************************************
C  LICENSED MATERIAL.  PROPERTY OF ASPEN TECHNOLOGY, INC.  TO BE       *
C  TREATED AS ASPEN TECH PROPRIETARY INFORMATION UNDER THE TERMS       *
C  OF THE ASPEN PLUS SUBSCRIPTION AGREEMENT.                           *
C***********************************************************************
C-----------------------------------------------------------------------
C         COPYRIGHT (C) 2004
C          ASPEN TECHNOLOGY, INC.
C          CAMBRIDGE, MA
C-----------------------------------------------------------------------
C
C     DESCRIPTION: User provided RateSep routine to calculate the
C                  specific interface area AREAIF (see NOTE-1).
C
C     VARIABLES IN ARGUMENT LIST
C
C     VARIABLE I/O  TYPE   DIMENSION   DESCRIPTION AND RANGE
C     -------- ---  ----   ---------   --------------------------------
C     KSTG      I    I         -       SEGMENT NUMBER
C     NCOMPS    I    I         -       NUMBER OF COMPONENTS
C     IDX       I    I       NCOMPS    COMPONENT INDEX VECTOR
C     NBOPST    I    I         6       PHYSICAL PROPERTY OPTION
```

```
C                                                SET BEAD POINTER
C     KPDIAG    I    I         -                 PHYSICAL PROPERTY
C                                                DIAGOSTIC CODE
C     XCOMPB    I    R       NCOMPS              BULK LIQUID MOLE FRACTION
C     FRATEL    I    R         -                 FLOW OF LIQUID (KMOL/SEC)
C     YCOMPB    I    R       NCOMPS              BULK VAPOR MOLE FRACTION
C     FRATEV    I    R         -                 FLOW OF VAPOR (KMOL/SEC)
C     PRESS     I    R         -                 PRESSURE (N/SQ.M)
C     TLIQ      I    R         -                 LIQUID TEMPERATURE (K)
C     TVAP      I    R         -                 VAPOR TEMPERATURE (K)
C     AVMWLI    I    R         -                 AVERAGE MOLECULAR WEIGHT
C                                                OF LIQUID MIXTURE
C                                                (KG/KMOL)
C     AVMWVA    I    R         -                 AVERAGE MOLECULAR WEIGHT
C                                                OF VAPOR MIXTURE (KG/KMOL)
C     VISCML    I    R         -                 VISCOSITY OF LIQUID
C                                                (N-SEC/SQ.M)
C     DENMXL    I    R         -                 DENSITY OF LIQUID MIXTURE
C                                                (KMOL/CU.M)
C     SIGMAL    I    R         -                 SURFACE TENSION OF LIQUID
C                                                (N/M)
C     VISCMV    I    R         -                 VISCOSITY OF VAPOR MIXTURE
C                                                (N-SEC/SQ.M)
C     DENMXV    I    R         -                 DENSITY OF VAPOR MIXTURE
C                                                (KMOL/CU.M)
C     AREAIF    O    R         -                 INTERFACIAL AREA
C                                                (SEE NOTE-1 BELOW)
C     COLTYP    I    I         -                 TYPE OF COLUMN
C                                                1 = PACKED
C                                                2 = TRAY
C     USRCOR    I    I         -                 CALCULATION METHOD (I.E.
C                                                CHOICE OF USER CORRELATION)
C                                                  1 = USER1
C                                                  2 = USER2
C                                                  3 = USER3
C                                                  4 = USER4
C     TWRARA    I    R         -                 CROSS-SECTIONAL AREA OF
C                                                TOWER (SQ.M)
C     COLDIA    I    R         -                 COLUMN DIAMETER (M)
C     HTPACK    I    R         -                 HEIGHT OF PACKING IN THE
C                                                SEGMENT (M)
C     PACSIZ    I    R         -                 SIZE OF PACKING (M)
C     SPAREA    I    R         -                 SPECIFIC SURFACE AREA OF
C                                                PACKING (SQ.M/CU.M)
C     CSIGMA    I    R         -                 CRITICAL SURFACE TENSION
C                                                OF PACKING MATERIAL (N/M)
C     PFACT     I    R         -                 PACKING FACTOR (1/M)
C     PKPRMS    I    R        20                 PACKING PARAMETERS
C                                                PKPRMS(1) = STICHLMAIR CONSTANT C1
C                                                PKPRMS(2) = STICHLMAIR CONSTANT C2
C                                                PKPRMS(3) = STICHLMAIR CONSTANT C3
C                                                PKPRMS(4) = CL IN BILLET 93
C                                                PKPRMS(5) = CV IN BILLET 93
C                                                PKPRMS(6) = B IN BRF 85
C                                                PKPRMS(7) = S IN BRF 85
C                                                PKPRMS(8) = H IN BRF 85
C                                                PKPRMS(9) = Fse IN BRF 92
C                                                PKPRMS(10) = CE IN BRF 92
C                                                PKPRMS(11) = THETA IN BRF 92
C     VOIDFR    I    R         -                 VOID FRACTION OF PACKING
C     IPAKAR    I    I         -                 PACKING ARRANGEMENT
C                                                  1 = RANDOM
C                                                  2 = STRUCTURED
C     IPTYPE    I    I         -                 PACKING TYPE
C                                                See IPTYPE in packsr.f
C     IVENDR    I    I         -                 PACKING VENDOR CODE
C     IPMAT     I    I         -                 PACKING MATERIAL CODE
C     IPSIZE    I    I         -                 PACKING SIZE CODE
C     WEIRHT    I    R         -                 AVERAGE WEIR HEIGHT (M)
C     DCAREA    I    R         -                 TOTAL AREA OF DOWNCOMER
C                                                ON TRAY (SQ.M)
C     ARAACT    I    R         -                 TOTAL ACTIVE AREA AVAILABLE
C                                                ON TRAY (SQ.M)
C     FLOPTH    I    R         -                 AVERAGE FLOWPATH LENGTH (M)
C     NPASS     I    I         -                 NUMBER OF TRAY PASSES
```

```
C     WEIRL     I    R          -       AVERAGE WEIRH LENGTH (M)
C     IFMETH    I    I          -       FLOODING CALCULATION
C                                       METHOD; REQUIRED FOR SIEVE
C                                       TRAY
C     SYSFAC    I    R          -       SYSTEM FACTOR; REQUIRED FOR
C                                       SIEVE TRAY
C     HOLEAR    I    R          -       HOLE AREA/ACTIVE AREA; REQUIRED
C                                       FOR SIEVE TRAY
C     ITTYPE    I    I          -       TRAY TYPE
C                                         1 - BUBBLE CAPS
C                                         2 - SIEVE
C                                         3 - GLITSCH BALLAST
C                                         4 - KOCH FLEXITRAY
C                                         5 - NUTTER FLOAT VALVE
C     TRASPC    I    R          -       TRAY SPACING (M)
C     PITCH     I    R          -       SIEVE TRAY HOLE PITCH (M)
C     NINT      I    I          -       Size of INT
C     INT      I/O   I       NINT      User correlation INT array
C     NREAL     I    I          -       Size of REAL
C     REAL     I/O   I       NREAL     User correlation REAL array
C
C     NOTE-1:
C           SPECIFIC INTERFACIAL AREA "AREAIF" HAS THE FOLLOWING UNITS.
C            FOR PACKED COLUMNS, THE UNITS IS "SQ.M/CU.M OF PACKING"
C            FOR TRAY COLUMNS, THE UNITS IS "SQ.M/SQ.M ACTIVE TRAY AREA"
C
C***********************************************************************
C     Declare local variables used in the user correlations
C
      REAL*8 WeL,   dTemp,  uV,    rhoVms,
     +       uL,    rhoLms, ReL,   FrL,    uL2,
     +       ReV,    d,     Wprime, LP, Ca
C
C     Compute specific interface area as described above
C     Check COLTYP/USRCOR if providing multiple area correlations
C
      IF (COLTYP .EQ. 1) THEN
C
C**** PACKED COLUMN
C
         IF (USRCOR .EQ. 1) THEN
C           user subroutine example for packed column: Onda 68
C
C           Onda, K., Takeuchi, H. and Okumoto, Y., "Mass Transfer
C             Coefficients between Gas and Liquid Phases in Packed
C             Columns", J. Chem. Eng. Jap., 1, (1968) p. 56
C
            rhoLms = DENMXL * AVMWLI
            uL = FRATEL / TWRARA / DENMXL
            uL2 = uL * uL
            ReL = rhoLms * uL / VISCML / SPAREA
            FrL = SPAREA * uL2 / 9.81D0
C           WHERE 9.81D0 IS GRAVITY CONSTANT IN M/S^2
            WeL    = rhoLms * uL2 / SIGMAL / SPAREA
            dTemp = -1.45D0*((CSIGMA/SIGMAL)**0.75D0)
     +                     *(ReL**0.1D0)*(FrL**(-0.05D0))
     +                     *(WeL**0.2D0)
            dTemp = 1.D0 - DEXP(dTemp)

            AREAIF = SPAREA*dTemp
C
           ELSE IF (USRCOR .EQ. 2) THEN
C**** *  Correlation for Packing based on Tsai but modified by Wang for hybrid and random packing. Q4-
2010
C          ae/ap = Ca*(rhoL/sigmaL)**0.116 ULs**0.115
C
            rhoLms = AVMWLI * DENMXL
            uL = FRATEL / TWRARA / DENMXL

C     The Ca constant in Wang is entered in the subroutine space in Aspen and manipulated here
C
            Ca = Real(1)
            dTemp = Ca*((rhoLms/SIGMAL)**0.116D0)*uL**0.155D0
               AREAIF = (SPAREA*dTemp)
           END IF
```

```
C        END OF IF (USRCOR)
C
      ELSE IF (COLTYP .EQ. 2) THEN
C
C**** TRAY COLUMN
C
         IF (USRCOR .EQ. 1) THEN
C           user subroutine example for tray column: Scheffe-Weiland 87
C
C           Scheffe, R.D. and Weiland, R.H., "Mass Transfer
C           Characteristics of Valve Trays." Ind. Eng. Chem. Res.
C           26, (1987) p. 228
C
C           The original paper only mentioned valve tray.
C           It is also used for bubble-cap tray and sieve tray.
C
C           CHARACTERISTIC LENGTH IS ALWAYS 1 METER.
            d = 1.0D0
            rhoLms = DENMXL * AVMWLI
            rhoVms = DENMXV * AVMWVA
            uL = FRATEL / TWRARA / DENMXL
            uV = FRATEV / TWRARA / DENMXV
            ReL = rhoLms * uL * d / VISCML
            ReV = rhoVms * uV * d / VISCMV
            Wprime = WEIRHT / d
            AREAIF = 0.27D0 * ReV**0.375D0 * ReL**0.247D0
            AREAIF = AREAIF * Wprime**0.515
         END IF
C        END OF IF (USRCOR)
C
        END IF
C     END OF IF (COLTYP)
C
      RETURN
      END
```

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Vita

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